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Lee et al.

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(54) **ORGANOMETALLIC COMPOUND,
LIGHT-EMITTING DEVICE INCLUDING
THE SAME, AND ELECTRONIC APPARATUS
INCLUDING THE LIGHT-EMITTING
DEVICE**

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H10K 50/11 (2023.01)
(Continued)

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CPC **C07F 15/0086** (2013.01); **H10K 85/346**
(2023.02); **H10K 50/11** (2023.02);
(Continued)

(58) **Field of Classification Search**
CPC H10K 85/361
See application file for complete search history.

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(57) **ABSTRACT**
Provided are an organometallic compound represented by
Formula 1, a light-emitting device including the organome-
tallic compound, and an electronic apparatus including the
light-emitting device:
(Continued)

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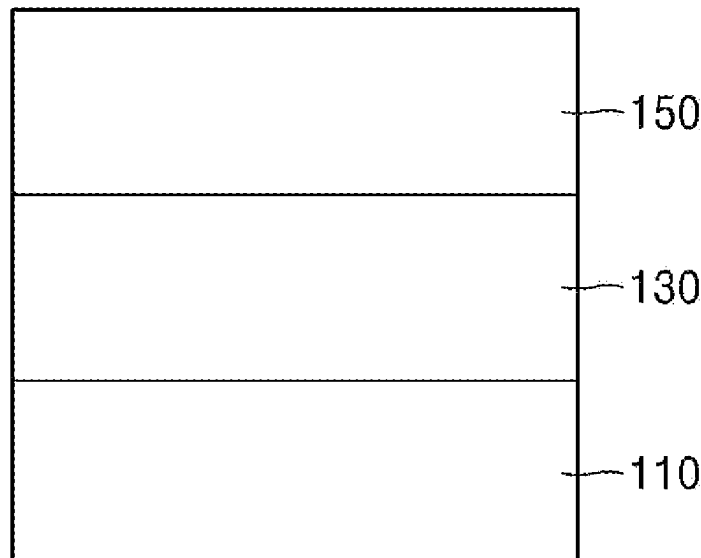


FIG. 1

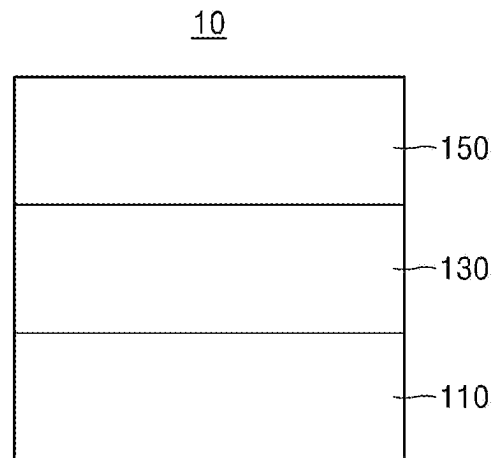


FIG. 2

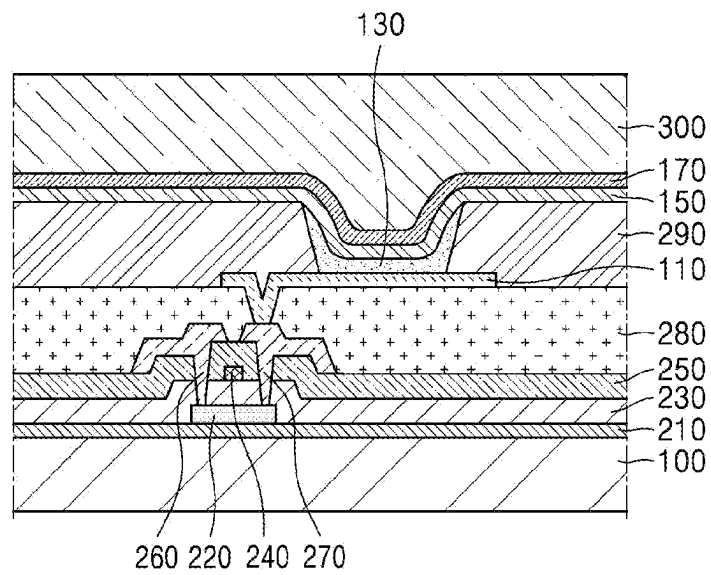
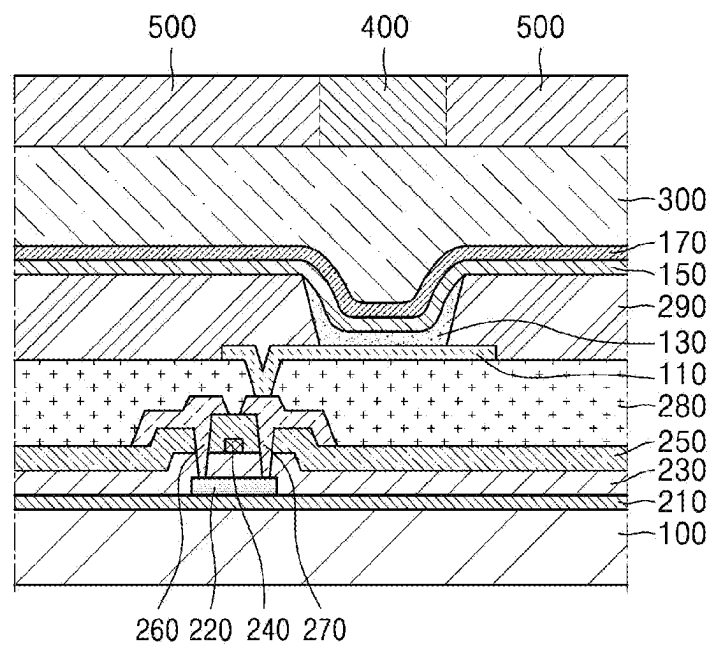


FIG. 3



1

**ORGANOMETALLIC COMPOUND,
LIGHT-EMITTING DEVICE INCLUDING
THE SAME, AND ELECTRONIC APPARATUS
INCLUDING THE LIGHT-EMITTING
DEVICE**

**CROSS-REFERENCE TO RELATED
APPLICATION(S)**

This application claims priority to and benefits of Korean Patent Application No. 10-2021-0037436 under 35 U.S.C. § 119, filed on Mar. 23, 2021 in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

Embodiments relate to an organometallic compound, a light-emitting device including the same, and an electronic apparatus including the light-emitting device.

2. Description of the Related Art

Organic light-emitting devices (OLEDs) are self-emissive devices that, as compared with devices of the related art, have wide viewing angles, high contrast ratios, short response times, and excellent characteristics in terms of luminance, driving voltage, and response speed, and produce full-color images.

Organic light-emitting devices may include a first electrode located on a substrate, and a hole transport region, an emission layer, an electron transport region, and a second electrode sequentially stacked on the first electrode. Holes provided from the first electrode may move toward the emission layer through the hole transport region, and electrons provided from the second electrode may move toward the emission layer through the electron transport region. Carriers, such as holes and electrons, recombine in the emission layer to produce excitons. These excitons transition from an excited state to a ground state, thereby generating light.

It is to be understood that this background of the technology section is, in part, intended to provide useful background for understanding the technology. However, this background of the technology section may also include ideas, concepts, or recognitions that were not part of what was known or appreciated by those skilled in the pertinent art prior to a corresponding effective filing date of the subject matter disclosed herein.

SUMMARY

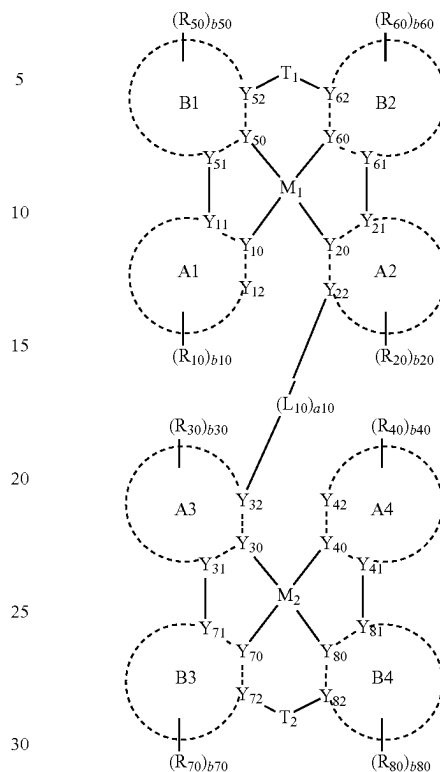
Provided are an organometallic compound, a light-emitting device having a low driving voltage, high luminance, high efficiency, and long lifespan characteristics due to the inclusion of the organometallic compound, and an electronic apparatus including the light-emitting device.

Additional aspects will be set forth in part in the description, which follows and, in part, will be apparent from the description, or may be learned by practice of the embodiments of the disclosure.

An organometallic compound according to an embodiment of the disclosure may be represented by Formula 1.

2

[Formula 1]



In Formula 1,

M_1 and M_2 may each independently be platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm),

A1 to A4 and B1 to B4 may each independently be a C_5 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group,

Y_{10} to Y_{12} , Y_{20} to Y_{22} , Y_{30} to Y_{32} , Y_{40} to Y_{42} , Y_{50} to Y_{52} , Y_{60} to Y_{62} , Y_{70} to Y_{72} , and Y_{80} to Y_{82} may each independently be C or N,

T_1 and T_2 may each independently be a single bond, $^*O^*$, $^*S^*$, $^*C(R_1)(R_2)^*$, $^*C(R_1)=^*$, $^*=C(R_1)^*$, $^*C(R_1)=C(R_2)^*$, $^*C(=O)^*$, $^*C(=S)^*$, $^*C\equiv C^*$, $^*B(R_1)^*$, $^*N(R_1)^*$, $^*P(R_1)^*$, $^*Si(R_1)(R_2)^*$, or $^*Ge(R_1)(R_2)^*$,

L_{10} may be a C_3 - C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} ,

a10 may be 1, 2, 3, 4, or 5,

R_1 , R_2 , R_{10} , R_{20} , R_{30} , R_{40} , R_{50} , R_{60} , R_{70} , and R_{80} may each independently be hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{20} alkoxy group unsubstituted or substituted with at least one R_{10a} , a C_3 - C_{10} cycloalkyl group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{10} heterocycloalkyl group unsubstituted or substituted with at least one R_{10a} , a C_3 - C_{10} cycloalkenyl group unsubstituted or

substituted with at least one R_{10a} , a C_1 - C_{10} heterocycloalkenyl group unsubstituted or substituted with at least one R_{10a} , a C_6 - C_{60} aryl group unsubstituted or substituted with at least one R_{10a} , a C_6 - C_{60} aryloxy group unsubstituted or substituted with at least one R_{10a} , a C_6 - C_{60} arylthio group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{60} heteroaryl group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{60} heteroaryloxy group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{60} heteroarylthio group unsubstituted or substituted with at least one R_{10a} , a monovalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R_{10a} , a monovalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R_{10a} , $-\text{Si}(\text{Q}_{41})(\text{Q}_{42})(\text{Q}_{43})$, $-\text{N}(\text{Q}_{41})(\text{Q}_{42})$, $-\text{B}(\text{Q}_{41})(\text{Q}_{42})$, $-\text{C}(=\text{O})(\text{Q}_{41})$, $-\text{S}(=\text{O})_2(\text{Q}_{41})$, or $-\text{P}(=\text{O})(\text{Q}_{41})(\text{Q}_{42})$,

two neighboring substituents among R_1 , R_2 , R_{10} , R_{20} , R_{30} , R_{40} , R_{50} , R_{60} , R_{70} , and R_{80} may optionally be linked to each other to form a C_3 - C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} ,

b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1, 2, 3, 4, 5, 6, 7, or 8,

* and *' each indicate a binding site to a neighboring atom, and

R_{10a} may be:

deuterium (-D), -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, or a nitro group;

a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkenyl group, or a C_1 - C_{60} alkoxy group, each unsubstituted or substituted with deuterium, -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, a nitro group, a C_3 - C_{60} carbocyclic group, a C_1 - C_{60} heterocyclic group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryloxy group, a C_1 - C_{60} heteroarylthio group, $-\text{Si}(\text{Q}_{11})(\text{Q}_{12})(\text{Q}_{13})$, $-\text{N}(\text{Q}_{11})(\text{Q}_{12})$, $-\text{B}(\text{Q}_{11})(\text{Q}_{12})$, $-\text{C}(=\text{O})(\text{Q}_{11})$, $-\text{S}(=\text{O})_2(\text{Q}_{11})$, $-\text{P}(=\text{O})(\text{Q}_{11})(\text{Q}_{12})$, or any combination thereof;

a C_3 - C_{60} carbocyclic group, a C_1 - C_{60} heterocyclic group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryloxy group, or a C_1 - C_{60} heteroarylthio group, each unsubstituted or substituted with deuterium, -F, -Cl, -Br, -I, a hydroxyl group, a cyano group, a nitro group, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, a C_3 - C_{60} carbocyclic group, a C_1 - C_{60} heterocyclic group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryloxy group, a C_1 - C_{60} heteroarylthio group, $-\text{Si}(\text{Q}_{21})(\text{Q}_{22})(\text{Q}_{23})$, $-\text{N}(\text{Q}_{21})(\text{Q}_{22})$, $-\text{B}(\text{Q}_{21})(\text{Q}_{22})$, $-\text{C}(=\text{O})(\text{Q}_{21})$, $-\text{S}(=\text{O})_2(\text{Q}_{21})$, $-\text{P}(=\text{O})(\text{Q}_{21})(\text{Q}_{22})$, or any combination thereof; or

$-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, or $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$,

wherein Q_{11} to Q_{13} , Q_{21} to Q_{23} , Q_{31} to Q_{33} , and Q_{41} to Q_{43} may each independently be hydrogen; deuterium; -F; -Cl; -Br; -I; a hydroxyl group; a cyano group; a nitro group; a C_1 - C_{60} alkyl group; a C_2 - C_{60} alkenyl group; a C_2 - C_{60} alkynyl group; a C_1 - C_{60} alkoxy group; or a C_3 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group, each unsubstituted or substituted with deuterium, -F, a cyano group, a C_1 - C_{60} alkyl group, a

C_1 - C_{60} alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

In an embodiment, M_1 and M_2 may each independently be Pt or Pd.

In an embodiment, A1 to A4 may each independently be a 5-membered heterocyclic group or may each independently be a polycyclic C_1 - C_{30} heterocyclic group comprising a 5-membered heterocyclic group.

In an embodiment, A1 and A4 may each independently be a group represented by one of Formulae 3-1 to 3-3, which are explained below, and A2 and A3 may each independently be a group represented by one of Formulae 3-4 and 3-5, which are explained below.

In an embodiment, a bond between M_1 and Y_{10} may be a coordinate bond, a bond between M_1 and Y_{20} may be a coordinate bond, a bond between M_2 and Y_{30} may be a coordinate bond, a bond between M_2 and Y_{40} may be a coordinate bond, a bond between M_1 and Y_{50} may be a covalent bond, a bond between M_1 and Y_{60} may be a covalent bond, a bond between M_2 and Y_{70} may be a covalent bond, and a bond between M_2 and Y_{80} may be a covalent bond.

In an embodiment, T_1 and T_2 may each independently be $^*\text{—O—}^*$ or $^*\text{—S—}^*$.

In an embodiment, L_{10} may be a C_1 - C_{20} alkylene group unsubstituted or substituted with at least one R_{10a} , a C_2 - C_{20} alkenylene group unsubstituted or substituted with at least one R_{10a} , a C_2 - C_{20} alkynylene group unsubstituted or substituted with at least one R_{10a} , a C_3 - C_{10} cycloalkylene group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{10} heterocycloalkylene group unsubstituted or substituted with at least one R_{10a} , a C_3 - C_{10} cycloalkenylene group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{10} heterocycloalkenylene group unsubstituted or substituted with at least one R_{10a} , a C_6 - C_{60} arylene group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{60} heteroarylene group unsubstituted or substituted with at least one R_{10a} , a divalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R_{10a} , or a divalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R_{10a} .

In an embodiment, L_{10} may be a group represented by one of Formulae 4-1 to 4-3, which are explained below.

In an embodiment, R_1 , R_2 , R_{10} , R_{20} , R_{30} , R_{40} , R_{50} , R_{60} , R_{70} , and R_{80} may each independently be: hydrogen, deuterium, -F, -Cl, -Br, -I, a cyano group, a C_1 - C_{20} alkyl group, or a C_1 - C_{20} alkoxy group; or a C_1 - C_{20} alkyl group or a C_1 - C_{20} alkoxy group, each substituted with deuterium, -F, -Cl, -Br, -I, a cyano group, a phenyl group, a biphenyl group, or any combination thereof; or a group represented by one of Formulae 5-1 to 5-26 and 6-1 to 6-55, which are explained below. Neighboring two or more of R_1 , R_2 , R_{10} , R_{20} , R_{30} , R_{40} , R_{50} , R_{60} , R_{70} , and R_{80} may be optionally linked to each other to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthylene group, a fluorene group, a pyridine group, a pyrimidine group, a triazine group, or a carbazole group.

In an embodiment, the organometallic compound may be represented by one of Formulae 11-1 to 11-3, which are explained below.

In an embodiment, the organometallic compound may be electrically neutral.

In an embodiment, the organometallic compound may be one selected from Compounds BD1 to BD104, which are explained below.

Another aspect of the disclosure provides a light-emitting device which may include a first electrode, a second elec-

trode facing the first electrode, an interlayer disposed between the first electrode and the second electrode and including an emission layer, and the organometallic compound.

In an embodiment, the first electrode may be an anode, the second electrode may be a cathode, and the interlayer may further include a hole transport region disposed between the first electrode and the emission layer and an electron transport region disposed between the emission layer and the second electrode. The hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

In an embodiment, the emission layer may include the organometallic compound.

In an embodiment, the emission layer may include a host and a dopant, and the dopant may include the organometallic compound.

In an embodiment, the hole transport region may further include a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level equal to or less than about -3.5 eV.

In an embodiment, the electron transport region may include the electron transport layer and the electron injection layer, and at least one layer of the electron transport layer and the electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

Another aspect of the disclosure provides an electronic apparatus which may include the light-emitting device.

In an embodiment, the electronic apparatus may further include a thin-film transistor. The thin-film transistor may include a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically connected to the source electrode or the drain electrode.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects and features of the disclosure will become more apparent by describing in detail embodiments thereof with reference to the accompanying drawings, in which:

FIG. 1 is a schematic cross-sectional view of a light-emitting device according to an embodiment;

FIG. 2 is a schematic cross-sectional view of an electronic apparatus according to an embodiment; and

FIG. 3 is a schematic cross-sectional view of an electronic apparatus according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

The disclosure will now be described more fully herein-after with reference to the accompanying drawings, in which embodiments are shown. This disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art.

In the drawings, the sizes, thicknesses, ratios, and dimensions of the elements may be exaggerated for ease of description and for clarity. Like numbers refer to like elements throughout.

In the description, it will be understood that when an element (or region, layer, part, etc.) is referred to as being “on”, “connected to”, or “coupled to” another element, it can be directly on, connected to, or coupled to the other element, or one or more intervening elements may be present therebetween. In a similar sense, when an element (or region, layer, part, etc.) is described as “covering” another element, it can directly cover the other element, or one or more intervening elements may be present therebetween.

In the description, when an element is “directly on”, “directly connected to”, or “directly coupled to” another element, there are no intervening elements present. For example, “directly on” may mean that two layers or two elements are disposed without an additional element such as an adhesion element therebetween.

As used herein, the expressions used in the singular such as “a,” “an,” and “the,” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. For example, “A and/or B” may be understood to mean “A, B, or A and B.” The terms “and” and “or” may be used in the conjunctive or disjunctive sense and may be understood to be equivalent to “and/or”.

In the specification and the claims, the term “at least one of” is intended to include the meaning of “at least one selected from the group of” for the purpose of its meaning and interpretation. For example, “at least one of A and B” may be understood to mean “A, B, or A and B.” When preceding a list of elements, the term, “at least one of,” modifies the entire list of elements and does not modify the individual elements of the list.

When an embodiment is implemented differently, a specific process order may be performed differently from the described order. For example, two processes described in succession may be performed substantially simultaneously, or may be performed in an order opposite to the described order.

It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. Thus, a first element could be termed a second element without departing from the teachings of the disclosure. Similarly, a second element could be termed a first element, without departing from the scope of the disclosure.

The term “interlayer” as used herein refers to a single layer and/or all layers between a first electrode and a second electrode of a light-emitting device.

The spatially relative terms “below”, “beneath”, “lower”, “above”, “upper”, or the like, may be used herein for ease of description to describe the relations between one element or component and another element or component as illustrated in the drawings. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation, in addition to the orientation depicted in the drawings. For example, in the case where a device illustrated in the drawing is turned over, the device positioned “below” or “beneath” another device may be placed “above” another device. Accordingly, the illustrative term “below” may include both the lower and upper positions. The device may also be oriented in other

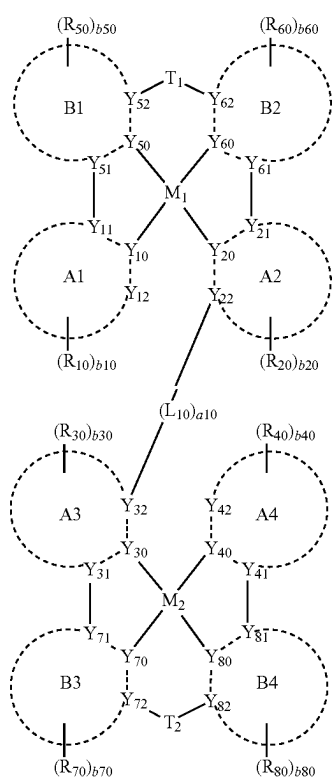
directions and thus the spatially relative terms may be interpreted differently depending on the orientations.

The terms “about” or “approximately” as used herein is inclusive of the stated value and means within an acceptable range of deviation for the recited value as determined by one of ordinary skill in the art, considering the measurement in question and the error associated with measurement of the recited quantity (i.e., the limitations of the measurement system). For example, “about” may mean within one or more standard deviations, or within +20%, 10%, or $\pm 5\%$ of the stated value.

It should be understood that the terms “comprises,” “comprising,” “includes,” “including,” “have,” “having,” “contains,” “containing,” and the like are intended to specify the presence of stated features, integers, steps, operations, elements, components, or combinations thereof in the disclosure, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, or combinations thereof.

Unless otherwise defined or implied herein, all terms (including technical and scientific terms) used have the same meaning as commonly understood by those skilled in the art to which this disclosure pertains. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an ideal or excessively formal sense unless clearly defined in the specification.

According to an aspect of the disclosure, an organometallic compound may be represented by Formula 1:



[Formula 1]

In Formula 1, M_1 and M_2 may each independently be platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium

(Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm).

In an embodiment, M_1 and M_2 may each independently be Pt or Pd.

In Formula 1, A1 to A4 and B1 to B4 may each independently be a C_5 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group.

In an embodiment, A1 to A4 may each independently be a 5-membered heterocyclic group or may each independently be a polycyclic C_1 - C_{30} heterocyclic group including a 5-membered heterocyclic group.

In an embodiment, A1 to A4 and B1 to B4 may each independently be a benzene group, a naphthalene group, an anthracene group, a phenanthrene group, an azulene group, a triphenylene group, a pyrene group, a chrysene group, a cyclopentadiene group, a 1,2,3,4-tetrahydronaphthalene group, a furan group, a thiophene group, a silole group, an indene group, a fluorene group, an indole group, a carbazole group, a benzofuran group, a dibenzofuran group, a benzothiophene group, a dibenzothiophene group, a benzosilole group, a dibenzosilole group, an indenopyridine group, an indolopyridine group, a benzofuropyridine group, a benzothienopyridine group, a benzosilolopyridine group, an indenopyrimidine group, an indolopyrimidine group, a benzofuropyrimidine group, a benzothienopyrimidine group, a benzosilolopyrimidine group, a dihydropyridine group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a quinoxaline group, a quinazoline group, a phenanthroline group, a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a pyrazolopyridine group, a furopyrazole group, a thienopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an imidazopyridine group, a 2,3-dihydroimidazopyridine group, a furoimidazole group, a thienoimidazole group, an imidazopyrimidine group, a 2,3-dihydroimidazopyrimidine group, an imidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, a benzothiadiazole group, a 5,6,7,8-tetrahydroisoquinoline group, or a 5,6,7,8-tetrahydroquinoline group.

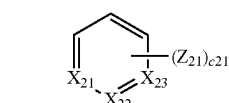
In an embodiment, A1 to A4 may each independently be a pyrrole group, a pyrazole group, an imidazole group, a 2,3-dihydroimidazole group, a triazole group, a 2,3-dihydrotriazole group, an oxazole group, an isoxazole group, a thiazole group, an isothiazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a pyrazolopyridine group, a furopyrazole group, a thienopyrazole group, a benzimidazole group, a 2,3-dihydrobenzimidazole group, an imidazopyridine group, a 2,3-dihydroimidazopyridine group, a furoimidazole group, a thienoimidazole group, an imidazopyrimidine group, a 2,3-dihydroimidazopyrimidine group, an imidazopyrazine group, a benzoxazole group, a benzothiazole group, a benzoxadiazole group, or a benzothiadiazole group.

In an embodiment, B1 to B4 may each independently be a benzene group, a naphthalene group, a phenanthrene group, a fluoranthene group, a triphenylene group, a pyrene group, a chrysene group, an imidazole group, a 2,3-dihydroimidazole group, an imidazopyrimidine group, a benzimidazole group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, a dibenzofluorene group, an indole group, a pyridine group, a pyrimidine group, a carbazole group, a benzocarbazole group, a

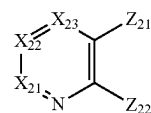
9

dibenzocarbazole group, an azacarbazole group, a furan group, a benzofuran group, a dibenzofuran group, an azadibenzofuran group, a naphthofuran group, a benzonaphthofuran group, a dinaphthofuran group, a thiophene group, a benzothiophene group, a dibenzothiophene group, an azadibenzothiophene group, a naphthothiophene group, a benzonaphthothiophene group, or a dinaphthothiophene group.

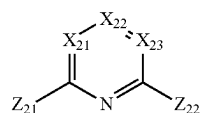
In an embodiment, A1 to A4 and B1 to B4 may each independently be a group represented by one of Formulae 2-1 to 2-43:



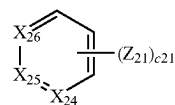
2-1 15



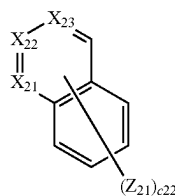
2-2 20



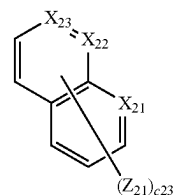
2-3 25



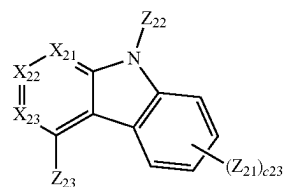
2-4 30



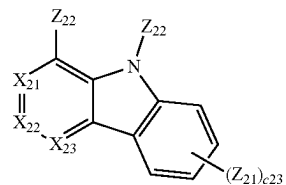
2-5 35



2-6 40



2-7 45

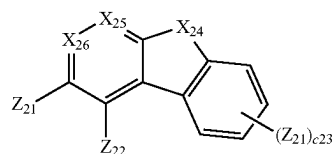


2-8 50

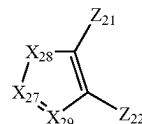
10

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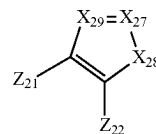
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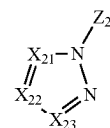
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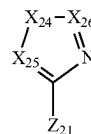
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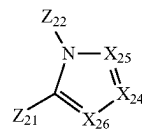
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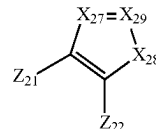
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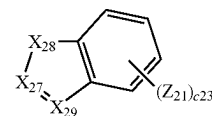
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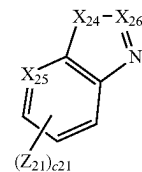
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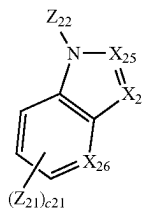
2-16



2-17



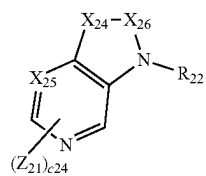
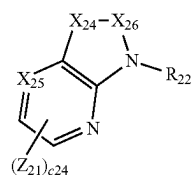
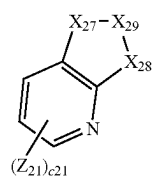
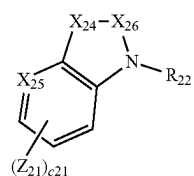
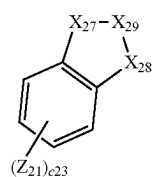
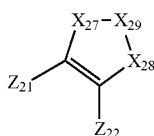
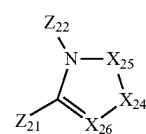
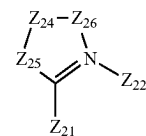
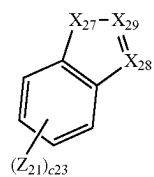
2-18



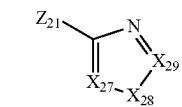
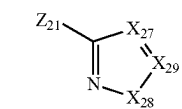
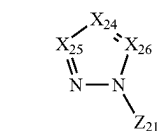
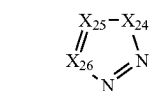
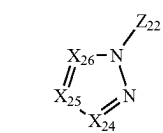
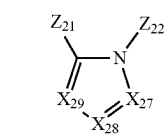
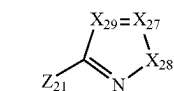
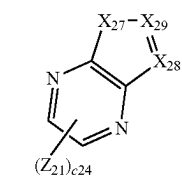
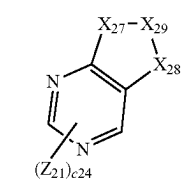
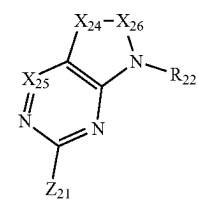
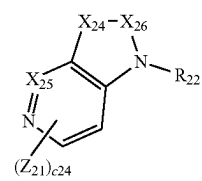
65

11

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**12**

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2-19

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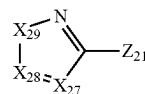
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2-37

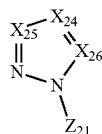
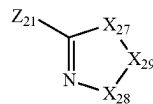
2-38

13

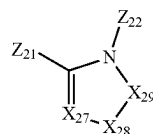
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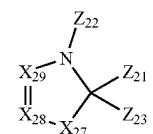
2-39

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2-40

2-41



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2-43

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In Formulae 2-1 to 2-43,

X₂₁ to X₂₃ may each independently be C(Z₂₄) or C—*,
wherein at least two of X₂₁ to X₂₃ may be C—*,X₂₄ may be N—*, and X₂₅ and X₂₆ may each independently be C(Z₂₄) or C—*, wherein at least one of X₂₅
and X₂₆ may be C—*,X₂₇ and X₂₈ may each independently be N, N(Z₂₅), or
N—*, and X₂₉ may be C(Z₂₄) or C—*. At least one of
X₂₇ and X₂₈ may be N—*, and X₂₉ may be C—*; or
X₂₇ and X₂₈ may each be N—*, and X₂₉ may be
C(Z₂₄),Z₂₁ to Z₂₅ may each independently be the same as
described in connection with R₁₀,

c21 may be 1, 2, or 3,

c22 may be 1, 2, 3, 4, or 5,

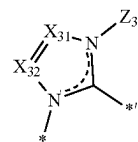
c23 may be 1, 2, 3, or 4,

c24 may be 1 or 2, and

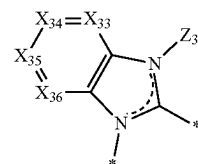
* indicates a binding site to a neighboring atom.

In an embodiment, Z₂₁ to Z₂₅ in Formulae 2-1 to 2-43 may
each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C₁-C₂₀ alkyl group, or a C₁-C₂₀
alkoxy group;a C₁-C₂₀ alkyl group or a C₁-C₂₀ alkoxy group, each
substituted with deuterium, —F, —Cl, —Br, —I, a
cyano group, a phenyl group, a biphenyl group, or any
combination thereof;a phenyl group, a biphenyl group, a terphenyl group, a
pentalenyl group, an indenyl group, a naphthyl group,
an azulenyl group, an indacenyl group, an acenaphthyl
group, a fluorenyl group, a spiro-bifluorenyl group, a
benzofluorenyl group, a dibenzofluorenyl group, a car-
bazolyl group, an acridinyl group, a dibenzofuranyl
group, a dibenzothiophenyl group, a benzocarbazolyl
group, or a dibenzocarbazolyl group;a phenyl group, a biphenyl group, a terphenyl group, a
pentalenyl group, an indenyl group, a naphthyl group,
an azulenyl group, an indacenyl group, an acenaphthyl

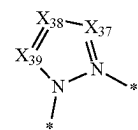
14

group, a fluorenyl group, a spiro-bifluorenyl group, a
benzofluorenyl group, a dibenzofluorenyl group, a car-
bazolyl group, an acridinyl group, a dibenzofuranyl
group, a dibenzothiophenyl group, a benzocarbazolyl
group, or a dibenzocarbazolyl group, each substituted
with deuterium, —F, —Cl, —Br, —I, a cyano group, a
C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl
group, a biphenyl group, or any combination thereof; or
—Si(Q₄₁)(Q₄₂)(Q₄₃) or —N(Q₄₁)(Q₄₂). Q₄₁, Q₄₂, and
Q₄₃ may each independently be the same as described
in the Specification.In an embodiment, A1 to A4 may each independently be
a group represented by one of Formulae 2-10 to 2-43.In an embodiment, A1 and A4 may each independently be
a group represented by one of Formulae 3-1 to 3-3, and A2
and A3 may each independently be a group represented by
one of Formulae 3-4 and 3-5:

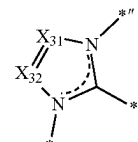
3-1



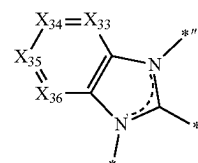
3-2



3-3



3-4



3-5

In Formulae 3-1 to 3-5,

X₃₁ to X₃₉ may each independently be C(Z₃₂) or N,
Z₃₁ and Z₃₂ may each independently be the same as
described in connection with R₁₀, and
*, **, and *** each indicate a binding site to a neighboring
atom.In an embodiment, Z₃₁ and Z₃₂ in Formulae 3-1 to 3-5
may each independently be: hydrogen, deuterium, —F,
—Cl, —Br, —I, a cyano group, a C₁-C₂₀ alkyl group, or a
C₁-C₂₀ alkoxy group,a C₁-C₂₀ alkyl group or a C₁-C₂₀ alkoxy group, each
substituted with deuterium, —F, —Cl, —Br, —I, a
cyano group, a phenyl group, a biphenyl group, or any
combination thereof;

15

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenylyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazoyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, or a dibenzocarbazoyl group;

a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenylyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a carbazoyl group, an acridinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazoyl group, or a dibenzocarbazoyl group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a phenyl group, a biphenyl group, or any combination thereof; or —Si(Q₄₁)(Q₄₂)(Q₄₃) or —N(Q₄₁)(Q₄₂), wherein Q₄₁, Q₄₂, and Q₄₃ may each independently be the same as described in the Specification, and

*, **, and *** each indicate a binding site to a neighboring atom.

In Formula 1, Y₁₀ to Y₁₂, Y₂₀ to Y₂₂, Y₃₀ to Y₃₂, Y₄₀ to Y₄₂, Y₅₀ to Y₅₂, Y₆₀ to Y₆₂, Y₇₀ to Y₇₂, and Y₈₀ to Y₈₂ may each independently be C or N.

In an embodiment, Y₅₀, Y₆₀, Y₇₀, and Y₈₀ may each be C.

In an embodiment, Y₁₀ and Y₂₀ may each be C.

In an embodiment, a bond between M₁ and Y₁₀ may be a coordinate bond, a bond between M₁ and Y₂₀ may be a coordinate bond, a bond between M₂ and Y₃₀ may be a coordinate bond, a bond between M₂ and Y₄₀ may be a coordinate bond, a bond between M₁ and Y₅₀ may be a covalent bond, a bond between M₁ and Y₆₀ may be a covalent bond, a bond between M₂ and Y₇₀ may be a covalent bond, and a bond between M₂ and Y₈₀ may be a covalent bond.

In Formula 1, T₁ and T₂ may each independently be a single bond, *—O—*, *—S—*, *—C(R₁)(R₂)—*, *—C(R₁)=*, *—C(R₁)—*, *—C(R₁)=C(R₂)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B(R₁)—*, *—N(R₁)—*, *—P(R₁)—*, *—Si(R₁)(R₂)—*, or *—Ge(R₁)(R₂)—*.

In an embodiment, T₁ and T₂ may each independently be *—O—* or *—S—*.

In Formula 1, L₁₀ may be a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a}.

In an embodiment, L₁₀ may be a C₁-C₂₀ alkylene group unsubstituted or substituted with at least one R_{10a}; a C₂-C₂₀ alkenylene group unsubstituted or substituted with at least one R_{10a}; a C₂-C₂₀ alkynylene group unsubstituted or substituted with at least one R_{10a}; a C₃-C₁₀ cycloalkylene group unsubstituted or substituted with at least one R_{10a}; a C₁-C₁₀ heterocycloalkylene group unsubstituted or substituted with at least one R_{10a}; a C₃-C₁₀ cycloalkenylene group unsubstituted or substituted with at least one R_{10a}; a C₁-C₁₀ heterocycloalkenylene group unsubstituted or substituted with at least one R_{10a}; a C₆-C₆₀ arylene group unsubstituted or substituted with at least one R_{10a}; a C₁-C₆₀ heteroarylene group unsubstituted or substituted with at least one R_{10a}; a divalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R_{10a}; or a divalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R_{10a}.

16

In an embodiment, L₁₀ may be: a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylenylene group, a fluorenylene group, a spiro-bifluorenylene group, a fluorenylene group, a spiro-bifluorenylene group, a spiro-fluorene-benzofluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a pyrrolylene group, a thiophenylene group, a furanylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a pyridinylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, an isoindolylenylene group, an indolylenylene group, an indazolylenylene group, a purinylenylene group, a quinoxalinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylene group, a cinnolinylenylene group, a carbazolylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylene group, a benzimidazolylenylene group, a benzofuranylenylene group, a benzothiophenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an oxadiazolylenylene group, a triazinylene group, a dibenzofuranylenylene group, a dibenzothiophenylene group, a dibenzosilolylenylene group, a benzocarbazolylenylene group, a dibenzocarbazolylenylene group, a thiadiazolylenylene group, an imidazopyridinylenylene group, or an imidazopyrimidinylene group; or

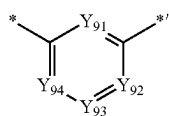
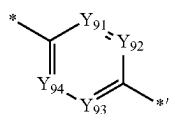
a phenylene group, a pentalenylene group, an indenylene group, a naphthylene group, an azulenylene group, a heptalenylene group, an indacenylene group, an acenaphthylenylene group, a fluorenylene group, a spiro-bifluorenylene group, a spiro-fluorene-benzofluorenylene group, a benzofluorenylene group, a dibenzofluorenylene group, a phenalenylene group, a phenanthrenylene group, an anthracenylene group, a fluoranthrenylene group, a triphenylenylene group, a pyrenylene group, a chrysenylene group, a naphthacenylene group, a picenylene group, a perylenylene group, a pentaphenylenylene group, a hexacenylene group, a pentacenylene group, a rubicenylene group, a coronenylene group, an ovalenylene group, a pyrrolylene group, a thiophenylene group, a furanylene group, an imidazolylenylene group, a pyrazolylenylene group, a thiazolylenylene group, an isothiazolylenylene group, an oxazolylenylene group, an isoxazolylenylene group, a pyridinylenylene group, a pyrazinylenylene group, a pyrimidinylenylene group, a pyridazinylenylene group, an isoindolylenylene group, an indolylenylene group, an indazolylenylene group, a purinylenylene group, a quinoxalinylenylene group, an isoquinolinylenylene group, a benzoquinolinylenylene group, a phthalazinylene group, a naphthyridinylenylene group, a quinoxalinylenylene group, a quinazolinylene group, a cinnolinylenylene group, a carbazolylenylene group, a phenanthridinylenylene group, an acridinylenylene group, a phenanthrolinylenylene group, a phenazinylene group, a benzimidazolylenylene group, a benzofuranylenylene group, a benzothiophenylene group, an isobenzothiazolylenylene group, a benzoxazolylenylene group, an isobenzoxazolylenylene group, a triazolylenylene group, a tetrazolylenylene group, an oxadiazolylenylene group, a triazinylene group, a

17

dibenzofuranylene group, a dibenzothiophenylene group, a dibenzosilolylene group, a benzocarbazolyene group, a dibenzocarbazolyene group, a thiadiazolyene group, an imidazopyridinylene group, or an imidazopyrimidinylene group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclopentenyl group, a cyclohexenyl group, a phenyl group, a biphenyl group, a terphenyl group, a pentalenyl group, an indenyl group, a naphthyl group, an azulenyl group, a heptalenyl group, an indacenyl group, an acenaphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a naphthacenyl group, a picenyl group, a perylenyl group, a pentaphenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, an ovalenyl group, a pyrrolyl group, a thiophenyl group, a furanyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a phenazinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a dibenzosilolyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, a thiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, —Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), —P(=O)(Q₃₁)(Q₃₂), or any combination thereof.

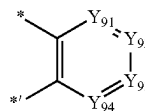
In an embodiment, Q₃₁ to Q₃₃ may each independently be a C₁-C₁₀ alkyl group, a C₁-C₁₀ alkoxy group, a phenyl group, a phenyl group substituted with a C₁-C₁₀ alkyl group, a biphenyl group, a terphenyl group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, a triazinyl group, a quinolinyl group, or an isoquinolinyl group.

In an embodiment, L₁₀ may be a group represented by one of Formulae 4-1 to 4-3:



18

-continued



4-3

In Formulae 4-1 to 4-3,

Y₉₁ may be C(R₉₁) or N, Y₉₂ may be C(R₉₂) or N, Y₉₃ may be C(R₉₃) or N, and Y₉₄ may be C(R₉₄) or N,

R₉₁ to R₉₄ may each independently be the same as described in connection with R_{10a}, and

* and *' each indicate a binding site to a neighboring atom. R_{10a} may be the same as described in the Specification.

In Formula 1, a10 may be 1, 2, 3, 4, or 5.

In an embodiment, a10 may be 1 or 2.

In an embodiment, a10 may be 1.

In Formula 1, R₁, R₂, R₁₀, R₂₀, R₃₀, R₄₀, R₅₀, R₆₀, R₇₀, and R₈₀ may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₂₀ alkoxy group unsubstituted or substituted with at least one R_{10a}, a C₃-C₁₀ cycloalkyl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₁₀ heterocycloalkyl group unsubstituted or substituted with at least one R_{10a}, a C₃-C₁₀ cycloalkenyl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₁₀ heterocycloalkenyl group unsubstituted or substituted with at least one R_{10a}, a C₆-C₆₀ aryl group unsubstituted or substituted with at least one R_{10a}, a C₆-C₆₀ aryloxy group unsubstituted or substituted with at least one R_{10a}, a C₆-C₆₀ arylthio group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heteroaryl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heteroaryloxy group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heteroarylthio group unsubstituted or substituted with at least one R_{10a}, a monovalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R_{10a}, a monovalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R_{10a}, —Si(Q₄₁)(Q₄₂)(Q₄₃), —N(Q₄₁)(Q₄₂), —B(Q₄₁)(Q₄₂), —C(=O)(Q₄₁), —S(=O)₂(Q₄₁), or —P(=O)(Q₄₁)(Q₄₂), and

two neighboring substituents among R₁, R₂, R₁₀, R₂₀, R₃₀, R₄₀, R₅₀, R₆₀, R₇₀, and R₈₀ may optionally be linked to each other to form a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a}.

In an embodiment, R₁, R₂, R₁₀, R₂₀, R₃₀, R₄₀, R₅₀, R₆₀, R₇₀, and R₈₀ may each independently be: hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, or a cyano group;

a C₁-C₂₀ alkyl group, a C₂-C₂₀ alkenyl group, a C₂-C₂₀ alkynyl group, or a C₁-C₂₀ alkoxy group;

a C₁-C₂₀ alkyl group, a C₂-C₂₀ alkenyl group, a C₂-C₂₀ alkynyl group, or a C₁-C₂₀ alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl

19

group, a naphthyl group, a pyridinyl group, a pyrimidinyl group, $-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$, or any combination thereof;

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a pentalenyl group, an indenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a furanyl group, a thiophenyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group; or

a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a pentalenyl group, an indenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a furanyl group, a thiophenyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, or an imidazopyrimidinyl group, each substituted with deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a $\text{C}_1\text{-C}_{20}$ alkyl

20

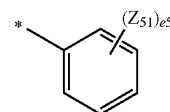
group, a $\text{C}_2\text{-C}_{20}$ alkenyl group, a $\text{C}_2\text{-C}_{20}$ alkynyl group, a $\text{C}_1\text{-C}_{20}$ alkoxy group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group, a norbornenyl group, a cyclopentenyl group, a cyclohexenyl group, a cycloheptenyl group, a phenyl group, a pentalenyl group, an indenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, a dibenzofluorenyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a pyrrolyl group, a furanyl group, a thiophenyl group, an imidazolyl group, a pyrazolyl group, a thiazolyl group, an isothiazolyl group, an oxazolyl group, an isoxazolyl group, a pyridinyl group, a pyrazinyl group, a pyrimidinyl group, a pyridazinyl group, an isoindolyl group, an indolyl group, an indazolyl group, a purinyl group, a quinolinyl group, an isoquinolinyl group, a benzoquinolinyl group, a phthalazinyl group, a naphthyridinyl group, a quinoxalinyl group, a quinazolinyl group, a cinnolinyl group, a carbazolyl group, a phenanthridinyl group, an acridinyl group, a phenanthrolinyl group, a benzimidazolyl group, a benzofuranyl group, a benzothiophenyl group, a benzothiazolyl group, an isobenzothiazolyl group, a benzoxazolyl group, an isobenzoxazolyl group, a triazolyl group, a tetrazolyl group, an oxadiazolyl group, a triazinyl group, a dibenzofuranyl group, a dibenzothiophenyl group, a benzocarbazolyl group, a dibenzocarbazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, $-\text{Si}(\text{Q}_{31})(\text{Q}_{32})(\text{Q}_{33})$, $-\text{N}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{B}(\text{Q}_{31})(\text{Q}_{32})$, $-\text{C}(=\text{O})(\text{Q}_{31})$, $-\text{S}(=\text{O})_2(\text{Q}_{31})$, $-\text{P}(=\text{O})(\text{Q}_{31})(\text{Q}_{32})$, or any combination thereof.

In an embodiment, Q_{31} to Q_{33} may each independently be hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a cyano group, a $\text{C}_1\text{-C}_{20}$ alkyl group, a $\text{C}_2\text{-C}_{20}$ alkenyl group, a $\text{C}_2\text{-C}_{20}$ alkynyl group, a $\text{C}_1\text{-C}_{20}$ alkoxy group, a $\text{C}_3\text{-C}_{10}$ cycloalkyl group, a $\text{C}_1\text{-C}_{10}$ heterocycloalkyl group, a $\text{C}_3\text{-C}_{10}$ cycloalkenyl group, a $\text{C}_1\text{-C}_{10}$ heterocycloalkenyl group, a $\text{C}_6\text{-C}_{20}$ aryl group, a $\text{C}_1\text{-C}_{20}$ heteroaryl group, a monovalent non-aromatic condensed polycyclic group, a monovalent non-aromatic condensed heteropolycyclic group, a biphenyl group, or a terphenyl group.

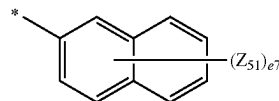
In an embodiment, R_1 , R_2 , R_{10} , R_{20} , R_{30} , R_{40} , R_{50} , R_{60} , R_{70} , and R_{80} may each independently be: hydrogen, deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a cyano group, a $\text{C}_1\text{-C}_{20}$ alkyl group, or a $\text{C}_1\text{-C}_{20}$ alkoxy group;

a $\text{C}_1\text{-C}_{20}$ alkyl group or a $\text{C}_1\text{-C}_{20}$ alkoxy group, each substituted with deuterium, $-\text{F}$, $-\text{Cl}$, $-\text{Br}$, $-\text{I}$, a cyano group, a phenyl group, a biphenyl group, or any combination thereof; or

a group represented by one of Formulae 5-1 to 5-26 and 6-1 to 6-55:



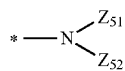
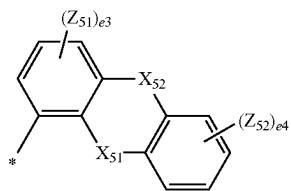
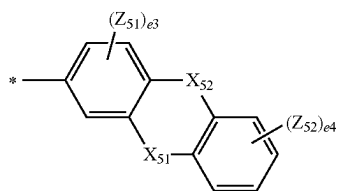
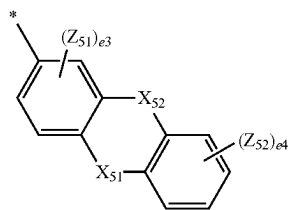
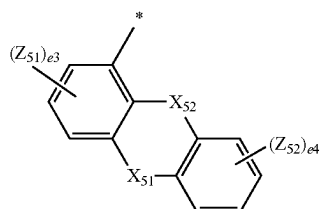
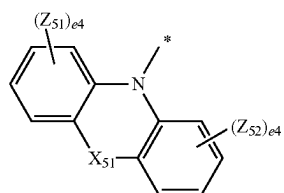
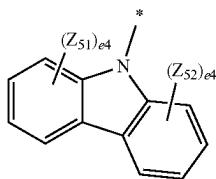
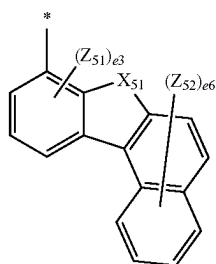
5-1



5-2

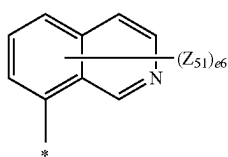
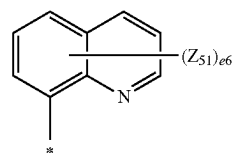
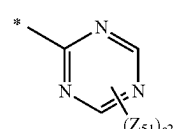
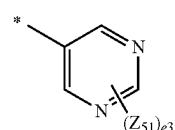
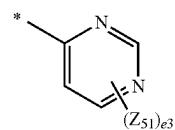
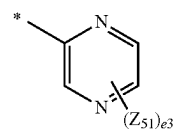
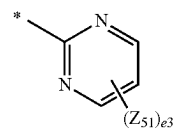
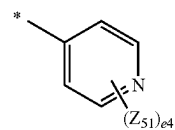
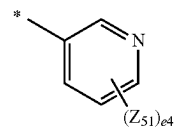
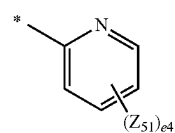
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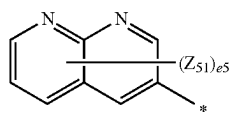
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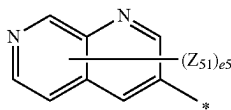
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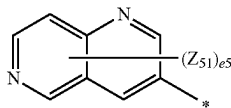
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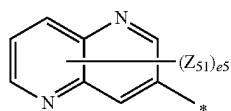
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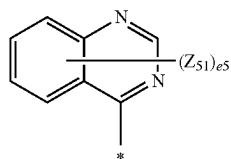
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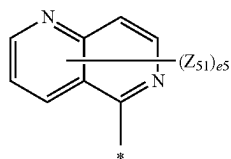
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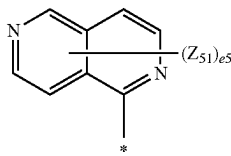
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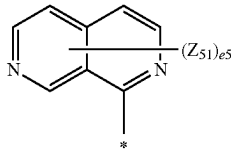
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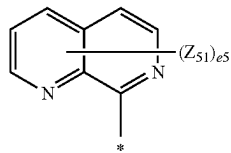
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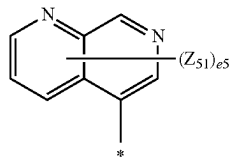
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6-41

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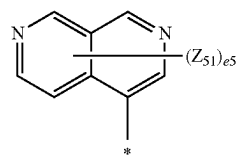


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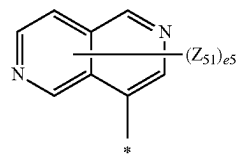
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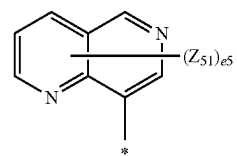
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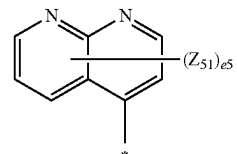
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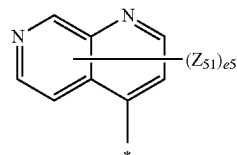
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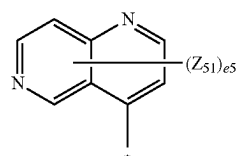
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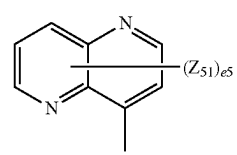
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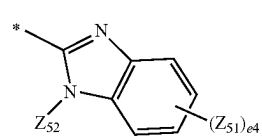
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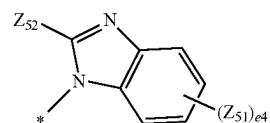
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6-49



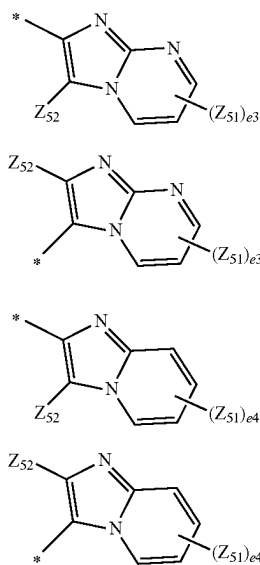
6-50



6-51

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In Formulae 5-1 to 5-26 and 6-1 to 6-55, X_{51} and X_{52} may each independently be O, S, $C(Z_{53})$ (Z_{54}), $N(Z_{53})$, or $Si(Z_{53})(Z_{54})$,

Z_{51} to Z_{54} may each independently be the same as described in connection with R_{10a} ,

e2 may be 1 or 2,

e3 may be 1, 2, or 3,

e4 may be 1, 2, 3, or 4,

e5 may be 1, 2, 3, 4, or 5,

e6 may be 1, 2, 3, 4, 5, or 6,

e7 may be 1, 2, 3, 4, 5, 6, or 7,

e9 may be 1, 2, 3, 4, 5, 6, 7, 8, or 9, and

* indicates a binding site to a neighboring atom.

In an embodiment, Z_{51} to Z_{54} in Formulae 5-1 to 5-26 and 6-1 to 6-55 may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C_1 - C_{20} alkyl group, a C_2 - C_{20} alkenyl group, a C_2 - C_{20} alkynyl group, a C_1 - C_{20} alkoxy group, a phenyl group, a biphenyl group, a terphenyl group, a naphthyl group, a fluorenyl group, a spiro-bifluorenyl group, a phenanthrenyl group, an anthracenyl group, a triphenylenyl group, a pyridinyl group, a pyrimidinyl group, a carbazolyl group, or a triazinyl group.

In an embodiment, two or more neighboring substituents among R_1 , R_2 , R_{10} , R_{20} , R_{30} , R_{40} , R_{50} , R_{70} , and R_{80} may optionally be linked to each other to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthylene group, a fluorene group, a pyridine group, a pyrimidine group, a triazine group, or a carbazole group.

In Formula 1, b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1, 2, 3, 4, 5, 6, 7, or 8.

In an embodiment, b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1, 2, 3, or 4.

In an embodiment, b10, b20, b30, b40, b50, b60, b70, and b80 may each independently be 1 or 2.

In Formula 1, * and *' each indicate a binding site to a neighboring atom.

R_{10a} may be:

deuterium (—D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group;

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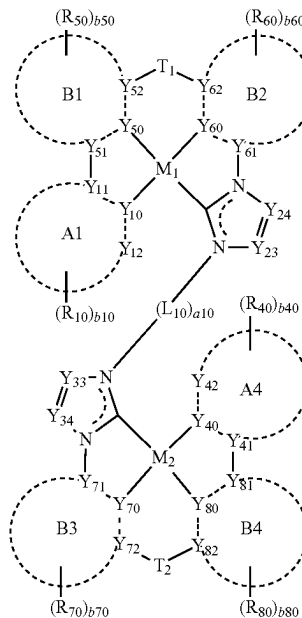
a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, or a C_1 - C_{60} alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C_3 - C_{60} carbocyclic group, a C_1 - C_{60} heterocyclic group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryloxy group, a C_1 - C_{60} heteroarylthio group, —Si(Q_{11})(Q_{12})(Q_{13}), —N(Q_{11})(Q_{12}), —B(Q_{11})(Q_{12}), —C(=O)(Q_{11}), —S(=O)₂(Q_{11}), —P(=O)(Q_{11})(Q_{12}), or any combination thereof;

a C_3 - C_{60} carbocyclic group, a C_1 - C_{60} heterocyclic group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryloxy group, or a C_1 - C_{60} heteroarylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C_1 - C_{60} alkyl group, a C_2 - C_{60} alkenyl group, a C_2 - C_{60} alkynyl group, a C_1 - C_{60} alkoxy group, a C_3 - C_{60} carbocyclic group, a C_1 - C_{60} heterocyclic group, a C_6 - C_{60} aryloxy group, a C_6 - C_{60} arylthio group, a C_1 - C_{60} heteroaryloxy group, a C_1 - C_{60} heteroarylthio group, —Si(Q_{21})(Q_{22})(Q_{23}), —N(Q_{21})(Q_{22}), —B(Q_{21})(Q_{22}), —C(=O)(Q_{21}), —S(=O)₂(Q_{21}), —P(=O)(Q_{21})(Q_{22}), or any combination thereof; or —Si(Q_{31})(Q_{32})(Q_{33}), —N(Q_{31})(Q_{32}), —B(Q_{31})(Q_{32}), —C(=O)(Q_{31}), —S(=O)₂(Q_{31}), or —P(=O)(Q_{31})(Q_{32}),

wherein Q_{11} to Q_{13} , Q_{21} to Q_{23} , Q_{31} to Q_{33} , and Q_{41} to Q_{43} may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C_1 - C_{60} alkyl group; a C_2 - C_{60} alkenyl group; a C_2 - C_{60} alkynyl group; a C_1 - C_{60} alkoxy group; or a C_3 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C_1 - C_{60} alkyl group, a C_1 - C_{60} alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

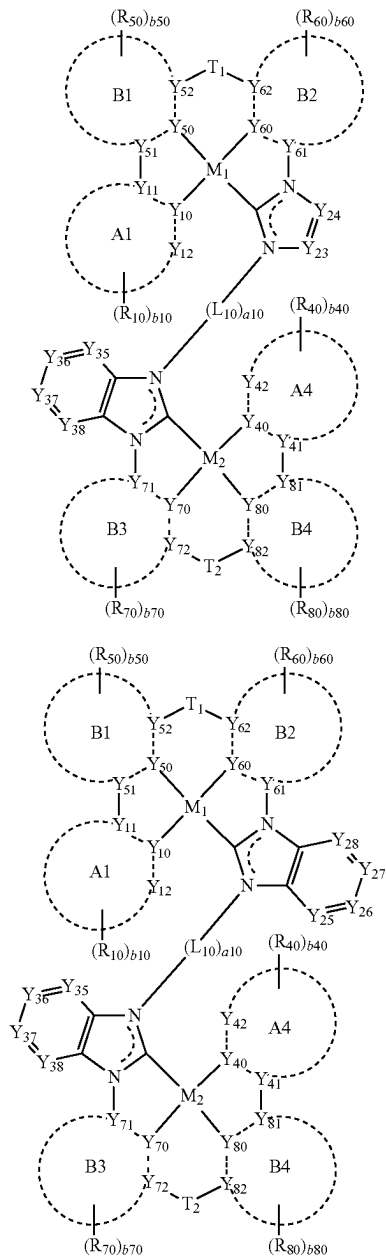
In an embodiment, the organometallic compound may be represented by one of Formulae 11-1 to 11-3:

[Formula 11-1]



31

-continued



In Formulae 11-1 to 11-3,

Y_{23} may be $C(R_{23})$ or N, Y_{24} may be $C(R_{24})$ or N, Y_{25} may be $C(R_{25})$ or N, Y_{26} may be $C(R_{26})$ or N, Y_{27} may be $C(R_{27})$ or N, Y_{28} may be $C(R_{28})$ or N, Y_{33} may be $C(R_{33})$ or N, Y_{34} may be $C(R_{34})$ or N, Y_{35} may be $C(R_{35})$ or N, Y_{36} may be $C(R_{36})$ or N, Y_{37} may be $C(R_{37})$ or N, and Y_{38} may be $C(R_{38})$ or N,

R_{23} to R_{28} may each independently be the same as described in connection with R_{20} in Formula 1, and

R_{33} to R_{38} may each independently be the same as described in connection with R_{30} in Formula 1, and Y_{10} to Y_{12} , Y_{40} to Y_{42} , Y_{50} to Y_{52} , Y_{60} to Y_{62} , Y_{70} to Y_{72} , Y_{80} to Y_{82} , R_{10} , R_{40} , R_{50} , R_{60} , R_{70} , R_{80} , A1, A4, B1 to B4, M_1 , M_2 , T_1 , T_2 , b10, b40, b50, b60, b70, b80, a10, and L_{10} may each independently be the same as described in the specification.

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In an embodiment, the organometallic compound may be one selected from Compounds BD1 to BD104:

[Formula 11-2]

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BD1

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[Formula 11-3]

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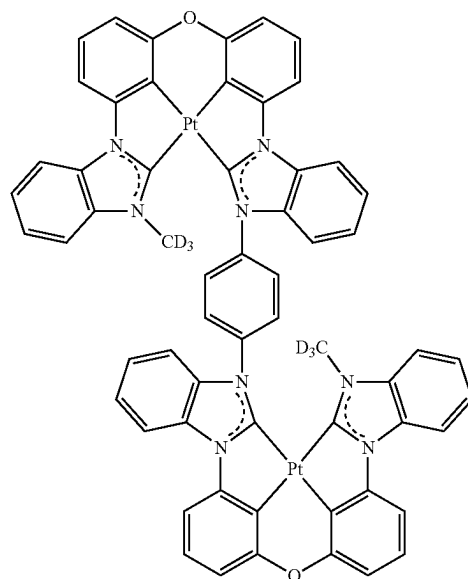
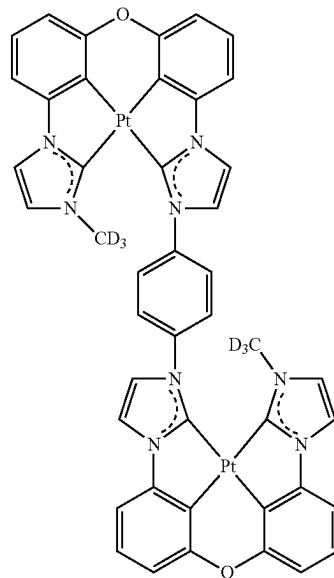
BD2

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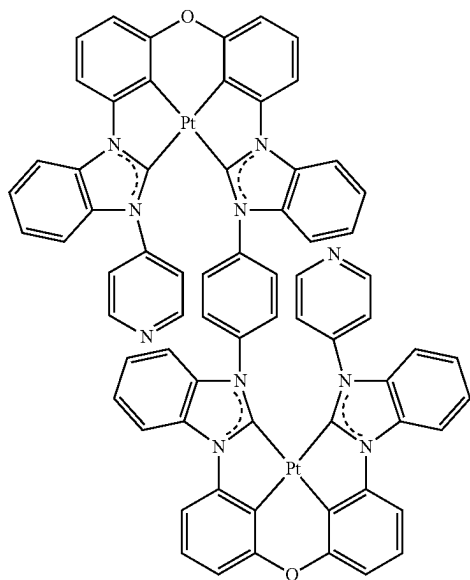
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34

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BD3

BD5

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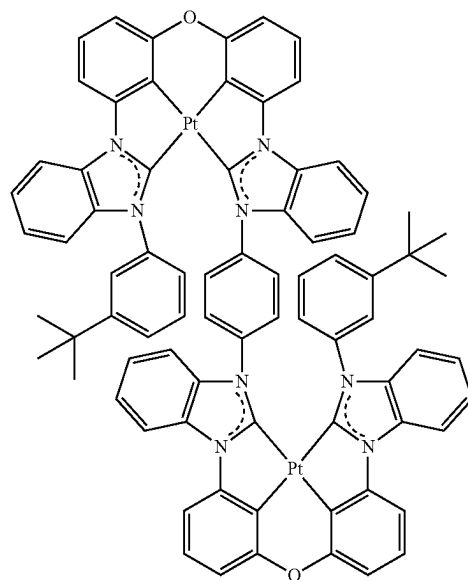
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BD4

BD6

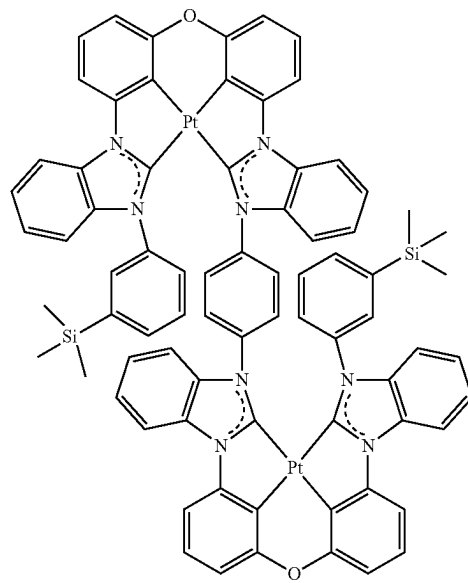
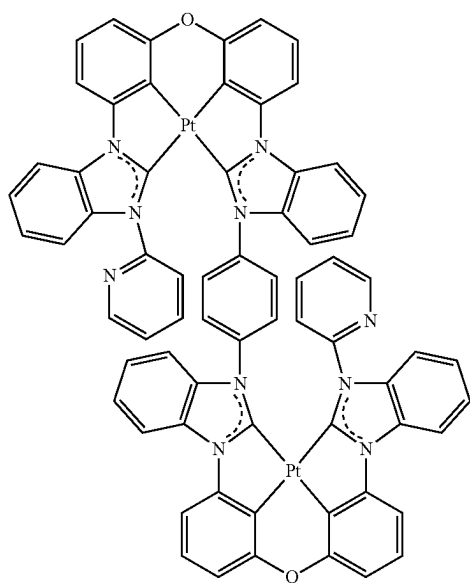
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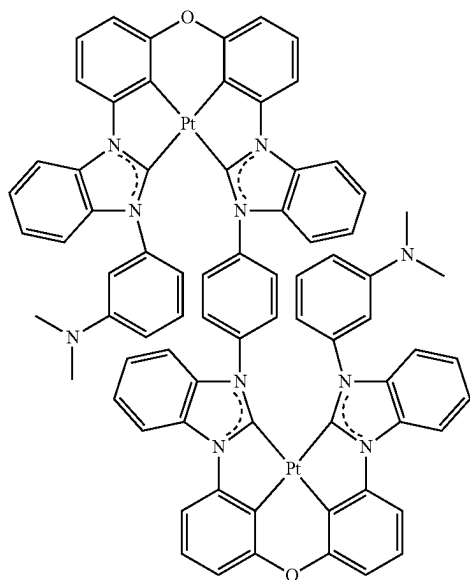
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**36**

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BD7

BD9

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BD8

BD10

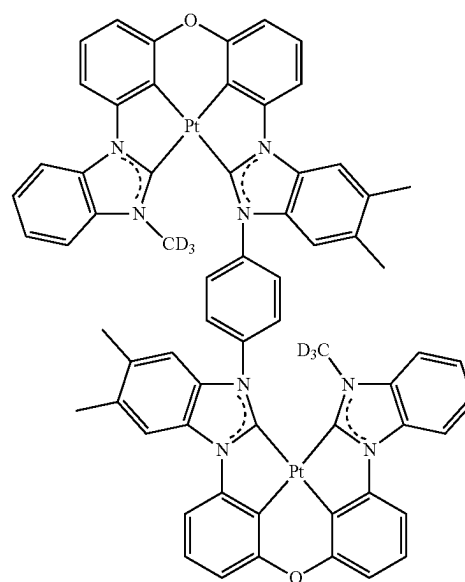
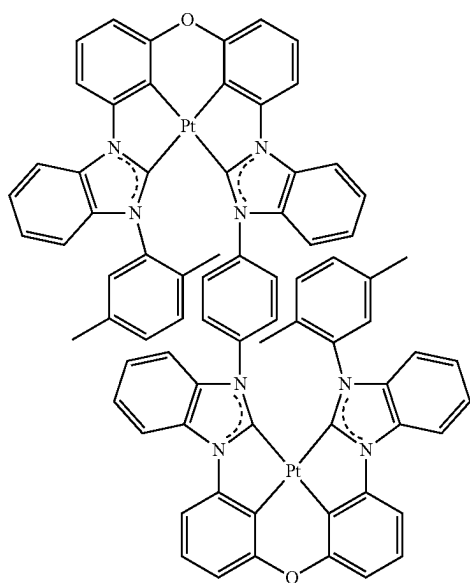
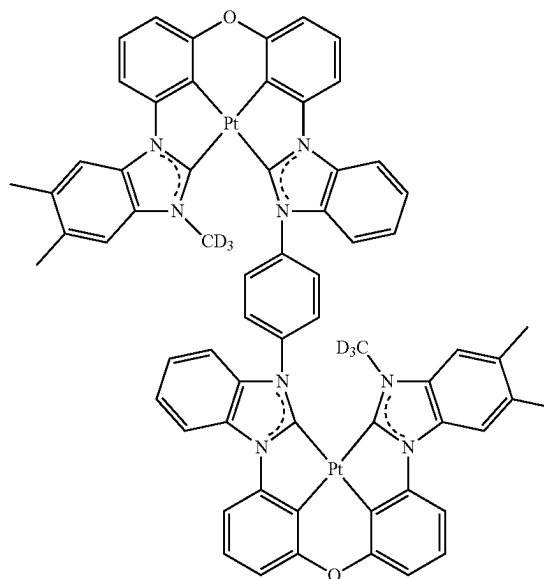
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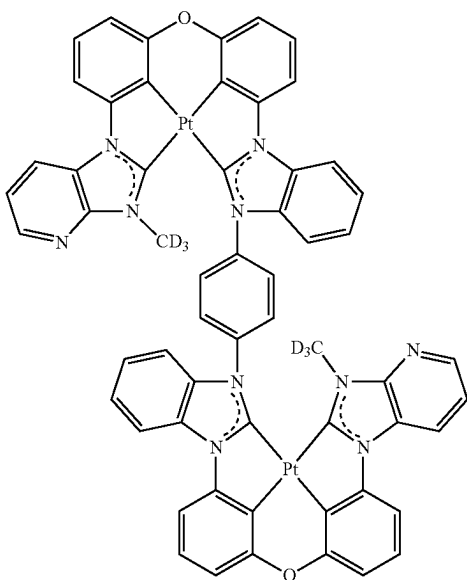
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37

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**38**

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BD11

BD13

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BD12

BD14

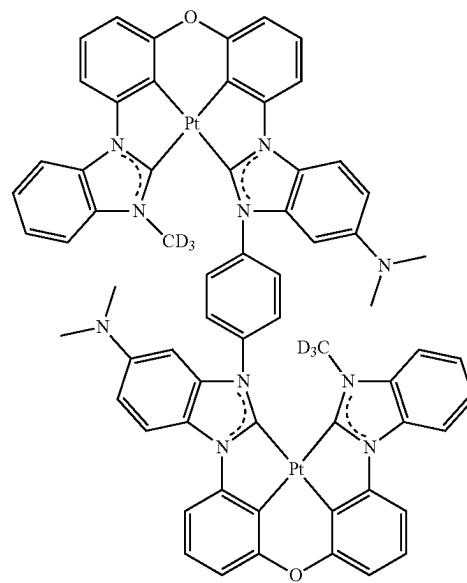
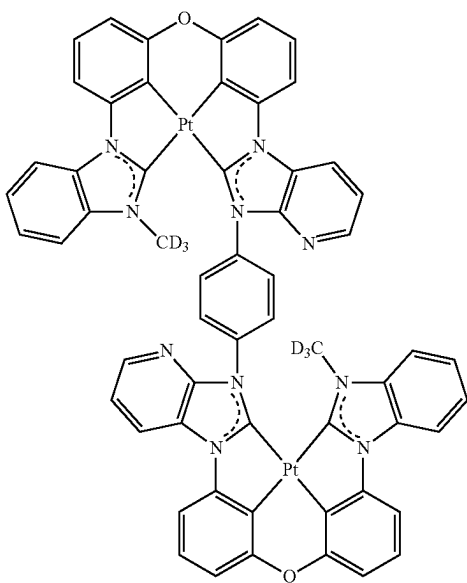
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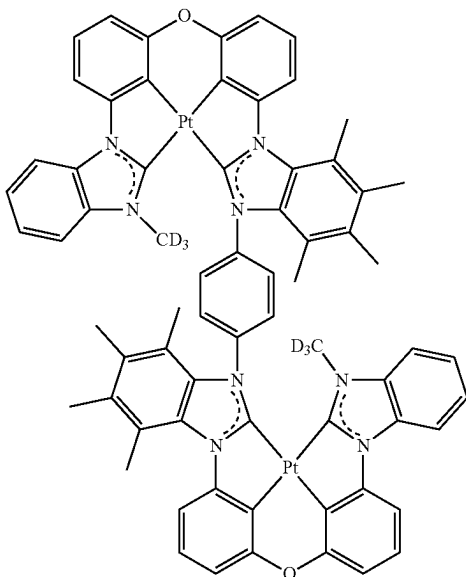
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**40**

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BD15

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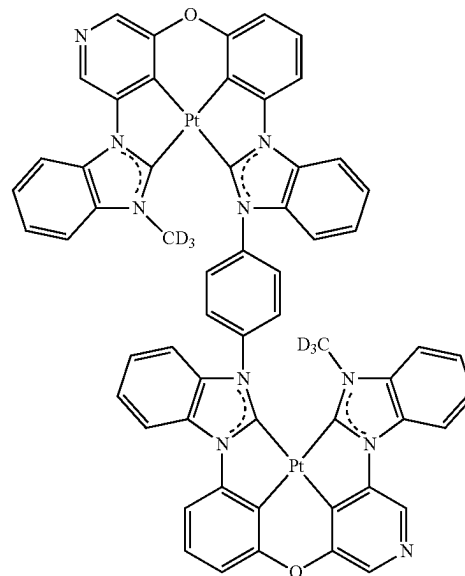
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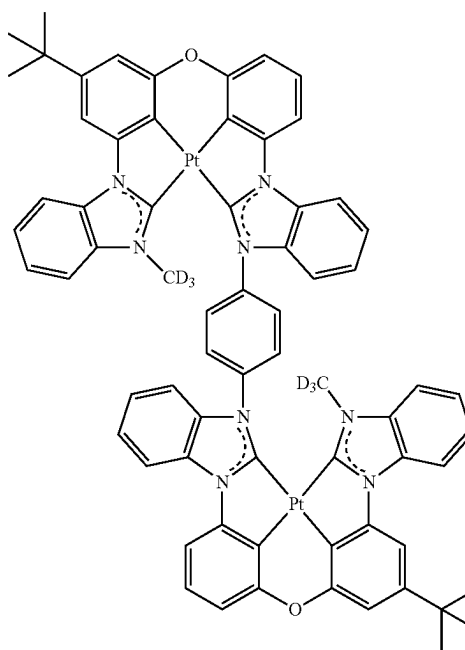
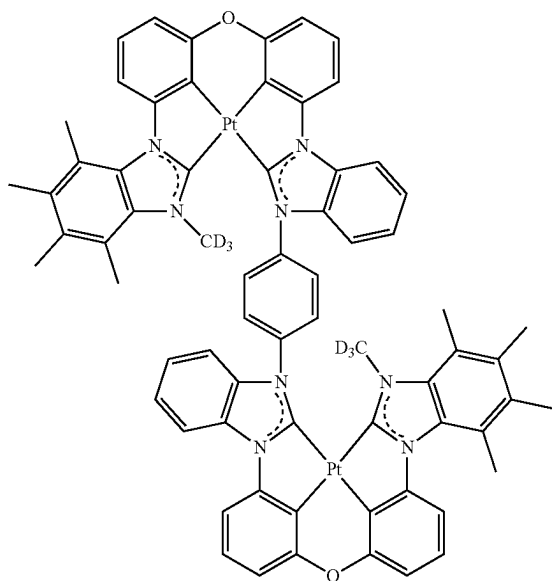
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BD17



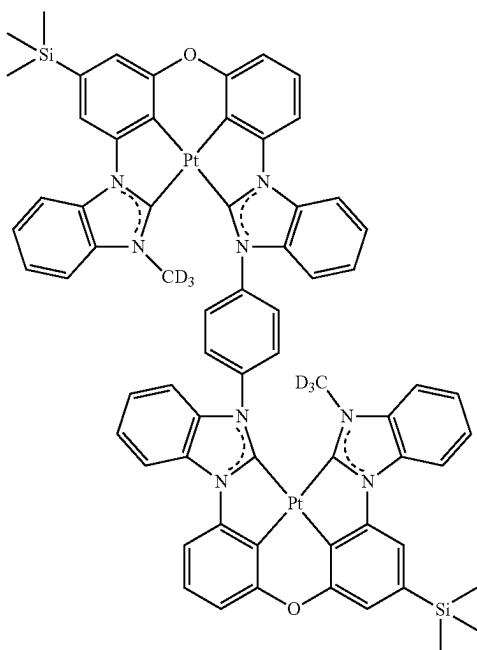
BD18

BD16



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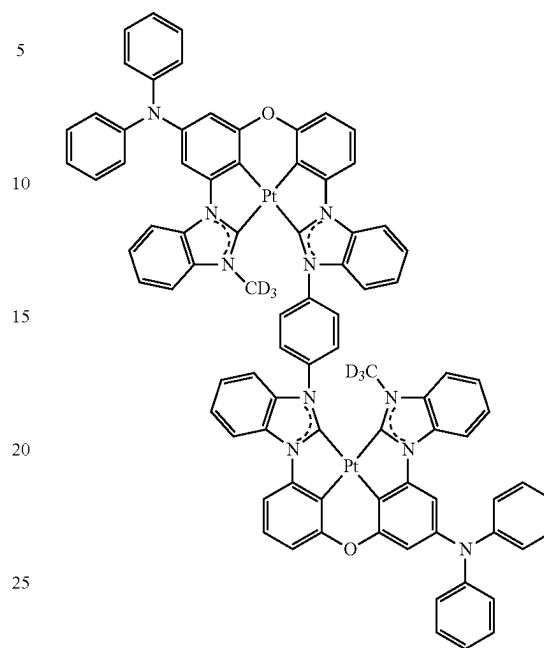
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BD19

42

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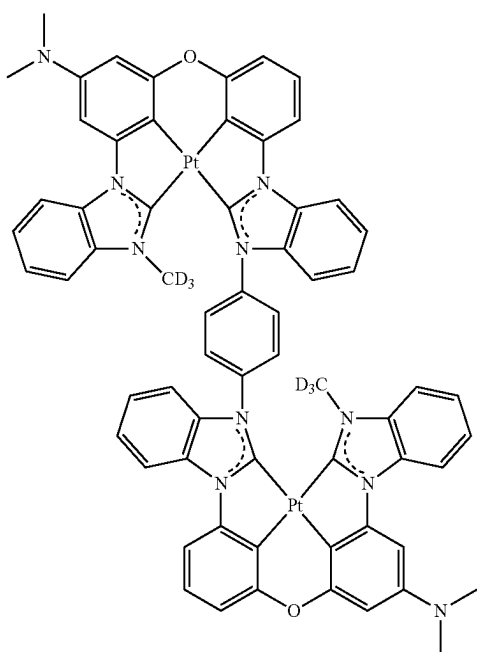
BD21

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BD20



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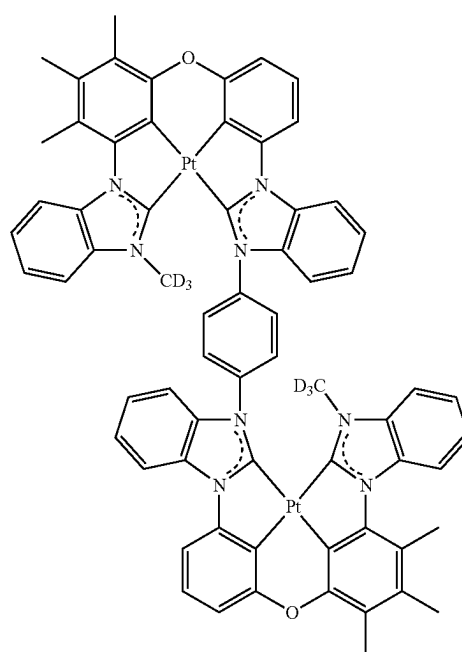
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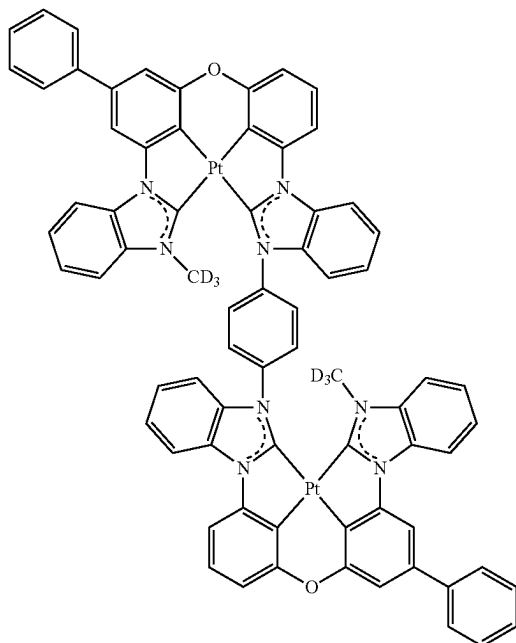
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BD22

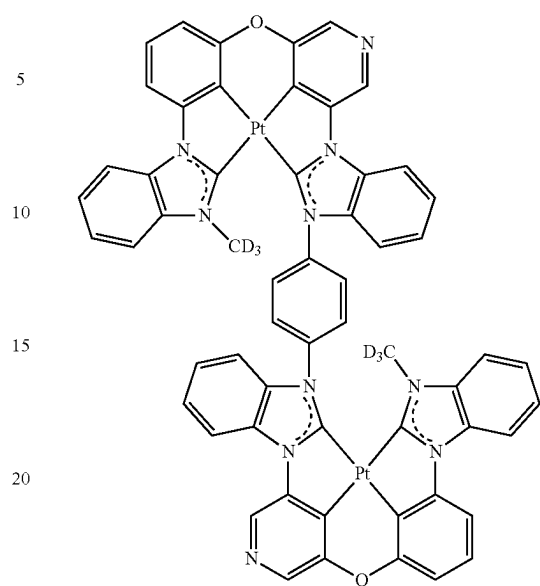


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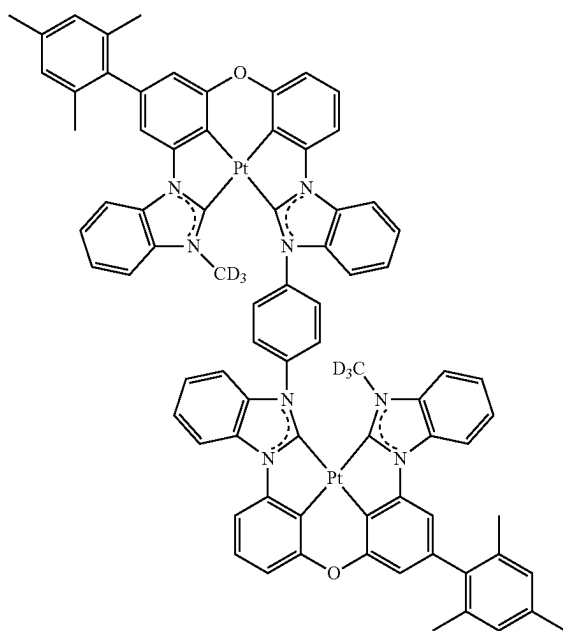
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**44**

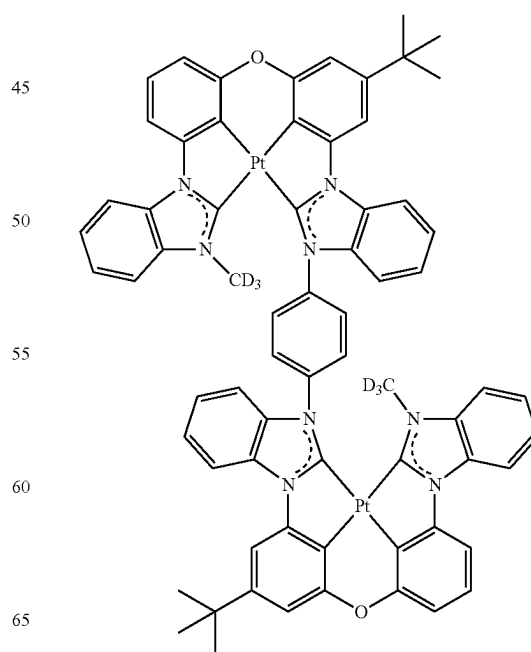
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BD24

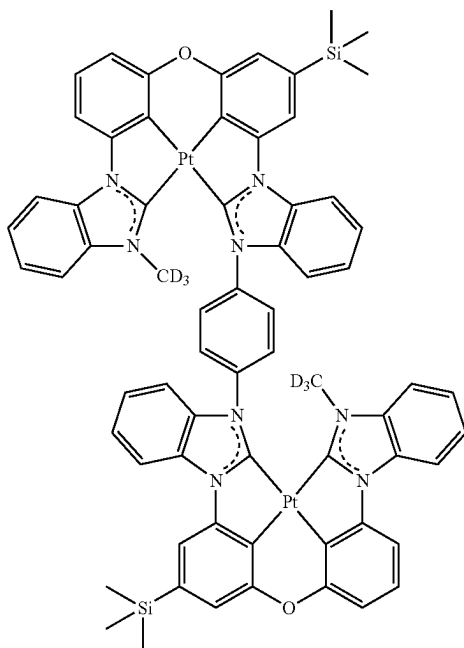


BD26



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**46**

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BD27

BD29

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BD28

BD30

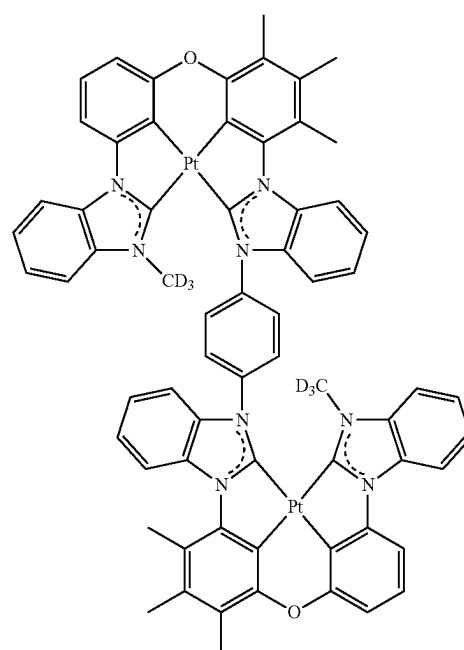
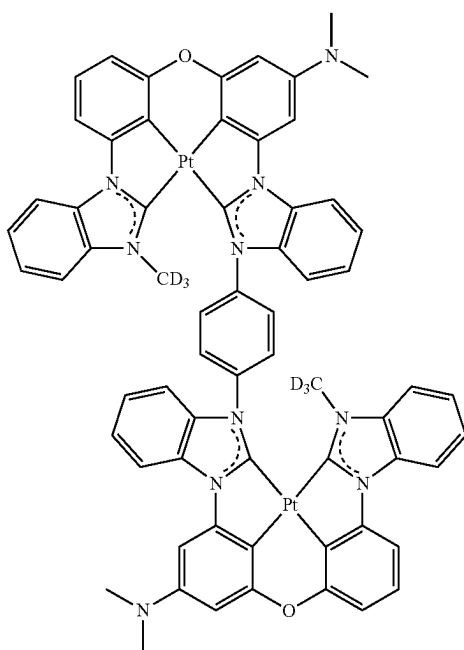
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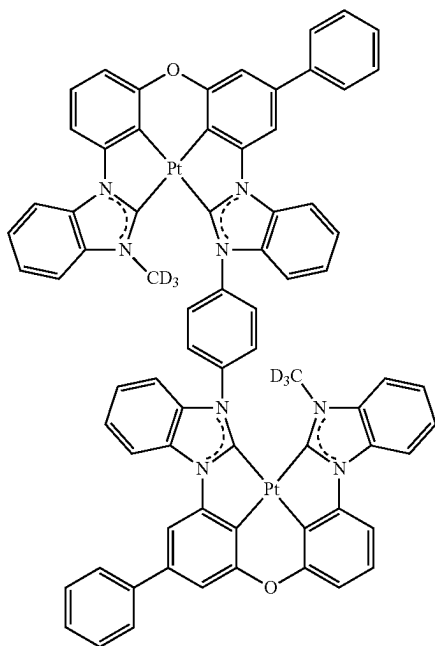
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**48**

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BD31

BD33

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BD32

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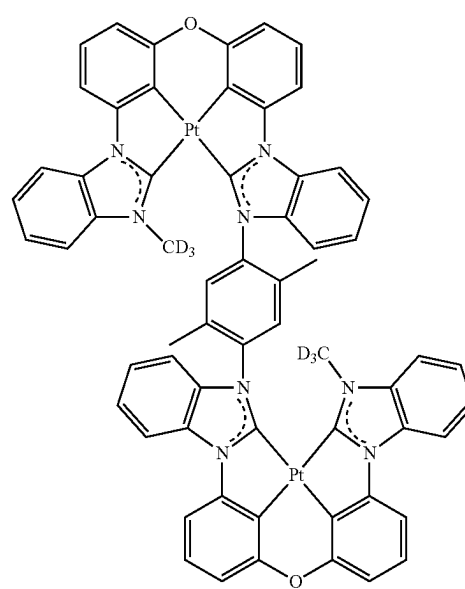
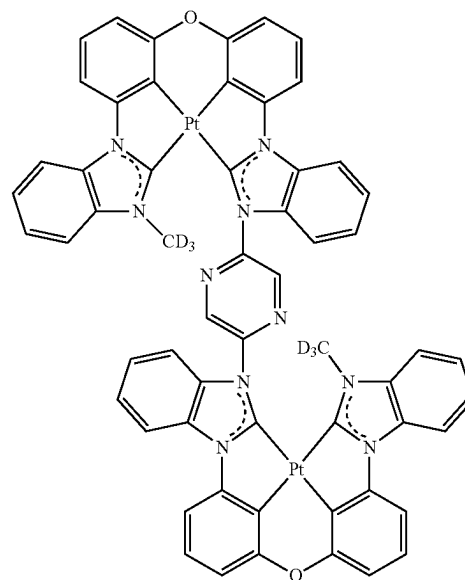
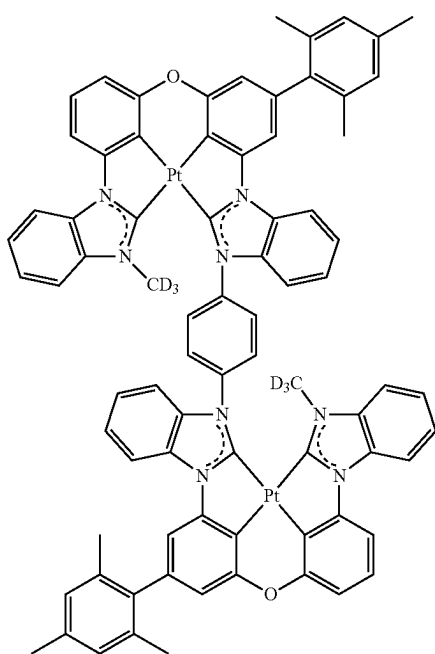
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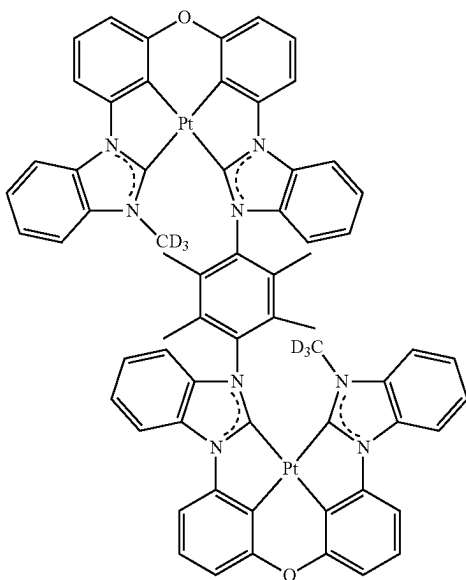
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BD34



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BD35

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BD36

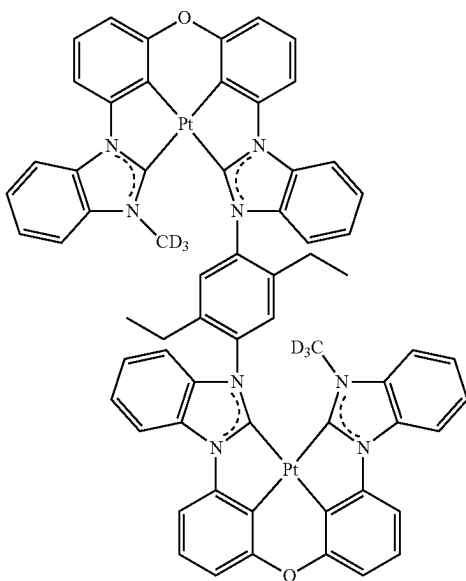
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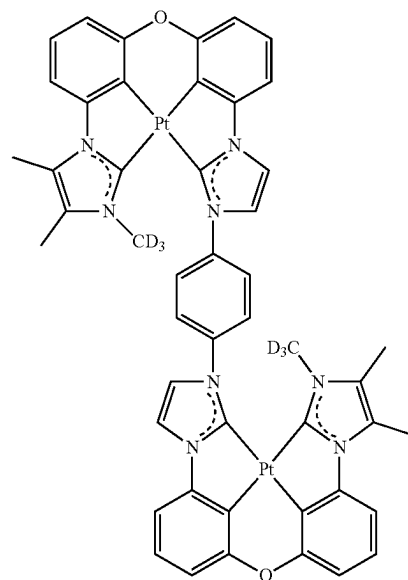
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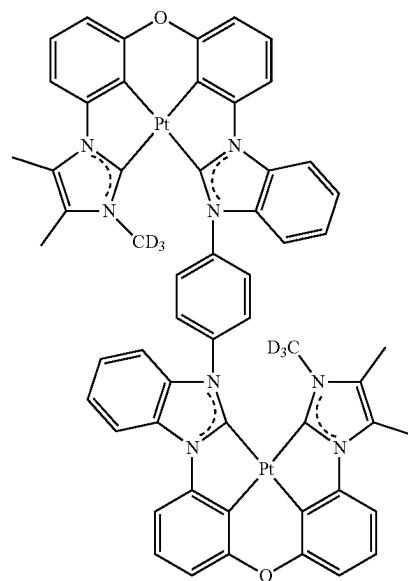
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BD37

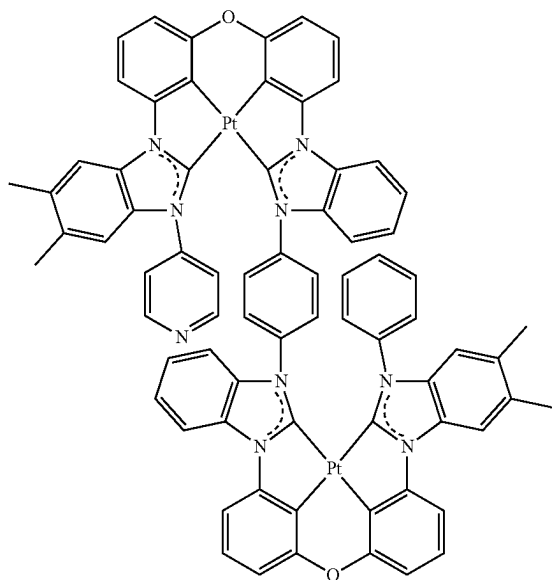


BD38

51

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BD39

**52**

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BD41

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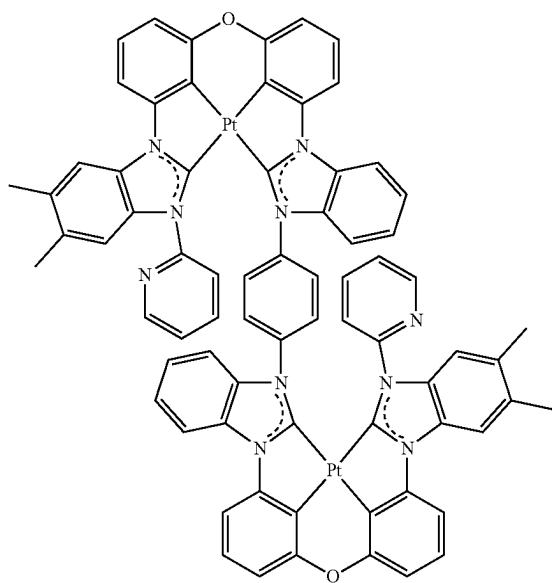
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BD40



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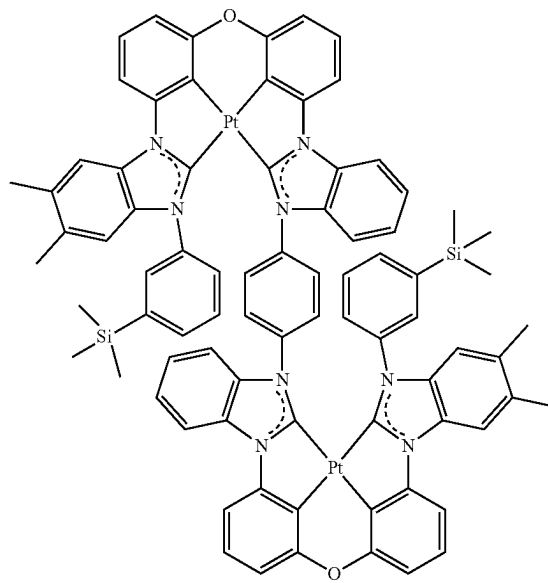
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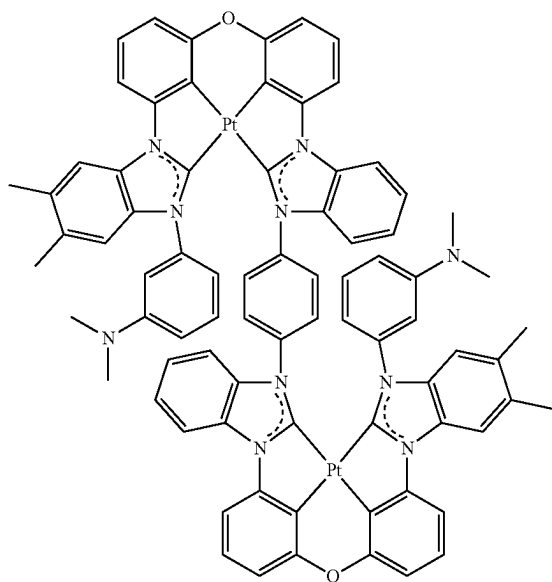
BD42



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BD43

**54**

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BD45

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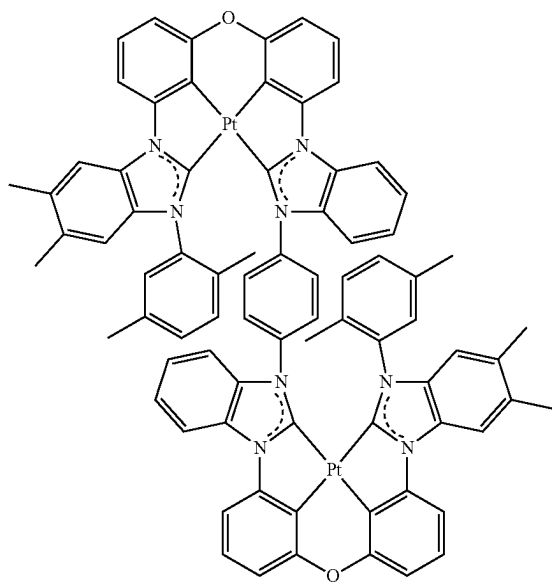
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BD44



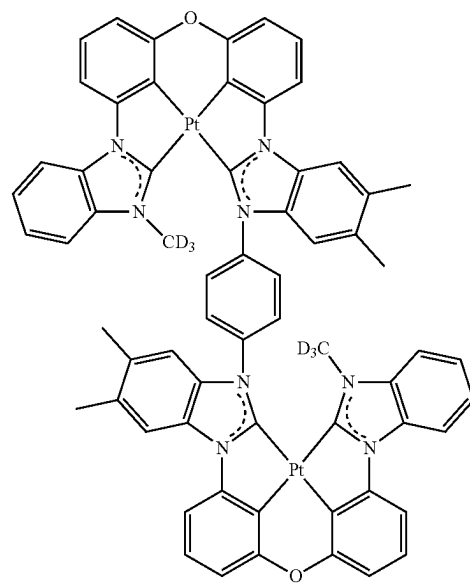
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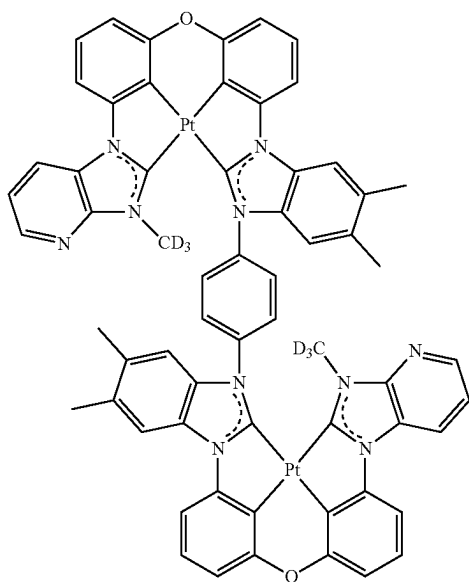
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BD46

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BD47

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BD48

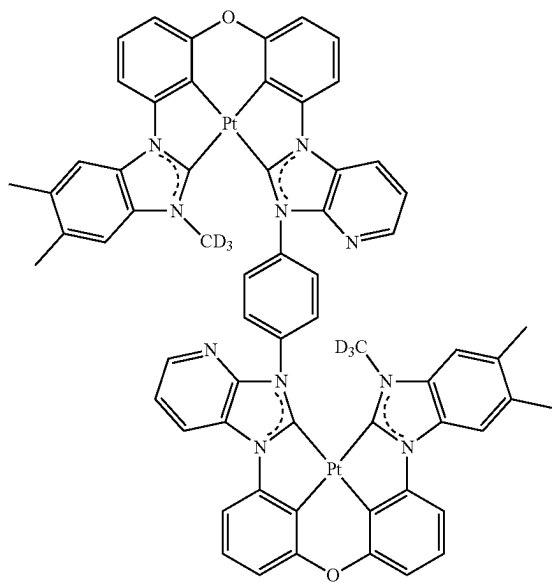
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**56**

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BD49

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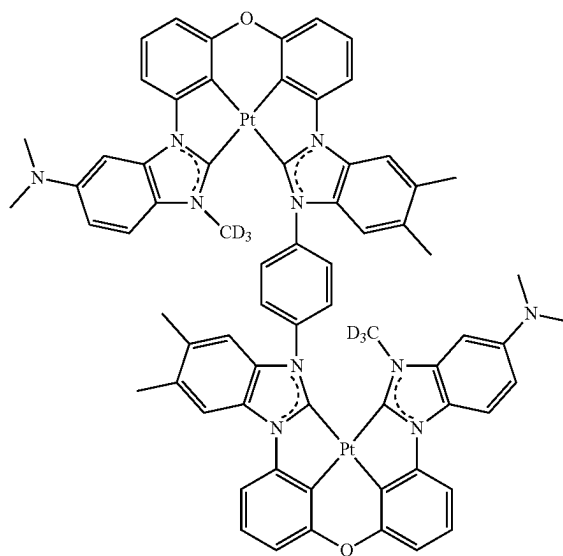
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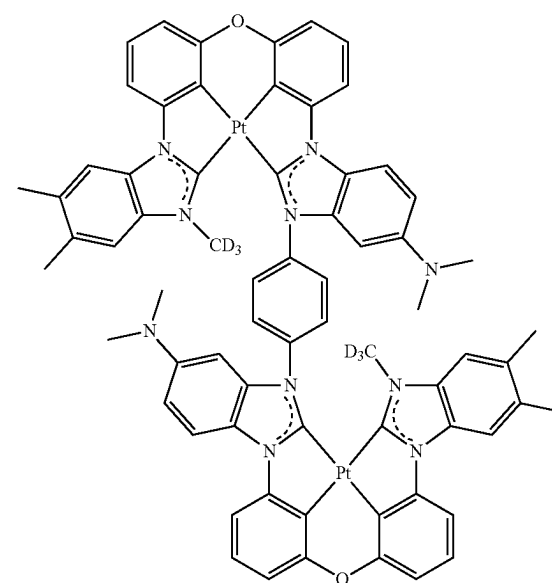
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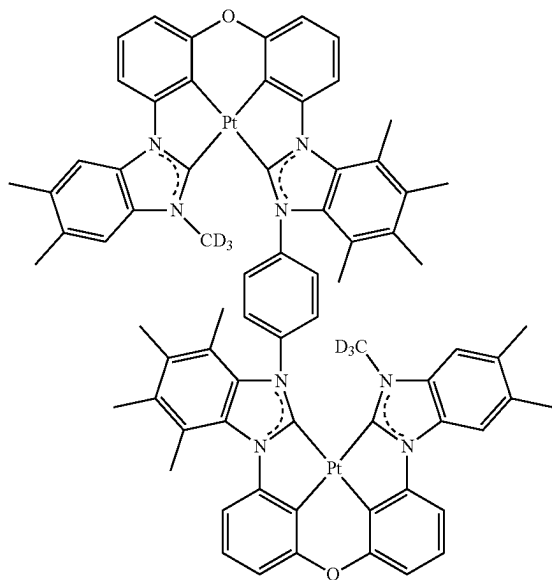
BD50



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-continued

BD51

**58**

-continued

BD53

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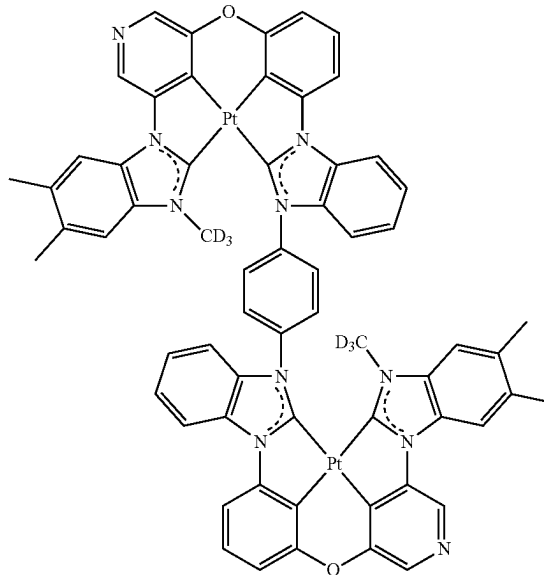
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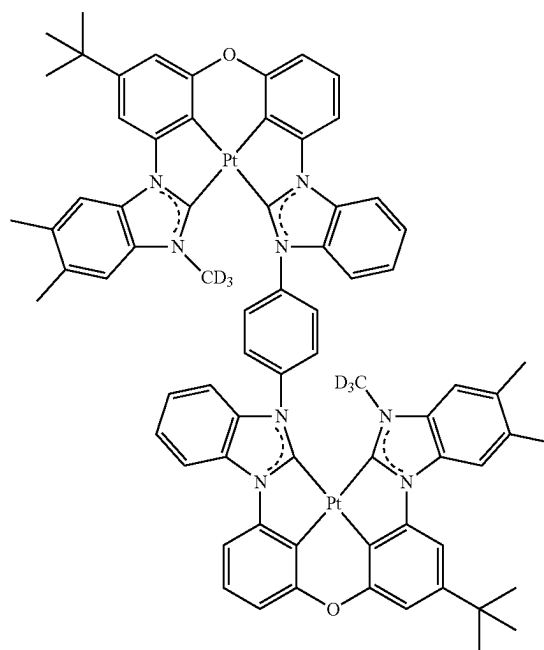
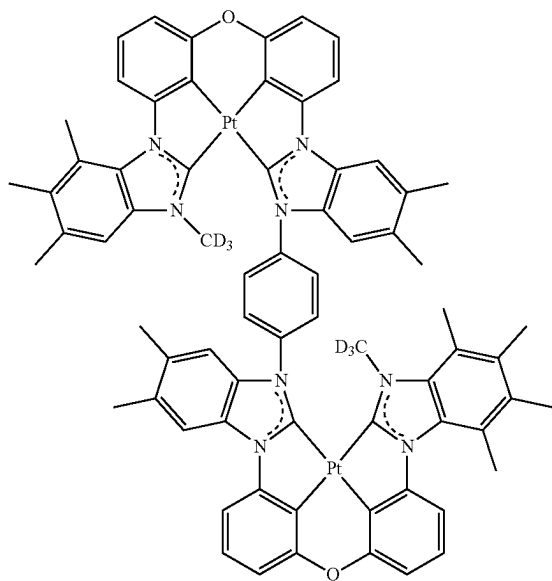
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BD54

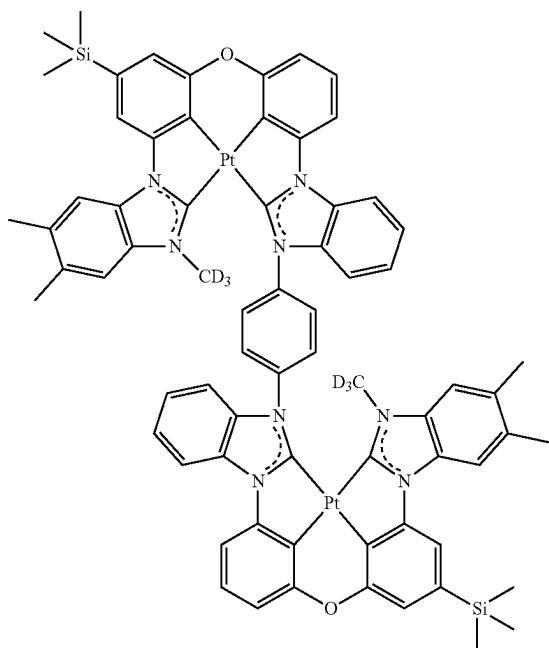
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BD55

**60**

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BD57

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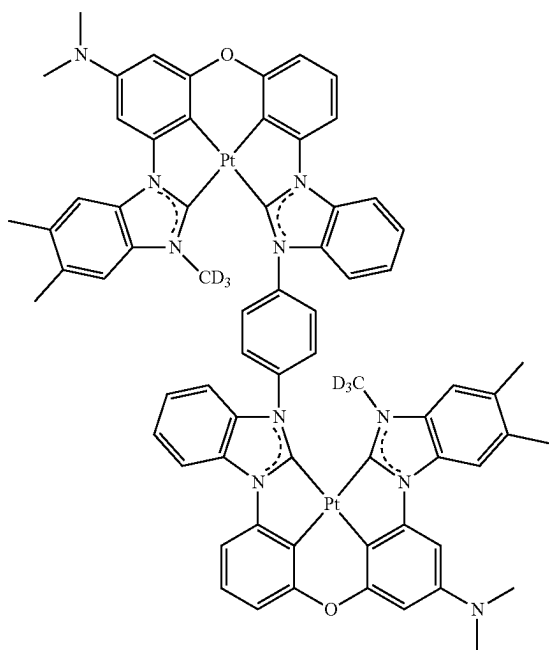
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BD56



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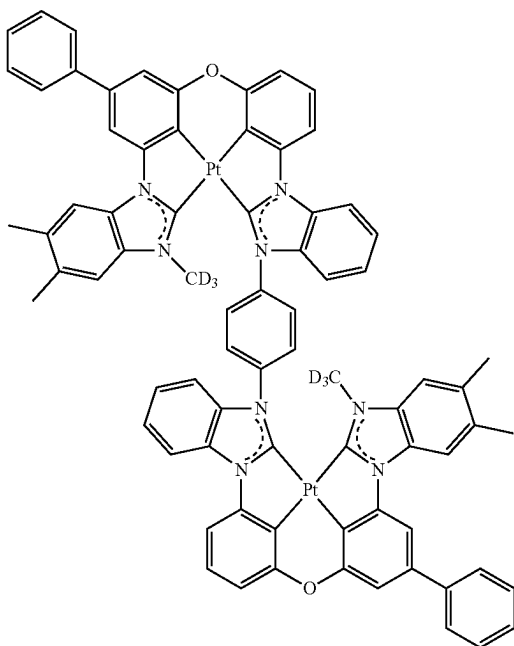
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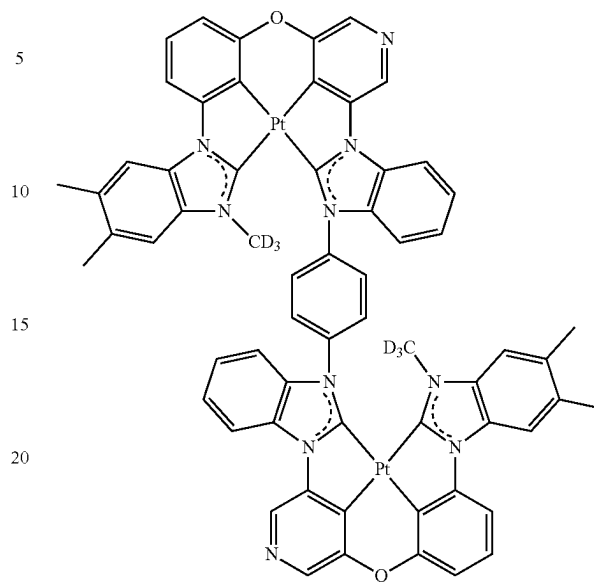
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BD59

**62**

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BD61

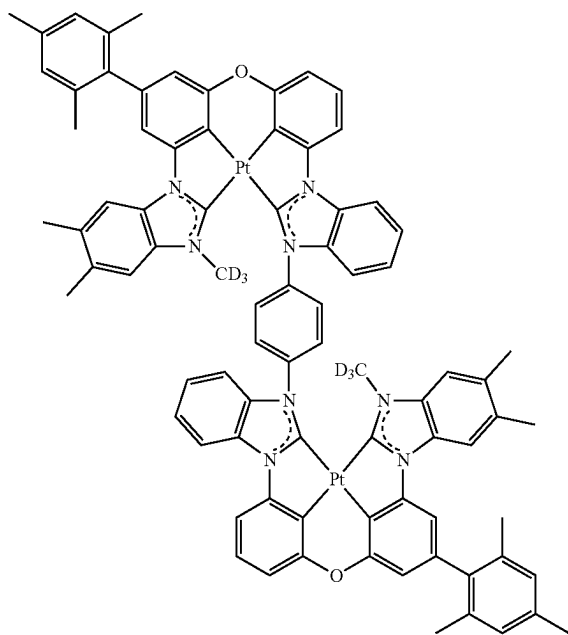


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BD60



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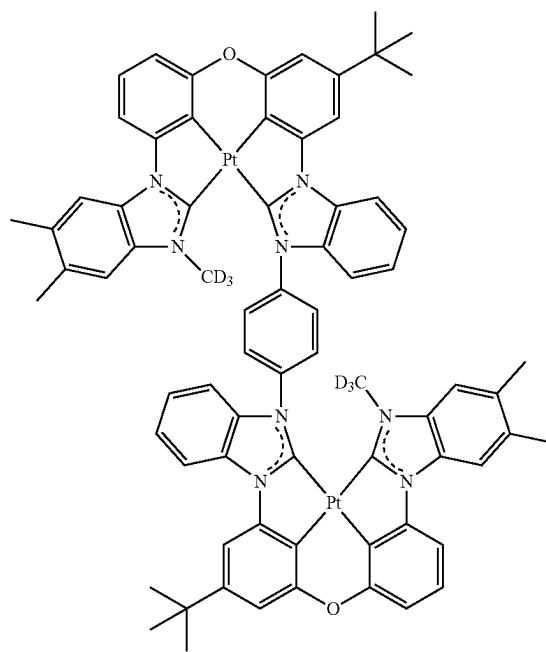
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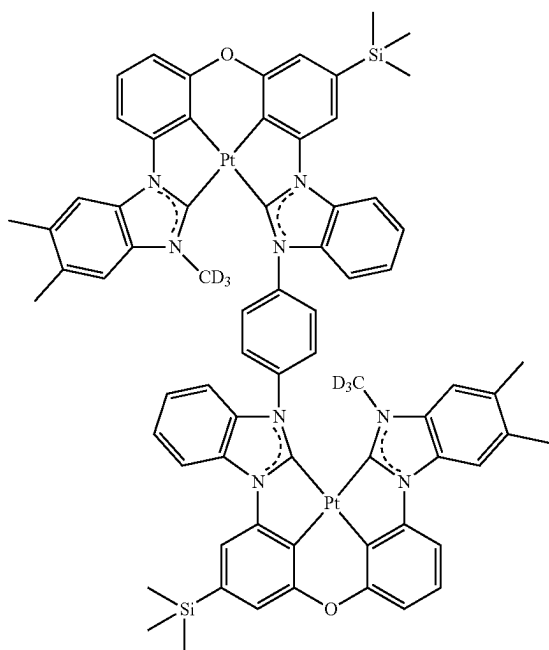
BD62



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BD63

**64**

-continued

BD65

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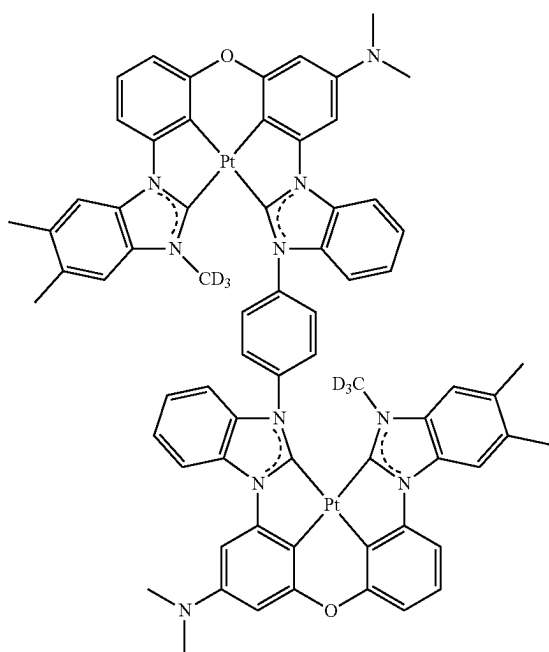
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BD64

BD66

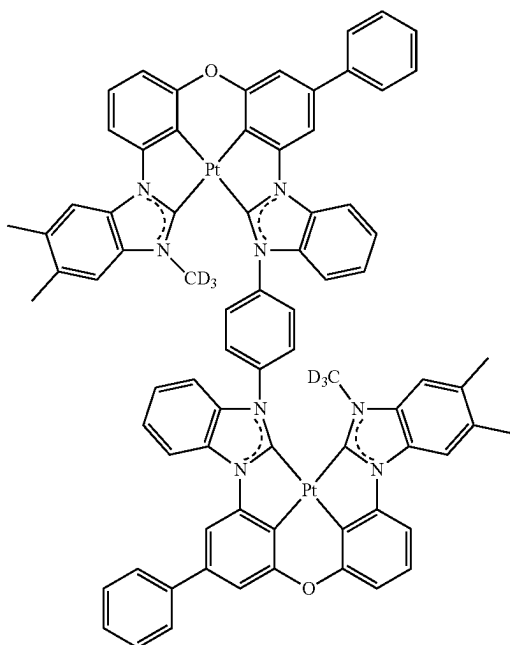


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BD67

**66**

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BD69

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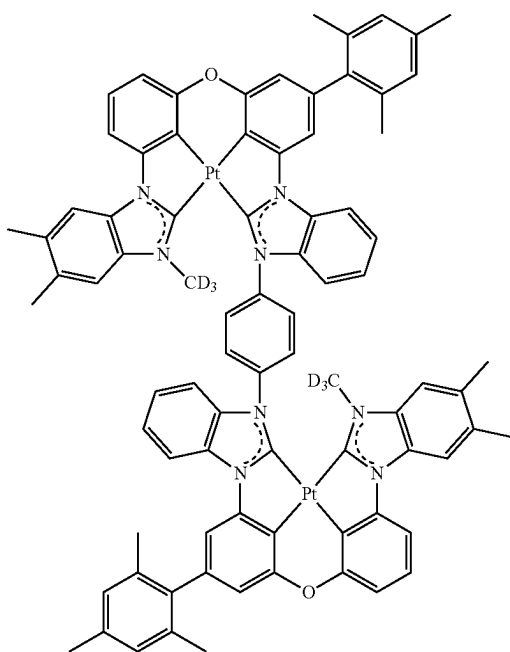
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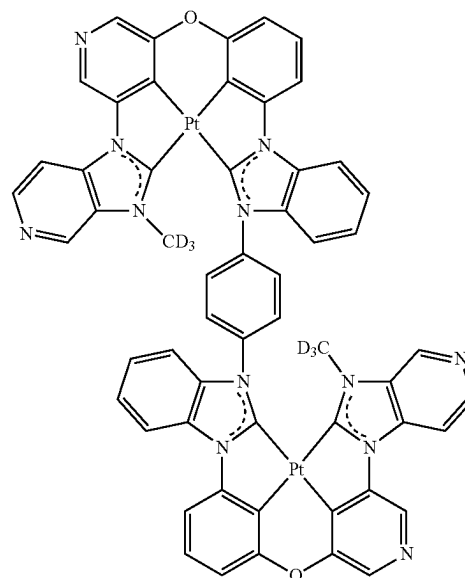
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BD68



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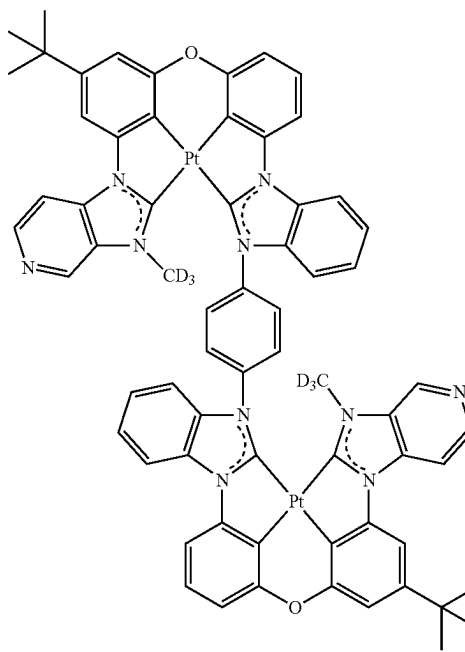
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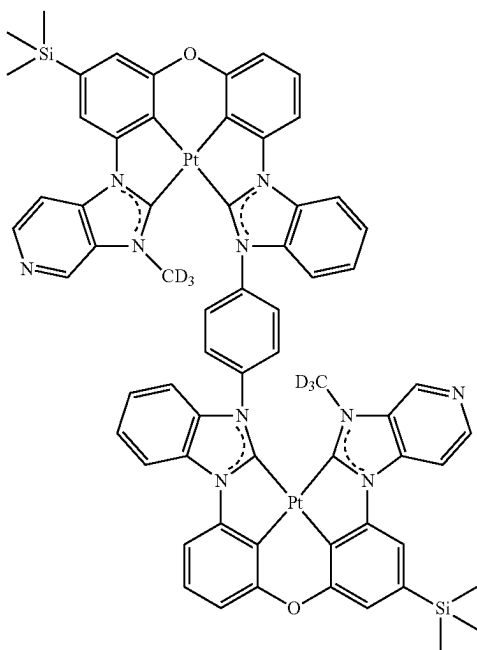
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BD70



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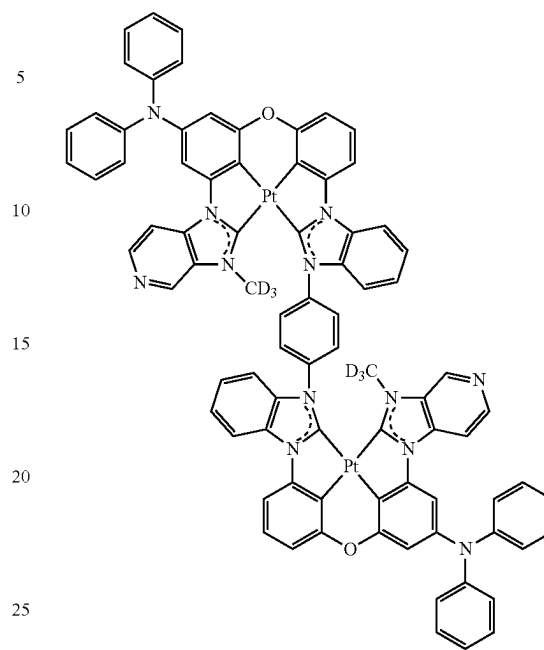
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BD71

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BD73

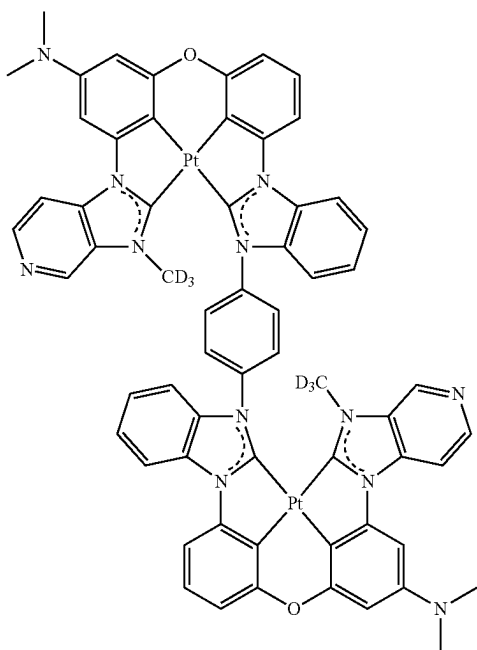
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BD72

BD74



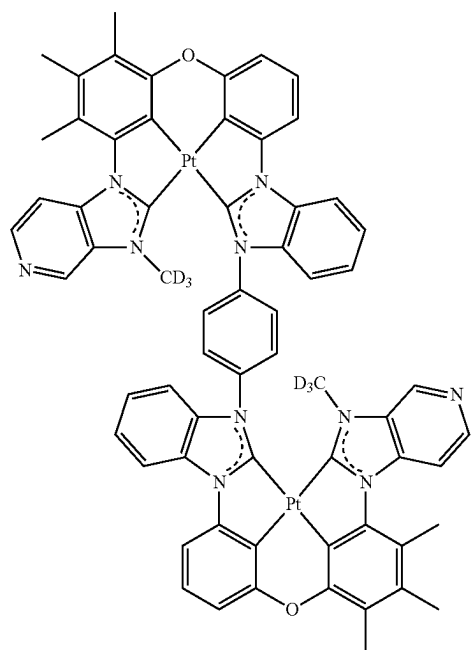
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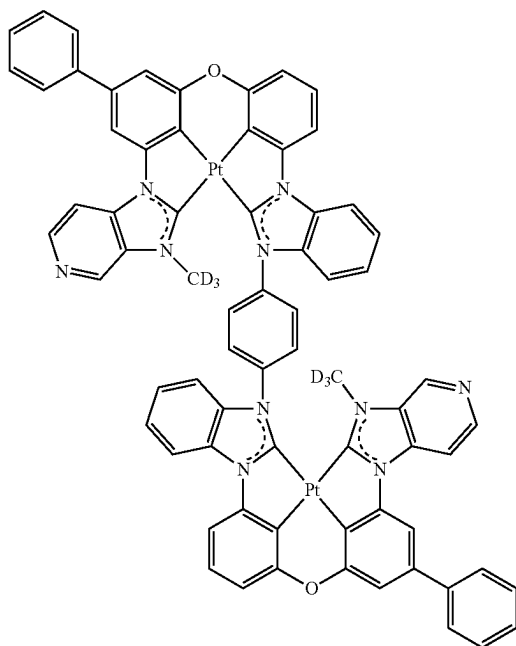
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69

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BD75

**70**

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BD77

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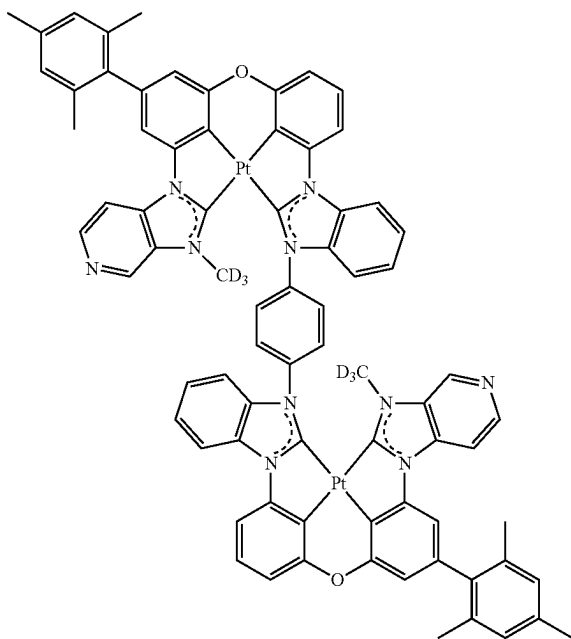
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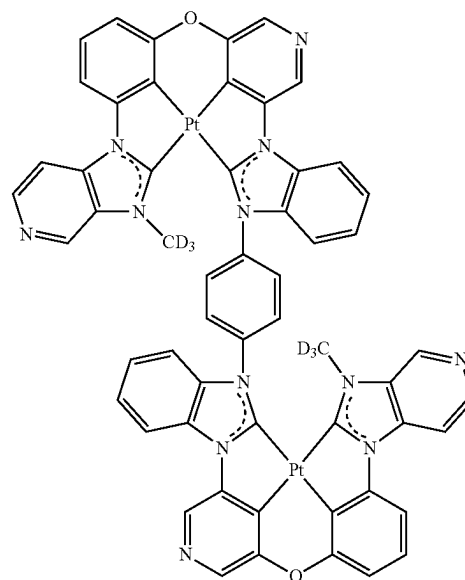
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BD76



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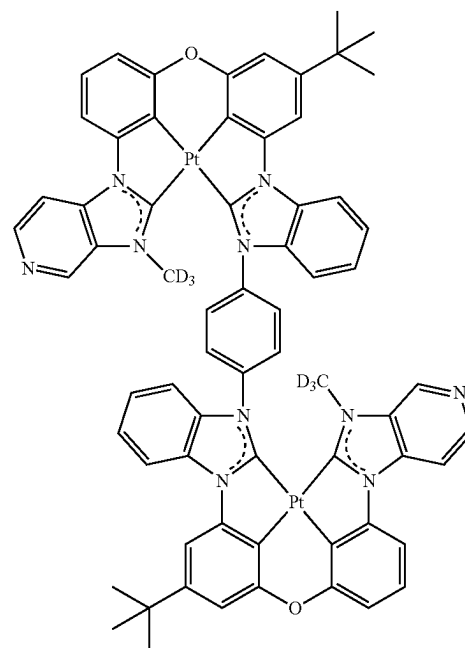
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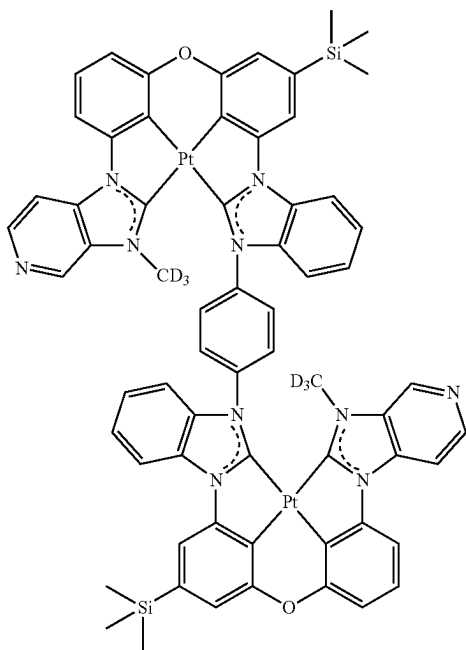
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BD78



71

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**72**

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BD79

BD81

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BD80

BD82

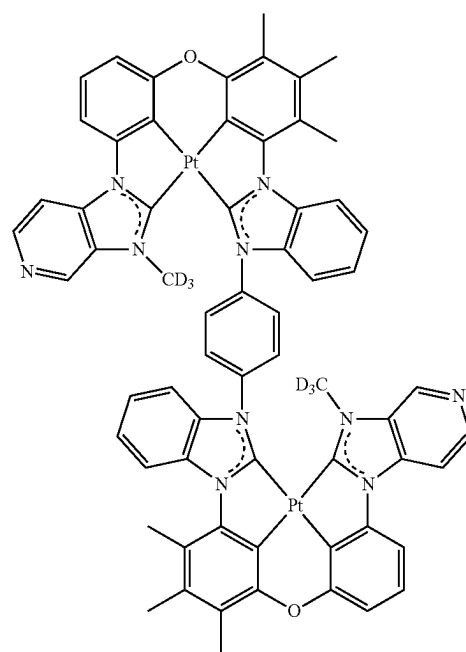
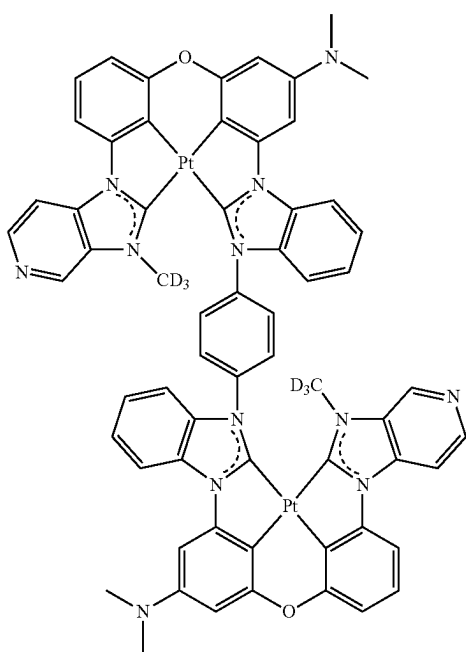
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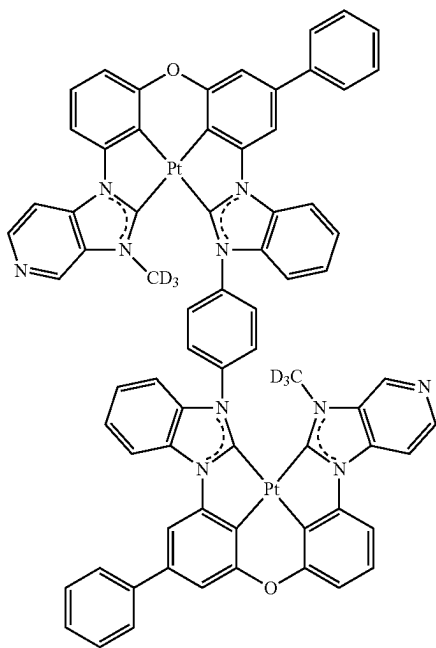
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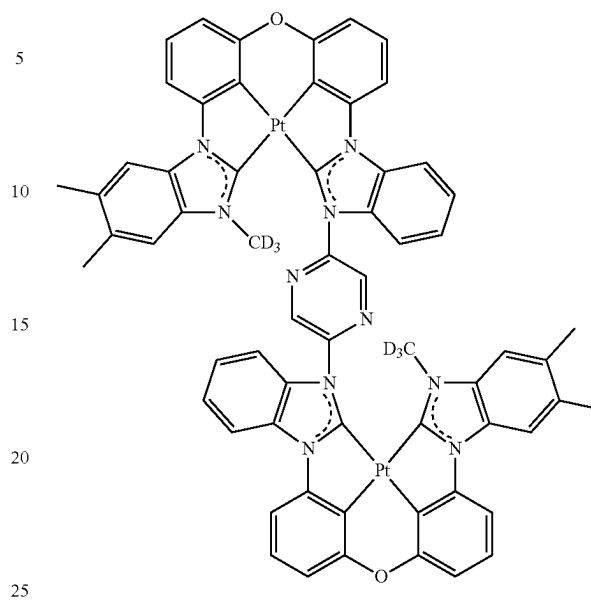
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**74**

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BD83

BD85



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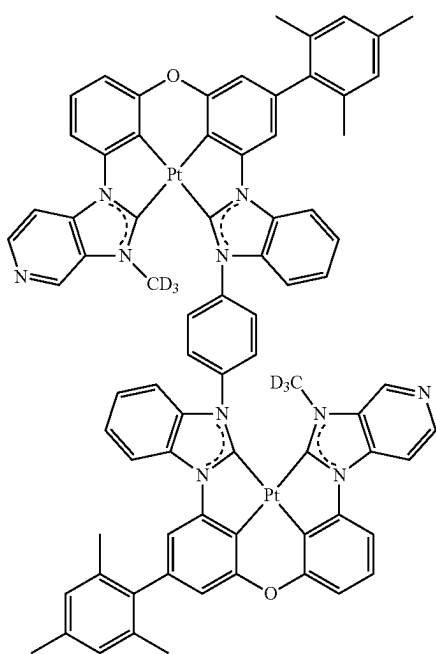
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BD84

BD86



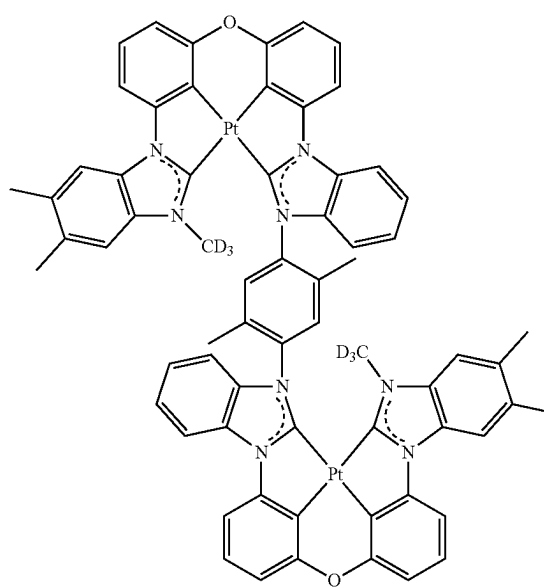
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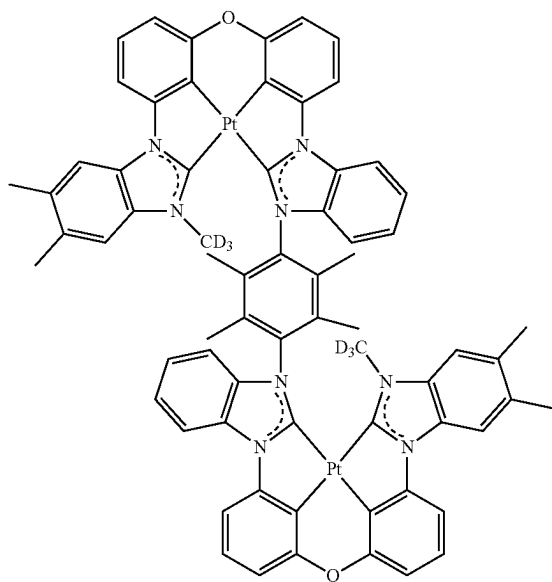
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BD87



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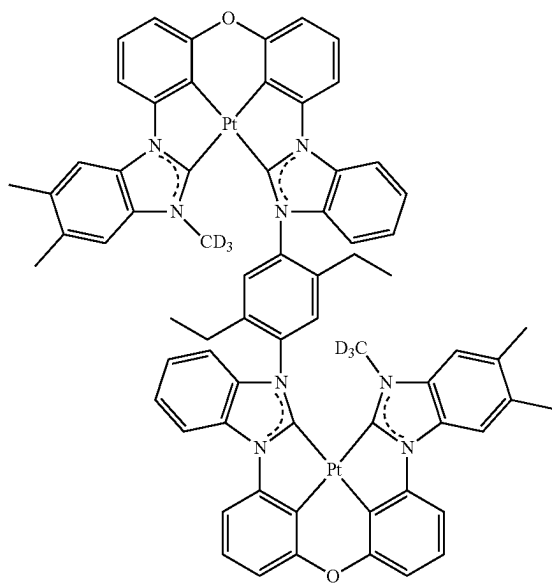
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BD88



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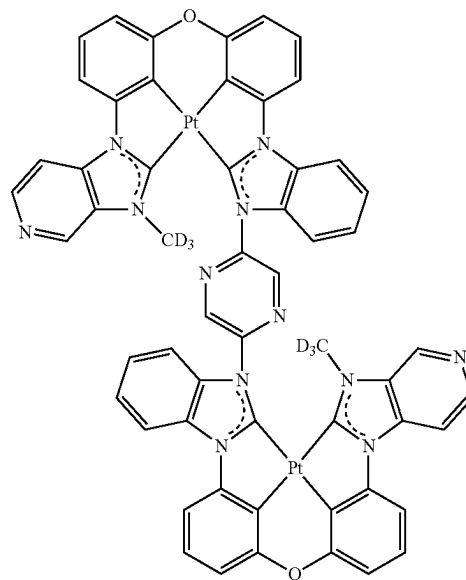
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BD89



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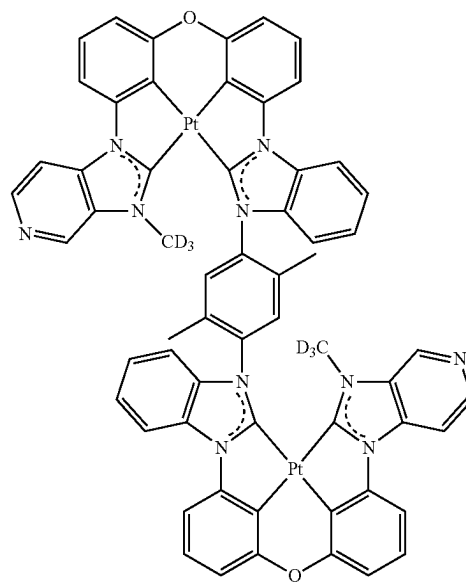
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BD90



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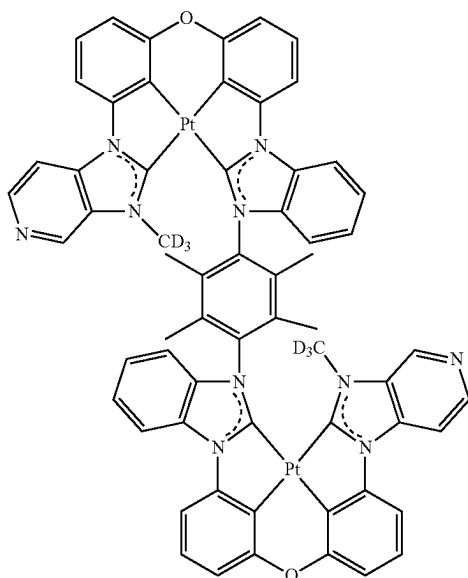
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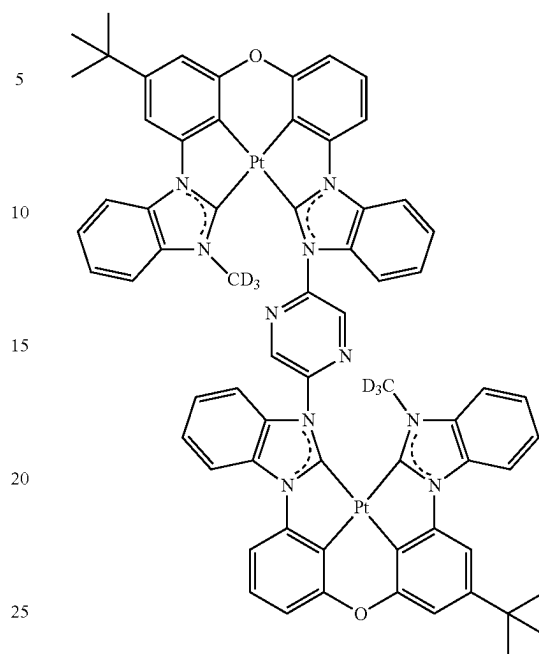
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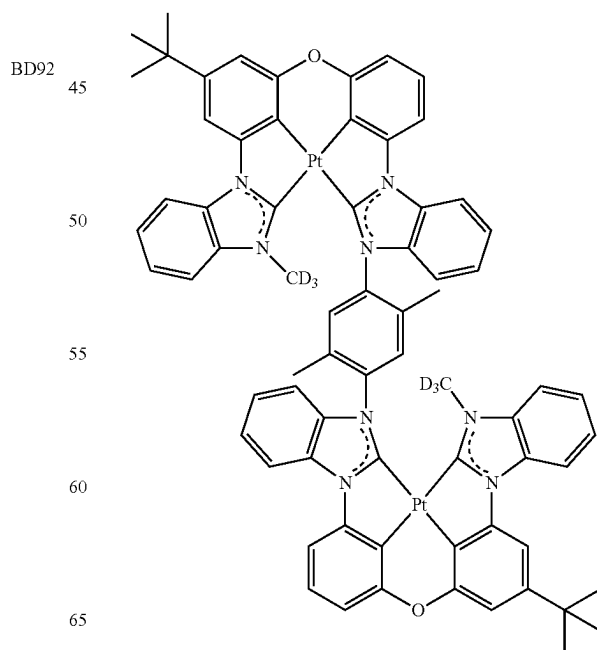
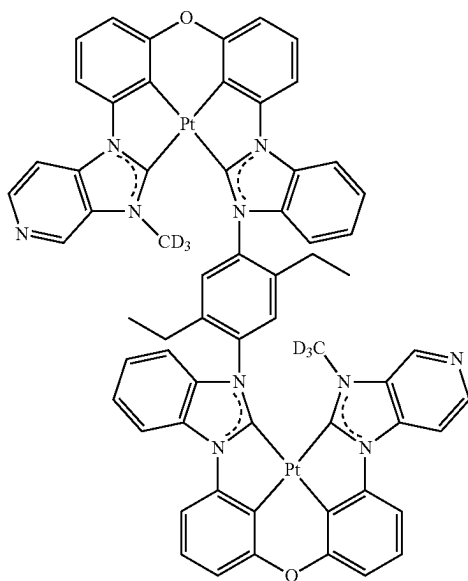
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BD91



BD93

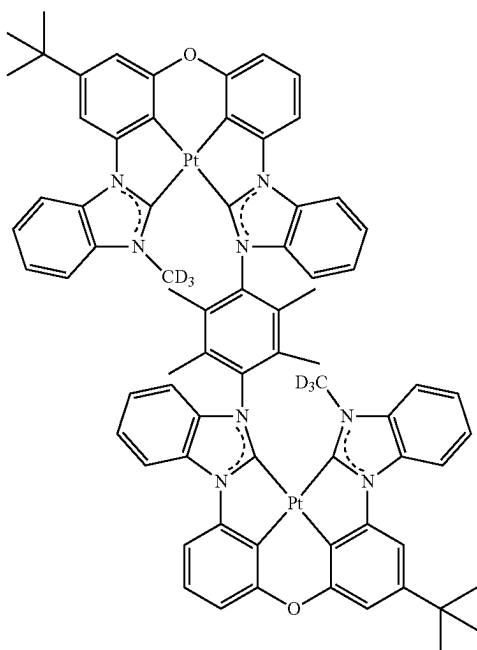
BD92



BD94

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**80**

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BD95

BD97

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BD96

BD98

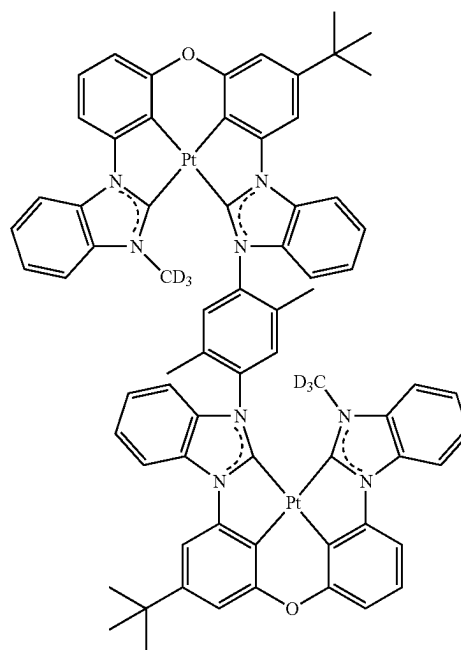
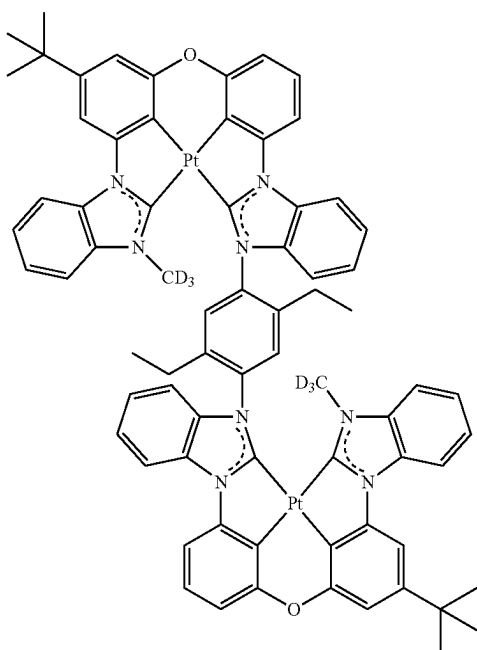
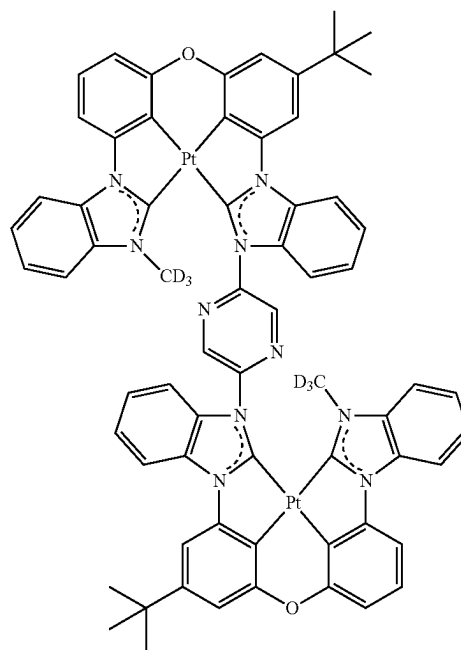
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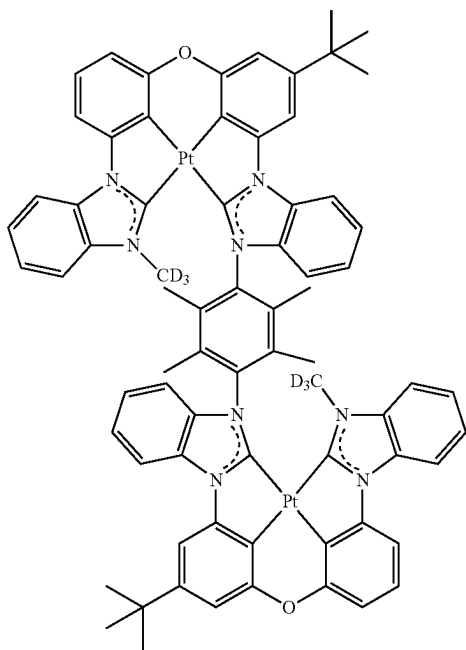
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**82**

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BD99

BD101

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BD100

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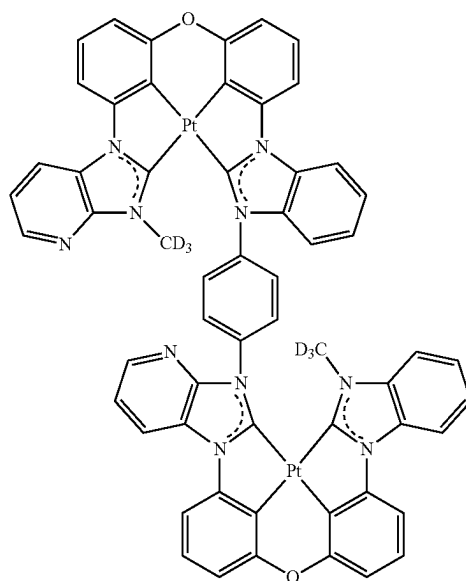
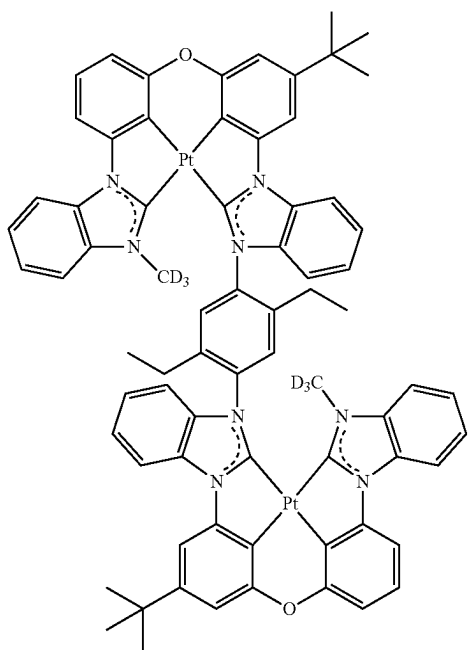
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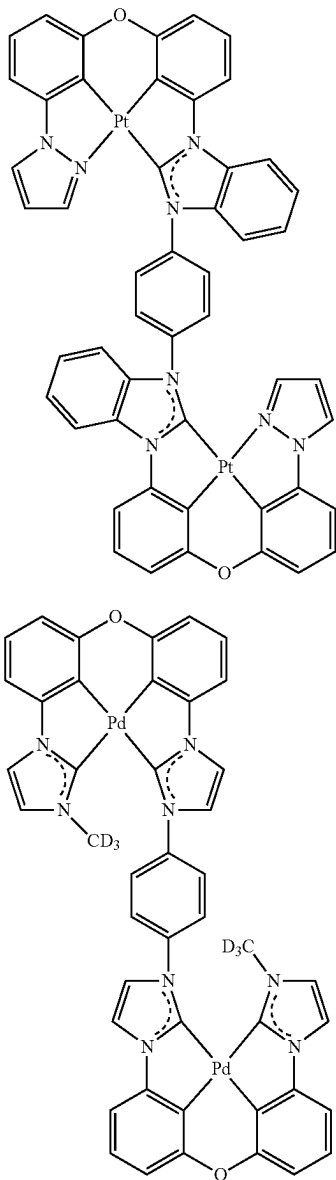
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BD102



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In an embodiment, the organometallic compound may be electrically neutral.

The organometallic compound represented by Formula 1 may include two metal atoms, and ligands coordinated with the metal atoms may be linked to each other, resulting in a rigid structure that leads to excellent molecular stability. Since the organometallic compound represented by Formula 1 has a structure in which angles of the ligands are tilted with respect to the metal atoms, excimer formation may be inhibited. Accordingly, a light-emitting device using the organometallic compound may have a low driving voltage and excellent luminance, efficiency, and lifespan.

The organometallic compound may emit blue light. In an embodiment, the organometallic compound may emit blue light (bottom emission CIE_{x,y} color coordinates 0.05 to 0.06) having a maximum emission wavelength in a range of about 430 nm to about 470 nm, but embodiments of the disclosure are not limited thereto. Accordingly, the organometallic compound represented by Formula 1 may be useful for the manufacture of a light-emitting device that emits blue light.

84

BD103

Synthesis methods of the organometallic compound represented by Formula 1 may be recognizable by one of ordinary skill in the art by referring to Examples provided below.

At least one of such organometallic compounds represented by Formula 1 may be used between a pair of electrodes of a light-emitting device. In an embodiment, the organometallic compound may be included in an emission layer. The organometallic compound included in the emission layer may act as a dopant. In embodiments, the organometallic compound represented by Formula 1 may be used as a material for a capping layer located outside a pair of electrodes of a light-emitting device.

Accordingly, provided is a light-emitting device which may include a first electrode, a second electrode facing the first electrode, an interlayer disposed between the first electrode and the second electrode and including an emission layer, and at least one organometallic compound represented by Formula 1.

BD104

The expression “(an interlayer) includes at least one of organometallic compounds” used herein may include a case in which “(an interlayer) includes identical organometallic compounds represented by Formula 1” and a case in which “(an interlayer) includes two or more different organometallic compounds represented by Formula 1.”

In an embodiment, the interlayer may include, as the organometallic compound, only Compound BD1. In this regard, Compound BD1 may exist in the emission layer of the light-emitting device. In embodiments, the interlayer may include, as the organometallic compound, Compound BD1 and Compound BD2. In this regard, Compound BD1 and Compound BD2 may exist in an identical layer (for example, both Compound BD1 and Compound BD2 may exist in an emission layer), or different layers (for example, Compound BD1 may exist in an emission layer and Compound BD2 may exist in an electron transport region).

In an embodiment, the first electrode of the light-emitting device may be an anode, the second electrode of the light-emitting device may be a cathode, the interlayer may further include a hole transport region located between the first electrode and the emission layer and an electron transport region located between the emission layer and the second electrode, the hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof, and the electron transport region may include a hole blocking layer, an electron transport layer, an electron injection layer, or any combination thereof.

The term “interlayer” as used herein refers to a single layer and/or all layers between a first electrode and a second electrode of a light-emitting device. A material included in the “interlayer” is not limited to an organic material.

In an embodiment, the emission layer may include the organometallic compound represented by Formula 1. In an embodiment, the emission layer may include a host and a dopant, and the dopant may include the organometallic compound represented by Formula 1. An amount of the host in the emission layer may be greater than an amount of the organometallic compound in the emission layer.

In an embodiment, an amount of the host may be in a range of about 0.1 parts by weight to about 99 parts by weight, based on 100 parts by weight of the emission layer. In an embodiment, an amount of the host may be in a range of about 0.5 parts by weight to about 95 parts by weight, based on 100 parts by weight of the emission layer. For example, an amount of the host may be in a range of about

85

0.1 parts by weight to about 90 parts by weight, based on 100 parts by weight of the emission layer.

The emission layer may emit red light, green light, blue light, and/or white light. In an embodiment, the emission layer may emit blue light having a maximum emission wavelength in a range of about 410 nm to about 500 nm.

In an embodiment, the hole transport region may include an electron blocking layer, and the electron blocking layer may include the organometallic compound; or the electron transport region may include a hole blocking layer, and the hole blocking layer may include the organometallic compound.

In an embodiment, the hole transport region may include a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level equal to or less than about -3.5 eV.

In an embodiment, the electronic transport region may include an electron transport layer and an electron injection layer, and at least one layer of the electron transport layer and the electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof.

Another aspect of the disclosure provides an electronic apparatus which may include the light-emitting device. The electronic apparatus may further include a thin-film transistor. In an embodiment, the electronic apparatus may further include a thin-film transistor including a source electrode and a drain electrode, and the first electrode of the light-emitting device may be electrically connected to the source electrode or the drain electrode. In an embodiment, the electronic apparatus may further include a color filter, a color conversion layer, a touch screen layer, a polarizing layer, or any combination thereof. More details on the electronic apparatus are the same as described in the specification.

[Description of FIG. 1]

FIG. 1 is a schematic cross-sectional view of a light-emitting device 10 according to an embodiment. The light-emitting device 10 includes a first electrode 110, an interlayer 130, and a second electrode 150.

Hereinafter, the structure of the light-emitting device 10 according to an embodiment and a method of manufacturing the light-emitting device 10 will be described in connection with FIG. 1.

[First Electrode 110]

In FIG. 1, a substrate may be further included under the first electrode 110 or above the second electrode 150. The substrate may be a glass substrate or a plastic substrate. In embodiments, the substrate may be a flexible substrate, and may include plastics with excellent heat resistance and durability, such as polyimide, polyethylene terephthalate (PET), polycarbonate, polyethylene naphthalate, polyarylate (PAR), polyetherimide, or any combination thereof.

The first electrode 110 may be formed by, for example, depositing or sputtering a material for forming the first electrode 110 on the substrate. When the first electrode 110 is an anode, a material for forming the first electrode 110 may be a high work function material that facilitates injection of holes.

The first electrode 110 may be a reflective electrode, a semi-transmissive electrode, or a transmissive electrode. When the first electrode 110 is a transmissive electrode, a material for forming the first electrode 110 may include indium tin oxide (ITO), indium zinc oxide (IZO), tin oxide (SnO₂), zinc oxide (ZnO), or any combinations thereof. In

86

embodiments, when the first electrode 110 is a semi-transmissive electrode or a reflective electrode, magnesium (Mg), silver (Ag), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), or any combinations thereof may be used as a material for forming a first electrode 110.

The first electrode 110 may have a structure consisting of a single layer or a structure including multiple layers. In an embodiment, the first electrode 110 may have a three-layered structure of ITO/Ag/ITO.

[Interlayer 130]

The interlayer 130 may be disposed on the first electrode 110. The interlayer 130 may include an emission layer.

The interlayer 130 may further include a hole transport region between the first electrode 110 and the emission layer and an electron transport region between the emission layer and the second electrode 150.

The interlayer 130 may further include, in addition to various organic materials, metal-containing compounds such as organometallic compounds, inorganic materials such as quantum dots, and the like.

In embodiments, the interlayer 130 may include two or more light-emitting units sequentially stacked between the first electrode 110 and the second electrode 150 and a charge generation layer between the two or more light-emitting units. When the interlayer 130 includes the two or more light-emitting units and the charge generation layer as described above, the light-emitting device 10 may be a tandem light-emitting device.

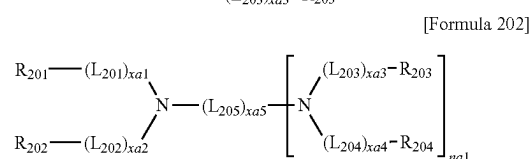
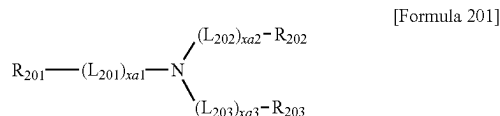
[Hole Transport Region in Interlayer 130]

The hole transport region may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer consisting of different materials, or a multi-layered structure including layers including different materials.

The hole transport region may include a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or any combination thereof.

In an embodiment, the hole transport region may have a multi-layered structure including a hole injection layer/hole transport layer structure, a hole injection layer/hole transport layer/emission auxiliary layer structure, a hole injection layer/emission auxiliary layer structure, a hole transport layer/emission auxiliary layer structure, or a hole injection layer/hole transport layer/emission auxiliary layer structure, wherein, in each structure, layers may be stacked from the first electrode 110 in its respective stated order, but embodiments are not limited thereto.

The hole transport region may include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof:



In Formulae 201 and 202,

L₂₀₁ to L₂₀₄ may each independently be a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least

87

one R_{10a} or a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} ,

L_{205} may be $^*-O-^*$, $^*-S-^*$, $^*-N(Q_{201})-^*$, a C_1 - C_{20} alkylene group unsubstituted or substituted with at least one R_{10a} , a C_2 - C_{20} alkenylene group unsubstituted or substituted with at least one R_{10a} , a C_3 - C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} , or a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} ,

$xa1$ to $xa4$ may each independently be an integer from 0 to 5,

$xa5$ may be an integer from 1 to 10,

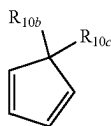
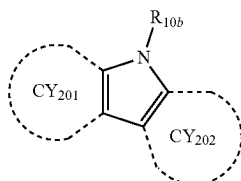
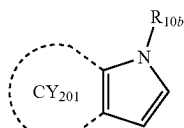
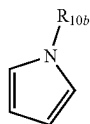
R_{201} to R_{204} and Q_{201} may each independently be a C_3 - C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} ,

R_{201} and R_{202} may optionally be linked to each other, via a single bond, a C_1 - C_5 alkylene group unsubstituted or substituted with at least one R_{10a} , or a C_2 - C_5 alkenylene group unsubstituted or substituted with at least one R_{10a} , to form a C_8 - C_{60} polycyclic group (for example, a carbazole group or the like) unsubstituted or substituted with at least one R_{10a} (for example, Compound HT16),

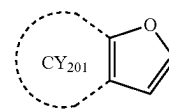
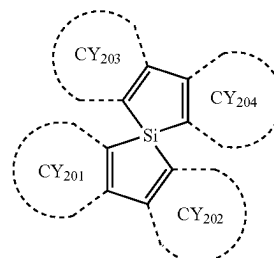
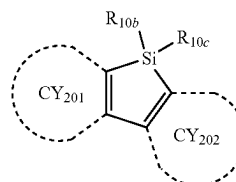
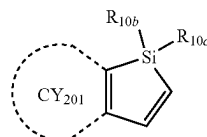
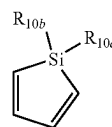
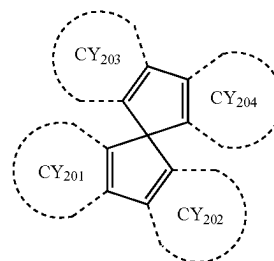
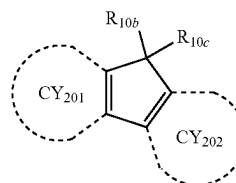
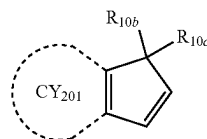
R_{203} and R_{204} may optionally be linked to each other, via a single bond, a C_1 - C_5 alkylene group unsubstituted or substituted with at least one R_{10a} , or a C_2 - C_5 alkenylene group unsubstituted or substituted with at least one R_{10a} , to form a C_8 - C_{60} polycyclic group unsubstituted or substituted with at least one R_{10a} ,

$na1$ may be an integer from 1 to 4, and * and * each indicate a binding site to a neighboring atom. R_{10a} may be the same as described in the specification.

In an embodiment, each of Formulae 201 and 202 may include at least one of groups represented by Formulae CY201 to CY217:

**88**

-continued



CY205

CY206

CY207

CY208

CY209

CY210

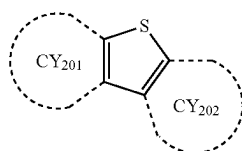
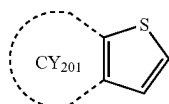
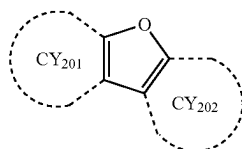
CY211

CY212

CY213

89

-continued



In Formulae CY201 to CY217, R_{10b} and R_{10c} may each independently be the same as described in connection with R_{10a} , ring CY₂₀₁ to ring CY₂₀₄ may each independently be a C₃-C₂₀ carbocyclic group or a C₁-C₂₀ heterocyclic group,

90

CY214

and at least one hydrogen in Formulae CY201 to CY217 may be unsubstituted or substituted with R_{10a} . R_{10a} may be the same as described in the specification.

In an embodiment, ring CY₂₀₁ to ring CY₂₀₄ in Formulae CY201 to CY217 may each independently be a benzene group, a naphthalene group, a phenanthrene group, or an anthracene group.

In embodiments, each of Formulae 201 and 202 may include at least one of groups represented by Formulae CY201 to CY203.

CY215

In embodiments, Formula 201 may include at least one of groups represented by Formulae CY201 to CY203 and at least one of groups represented by Formulae CY204 to CY217.

In embodiments, in Formula 201, xa_1 may be 1, R_{201} may be a group represented by one of Formulae CY201 to CY203, xa_2 may be 0, and R_{202} may be a group represented by one of Formulae CY204 to CY207.

CY216

In embodiments, each of Formulae 201 and 202 may not include a group represented by one of Formulae CY201 to CY203.

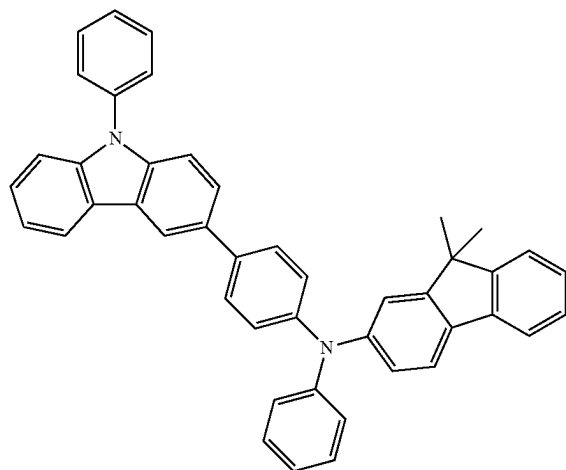
In embodiments, each of Formulae 201 and 202 may not include a group represented by one of Formulae CY201 to CY203, and may include at least one of groups represented by Formulae CY204 to CY217.

CY217

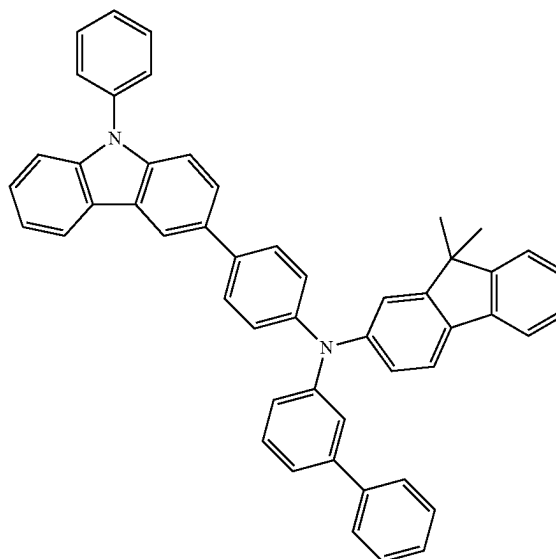
In embodiments, each of Formulae 201 and 202 may not include a group represented by one of Formulae CY201 to CY217.

In an embodiment, the hole transport region may include one of Compounds HT1 to HT46, m-MTDATA, TDATA, 2-TNATA, NPB(NPD), β -NPB, TPD, Spiro-TPD, Spiro-NPB, methylated-NPB, TAPC, HMTPD, 4,4',4"-tris(N-carbazolyl)triphenylamine (TCTA), polyaniline/dodecylbenzenesulfonic acid (PANI/DBSA), poly(3,4-ethylenedioxythiophene)/poly(4-styrenesulfonate) (PEDOT/PSS), polyaniline/camphor sulfonic acid (PANI/CSA), polyaniline/poly(4-styrenesulfonate) (PANI/PSS), or any combination thereof:

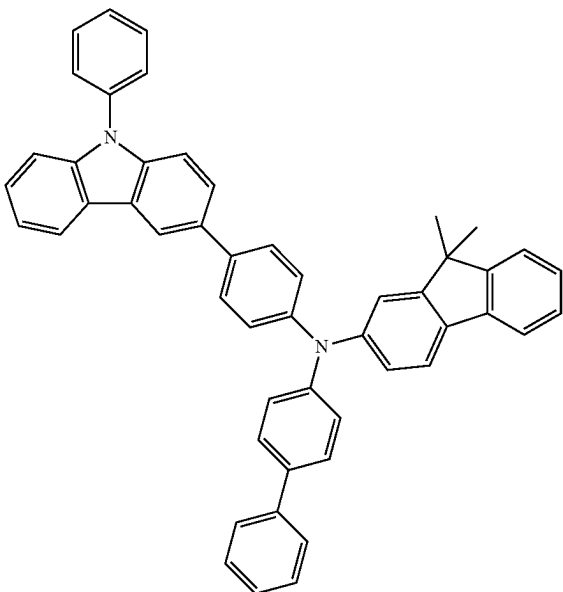
HT1



HT2

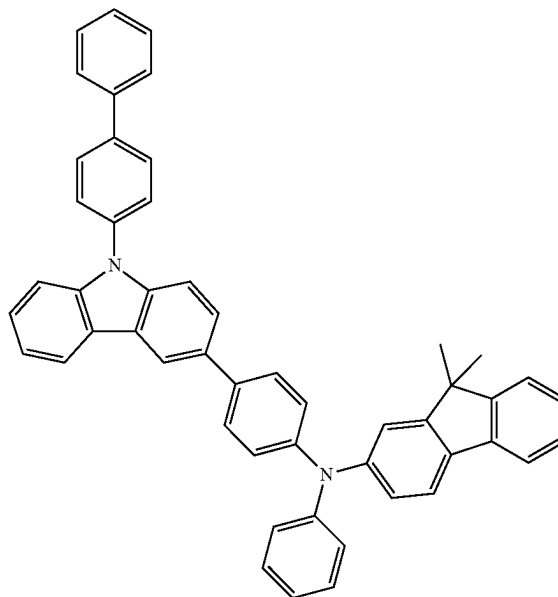


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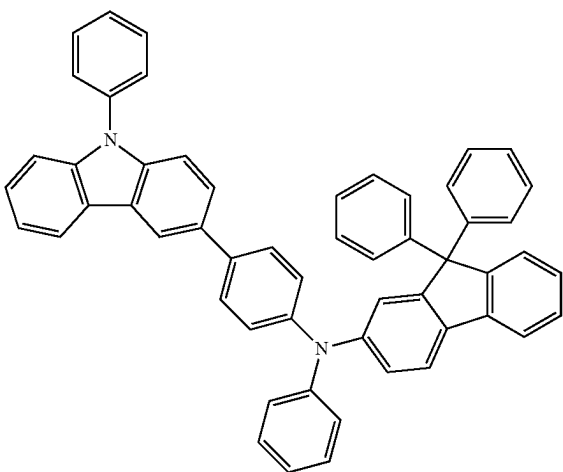
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HT3

92

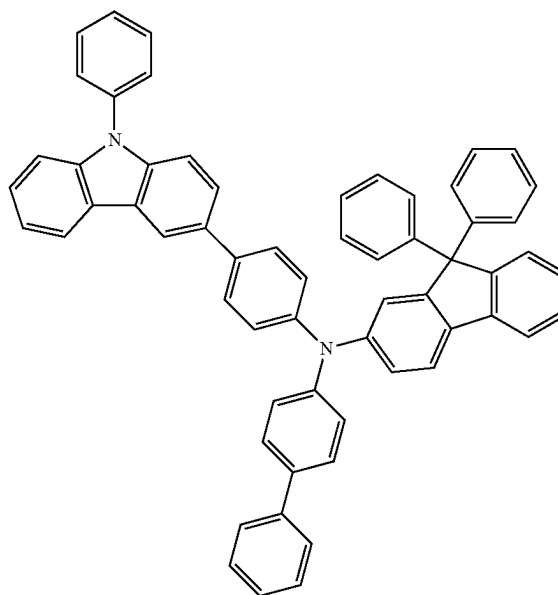
HT4



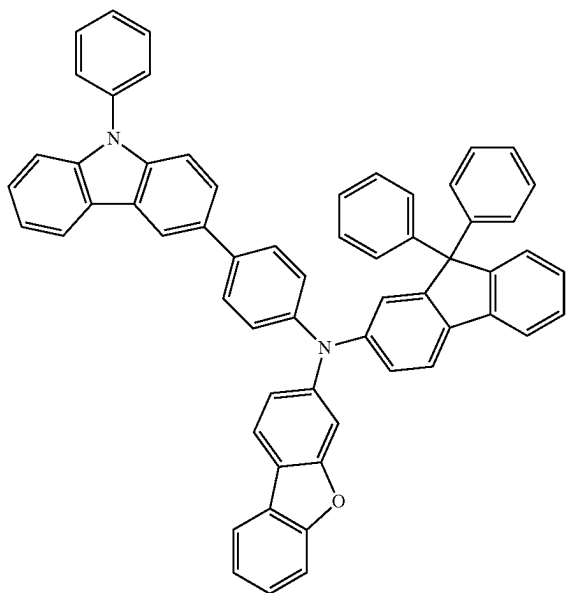
HT5



HT6

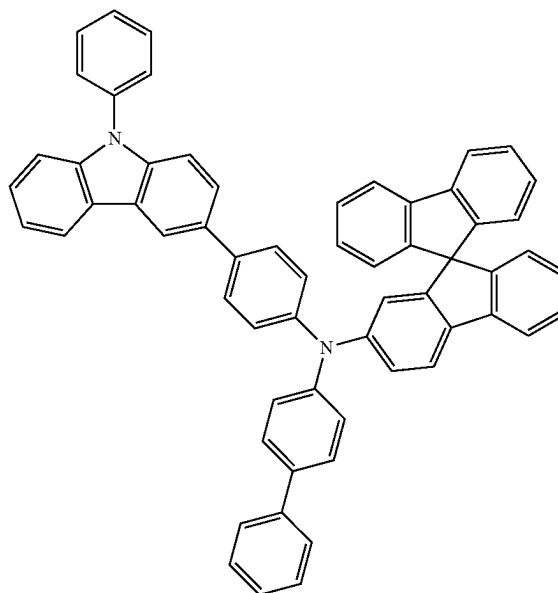


93

-continued
HT7

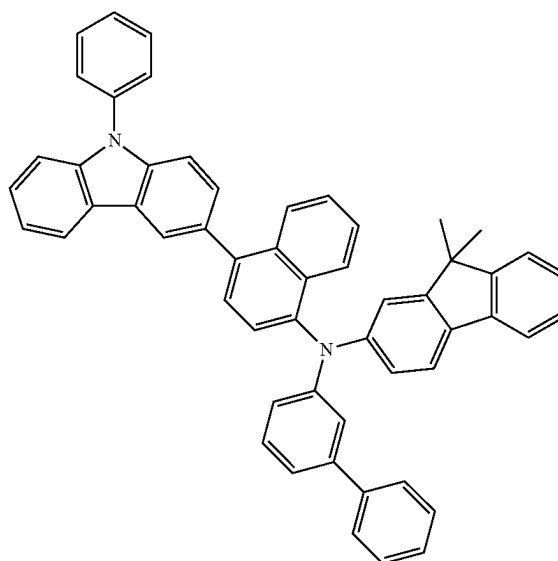
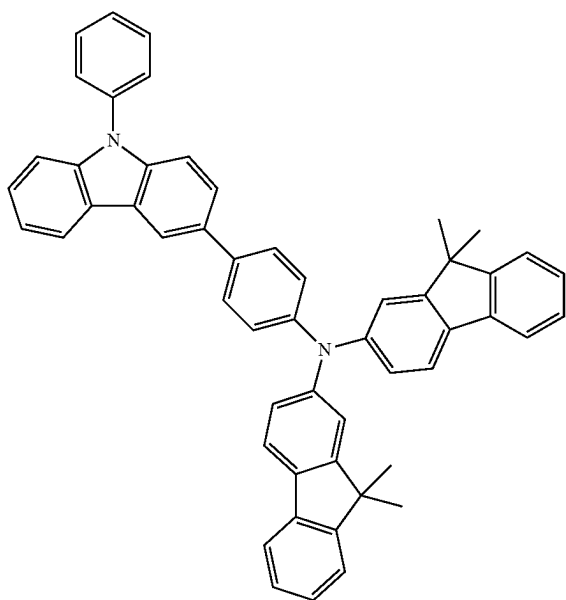
94

HT8

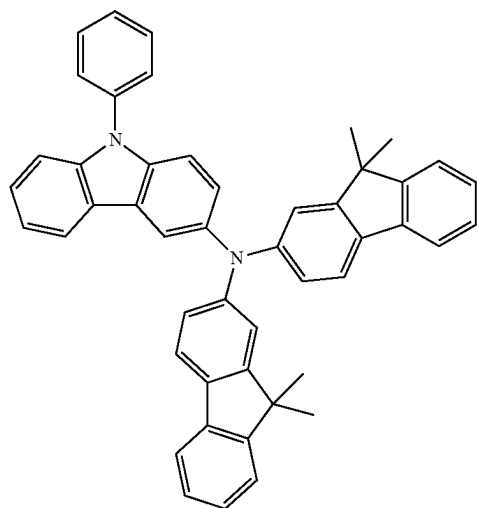


HT9

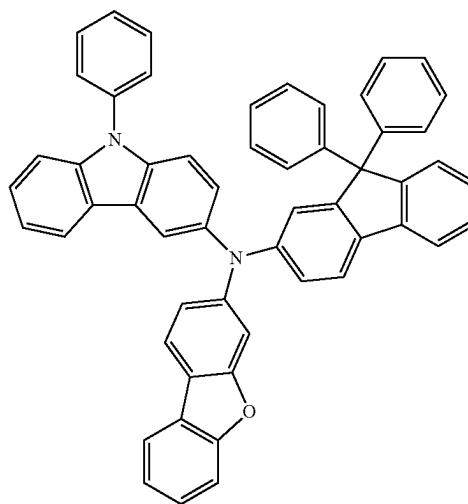
HT10



95

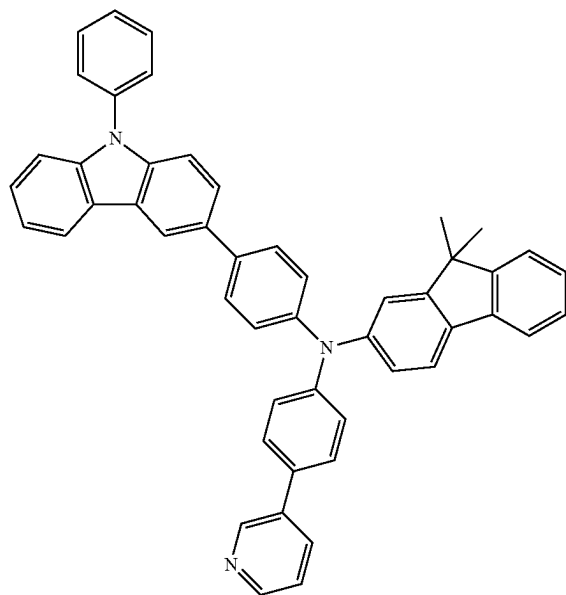
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HT11

96

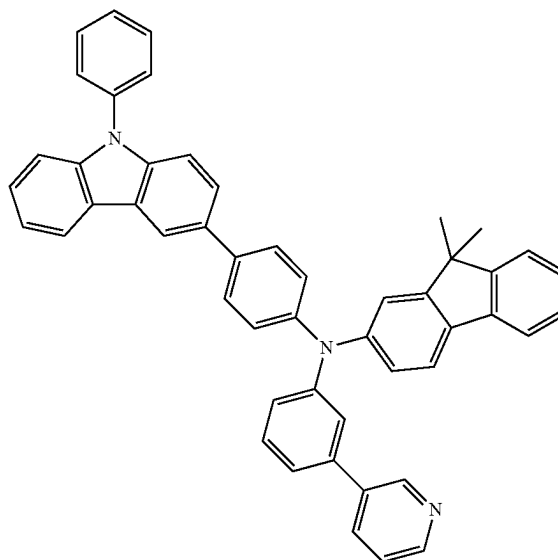


HT12

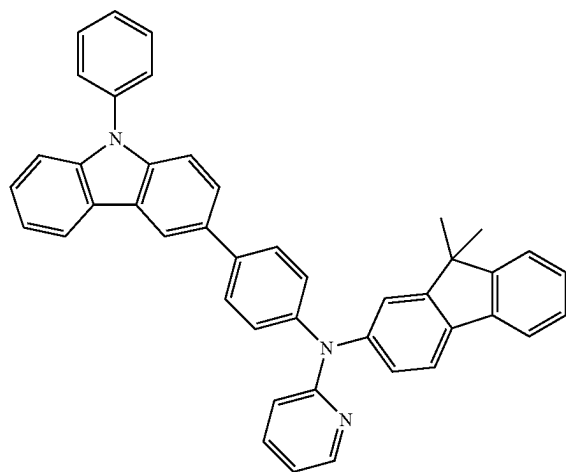
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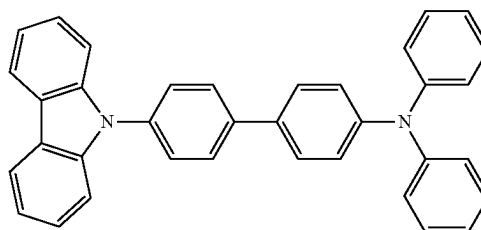
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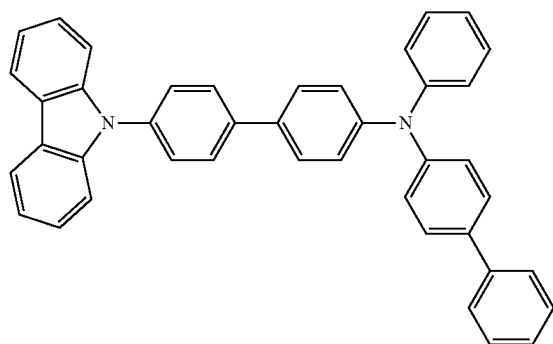
HT15



HT16

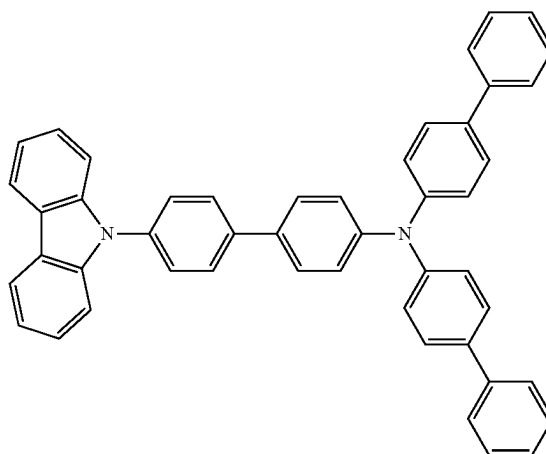


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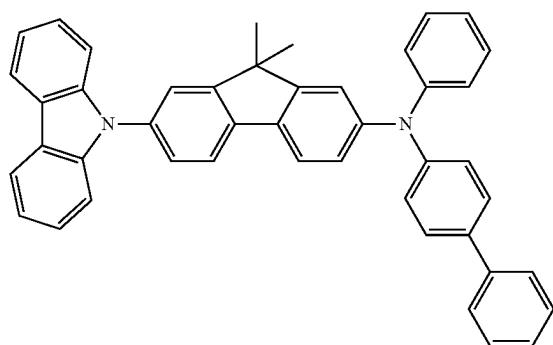
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HT17

98

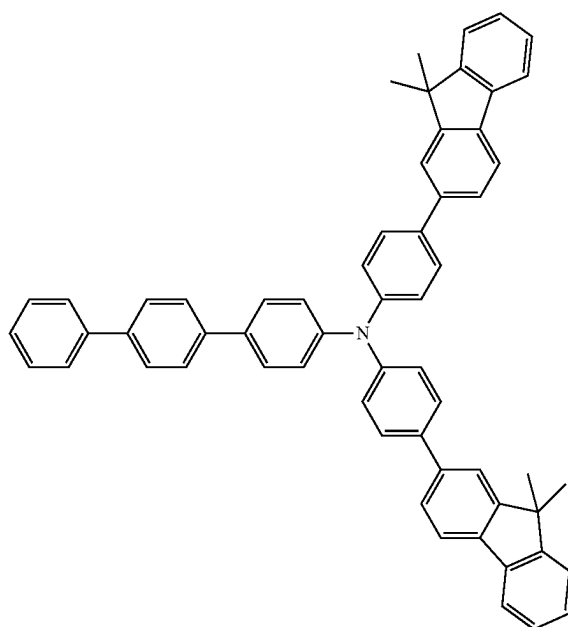
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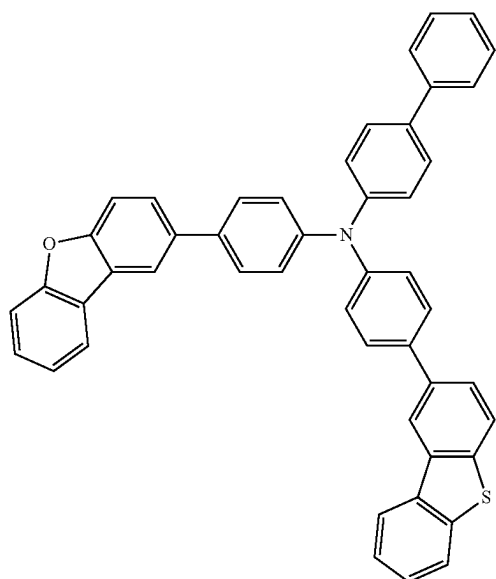
HT19



HT20

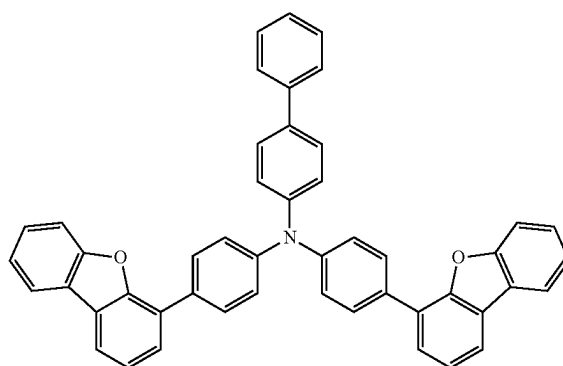


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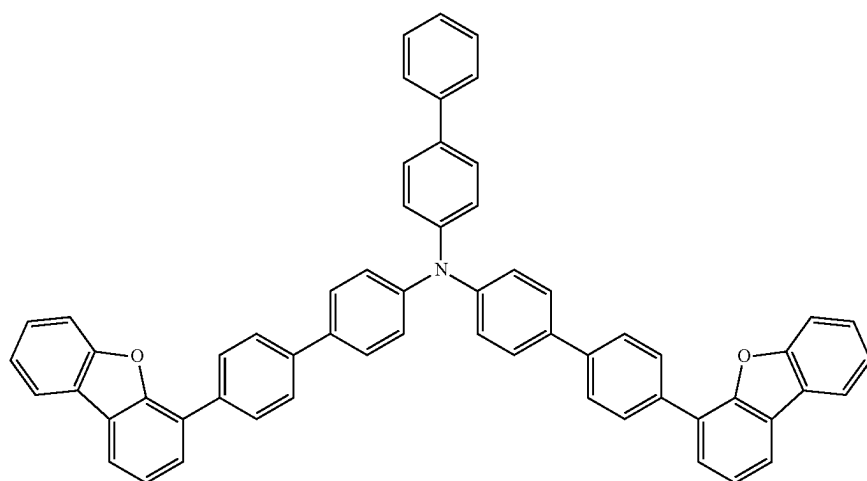


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HT21

100

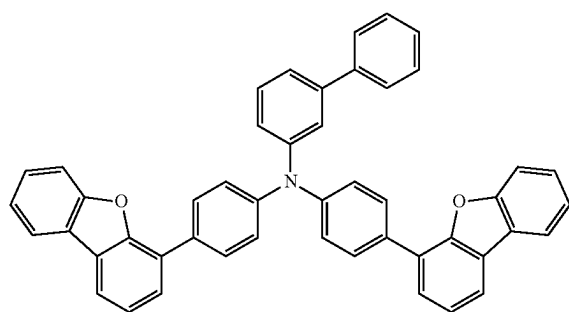


HT22

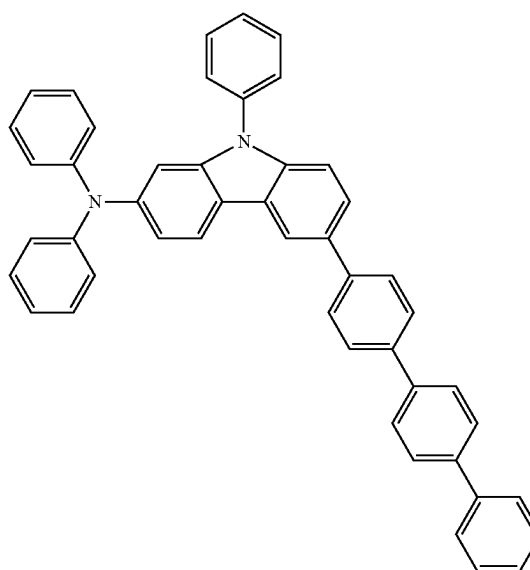


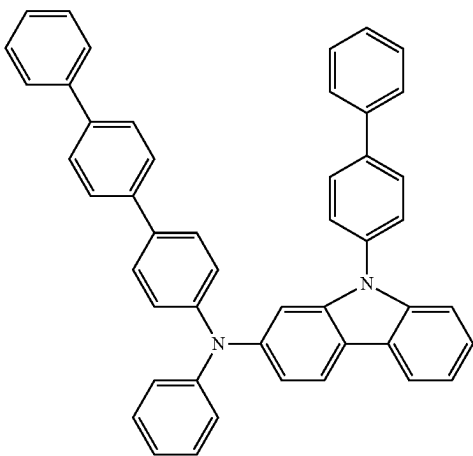
HT23

HT24

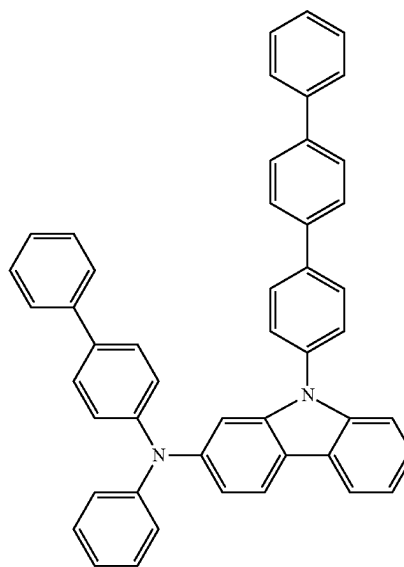


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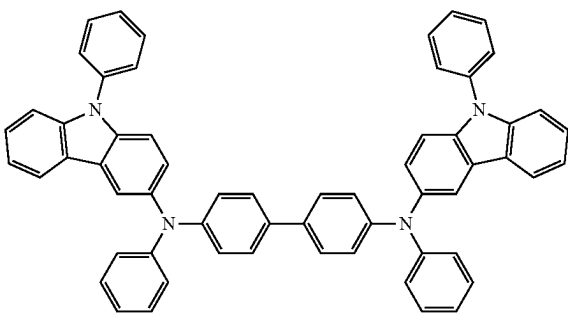


101-continued
HT26**102**

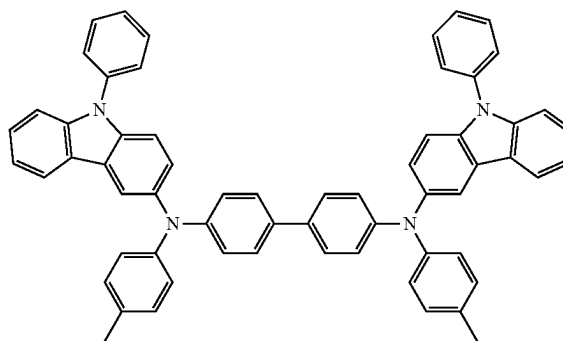
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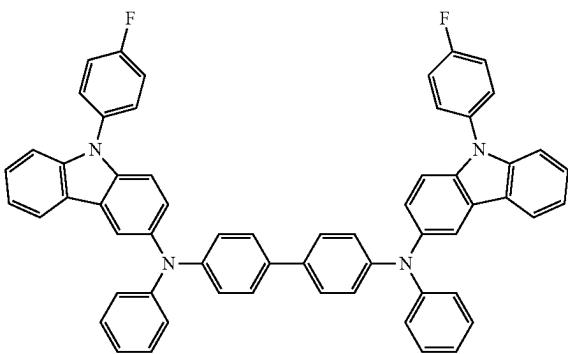
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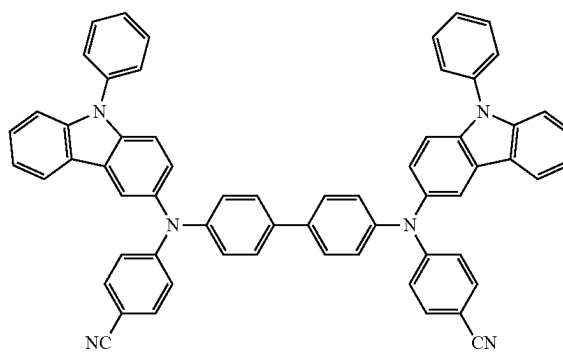
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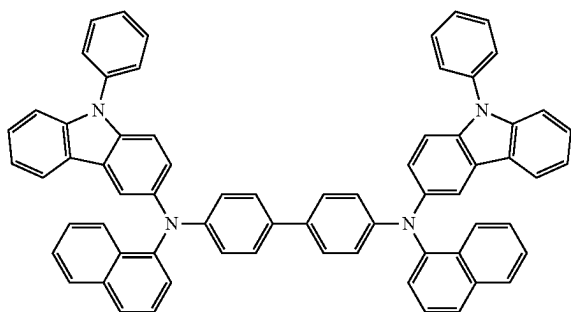
HT30



HT31

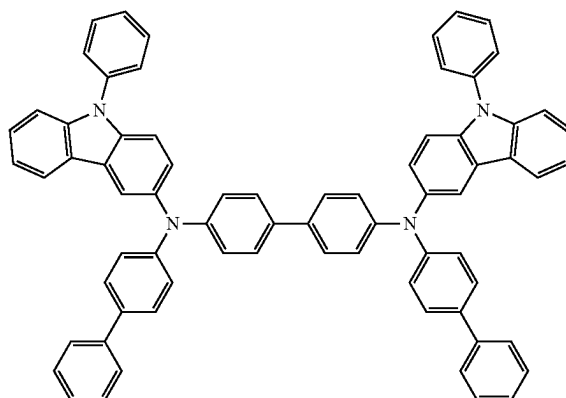


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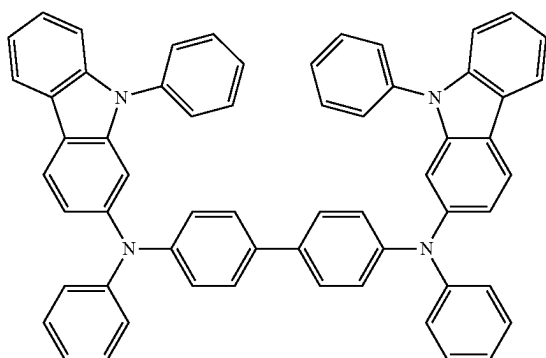
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HT32

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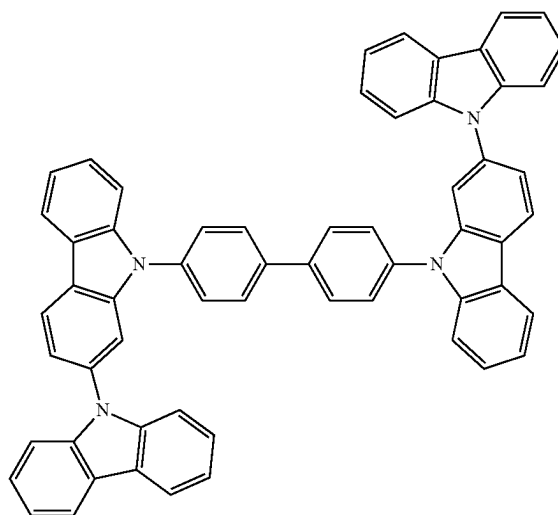
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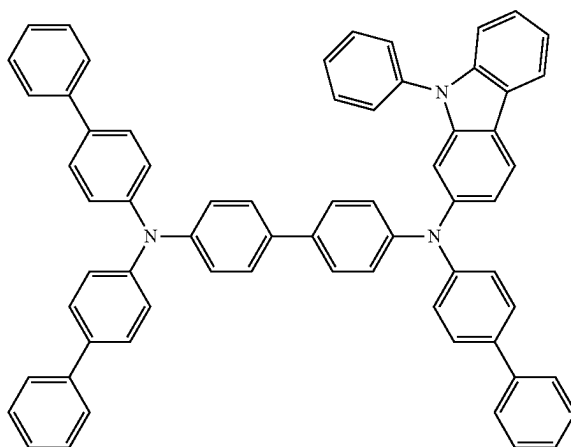
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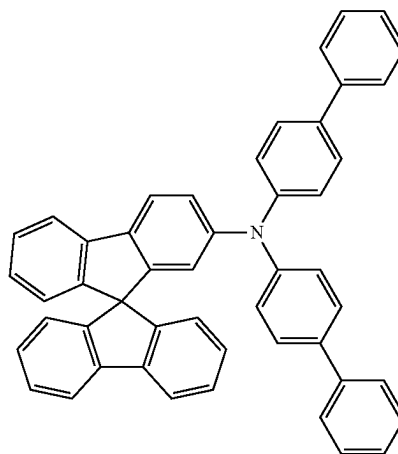
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HT36

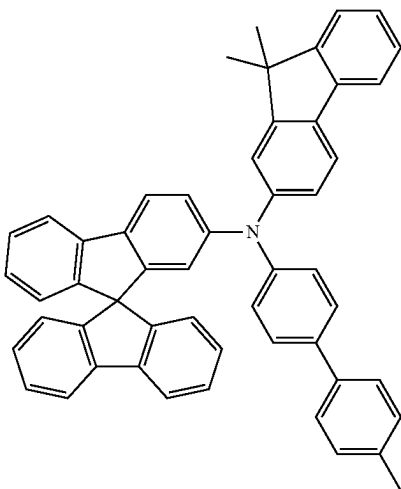


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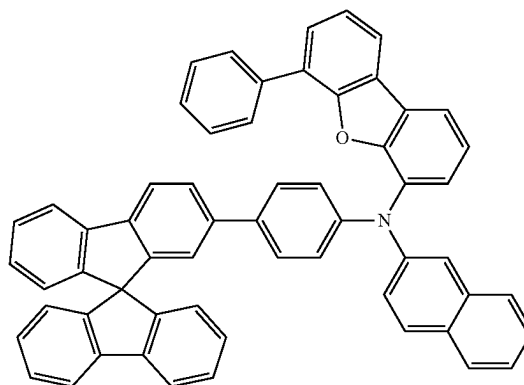
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HT38



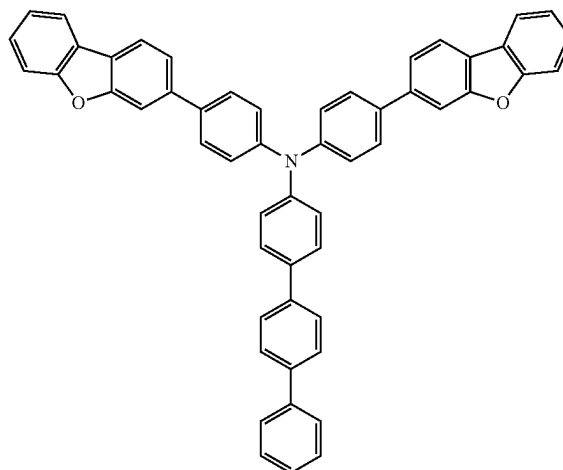
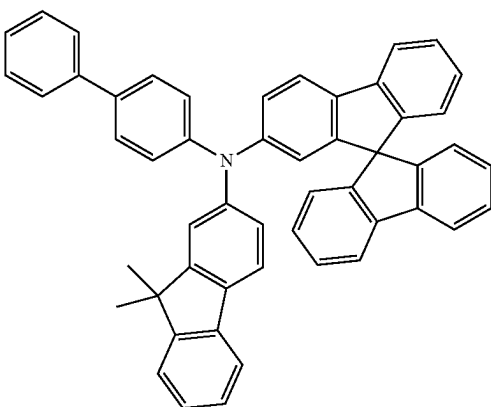
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HT39



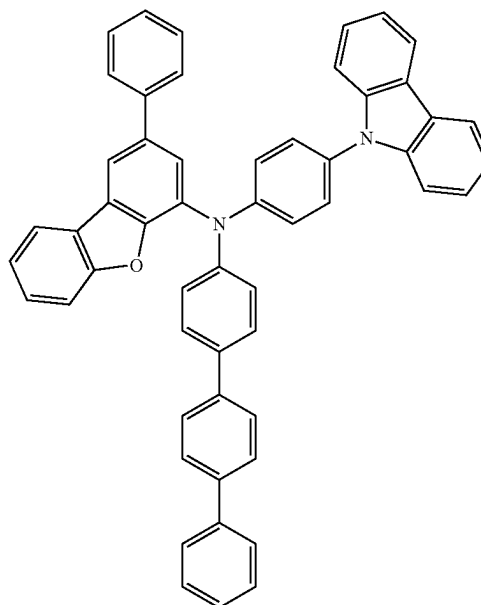
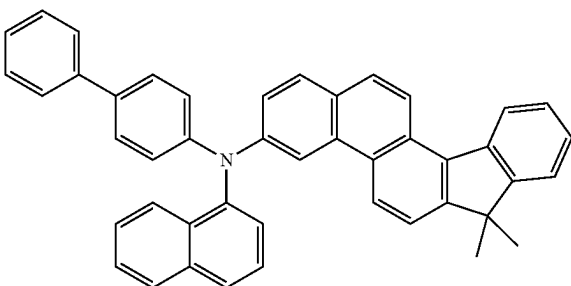
HT40

HT41

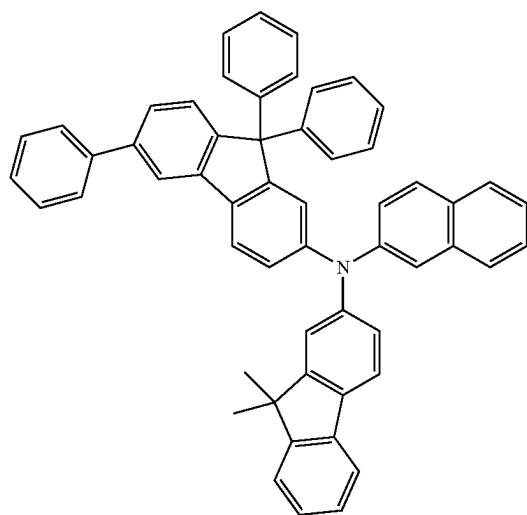


HT42

HT43

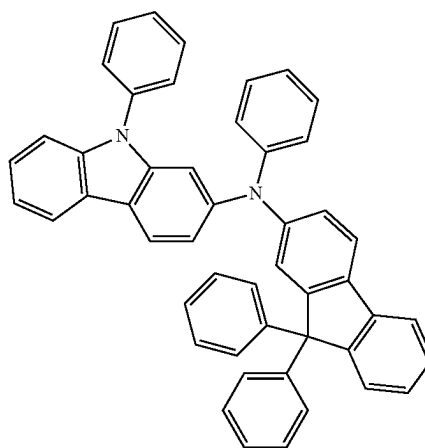


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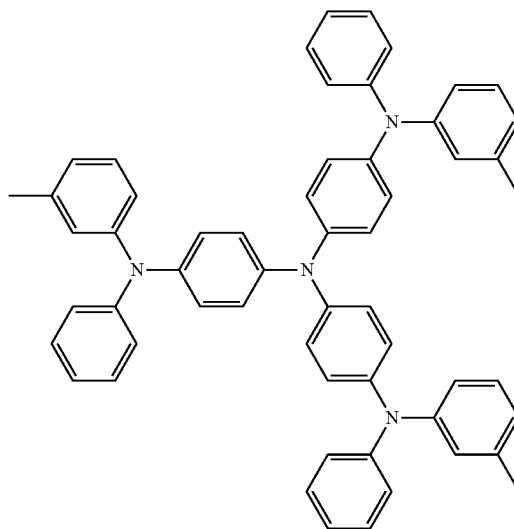
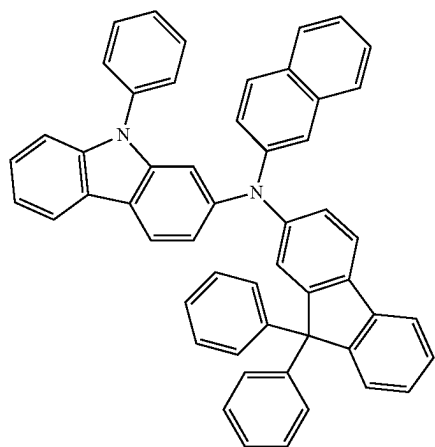
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HT44

108

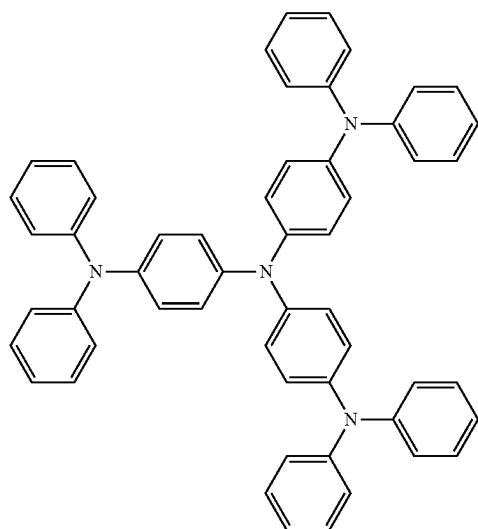


HT45

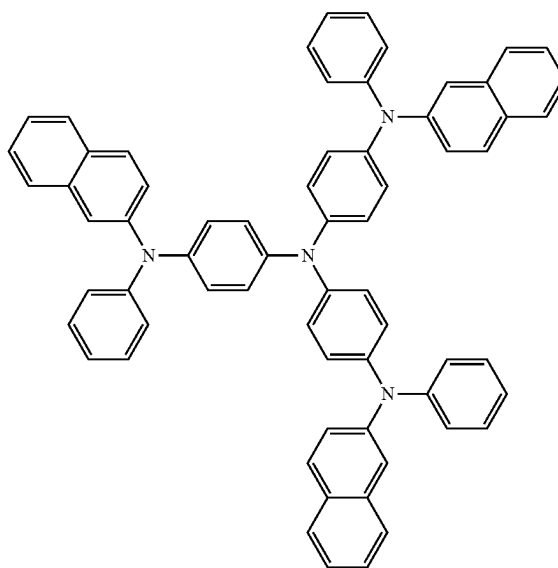
HT46



m-MTDATA



TDATA

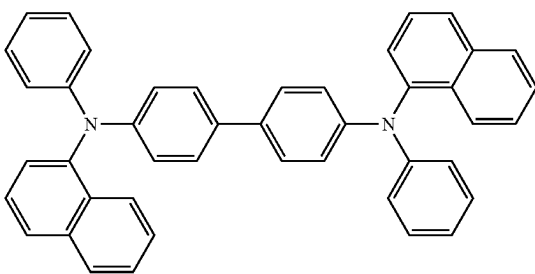


2-TNATA

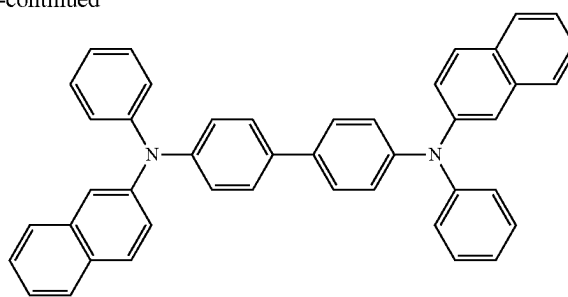
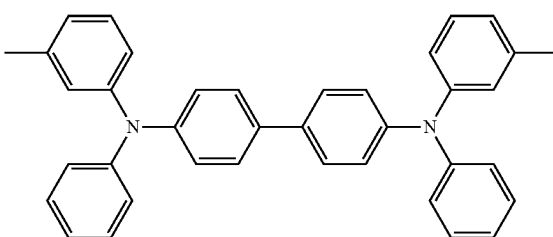
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110

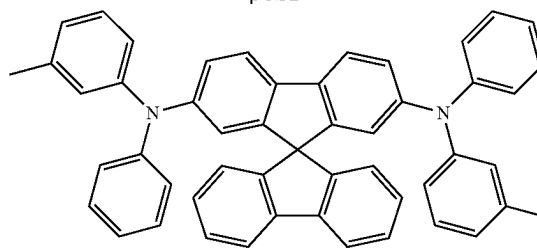
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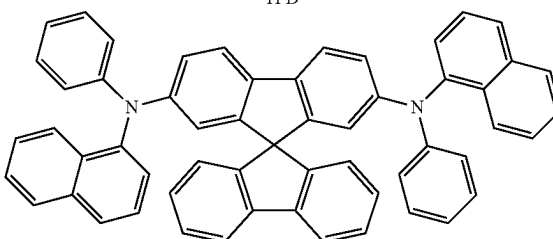
NPB

 β -NPB

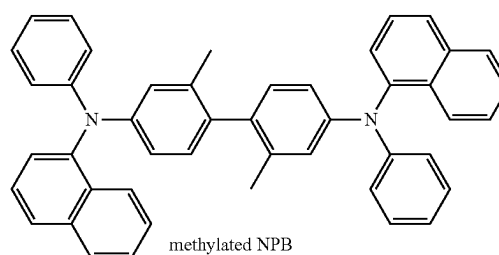
TPD



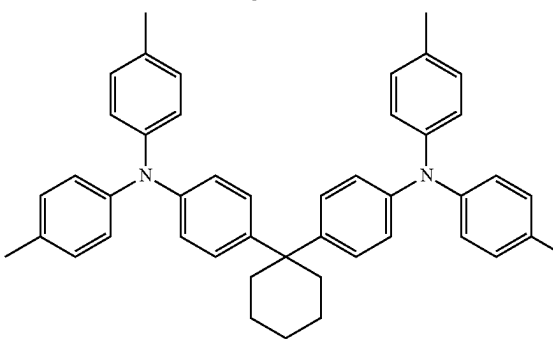
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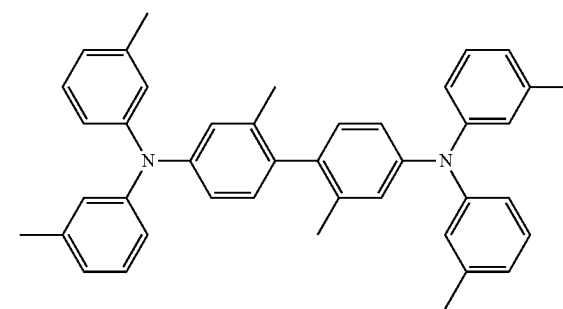
Spiro-NPB



methylated NPB



TAPC



HMTDP

A thickness of the hole transport region may be in a range of about 50 Å to about 10,000 Å. For example, the thickness of the hole transport region may be in a range of about 100 Å to about 4,000 Å. When the hole transport region includes a hole injection layer, a hole transport layer, or any combination thereof, a thickness of the hole injection layer may be in a range of about 100 Å to about 9,000 Å, and a thickness of the hole transport layer may be in a range of about 50 Å to about 2,000 Å. For example, the thickness of the hole injection layer may be in a range of 100 Å to about 1,000 Å. For example, the thickness of the hole transport layer may be in a range of about 100 Å to about 1,500 Å. When the thicknesses of the hole transport region, the hole injection layer, and the hole transport layer are within these ranges, satisfactory hole transporting characteristics may be obtained without a substantial increase in driving voltage.

The emission auxiliary layer may increase light-emission efficiency by compensating for an optical resonance distance

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according to a wavelength of light emitted by an emission layer, and the electron blocking layer may block the leakage of electrons from an emission layer to a hole transport region. Materials that may be included in the hole transport region may be included in the emission auxiliary layer and the electron blocking layer.

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[p-Dopant]

The hole transport region may further include, in addition to these materials, a charge-generation material for the improvement of conductive properties. The charge-generation material may be uniformly or non-uniformly dispersed in the hole transport region (for example, in the form of a single layer consisting of a charge-generation material).

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The charge-generation material may be, for example, a p-dopant.

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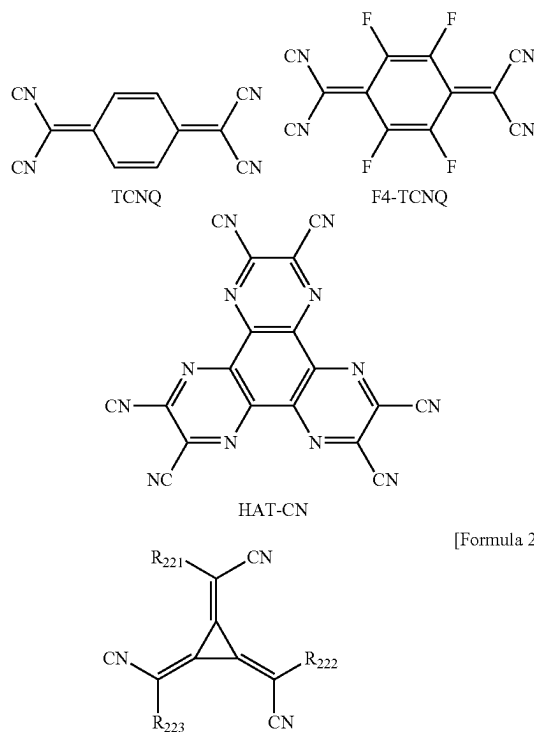
In an embodiment, a lowest unoccupied molecular orbital (LUMO) energy level of the p-dopant may be equal to or less than about -3.5 eV.

111

In an embodiment, the p-dopant may include a quinone derivative, a cyano group-containing compound, a compound containing element EL1 and element EL2, or any combination thereof.

Examples of the quinone derivative may include TCNQ and F4-TCNQ.

Examples of the cyano group-containing compound may include HAT-CN and a compound represented by Formula 221.



In Formula 221,

R₂₂₁ to R₂₂₃ may each independently be a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a}, wherein R_{10a} may be the same as described in the specification, and at least one of R₂₂₁ to R₂₂₃ may each independently be a C₃-C₆₀ carbocyclic group or a C₁-C₆₀ heterocyclic group, each substituted with: a cyano group; —F; —Cl; —Br; —I; a C₁-C₂₀ alkyl group substituted with a cyano group, —F, —Cl, —Br, —I, or any combination thereof; or any combination thereof.

In the compound containing element EL1 and element EL2, element EL1 may be a metal, a metalloid, or any combination thereof, and element EL2 may be a non-metal, a metalloid, or any combination thereof.

Examples of the metal may include an alkali metal (for example, lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), etc.); an alkaline earth metal (for example, beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), etc.); a transition metal (for example, titanium (Ti), zirconium (Zr), hafnium (Hf), vanadium (V), niobium (Nb), tantalum (Ta), chromium (Cr), molybdenum (Mo), tungsten (W), manganese (Mn), technetium (Tc), rhenium (Re), iron (Fe), ruthenium (Ru), osmium (Os), cobalt (Co), rhodium (Rh), iridium (Ir), nickel (Ni), palladium (Pd), platinum (Pt), copper (Cu), silver (Ag),

112

gold (Au), etc.); a post-transition metal (for example, zinc (Zn), indium (In), tin (Sn), etc.); and a lanthanide metal (for example, lanthanum (La), cerium (Ce), praseodymium (Pr), neodymium (Nd), promethium (Pm), samarium (Sm), europium (Eu), gadolinium (Gd), terbium (Tb), dysprosium (Dy), holmium (Ho), erbium (Er), thulium (Tm), ytterbium (Yb), lutetium (Lu), etc.).

Examples of the metalloid may include silicon (Si), antimony (Sb), and tellurium (Te).

Examples of the non-metal may include oxygen (O) and a halogen (for example, F, Cl, Br, I, etc.).

In an embodiment, examples of the compound containing element EL1 and element EL2 may include a metal oxide, a metal halide (for example, metal fluoride, metal chloride, metal bromide, or metal iodide), a metalloid halide (for example, metalloid fluoride, metalloid chloride, metalloid bromide, or metalloid iodide), a metal telluride, or any combination thereof.

Examples of the metal oxide may include tungsten oxide (for example, WO, W₂O₃, WO₂, WO₃, W₂O₅, etc.), vanadium oxide (for example, VO, V₂O₃, VO₂, V₂O₅, etc.), molybdenum oxide (for example, MoO, Mo₂O₃, MoO₂, MoO₃, Mo₂O₅, etc.), and rhenium oxide (for example, ReO₃, etc.).

Examples of the metal halide may include an alkali metal halide, an alkaline earth metal halide, a transition metal halide, a post-transition metal halide, and a lanthanide metal halide.

Examples of the alkali metal halide may include LiF, NaF, KF, RbF, CsF, LiCl, NaCl, KCl, RbCl, CsCl, LiBr, NaBr, KBr, RbBr, CsBr, LiI, NaI, KI, RbI, and CsI.

Examples of the alkaline earth metal halide may include BeF₂, MgF₂, CaF₂, SrF₂, BaF₂, BeCl₂, MgCl₂, CaCl₂, SrCl₂, BaCl₂, BeBr₂, MgBr₂, CaBr₂, SrBr₂, BaBr₂, BeI₂, MgI₂, CaI₂, SrI₂, and BaI₂.

Examples of the transition metal halide may include titanium halide (for example, TiF₄, TiCl₄, TiBr₄, TiI₄, etc.), zirconium halide (for example, ZrF₄, ZrCl₄, ZrBr₄, ZrI₄, etc.), hafnium halide (for example, HfF₄, HfCl₄, HfBr₄, HfI₄, etc.), vanadium halide (for example, VF₃, VCl₃, VBr₃, VI₃, etc.), niobium halide (for example, NbF₃, NbCl₃, NbBr₃, NbI₃, etc.), tantalum halide (for example, TaF₃, TaCl₃, TaBr₃, TaI₃, etc.), chromium halide (for example, CrF₃, CrCl₃, CrBr₃, CrI₃, etc.), molybdenum halide (for example, MoF₃, MoCl₃, MoBr₃, MoI₃, etc.), tungsten halide (for example, WF₃, WCl₃, WBr₃, WI₃, etc.), manganese halide (for example, MnF₂, MnCl₂, MnBr₂, MnI₂, etc.), technetium halide (for example, TcF₂, TcCl₂, TcBr₂, TcI₂, etc.), rhenium halide (for example, ReF₂, ReCl₂, ReBr₂, ReI₂, etc.), iron halide (for example, FeF₂, FeCl₂, FeBr₂, FeI₂, etc.), ruthenium halide (for example, RuF₂, RuCl₂, RuBr₂, RuI₂, etc.), osmium halide (for example, OsF₂, OsCl₂, OsBr₂, OsI₂, etc.), cobalt halide (for example, CoF₂, CoCl₂, CoBr₂, CoI₂, etc.), rhodium halide (for example, RhF₂, RhCl₂, RhBr₂, RhI₂, etc.), iridium halide (for example, IrF₂, IrCl₂, IrBr₂, IrI₂, etc.), nickel halide (for example, NiF₂, NiCl₂, NiBr₂, NiI₂, etc.), palladium halide (for example, PdF₂, PdCl₂, PdBr₂, PdI₂, etc.), platinum halide (for example, PtF₂, PtCl₂, PtBr₂, PtI₂, etc.), copper halide (for example, CuF, CuCl, CuBr, CuI, etc.), silver halide (for example, AgF, AgCl, AgBr, AgI, etc.), and gold halide (for example, AuF, AuCl, AuBr, AuI, etc.).

Examples of the post-transition metal halide may include zinc halide (for example, ZnF₂, ZnCl₂, ZnBr₂, ZnI₂, etc.), indium halide (for example, InI₃, etc.), and tin halide (for example, SnI₂, etc.).

Examples of the lanthanide metal halide may include YbF, YbF₂, YbF₃, SmF₃, YbCl, YbCl₂, YbCl₃, SmCl₃, YbBr, YbBr₂, YbBr₃, SmBr₃, YbI, YbI₂, YbI₃, and SmI₃.

113

Examples of the metalloid halide may include antimony halide (for example, SbCl_5 , etc.).

Examples of the metal telluride may include an alkali metal telluride (for example, Li_2Te , Na_2Te , K_2Te , Rb_2Te , Cs_2Te , etc.), an alkaline earth metal telluride (for example, BeTe , MgTe , CaTe , SrTe , BaTe , etc.), a transition metal telluride (for example, TiTe_2 , ZrTe_2 , HfTe_2 , V_2Te_3 , Nb_2Te_3 , Ta_2Te_3 , Cr_2Te_3 , Mo_2Te_3 , W_2Te_3 , MnTe , TcTe , ReTe , FeTe , RuTe , OsTe , CoTe , RhTe , IrTe , NiTe , PdTe , PtTe , CuTe , Ag_2Te , AgTe , Au_2Te , etc.), a post-transition metal telluride (for example, ZnTe , etc.), and a lanthanide metal telluride (for example, LaTe , CeTe , PrTe , NdTe , PmTe , EuTe , GdTe , TbTe , DyTe , HoTe , ErTe , TmTe , YbTe , LuTe , etc.).

[Emission Layer in Interlayer 130]

The light-emitting device according to an embodiment may include an emission layer in an interlayer. The emission layer may be the same as described above, or as described in connection with the light-emitting device below.

When the light-emitting device 10 is a full-color light-emitting device, the emission layer may be patterned into a red emission layer, a green emission layer, and/or a blue emission layer, according to a subpixel. In embodiments, the emission layer may have a stacked structure of two or more layers of a red emission layer, a green emission layer, and a blue emission layer, in which the two or more layers may contact each other or may be separated from each other. In embodiments, the emission layer may include two or more materials of a red light-emitting material, a green light-emitting material, and a blue light-emitting material, in which the two or more materials are mixed with each other in a single layer to emit white light.

The emission layer may include a host and a dopant. The dopant may include a phosphorescent dopant, a fluorescent dopant, or any combination thereof.

An amount of the dopant in the emission layer may be in a range of about 0.01 to about 15 parts by weight, based on 100 parts by weight of the host.

In embodiments, the emission layer may include a quantum dot.

114

A thickness of the emission layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the emission layer may be in a range of about 200 Å to about 600 Å. When the thickness of the emission layer is within this range, excellent light-emission characteristics may be obtained without a substantial increase in driving voltage. [Host]

The host may include a compound represented by Formula 301:



In Formula 301,

Ar_{301} and L_{301} may each independently be a C_3 - C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} ,

$xb11$ may be 1, 2, or 3,

$xb1$ may be an integer from 0 to 5,

R_{301} may be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C_1 - C_{60} alkyl group unsubstituted or substituted with at least one R_{10a} , a C_2 - C_{60} alkenyl group unsubstituted or substituted with at least one R_{10a} , a C_2 - C_{60} alkynyl group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{60} alkoxy group unsubstituted or substituted with at least one R_{10a} , a C_3 - C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} , —Si(Q_{301})(Q_{302})(Q_{303}), —N(Q_{301})(Q_{302}), —B(Q_{301})(Q_{302}), —C(=O)(Q_{301}), —S(=O)₂(Q_{301}), or —P(=O)(Q_{301})(Q_{302}),

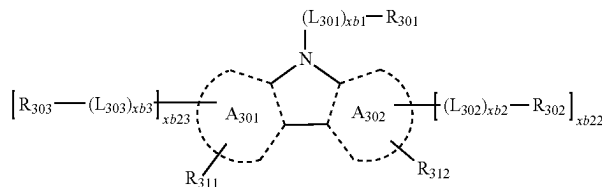
$xb21$ may be an integer from 1 to 5, and

Q_{301} to Q_{303} may each independently be the same as described in connection with Q_1 . R_{10a} may be the same as described in the specification.

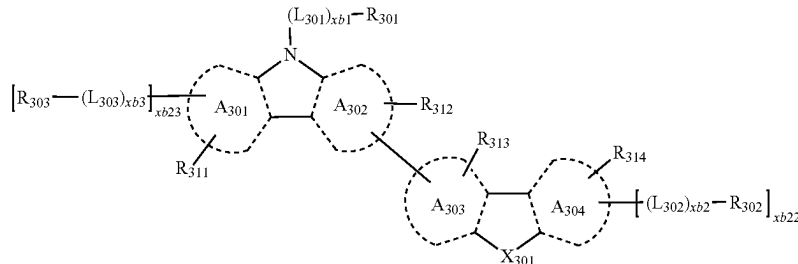
In an embodiment, when $xb11$ in Formula 301 is 2 or more, two or more of $\text{Ar}_{301}(\text{s})$ may be linked to each other via a single bond.

In embodiments, the host may include a compound represented by Formula 301-1, a compound represented by Formula 301-2, or any combination thereof:

[Formula 301-1]



[Formula 301-2]



The emission layer may include a delayed fluorescence material. The delayed fluorescence material may act as a host or a dopant in the emission layer.

In Formulae 301-1 and 301-2,

ring A_{301} to ring A_{304} may each independently be a C_3 - C_{60} carbocyclic group unsubstituted or substituted with

115

at least one R_{10a} or a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} .

X_{301} may be O, S, N— $[(L_{304})_{xb4}-R_{304}]$, $C(R_{304})(R_{305})$, or $Si(R_{304})(R_{305})$.

$xb22$ and $xb23$ may each independently be 0, 1, or 2, 5

L_{301} , $xb1$, and R_{301} may each respectively be the same as described in the specification,

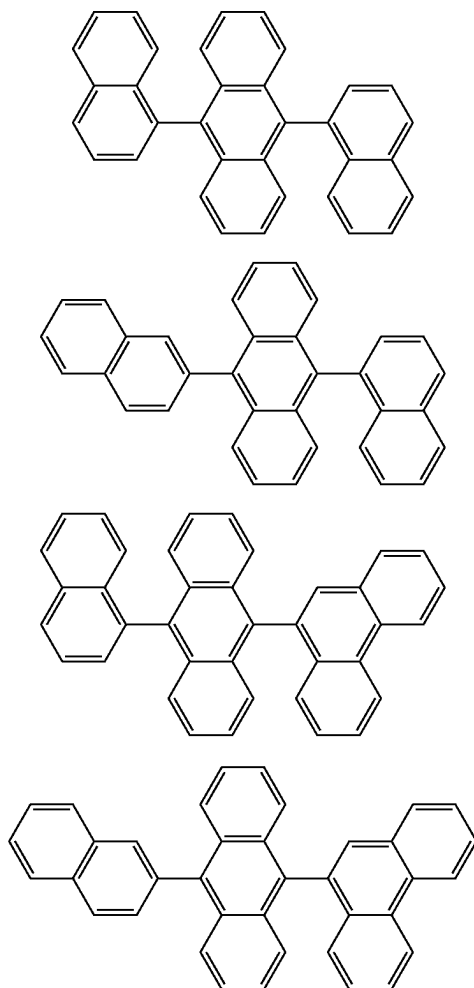
L_{302} to L_{304} may each independently be the same as described in connection with L_{301} ,

$xb2$ to $xb4$ may each independently be the same as described in connection with $xb1$, and 10

R_{302} to R_{305} and R_{311} to R_{314} may each independently be the same as described in connection with R_{301} . R_{10a} may be the same as described in the specification.

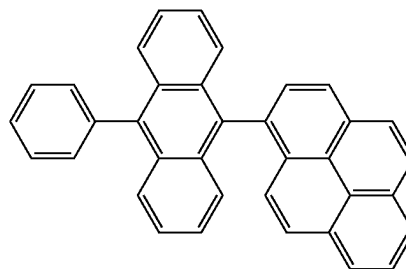
In embodiments, the host may include an alkali earth metal complex, a post-transition metal complex, or any combination thereof. In an embodiment, the host may include a Be complex (for example, Compound H55), a Mg complex, a Zn complex, or any combination thereof.

In embodiments, the host may include one of Compounds H1 to H124, 9,10-di(2-naphthyl)anthracene (ADN), 2-methyl-9,10-bis(naphthalen-2-yl)anthracene (MADN), 9,10-di(2-naphthyl)-2-t-butyl-anthracene (TBADN), 4,4'-bis(N-carbazolyl)-1,1'-biphenyl (CBP), 1,3-di-9-carbazolylbenzene (mCP), 1,3,5-tri(carbazol-9-yl)benzene (TCP), or any combination thereof: 25

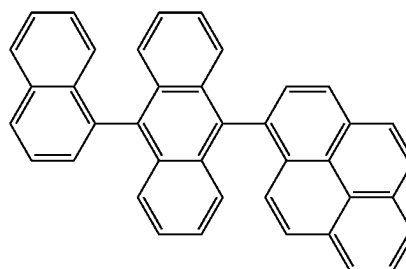


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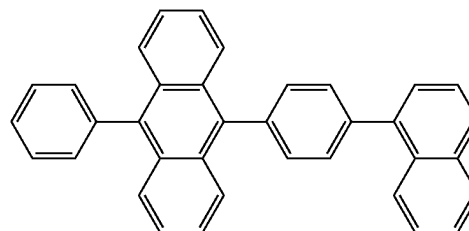
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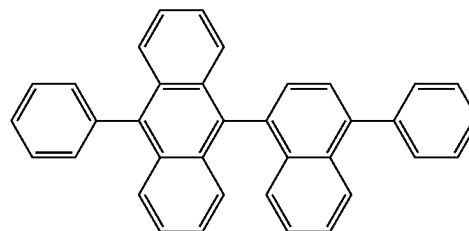
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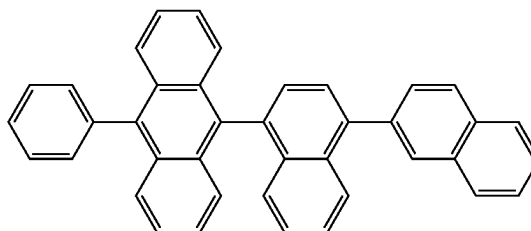
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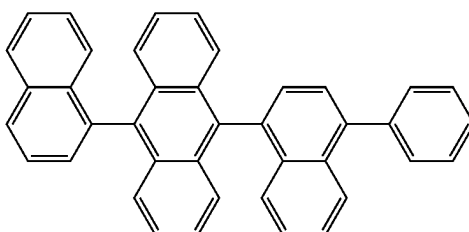
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H8



H9



H10

H1

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H2

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H3

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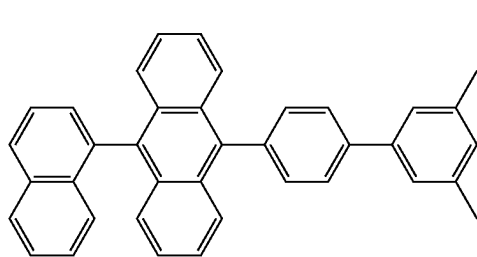
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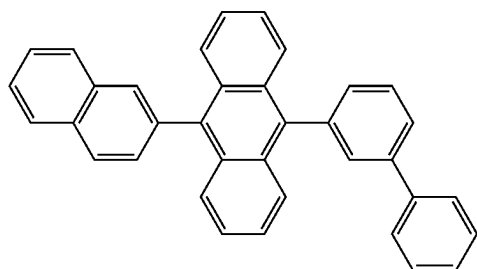
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H11

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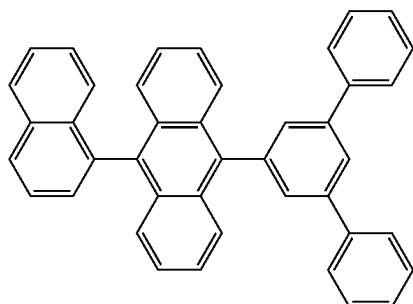


H12

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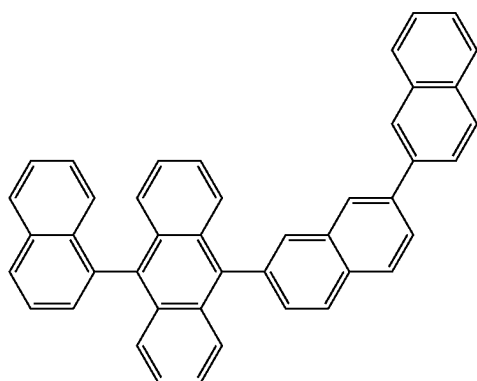


H13

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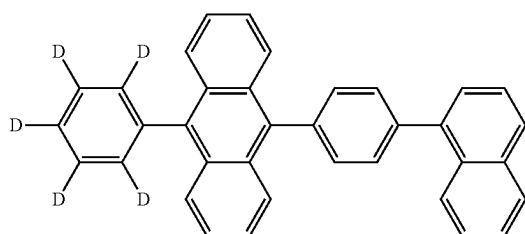
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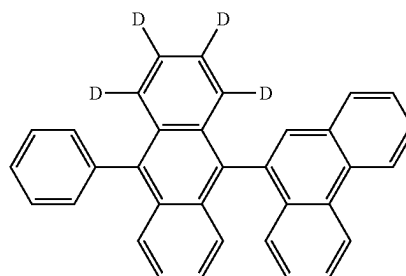
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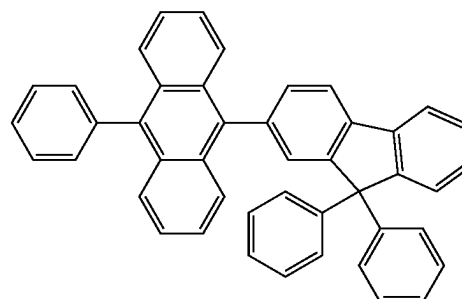
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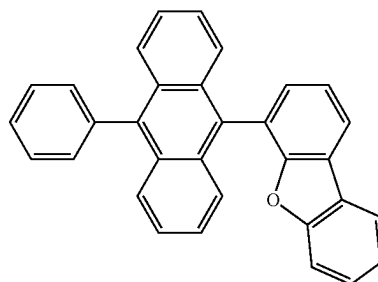
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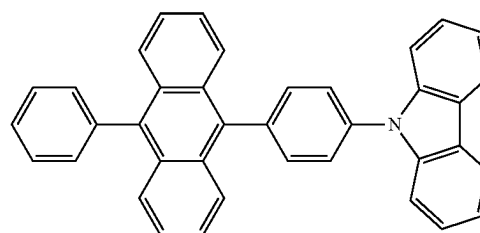
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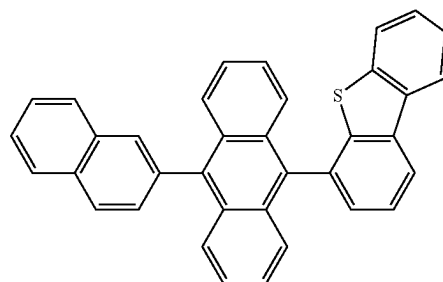
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H18



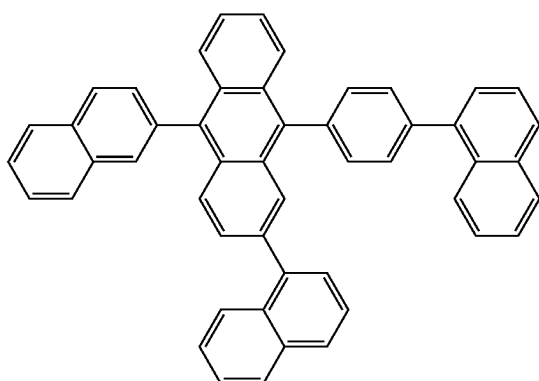
H19



H20

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H21

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H22

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H23

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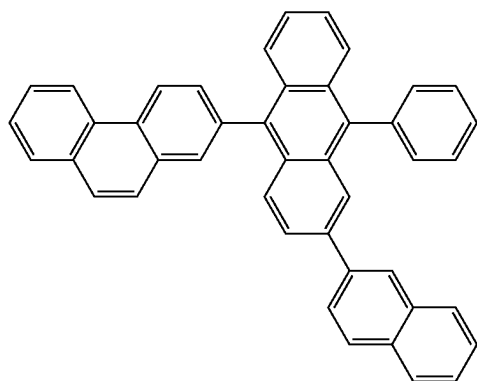
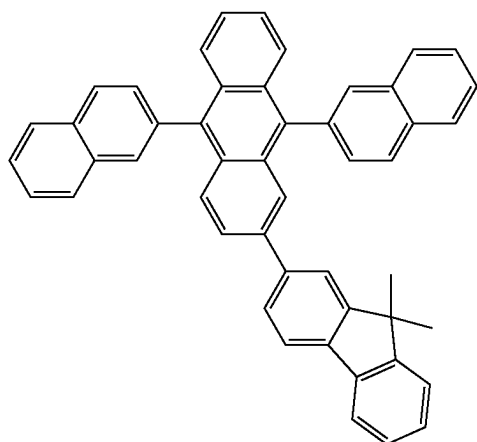
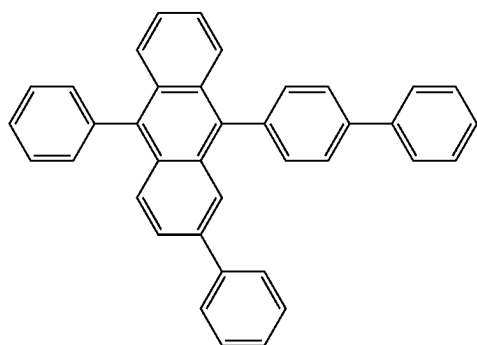
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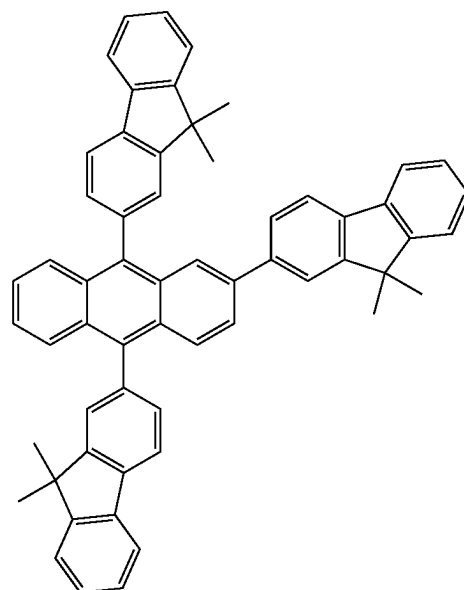
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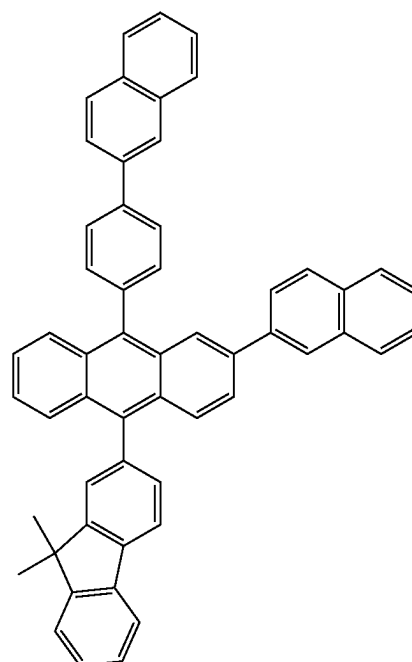
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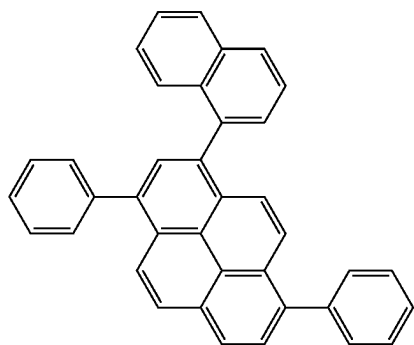
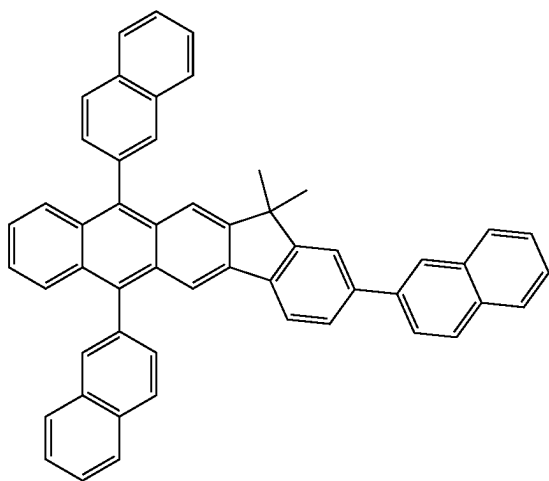
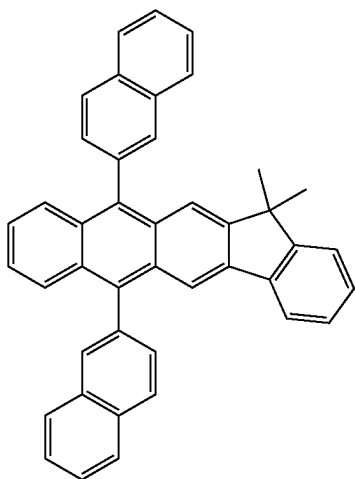
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H26

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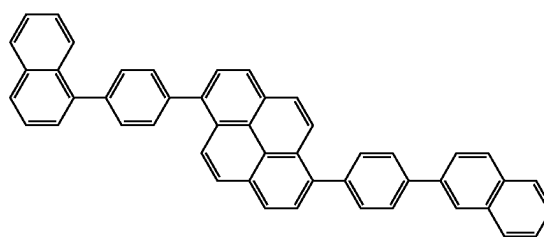
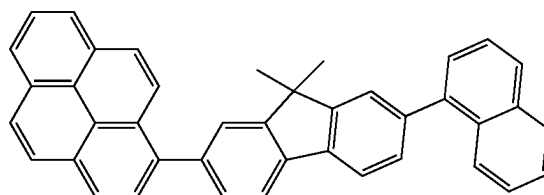
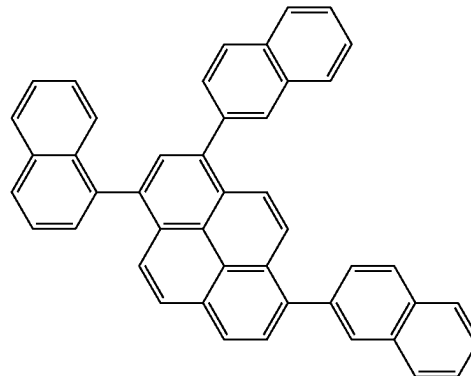
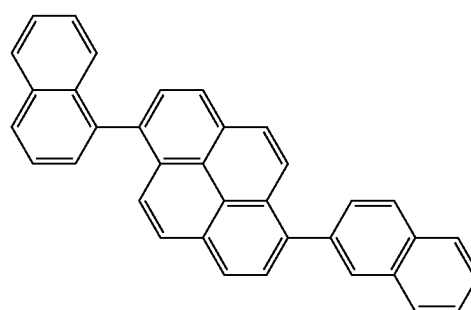
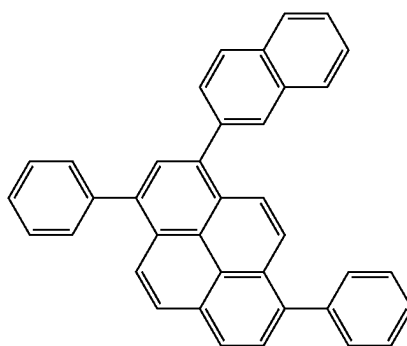
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H34

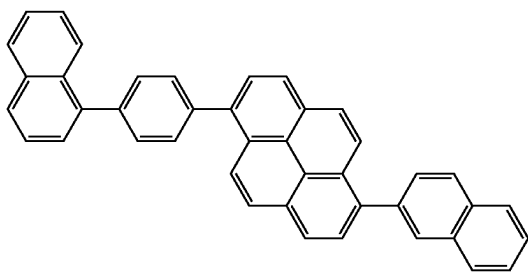
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H35

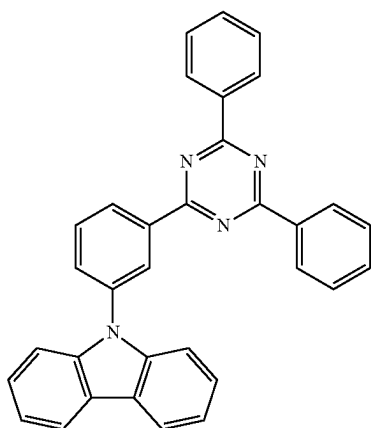
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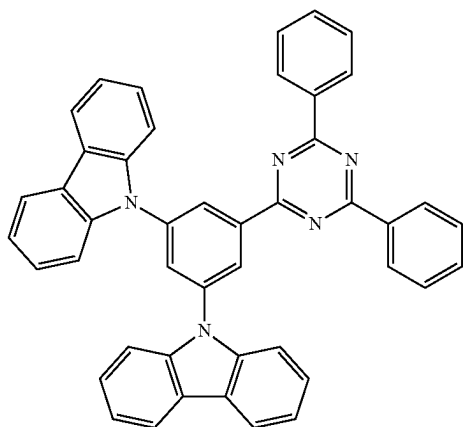
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H37

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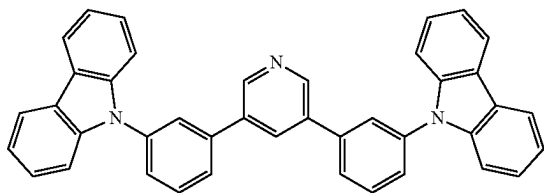
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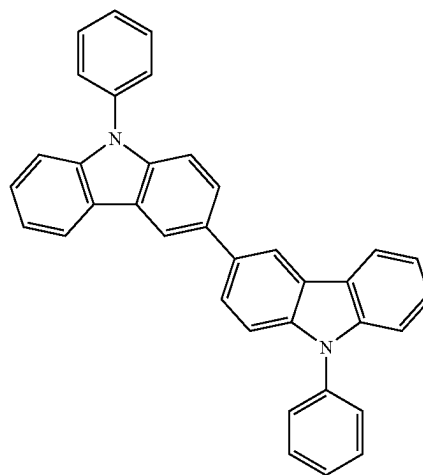
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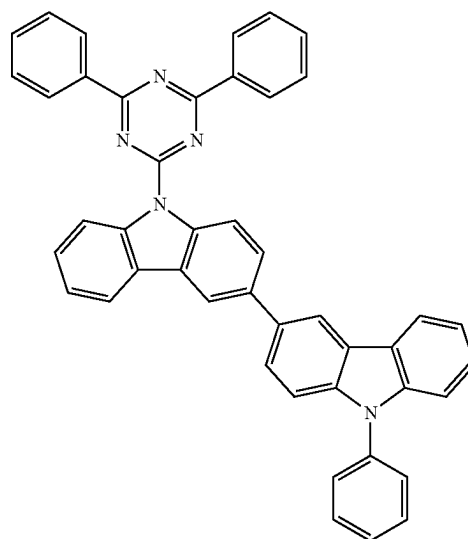
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**124**

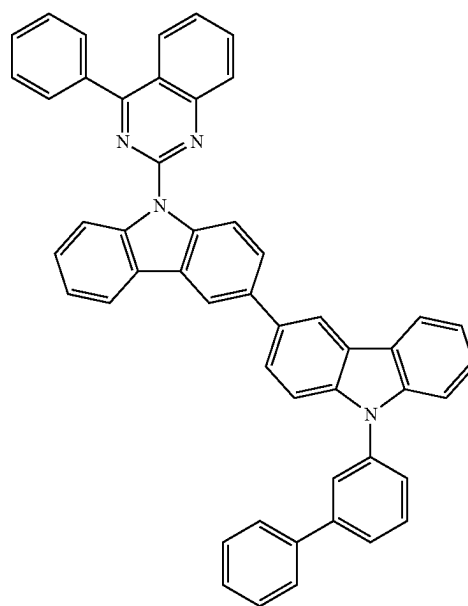
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H39



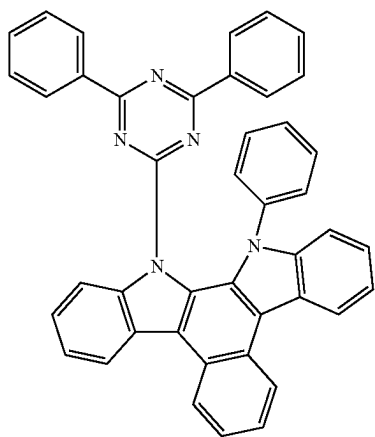
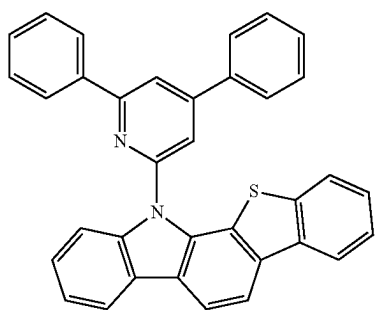
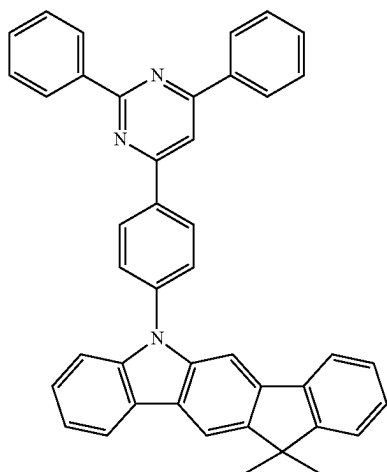
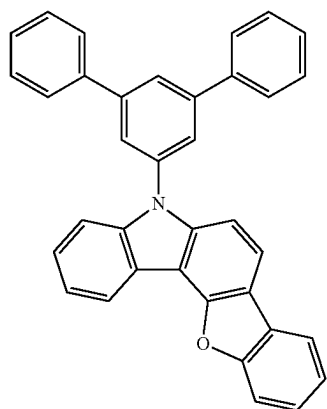
H40



H41

125

-continued

**126**

-continued

H42

H46

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H43

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H47

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H48

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H44

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H49

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H50

H45

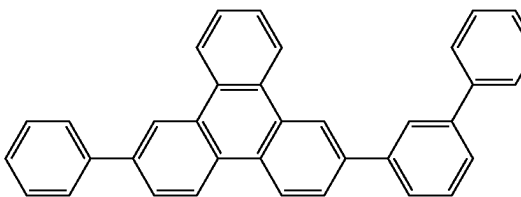
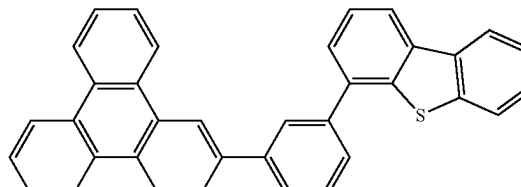
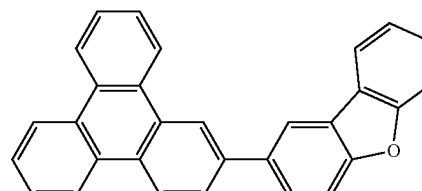
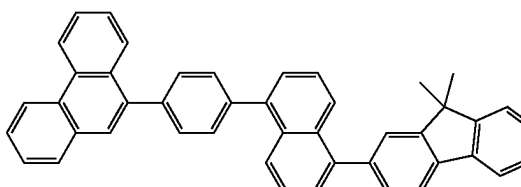
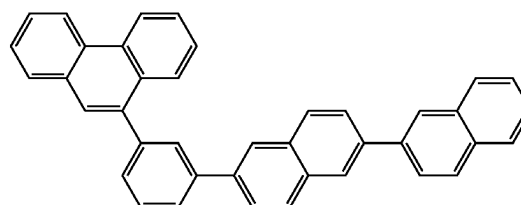
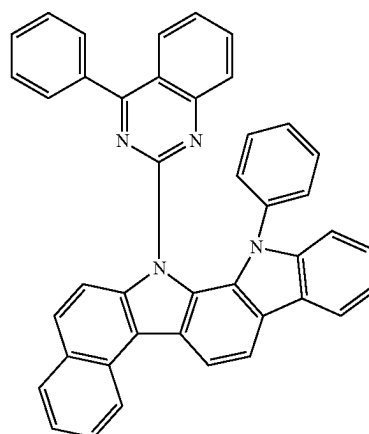
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H51

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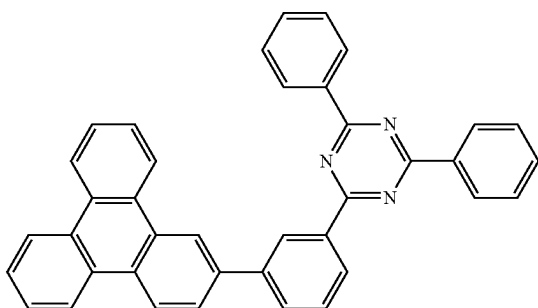
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127

-continued

H52



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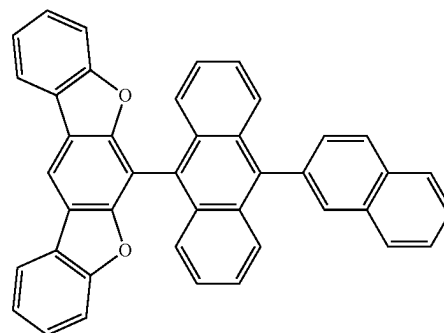
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128

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H56



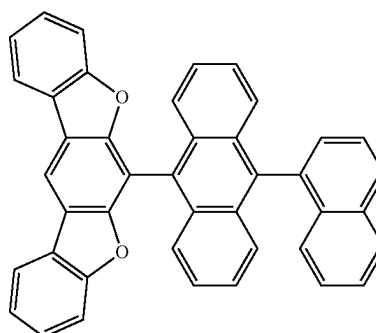
H53

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H57



H58

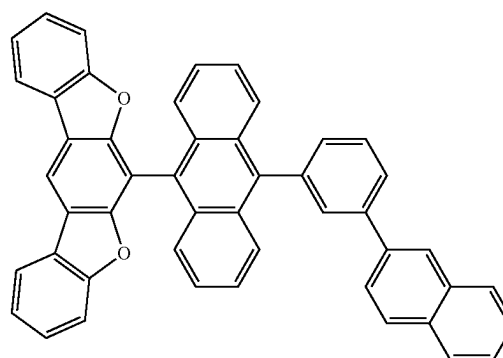
H54

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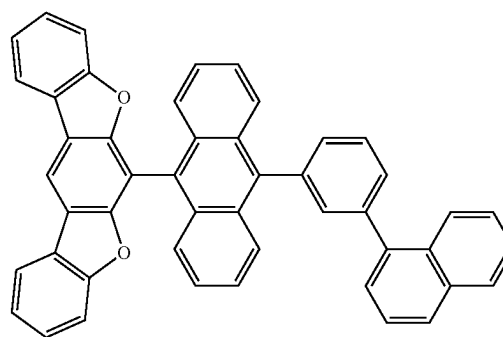
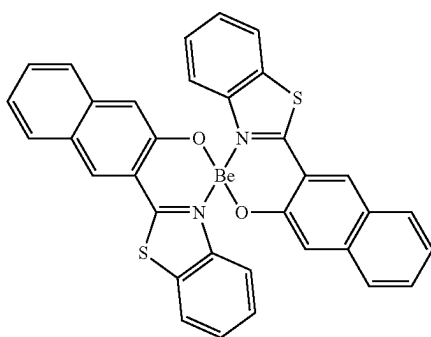
H59

H55

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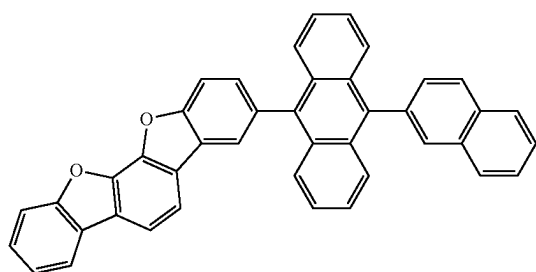
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129

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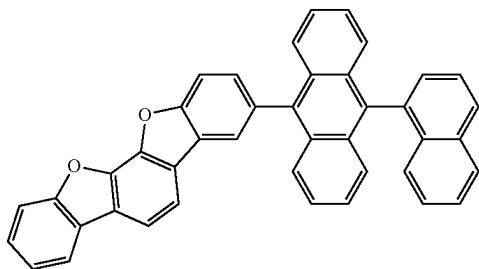
H60

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H61

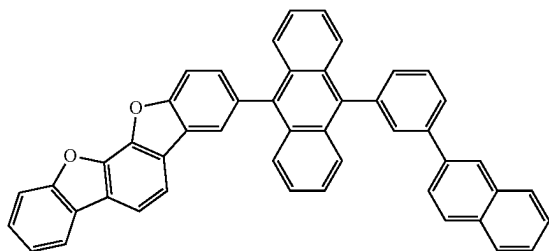


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H62

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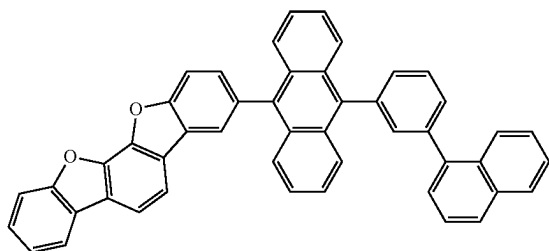


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H63

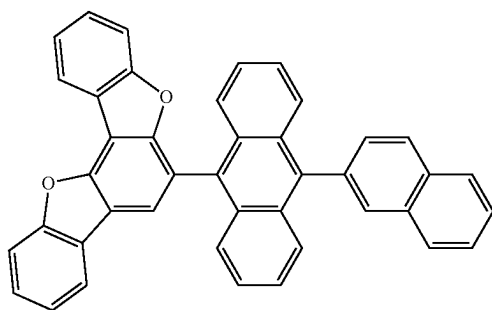
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H64

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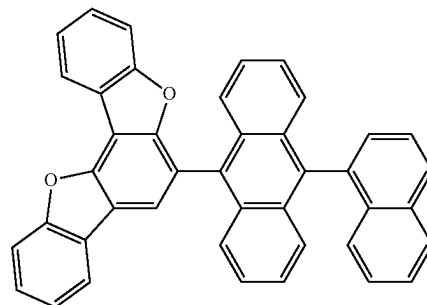
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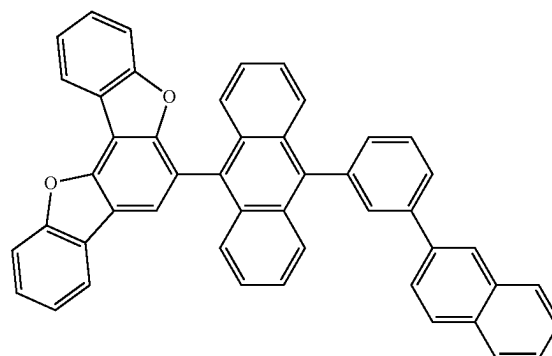
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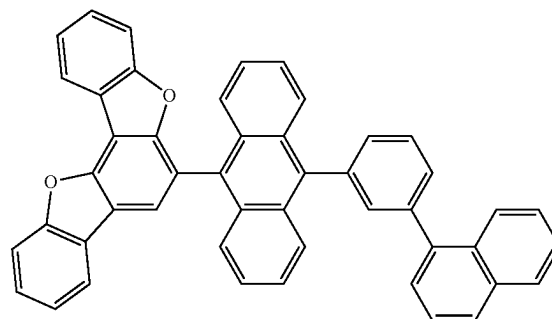
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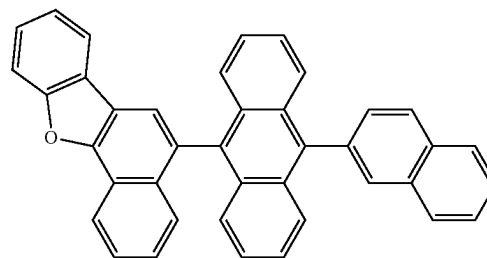
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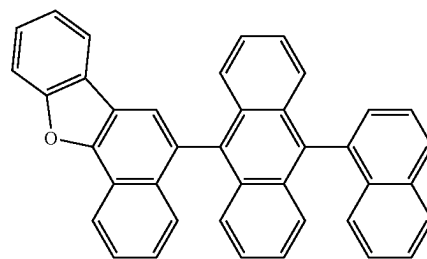
H67



H68



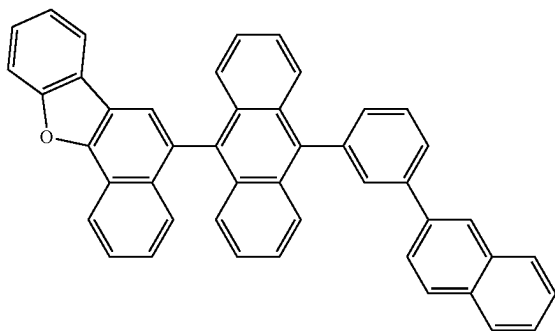
H69



131

-continued

H70



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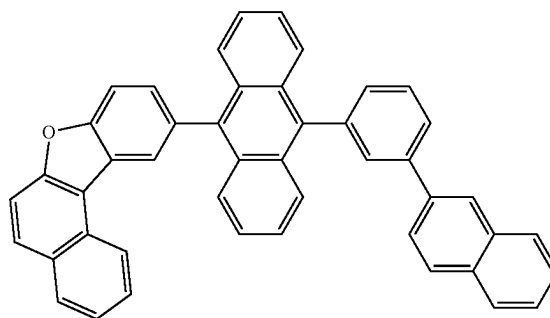
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132

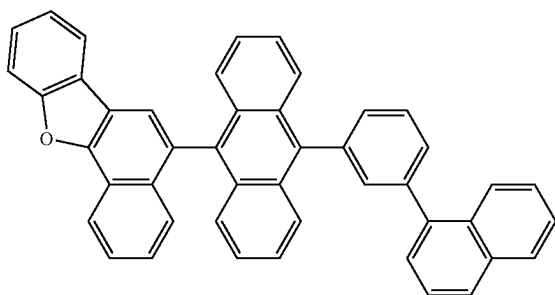
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H74



H75

H71

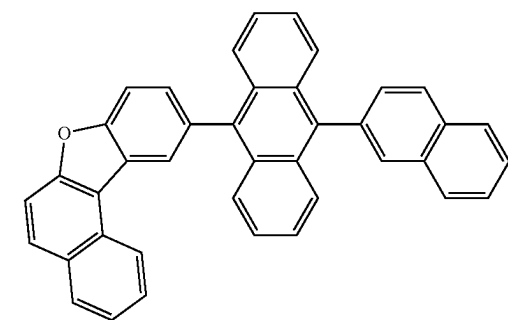


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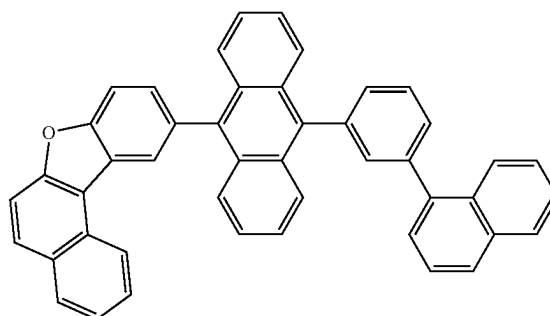
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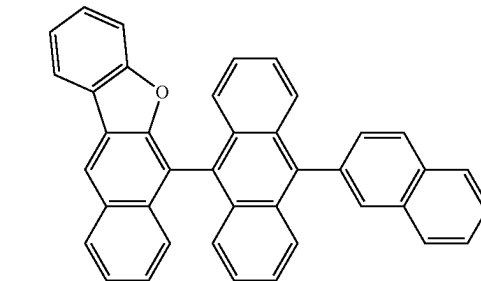
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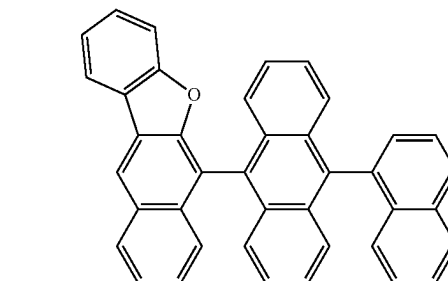
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H76

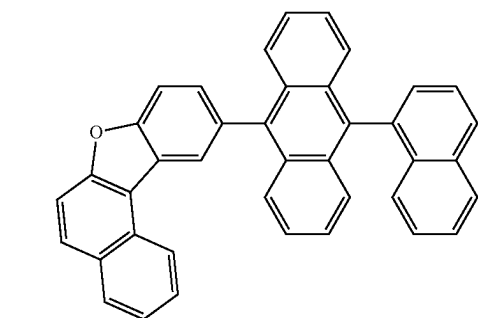


H77



H78

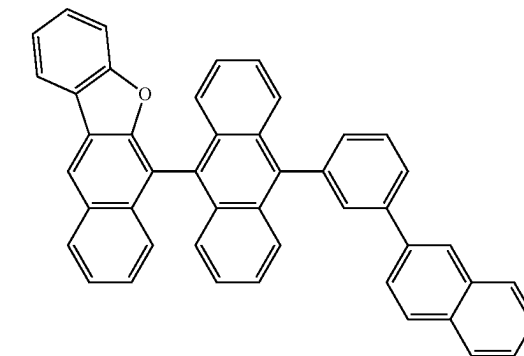
H73



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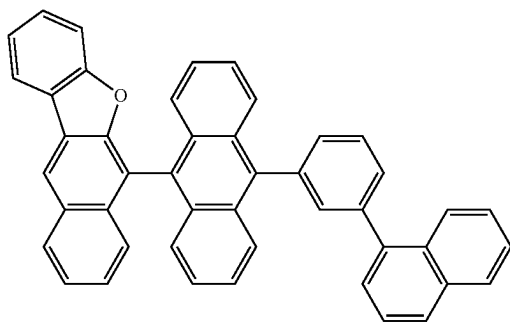
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133

-continued

H79



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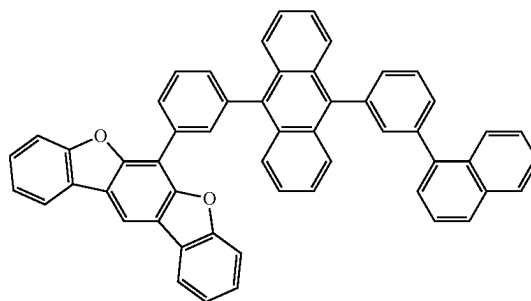
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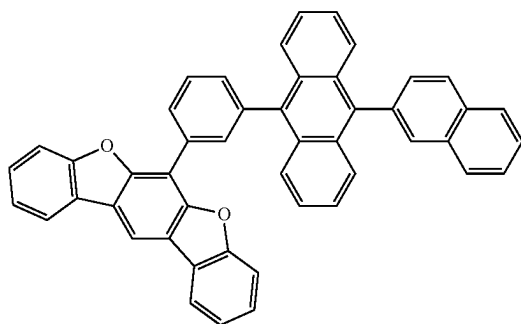
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H83



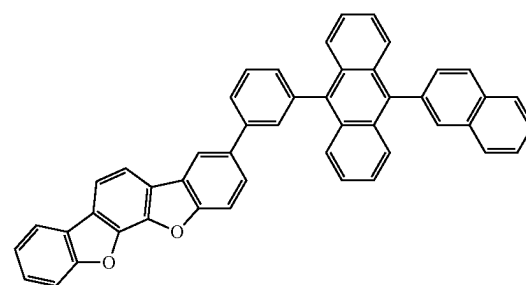
H80 20



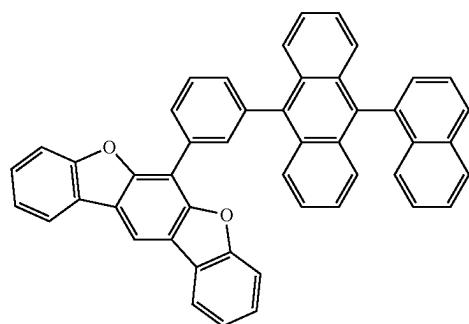
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H84



H81



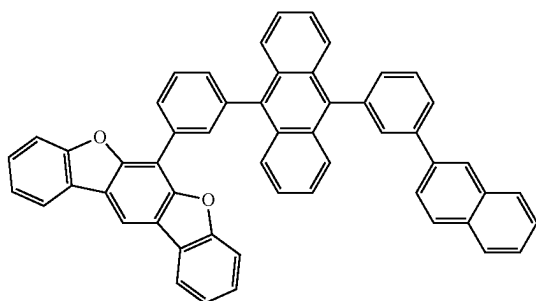
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H82

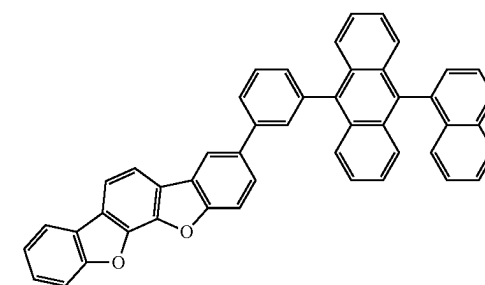
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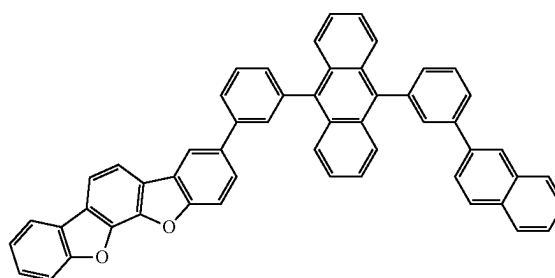
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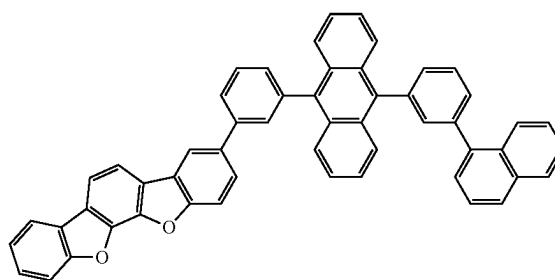
H85



H86



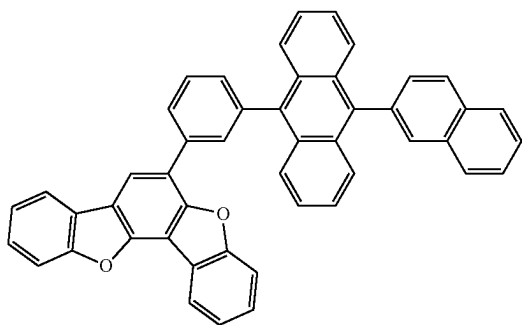
H87



135

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H88

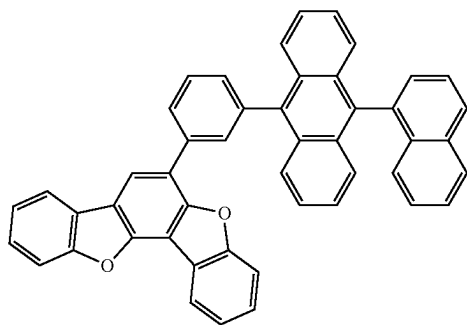


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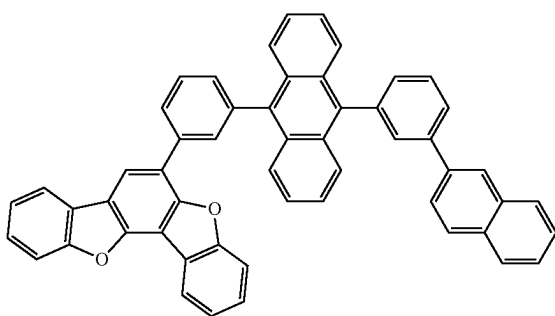
H89



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H90

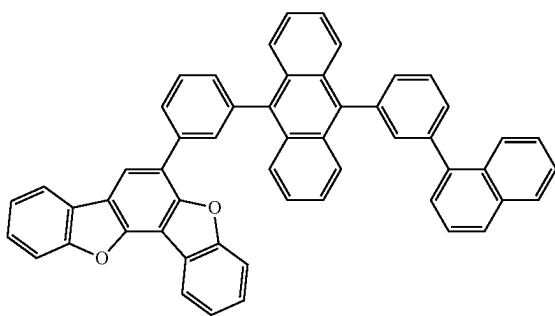


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H91



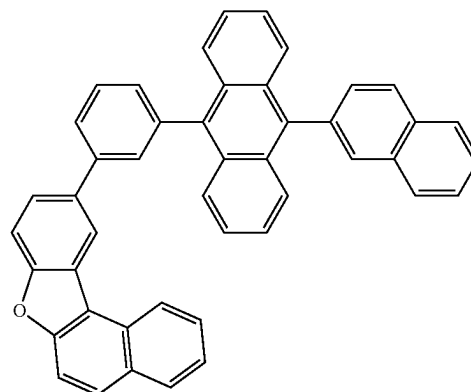
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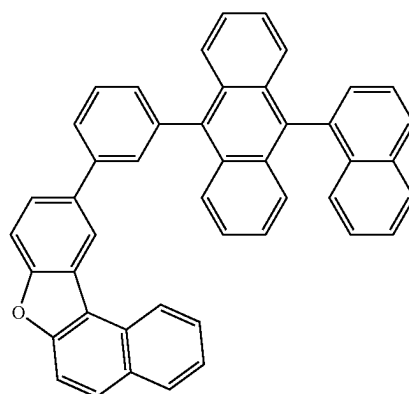
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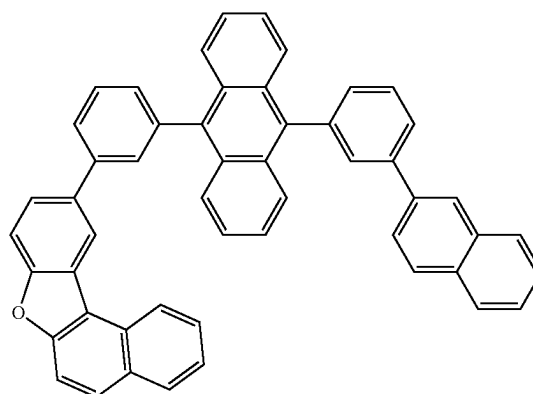
H92



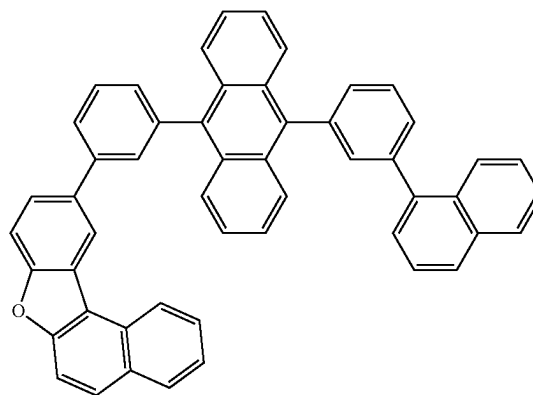
H93



H94



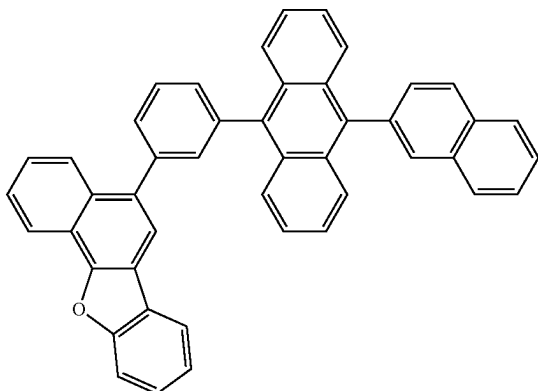
H95



137

-continued

H96



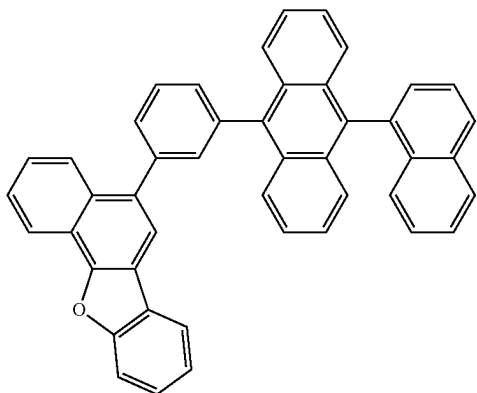
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H97

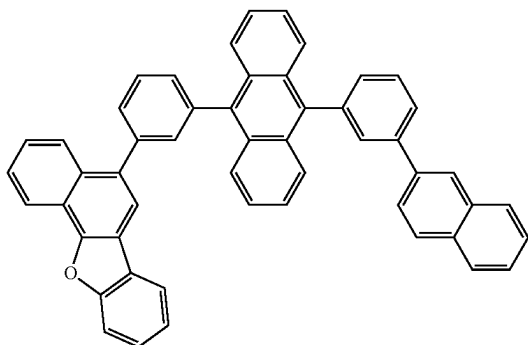


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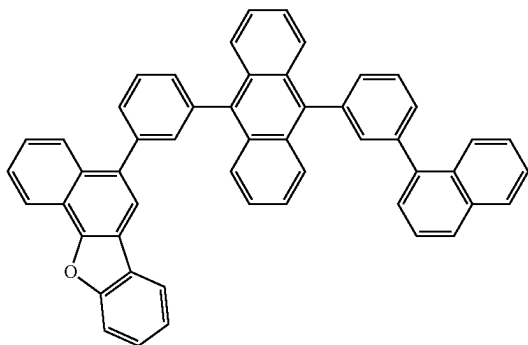
H98



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H99



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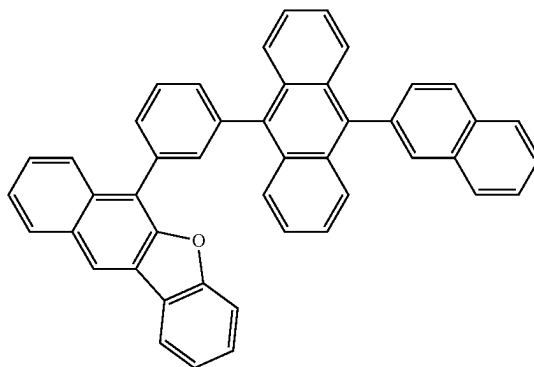
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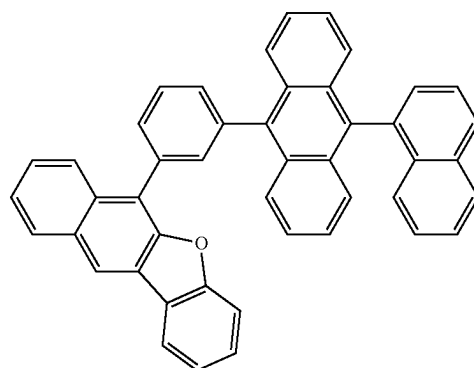
138

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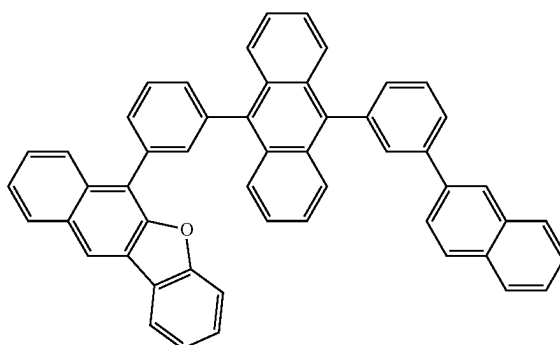
H100



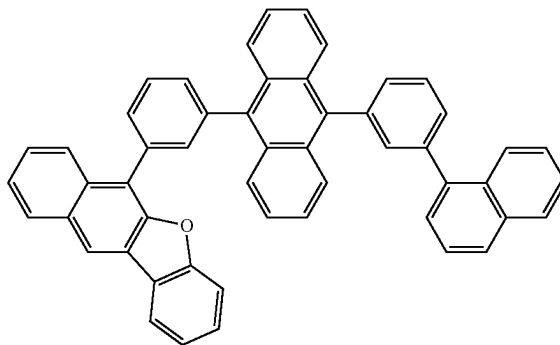
H101



H102

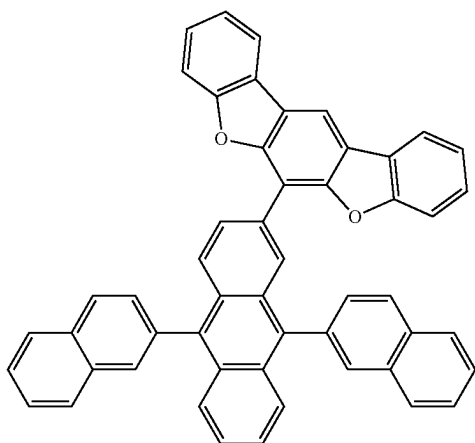


H103



139

-continued



H104

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H105

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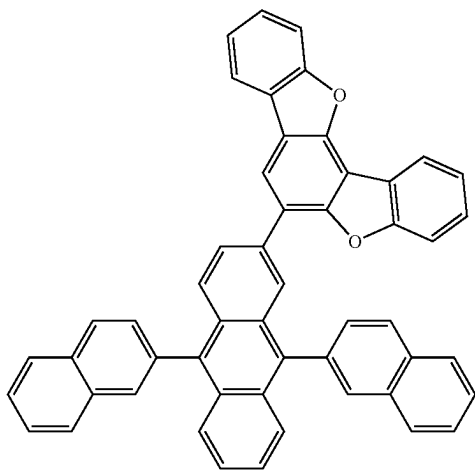
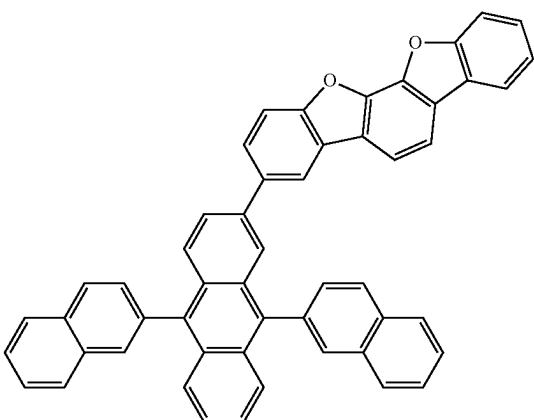
H106

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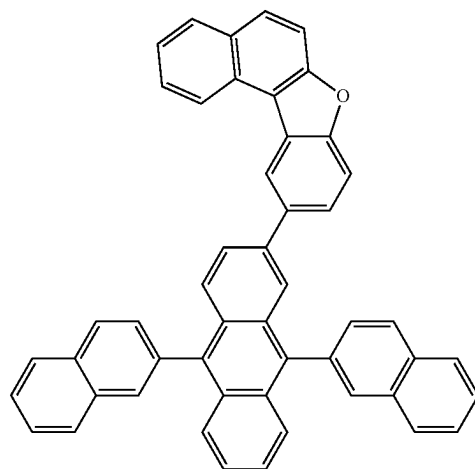
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**140**

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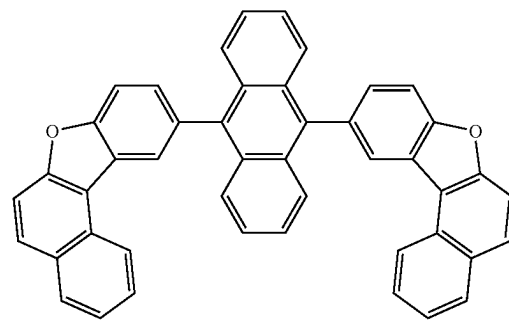
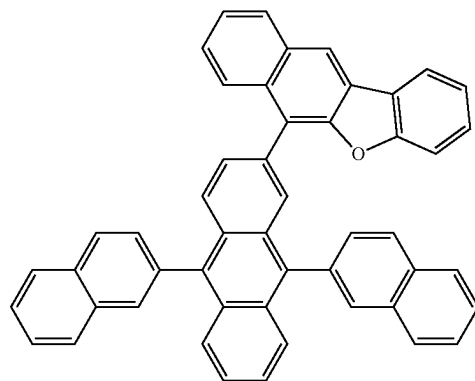
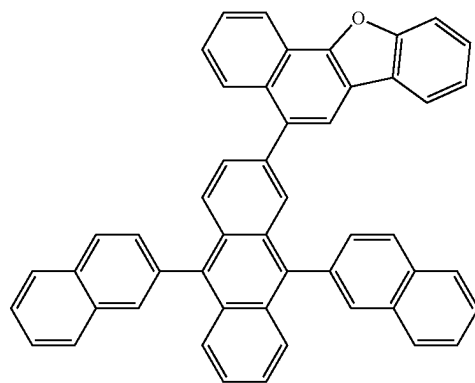


H107

H108

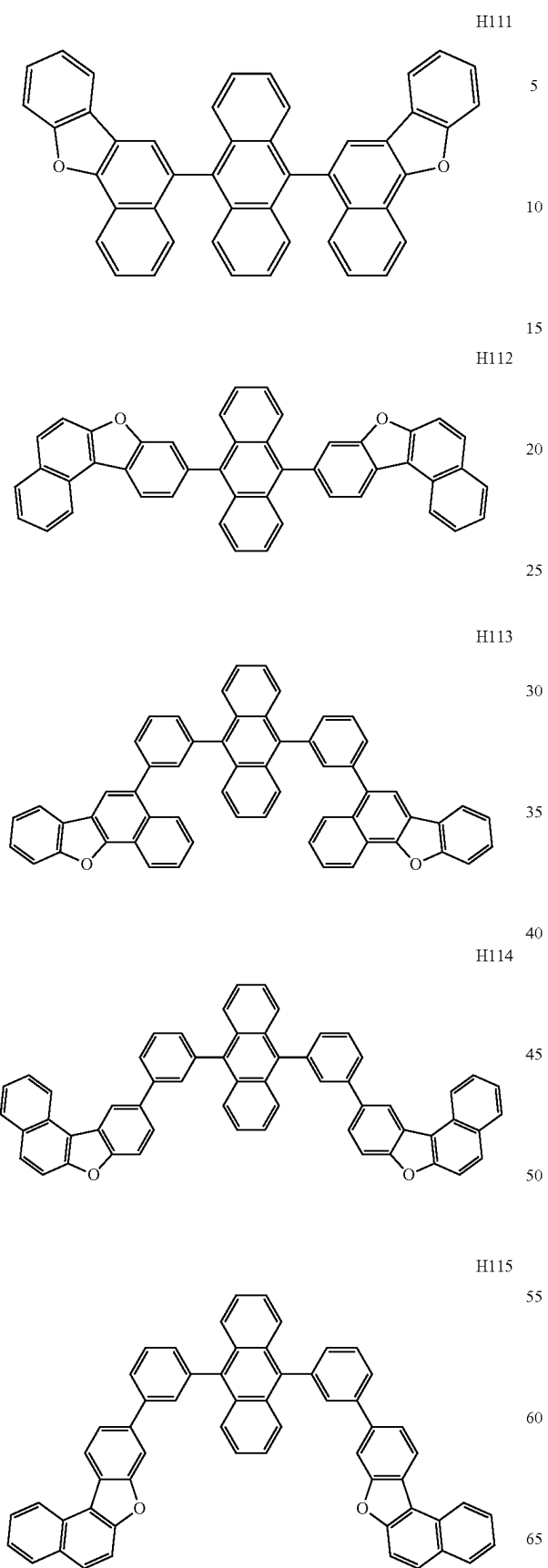
H109

H110

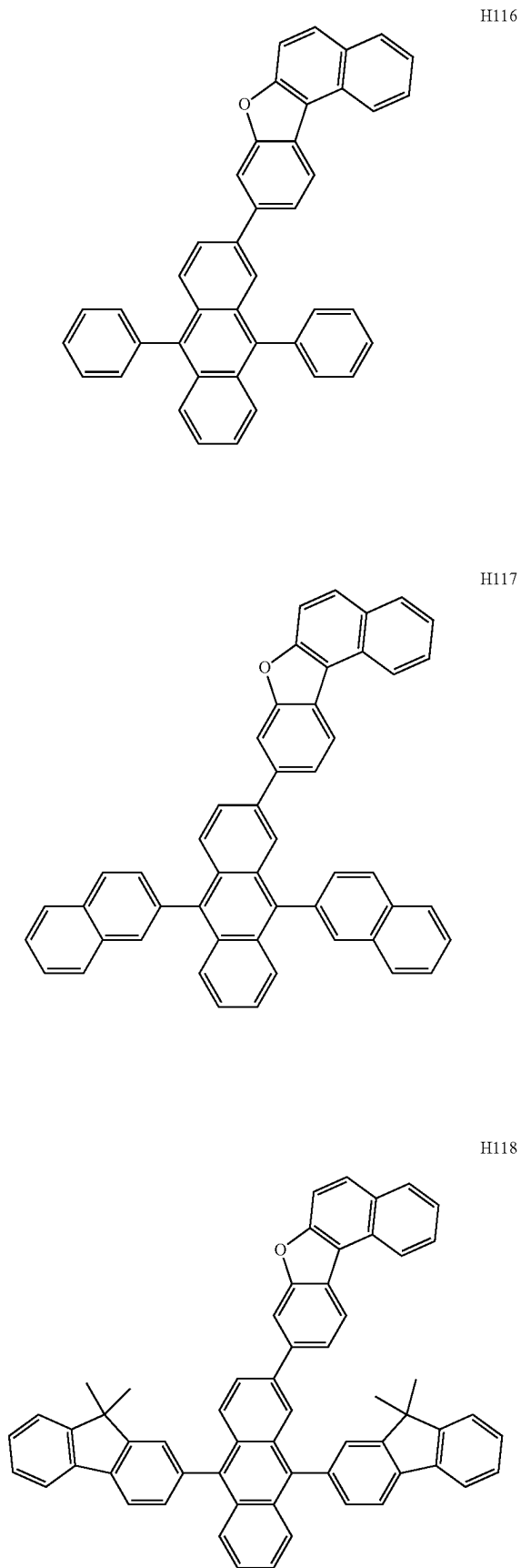


141

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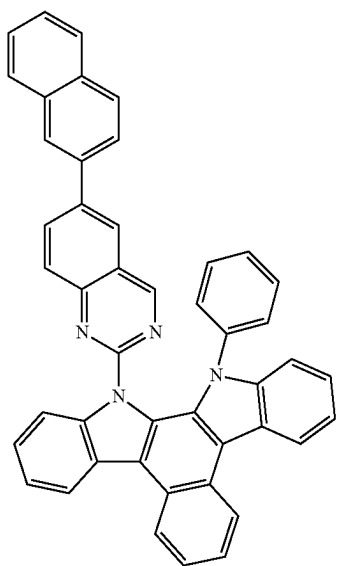
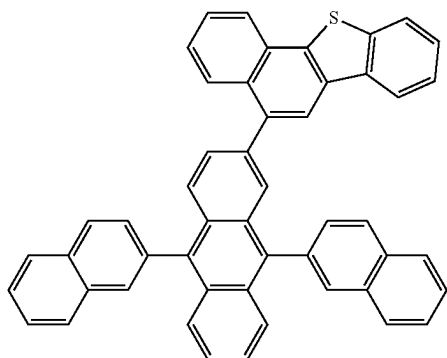
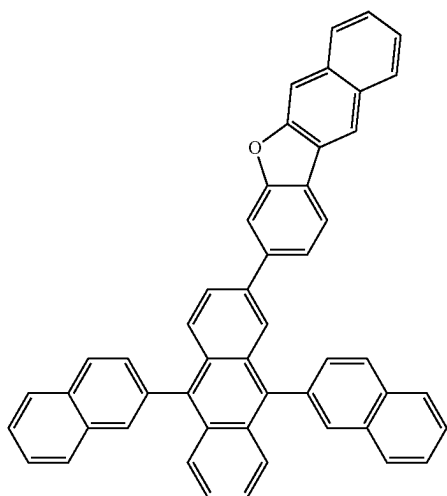
**142**

-continued



143

-continued

**144**

-continued

H119

H122

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H120

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H121 45

H123

H124

50 [Phosphorescent Dopant]

The phosphorescent dopant may include at least one transition metal as a central metal.

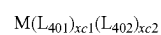
55 The phosphorescent dopant may include a monodentate ligand, a bidentate ligand, a tridentate ligand, a tetradentate ligand, a pentadentate ligand, a hexadentate ligand, or any combination thereof.

The phosphorescent dopant may be electrically neutral.

60 In an embodiment, the phosphorescent dopant may include an organometallic compound represented by Formula 401:

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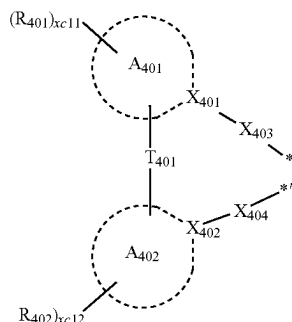
[Formula 401]



145

-continued

[Formula 402]



In Formulae 401 and 402,

M may be a transition metal (for example, iridium (Ir), platinum (Pt), palladium (Pd), osmium (Os), titanium (Ti), gold (Au), hafnium (Hf), europium (Eu), terbium (Tb), rhodium (Rh), rhenium (Re), or thulium (Tm)), L_{401} may be a ligand represented by Formula 402, and xc1 may be 1, 2, or 3, wherein when xc1 is 2 or more, two or more of L_{401} (s) may be identical to or different from each other,

L_{402} may be an organic ligand, and xc2 may be 0, 1, 2, 3, or 4, wherein when xc2 is 2 or more, two or more of L_{402} (s) may be identical to or different from each other, X_{401} and X_{402} may each independently be nitrogen (N) or carbon (C),

ring A_{401} and ring A_{402} may each independently be a C_3 - C_{60} carbocyclic group or a C_1 - C_{60} heterocyclic group,

T_{401} may be a single bond, $*-O-*$, $*-S-*$, $*-C(=O)-*$, $*-N(Q_{411})-*$, $*-C(Q_{411})(Q_{412})-*$, $*-C(Q_{411})=C(Q_{412})-*$, $*-C(Q_{411})=*$, or $*=C=*$,

X_{403} and X_{404} may each independently be a chemical bond (for example, a covalent bond or a coordination bond), O, S, $N(Q_{413})$, $B(Q_{413})$, $P(Q_{413})$, $C(Q_{413})(Q_{414})$, or $Si(Q_{413})(Q_{414})$,

Q_{411} to Q_{414} may each independently be the same as described in connection with Q_1 ,

R_{401} and R_{402} may each independently be hydrogen, deuterium, $-F$, $-Cl$, $-Br$, $-I$, a hydroxyl group, a cyano group, a nitro group, a C_1 - C_{20} alkyl group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{20} alkoxy group unsubstituted or substituted with at least one R_{10a} , a C_3 - C_{60} carbocyclic group unsubstituted or substituted with at least one R_{10a} , a C_1 - C_{60} heterocyclic group unsubstituted or substituted with at least one R_{10a} , $-Si(Q_{401})(Q_{402})(Q_{403})$, $-N(Q_{401})(Q_{402})$, $-B(Q_{401})(Q_{402})$, $-C(=O)(Q_{401})$, $-S(=O)_2(Q_{401})$, or $-P(=O)(Q_{401})(Q_{402})$,

Q_{401} to Q_{403} may each independently be the same as described in connection with Q_1 ,

xc11 and xc12 may each independently be an integer from 0 to 10, and

* and * in Formula 402 each indicates a binding site to M in Formula 401. R_{10a} may be the same as described in the specification.

In an embodiment, in Formula 402, X_{401} may be nitrogen and X_{402} may be carbon, or each of X_{401} and X_{402} may be nitrogen.

In embodiments, when xc1 in Formula 402 is 2 or more, two ring A_{401} (s) in two or more of L_{401} (s) may optionally be linked to each other via T_{402} , which is a linking group, and two ring A_{402} (s) may optionally be linked to each other via

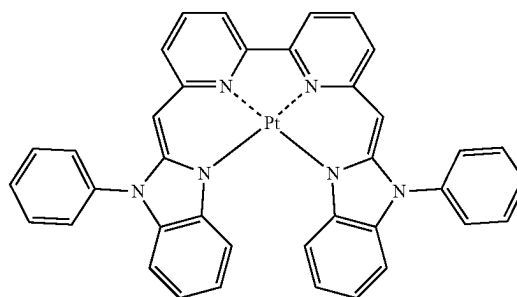
146

T_{403} , which is a linking group (see Compounds PD1 to PD4 and PD7). T_{402} and T_{403} may each independently be the same as described in connection with T_{401} .

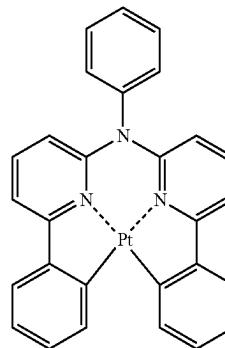
L_{402} in Formula 401 may be an organic ligand. In an embodiment, L_{402} may include a halogen group, a diketone group (for example, an acetylacetonate group), a carboxylic acid group (for example, a picolinate group), $-C(=O)$, an isonitrile group, $-CN$, a phosphorus-containing group (for example, a phosphine group, a phosphite group, etc.), or any combination thereof.

The phosphorescent dopant may include, for example, one of Compounds PD1 to PD25, or any combination thereof:

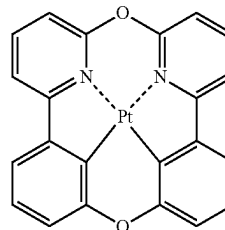
PD1



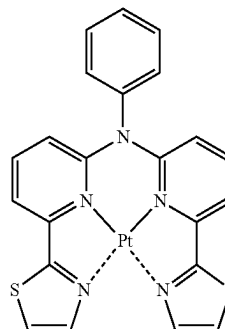
PD2



PD3

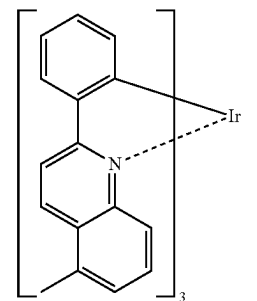
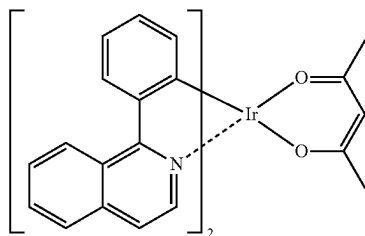
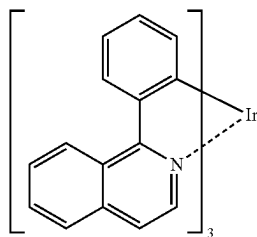
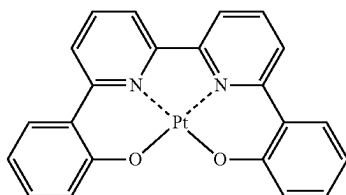
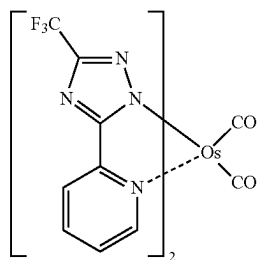
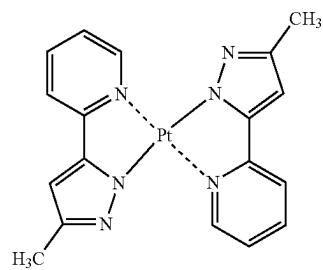


PD4

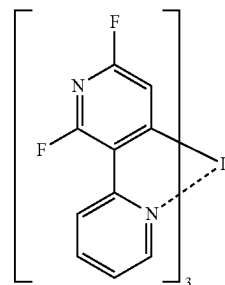
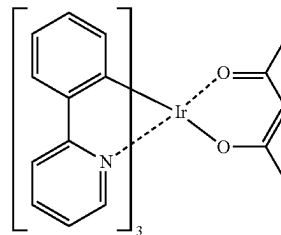
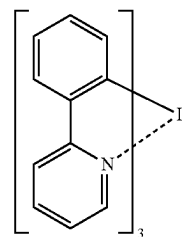
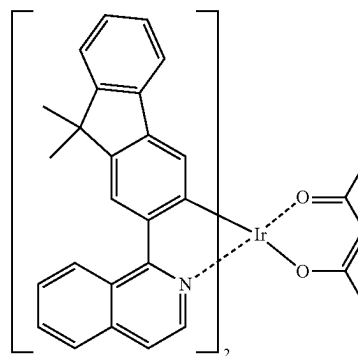
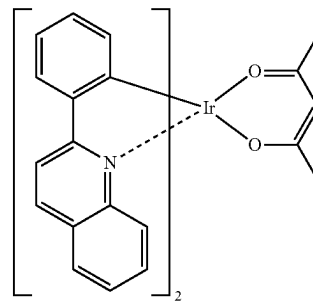


147

-continued

**148**

-continued



PD5

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PD6

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PD7

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PD8

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PD9

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PD10

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PD11

PD12

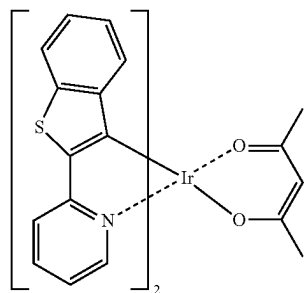
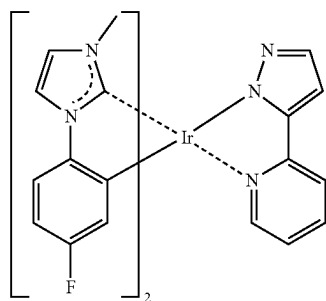
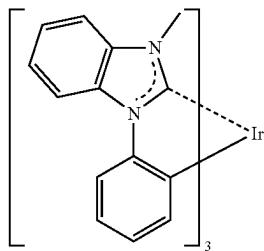
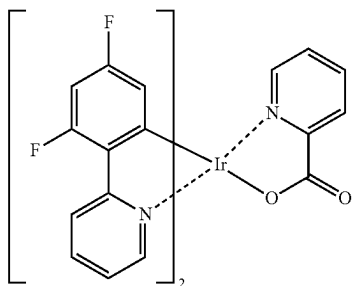
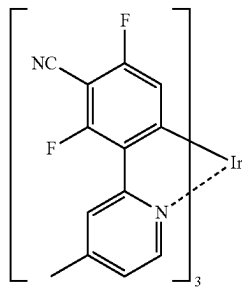
PD13

PD14

PD15

149

-continued

**150**

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PD16

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PD17

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PD18

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PD19

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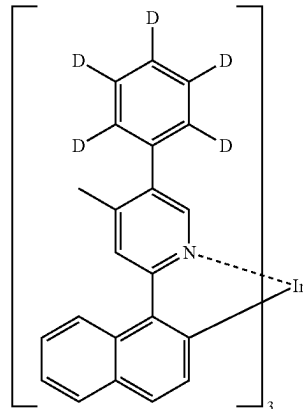
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PD20

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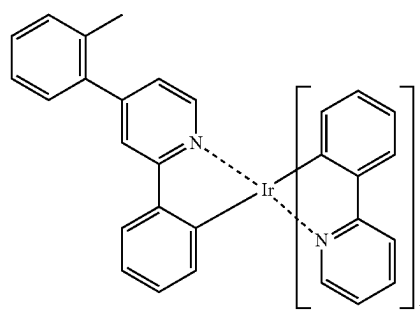


PD21

PD22

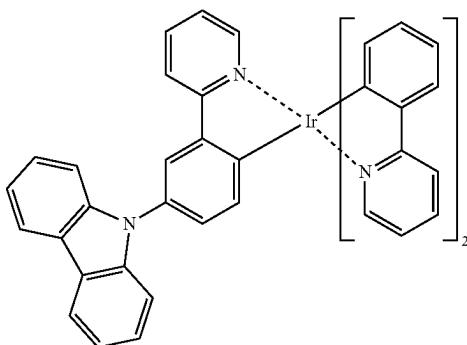
PD23

PD24



151

-continued



PD25

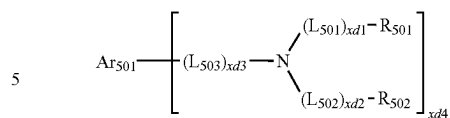
[Fluorescent Dopant]

The fluorescent dopant may include an amine group-containing compound, a styryl group-containing compound, or any combination thereof.

In an embodiment, the fluorescent dopant may include a compound represented by Formula 501:

152

[Formula 501]



In Formula 501,

Ar₅₀₁, L₅₀₁ to L₅₀₃, R₅₀₁, and R₅₀₂ may each independently be a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a},

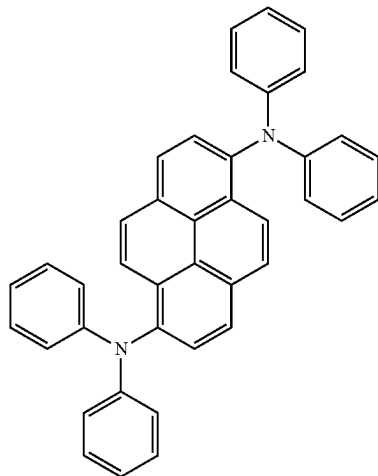
xd1 to xd3 may each independently be 0, 1, 2, or 3, and xd4 may be 1, 2, 3, 4, 5, or 6.

In an embodiment, Ar₅₀₁ in Formula 501 may be a condensed cyclic group (for example, an anthracene group, a chrysene group, or a pyrene group) in which three or more monocyclic groups are condensed together.

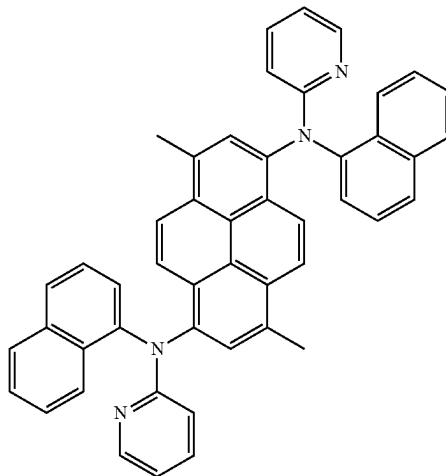
In embodiments, xd4 in Formula 501 may be 2.

In an embodiment, the fluorescent dopant may include one of Compounds FD1 to FD36, DPVBi, DPAVBi, or any combination thereof:

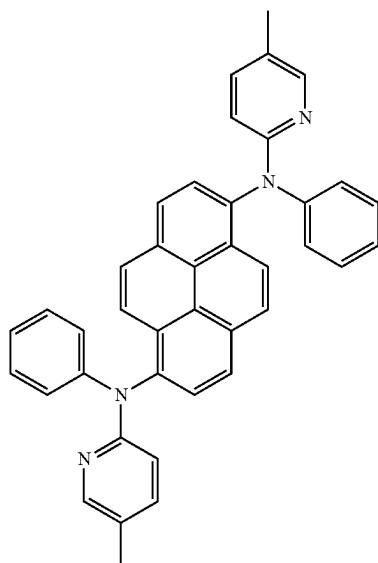
FD1



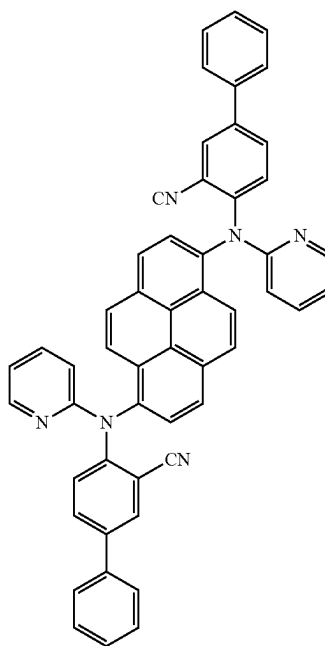
FD2



FD3

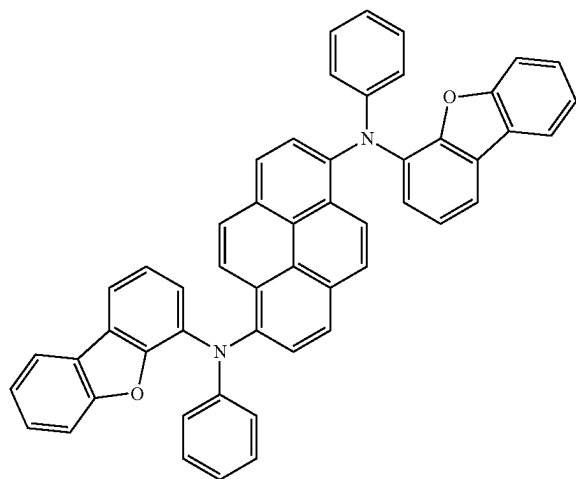


FD4



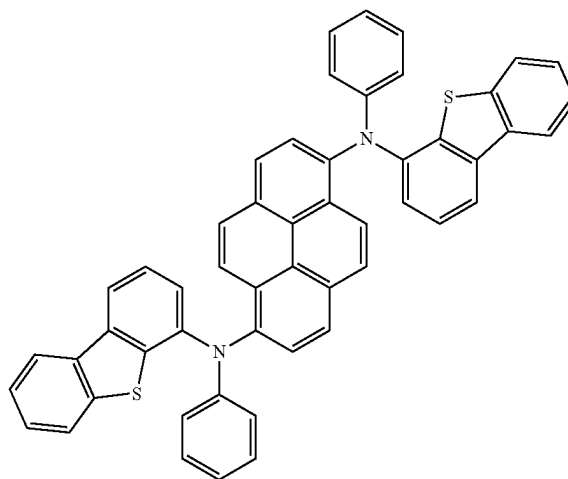
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FD5



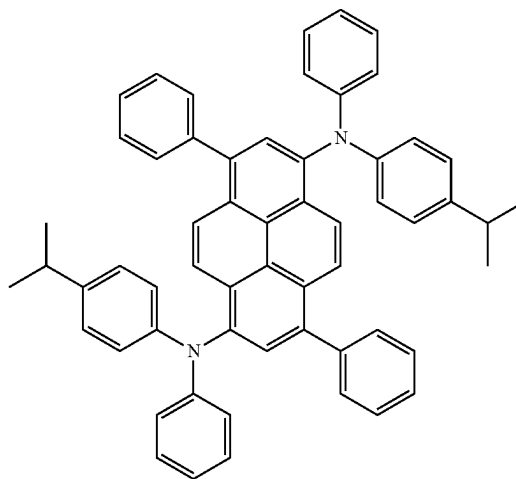
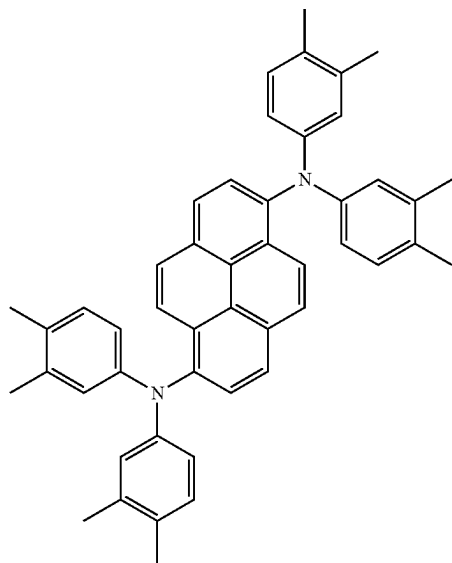
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FD6



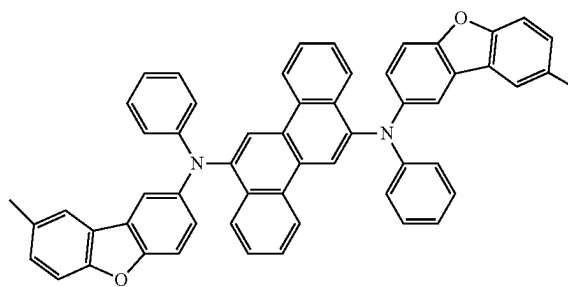
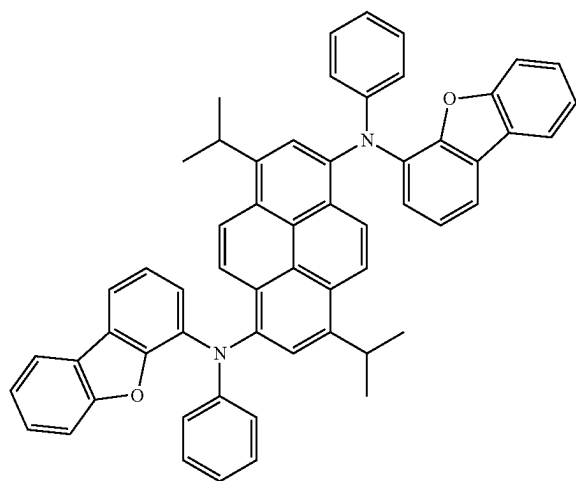
FD7

FD8



FD9

FD10

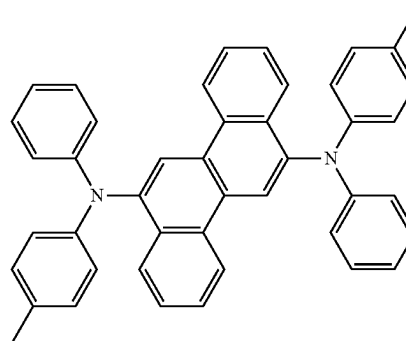
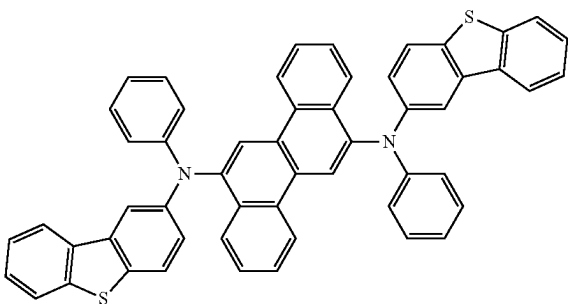


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156

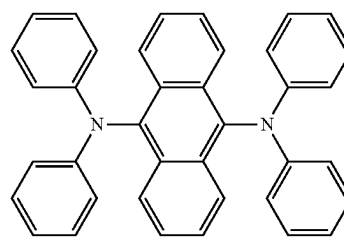
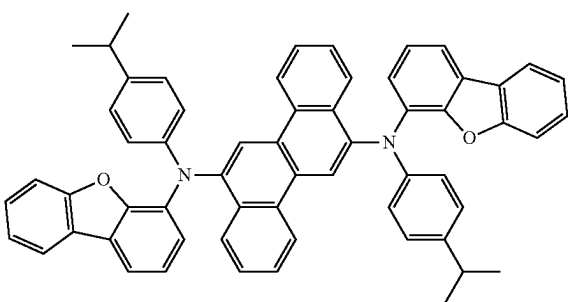
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FD11

FD12



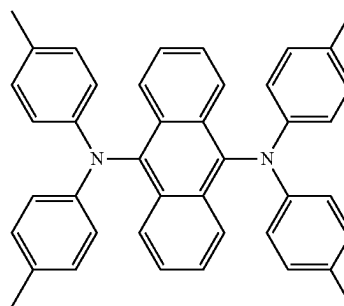
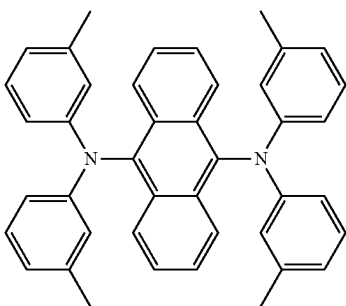
FD13

FD14



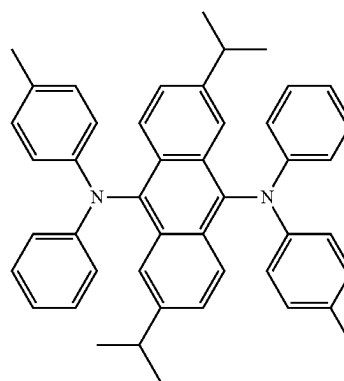
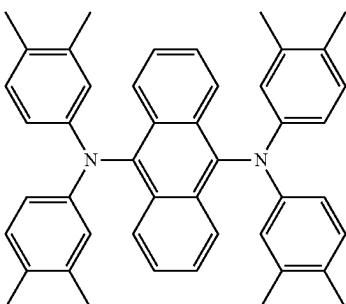
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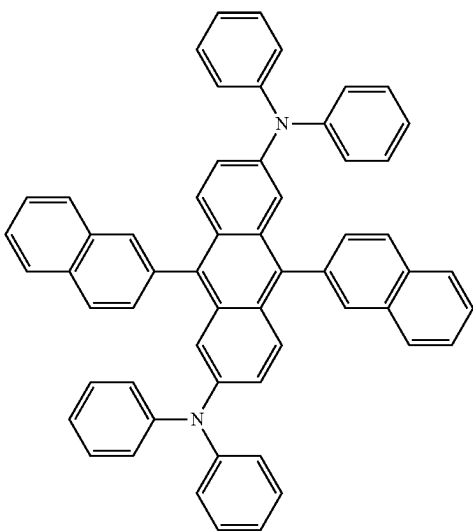
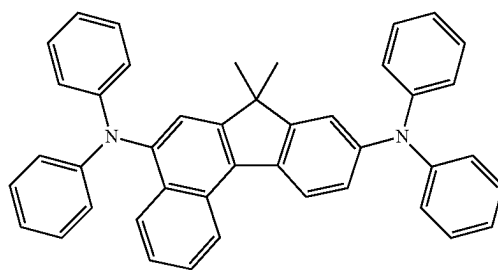
FD16



FD17

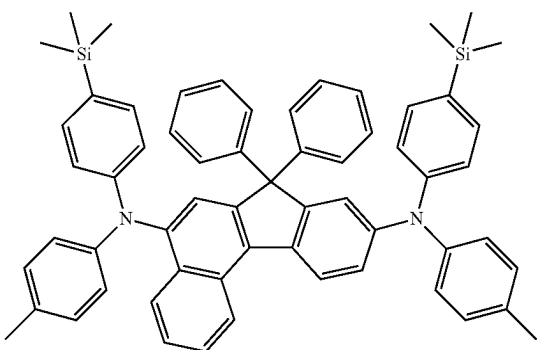
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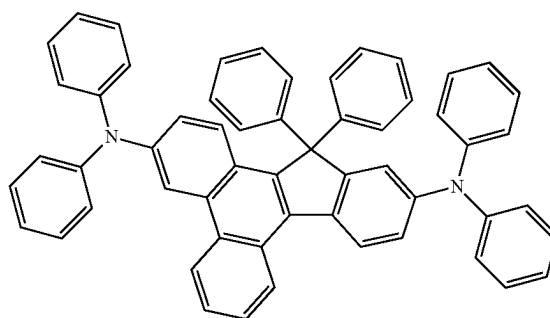
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FD19**158**

FD20

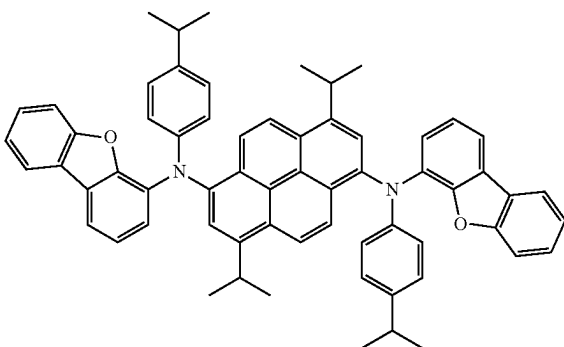
FD21



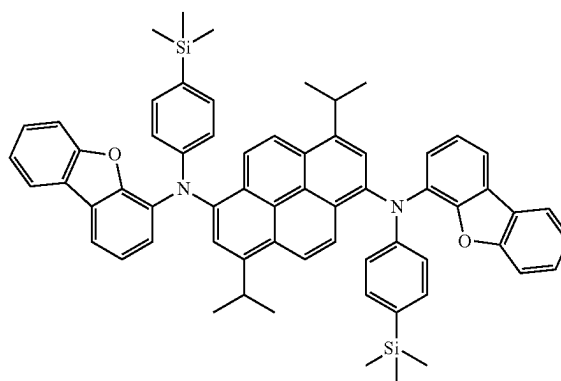
FD22



FD23



FD24

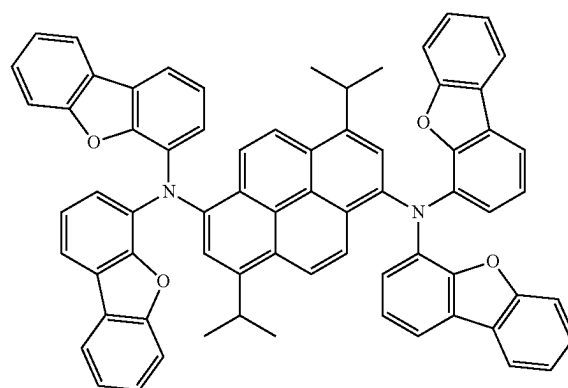
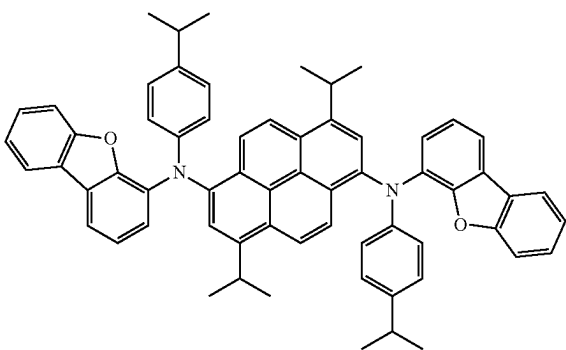


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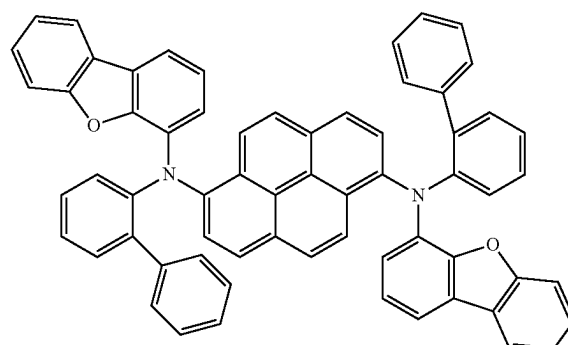
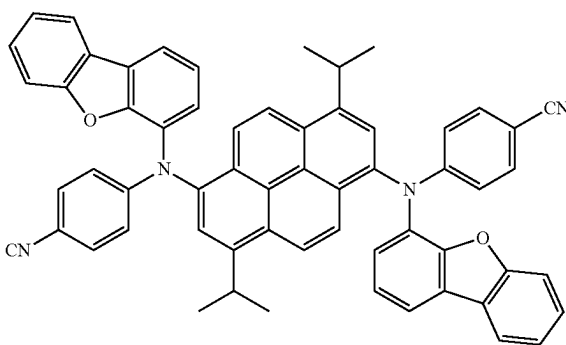
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FD25

FD26



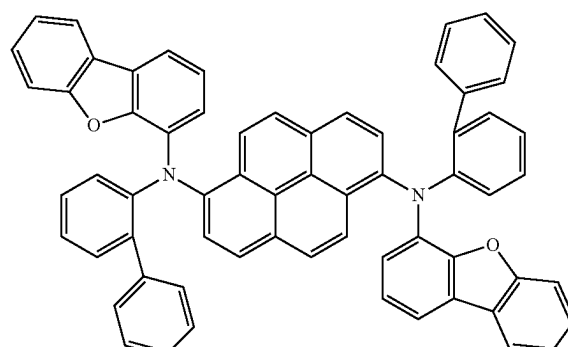
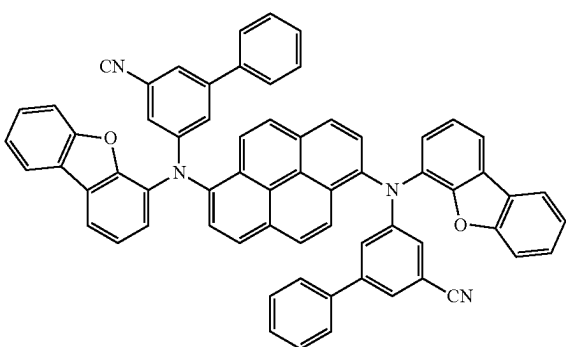
FD27

FD28



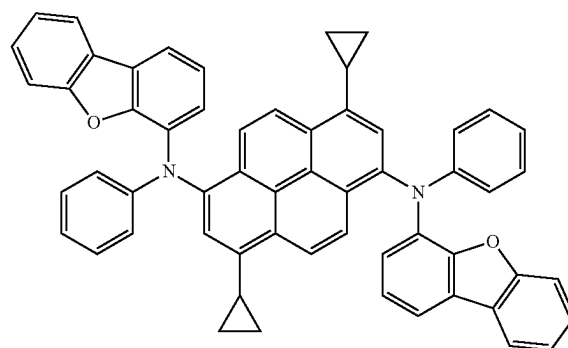
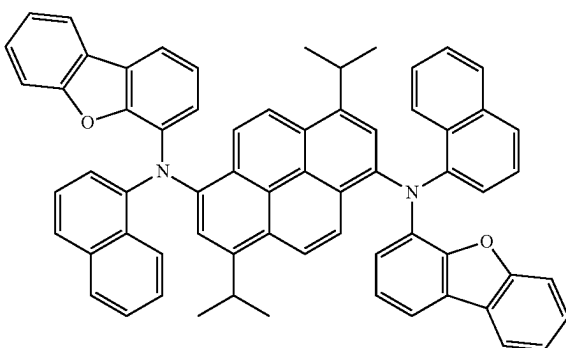
FD29

FD30



FD31

FD32

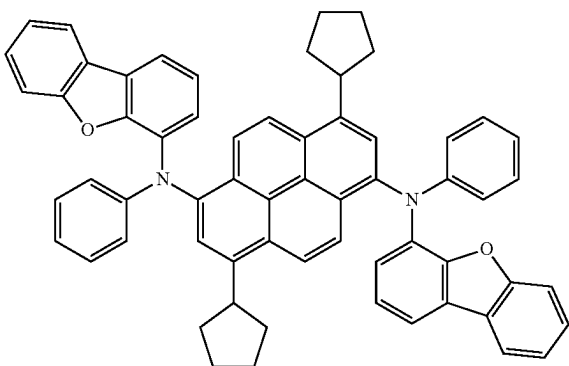


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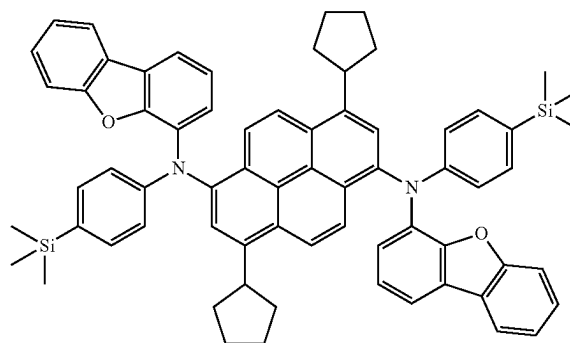
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FD33

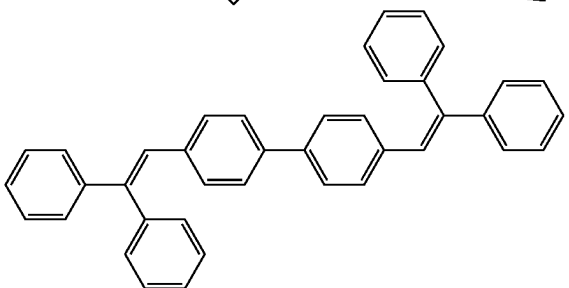
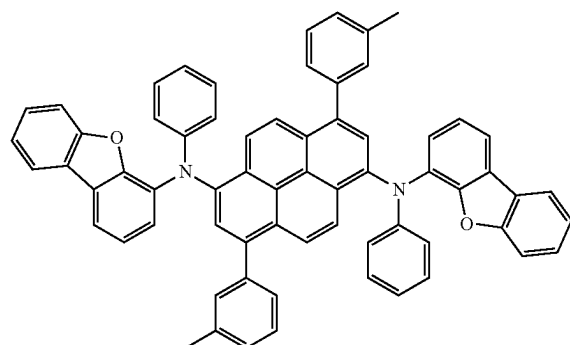
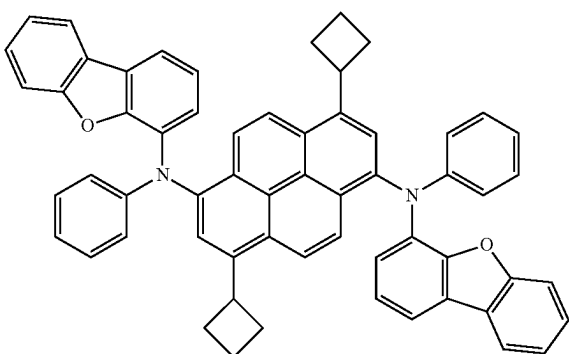
FD34



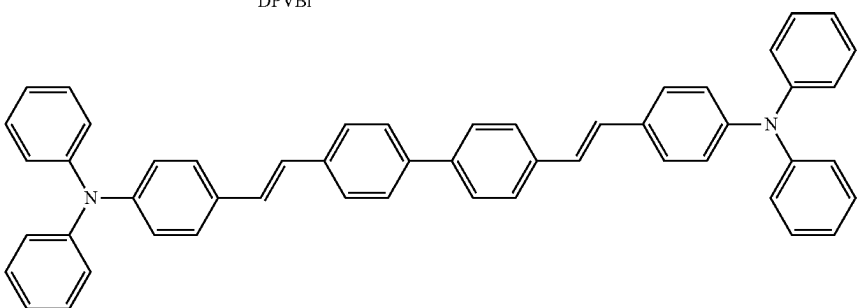
FD35



FD36



DPVBi



DPAVB

[Delayed Fluorescence Material]

The emission layer may include a delayed fluorescence material.

In the specification, the delayed fluorescence material may be selected from compounds capable of emitting delayed fluorescence based on a delayed fluorescence emission mechanism.

The delayed fluorescence material included in the emission layer may act as a host or a dopant depending on the type of other materials included in the emission layer.

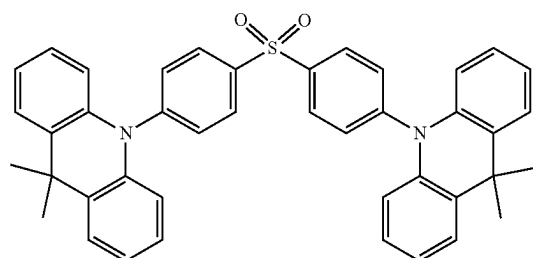
In an embodiment, a difference between a triplet energy level (eV) of the delayed fluorescence material and a singlet energy level (eV) of the delayed fluorescence material may be in a range of about 0 eV to about 0.5 eV. When the difference between the triplet energy level (eV) of the delayed fluorescence material and the singlet energy level

163

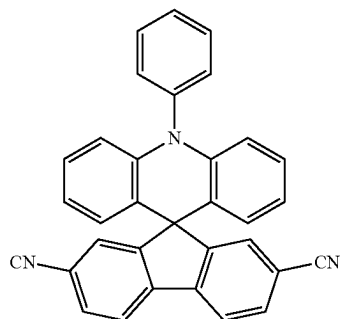
(eV) of the delayed fluorescence material satisfies the above-described range, up-conversion from the triplet state to the singlet state of the delayed fluorescence material may effectively occur, and thus, the luminescence efficiency of the light-emitting device **10** may be improved.

In an embodiment, the delayed fluorescence material may include a material including at least one electron donor (for example, a π electron-rich C_3 - C_{60} cyclic group, such as a carbazole group) and at least one electron acceptor (for example, a sulfoxide group, a cyano group, or a π electron-deficient nitrogen-containing C_1 - C_{60} cyclic group), and a material including a C_8 - C_{60} polycyclic group in which two or more cyclic groups are condensed while sharing boron (B).

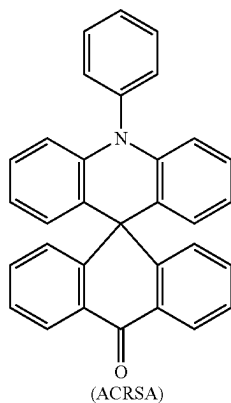
Examples of the delayed fluorescence material may include at least one of Compounds DF1 to DF9:



(DMAC-DPS)



(ACRFLCN)

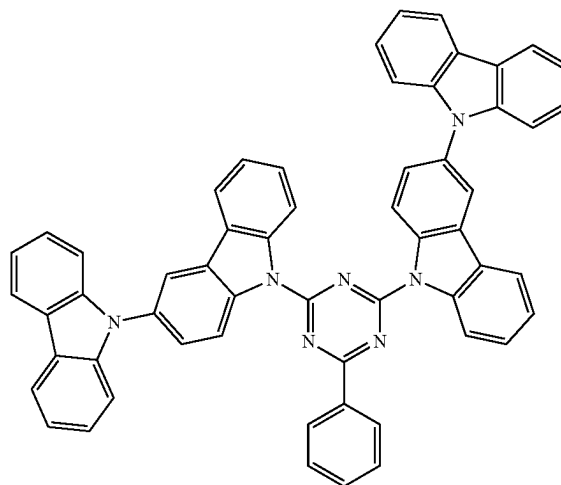


(ACRSA)

164

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DF4



(CC2TA)

DF1

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DF2

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DF3

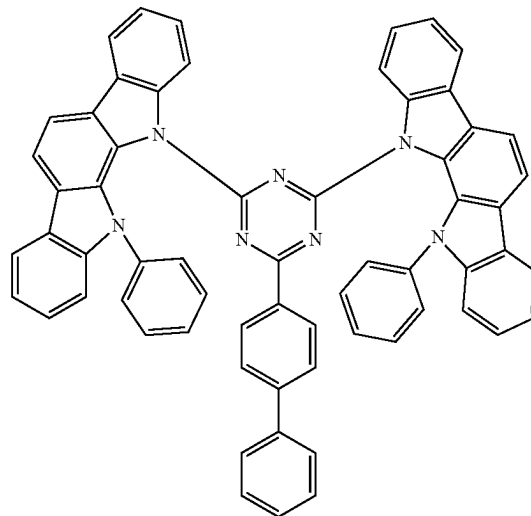
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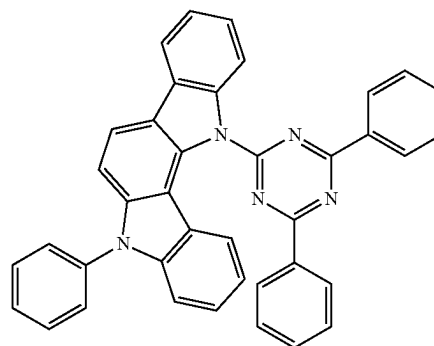
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DF5



(PIC-TRZ)

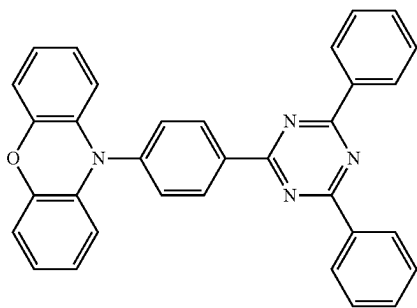
DF6



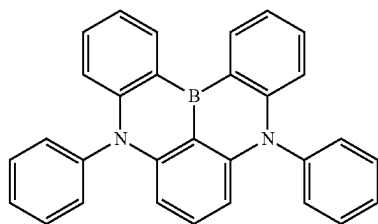
(PIC-TRZ2)

165

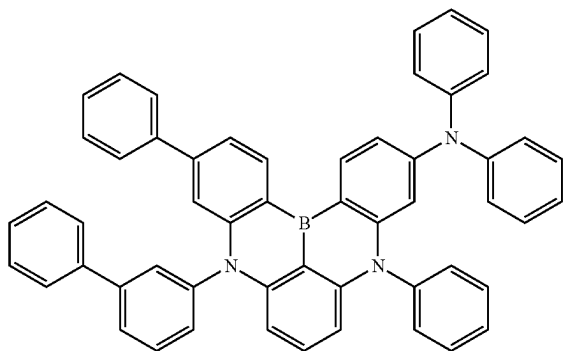
-continued



(PXZ-TRZ)



(DABNA-1)



(DABNA-2)

[Quantum Dot]

The emission layer may include a quantum dot.

In the specification, a quantum dot may be a crystal of a semiconductor compound, and may include any material capable of emitting light of various emission wavelengths according to the size of the crystal.

A diameter of the quantum dot may be, for example, in a range of about 1 nm to about 10 nm.

The quantum dot may be synthesized by a wet chemical process, a metal organic chemical vapor deposition process, a molecular beam epitaxy process, or any process similar thereto.

According to the wet chemical process, a precursor material is mixed with an organic solvent to grow a quantum dot particle crystal. When the crystal grows, the organic solvent naturally acts as a dispersant coordinated on the surface of the quantum dot crystal and controls the growth of the crystal so that the growth of quantum dot particles can be controlled through a process which is more easily performed than vapor deposition methods, such as metal organic chemical vapor deposition (MOCVD) or molecular beam epitaxy (MBE), and which requires low costs.

The quantum dot may include Group II-VI semiconductor compounds, Group III-V semiconductor compounds, Group

166

III-VI semiconductor compounds, Group I-III-VI semiconductor compounds, Group IV-VI semiconductor compounds, a Group IV element or compound, or any combination thereof.

5 Examples of the Group II-VI semiconductor compound may include a binary compound, such as CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnO, HgS, HgSe, HgTe, MgSe, or MgS; a ternary compound, such as CdSeS, CdSeTe, CdSeTe, ZnSeS, ZnSeTe, ZnSTe, HgSeS, HgSeTe, HgSTe, CdZnS, CdZnSe, CdZnTe, CdHgS, CdHgSe, CdHgTe, HgZnS, HgZnSe, HgZnTe, MgZnSe, or MgZnS; a quaternary compound, such as CdZnSeS, CdZnSeTe, CdZnSTe, CdHgSeS, CdHgSeTe, CdHgSTe, HgZnSeS, HgZnSeTe, or HgZnSTe; or any combination thereof.

15 Examples of the Group III-V semiconductor compound may include a binary compound, such as GaN, GaP, GaAs, GaSb, AlN, AlP, AlAs, AlSb, InN, InP, InAs, or InSb; a ternary compound, such as GaNP, GaNAs, GaNSb, GaPAS, GaPSb, AlNP, AlNAs, AlNSb, AlPAS, AlPSb, InGaP, InNP, InAlP, InNAs, InNSb, InPAS, or InPSb; a quaternary compound, such as GaAlNP, GaAlNAs, GaAlNSb, GaAlPAS, GaAlPSb, GaInNP, GaInNAs, GaInNSb, GaInPAS, GaInPSb, InAlNP, InAlNAs, InAlNSb, InAlPAS, or InAlPSb; or any combination thereof. The Group III-V semiconductor compound may further include a Group II element. Examples of the Group III-V semiconductor compound further including a Group II element may include InZnP, InGaZnP, or InAlZnP.

25 Examples of the Group III-VI semiconductor compound may include a binary compound, such as GaS, GaSe, Ga₂Se₃, GaTe, InS, InSe, In₂S₃, In₂Se₃, or InTe; a ternary compound, such as InGaS₃ or InGaSe₃; or any combination thereof.

30 Examples of the Group I-III-VI semiconductor compound may include: a ternary compound, such as AgInS, AgInS₂, CuInS, CuInS₂, CuGaO₂, AgGaO₂, or AgAlO₂; or any combination thereof.

35 Examples of the Group IV-VI semiconductor compound may include a binary compound, such as SnS, SnSe, SnTe, PbS, PbSe, or PbTe; a ternary compound, such as SnSeS, SnSeTe, SnSTe, PbSeS, PbSeTe, PbSTe, SnPbS, SnPbSe, or SnPbTe; a quaternary compound, such as SnPbSSe, SnPbSeTe, or SnPbSTe; or any combination thereof.

40 The Group IV element or compound may include a single element material, such as Si or Ge; a binary compound, such as SiC or SiGe; or any combination thereof.

Each element included in a multi-element compound such as the binary compound, the ternary compound, and the quaternary compound, may exist at a uniform concentration or at a non-uniform concentration in a particle.

45 The quantum dot may have a single structure or a core-shell structure. In the case of the quantum dot having a single structure, the concentration of each element included in the corresponding quantum dot may be uniform. In an embodiment, the material included in the core and the material included in the shell may be different from each other.

50 The shell of the quantum dot may be a protective layer that prevents chemical degeneration of the core to maintain semiconductor characteristics and/or may be a charging layer that imparts electrophoretic characteristics to the quantum dot. The shell may be a single layer or a multi-layer. An interface between the core and the shell may have a concentration gradient in which the concentration of an element existing in the shell decreases toward the center of the core.

65 Examples of the shell of the quantum dot may include an oxide of metal, an oxide of a metalloid, an oxide of a non-metal, a semiconductor compound, or a combination

167

thereof. Examples of the oxide of metal, metalloid, or non-metal may include a binary compound, such as SiO₂, Al₂O₃, TiO₂, ZnO, MnO, Mn₂O₃, Mn₃O₄, CuO, FeO, Fe₂O₃, Fe₃O₄, CoO, Co₃O₄, or NiO; a ternary compound, such as MgAl₂O₄, CoFe₂O₄, NiFe₂O₄, or CoMn₂O₄; or any combination thereof. Examples of the semiconductor compound may include, as described herein, Group II-VI semiconductor compounds; Group III-V semiconductor compounds; Group III-VI semiconductor compounds; Group I-III-VI semiconductor compounds; Group IV-VI semiconductor compounds; or any combination thereof. In an embodiment, the semiconductor compound may include CdS, CdSe, CdTe, ZnS, ZnSe, ZnTe, ZnSeS, ZnTeS, GaAs, GaP, GaSb, HgS, HgSe, HgTe, InAs, InP, InGaP, InSb, AlAs, AlP, AlSb, or any combination thereof.

A full width at half maximum (FWHM) of an emission wavelength spectrum of the quantum dot may be equal to or less than about 45 nm. For example, a FWHM of an emission wavelength spectrum of the quantum dot may be equal to or less than about 40 nm. For example, a FWHM of an emission wavelength spectrum of the quantum dot may be equal to or less than about 30 nm. Within these ranges, color purity or color gamut may be increased. Light emitted through the quantum dot may be emitted in all directions, and a wide viewing angle may be improved.

The quantum dot may be a spherical particle, a pyramidal particle, a multi-arm particle, a cubic nanoparticle, a nanotube particle, a nanowire particle, a nanofiber particle, or a nanoplate particle.

Since the energy band gap may be adjusted by controlling the size of the quantum dot, light having various wavelength bands may be obtained from the quantum dot emission layer. Accordingly, by using quantum dots of different sizes, a light-emitting device that emits light of various wavelengths may be implemented. For example, the size of the quantum dot may be selected to emit red, green and/or blue light. The size of the quantum dot may be configured to emit white light by combining light of various colors.

[Electron Transport Region in Interlayer 130]

The electron transport region may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer consisting of different materials, or a multi-layered structure including layers including different materials.

The electron transport region may include a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, an electron injection layer, or any combination thereof.

In an embodiment, the electron transport region may have an electron transport layer/electron injection layer structure, a hole blocking layer/electron transport layer/electron injection layer structure, an electron control layer/electron transport layer/electron injection layer structure, or a buffer layer/electron transport layer/electron injection layer structure, wherein, for each structure, constituting layers may be stacked from an emission layer in its respective stated order, but embodiments are not limited thereto.

The electron transport region (for example, the buffer layer, the hole blocking layer, the electron control layer, or the electron transport layer in the electron transport region) may include a metal-free compound including at least one π electron-deficient nitrogen-containing C₁-C₆₀ cyclic group.

In an embodiment, the electron transport region may include a compound represented by Formula 601:



In Formula 601,

Ar₆₀₁ and L₆₀₁ may each independently be a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at

168

least one R_{10a} or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a},

xe11 may be 1, 2, or 3,

xe1 may be 0, 1, 2, 3, 4, or 5,

R₆₀₁ may be a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a}, —Si(Q₆₀₁)(Q₆₀₂)(Q₆₀₃), —C(=O)(Q₆₀₁), —S(=O)₂(Q₆₀₁), or —P(=O)(Q₆₀₁)(Q₆₀₂),

Q₆₀₁ to Q₆₀₃ may each independently be the same as described in connection with Q₁,

xe21 may be 1, 2, 3, 4, or 5, and

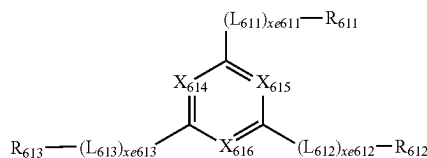
at least one of Ar₆₀₁, L₆₀₁, and R₆₀₁ may each independently be a π electron-deficient nitrogen-containing C₁-C₆₀ cyclic group unsubstituted or substituted with at least one R_{10a}. R_{10a} may be the same as described in the specification.

In an embodiment, when xe11 in Formula 601 is 2 or more, two or more of Ar₆₀₁(s) may be linked to each other via a single bond.

In embodiments, Ar₆₀₁ in Formula 601 may be a substituted or unsubstituted anthracene group.

In embodiments, the electron transport region may include a compound represented by Formula 601-1:

[Formula 601-1]



In Formula 601-1,

X₆₁₄ may be N or C(R₆₁₄), X₆₁₅ may be N or C(R₆₁₅), X₆₁₆ may be N or C(R₆₁₆), and at least one of X₆₁₄ to X₆₁₆ may be N,

L₆₁₁ to L₆₁₃ may each independently be the same as described in connection with L₆₀₁,

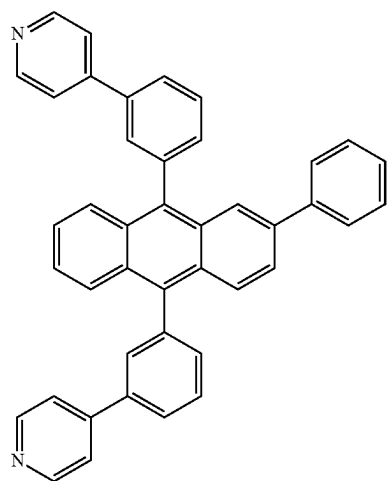
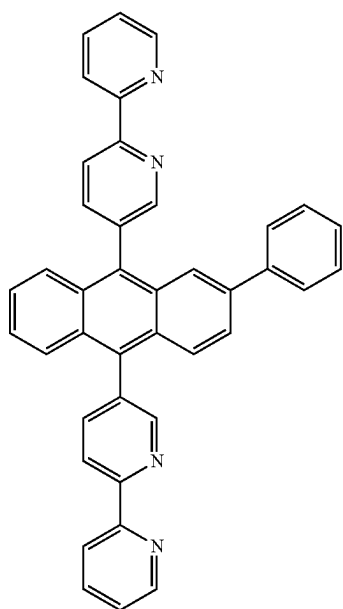
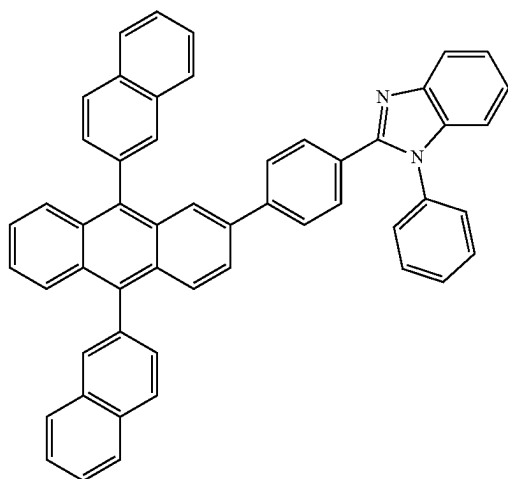
xe611 to xe613 may each independently be the same as described in connection with xe1,

R₆₁₁ to R₆₁₃ may each independently be the same as described in connection with R₆₀₁, and

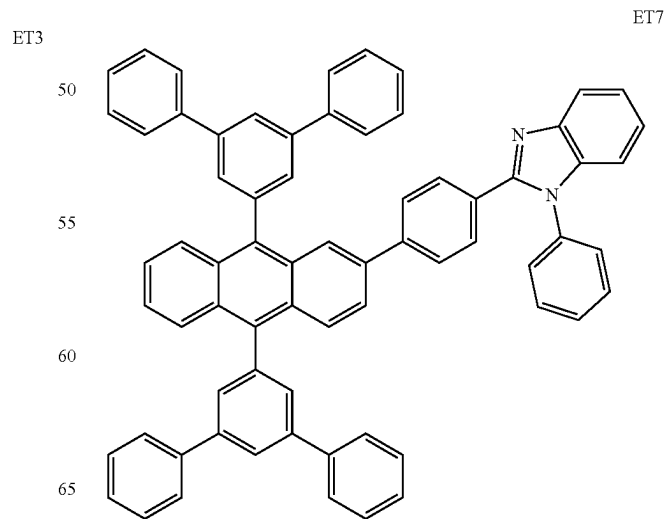
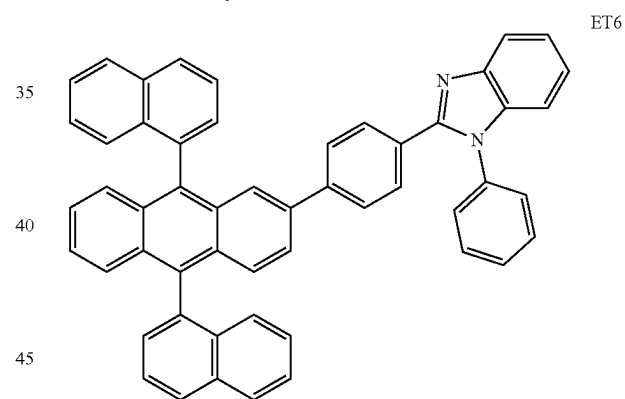
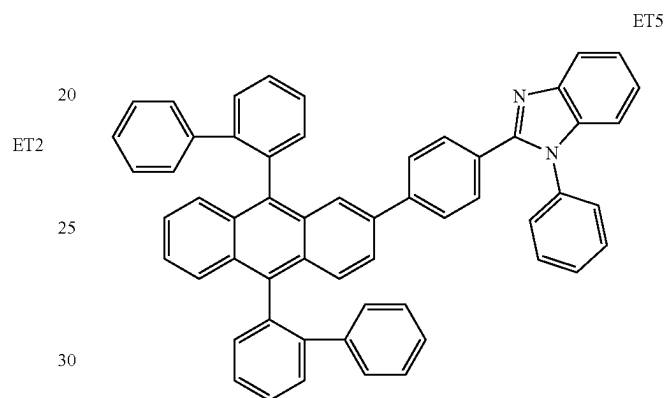
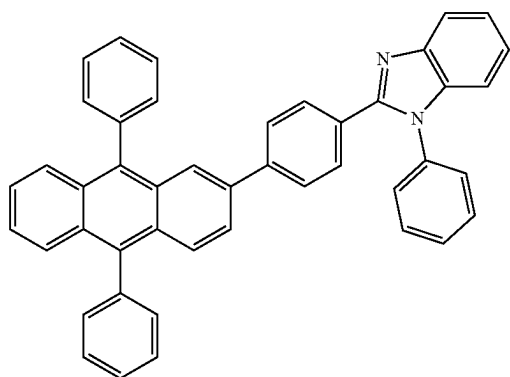
R₆₁₄ to R₆₁₆ may each independently be hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C₁-C₂₀ alkyl group, a C₁-C₂₀ alkoxy group, a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a}, or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a}. R_{10a} may be the same as described in the specification.

In an embodiment, xe1 and xe611 to xe613 in Formulae 601 and 601-1 may each independently be 0, 1, or 2.

The electron transport region may include one of Compounds ET1 to ET45, 2,9-dimethyl-4,7-diphenyl-1,10-phenanthroline (BCP), 4,7-diphenyl-1,10-phenanthroline (Bphen), Alq₃, BAlq, TAZ, NTAZ, or any combination thereof:

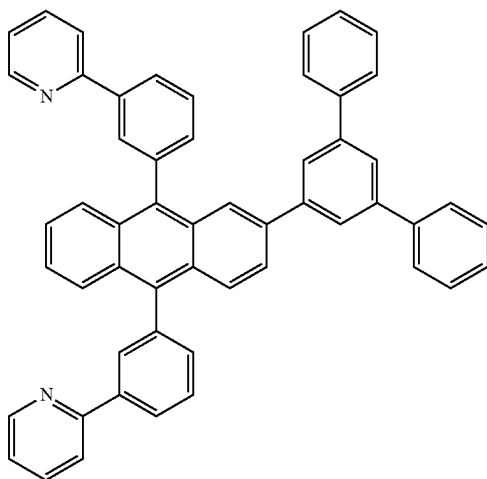
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ET8

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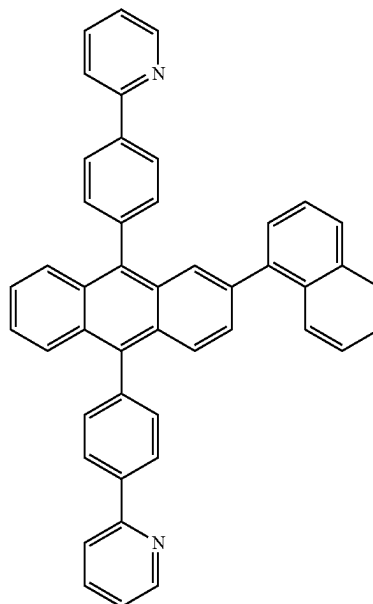
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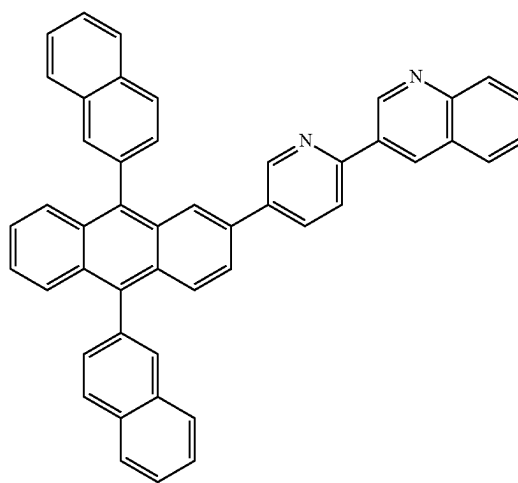
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ET9

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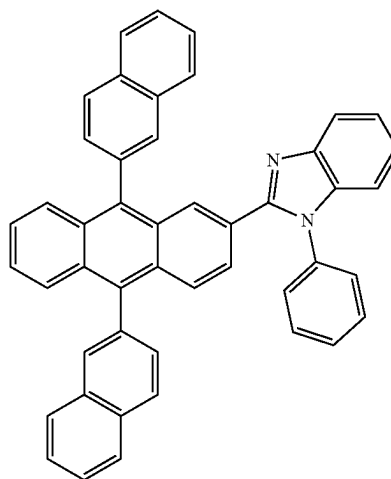
ET11

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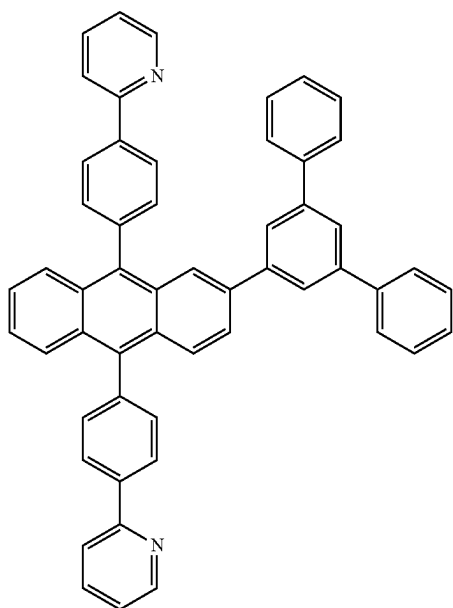
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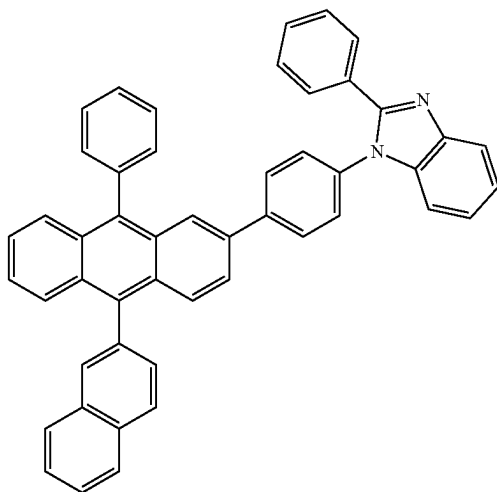
ET12



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ET13



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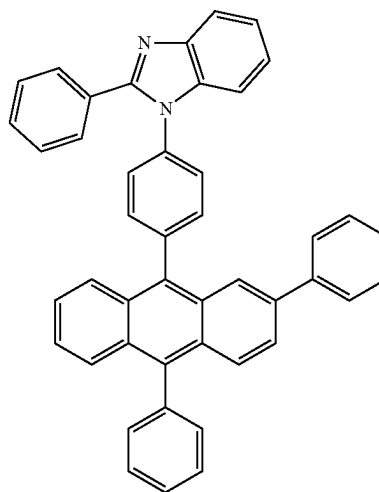
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ET16



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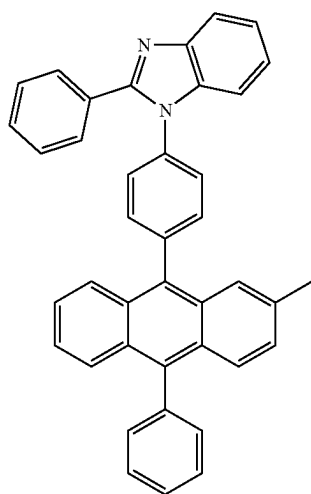
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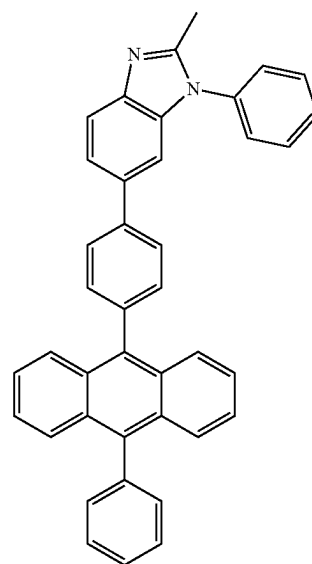


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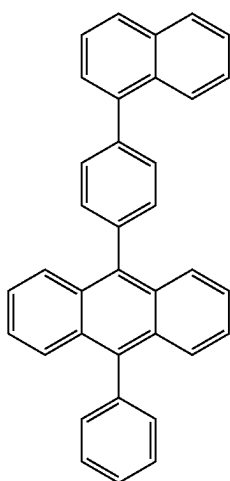
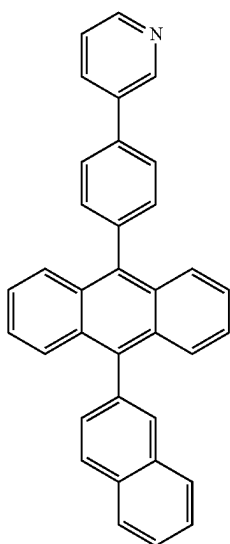
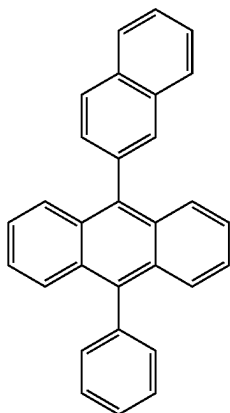


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ET21

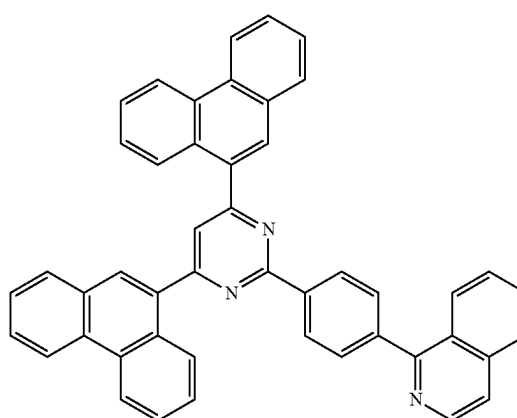
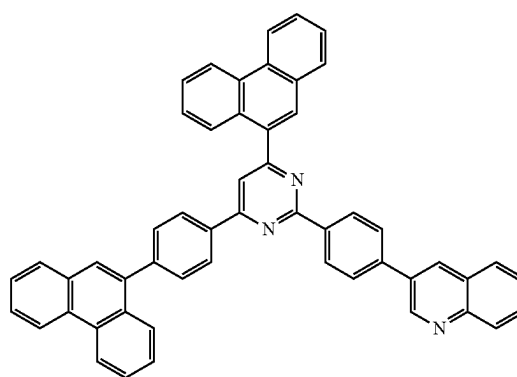
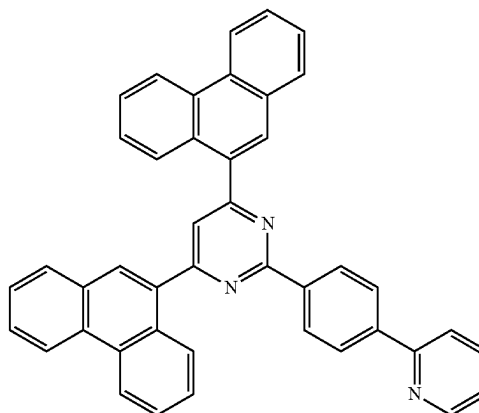
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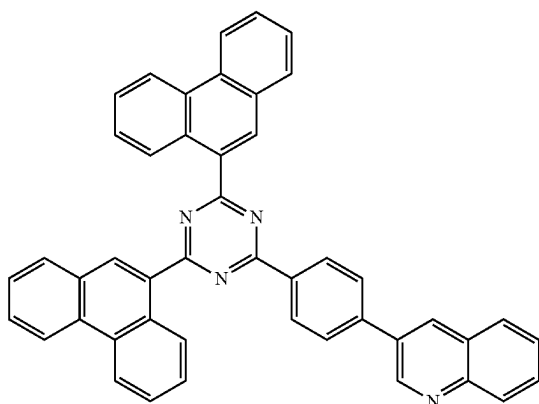
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ET25



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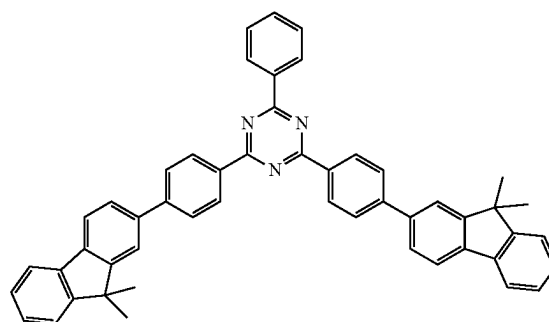
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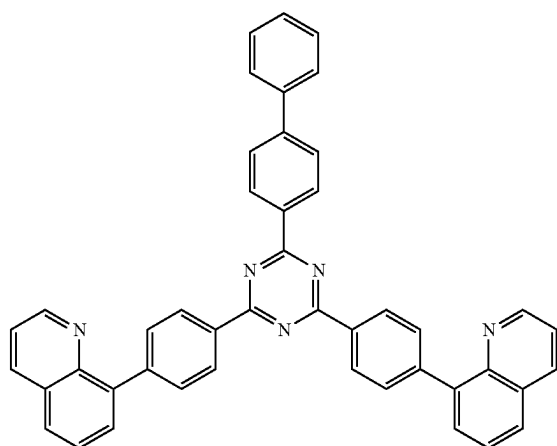
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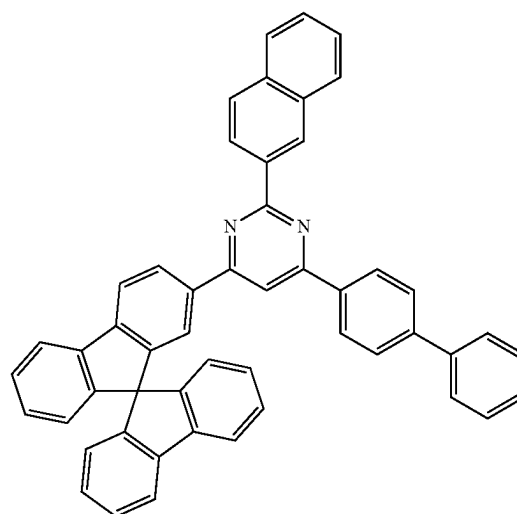
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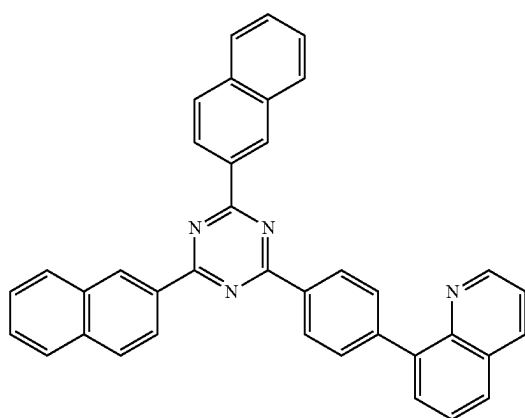


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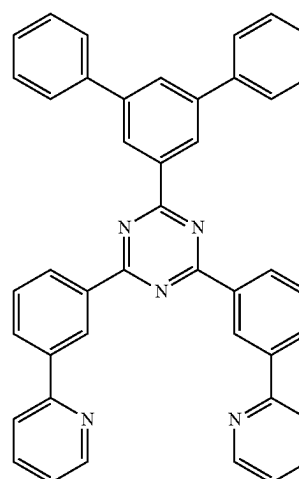
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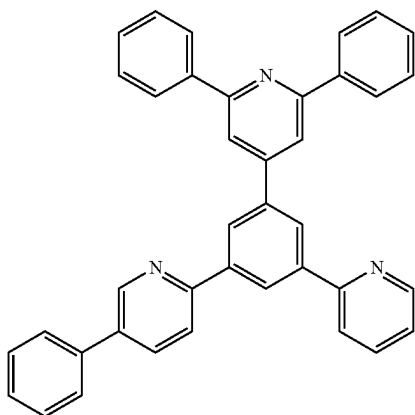
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ET31

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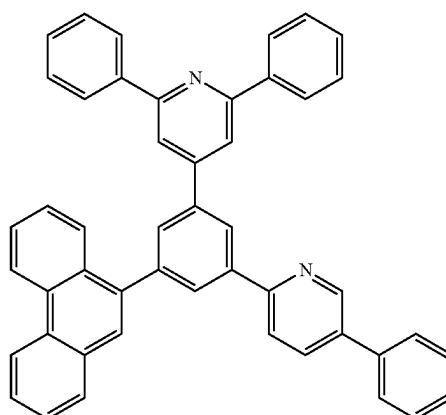
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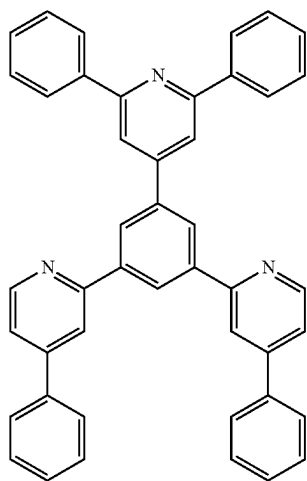
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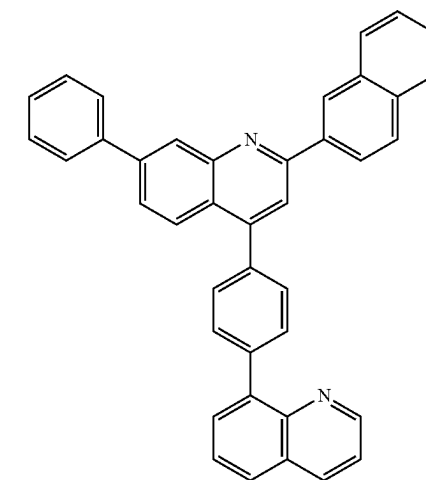
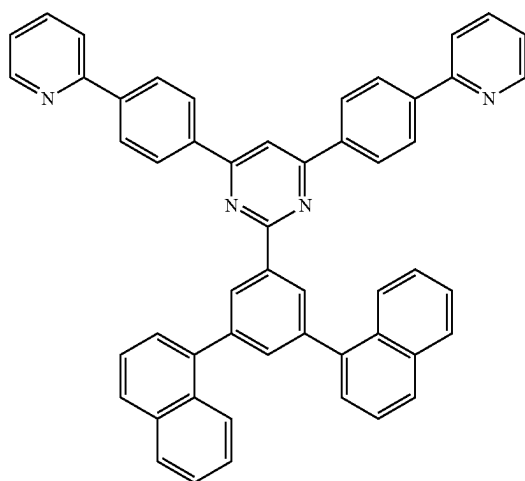
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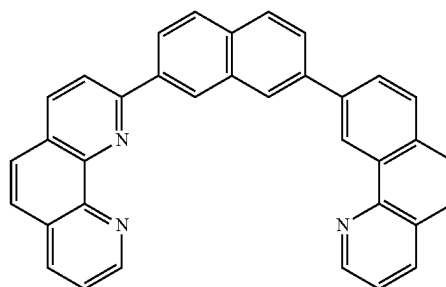
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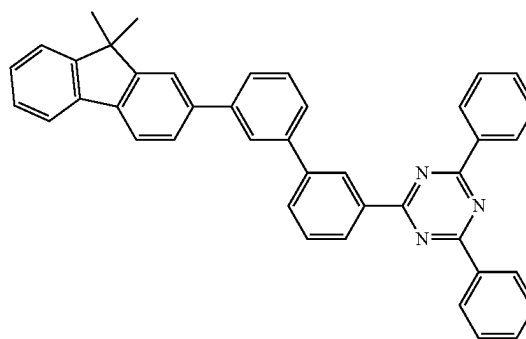
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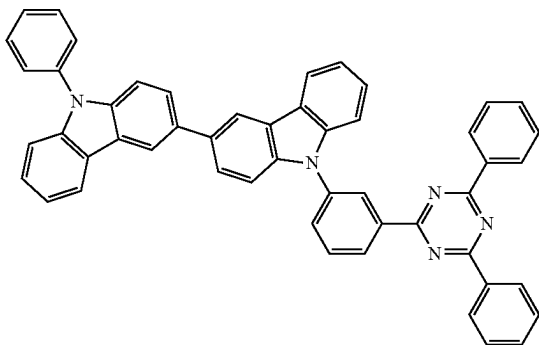
ET37



181

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ET38



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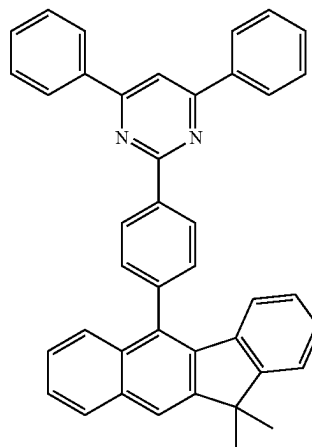
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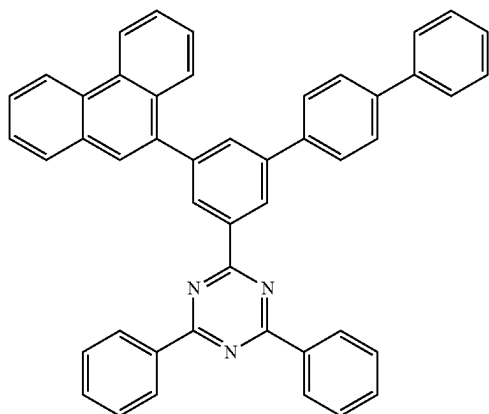
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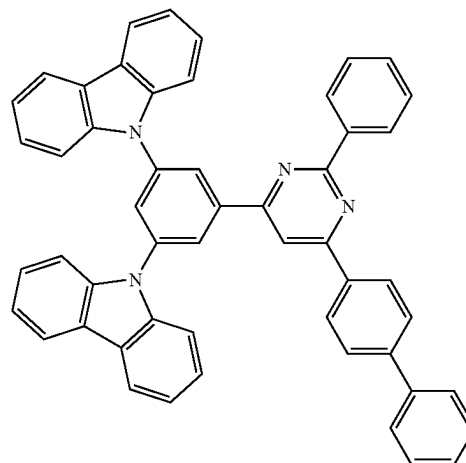
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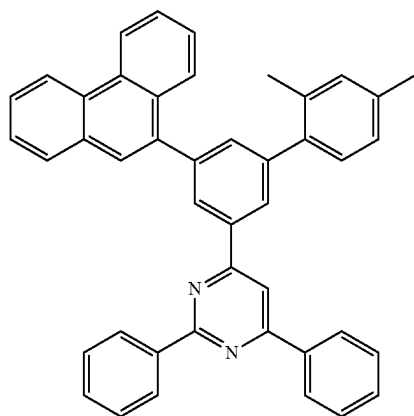
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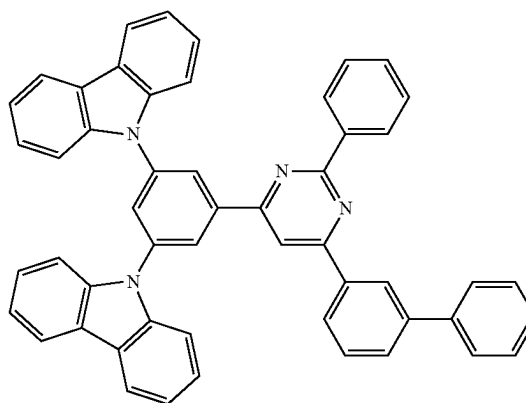


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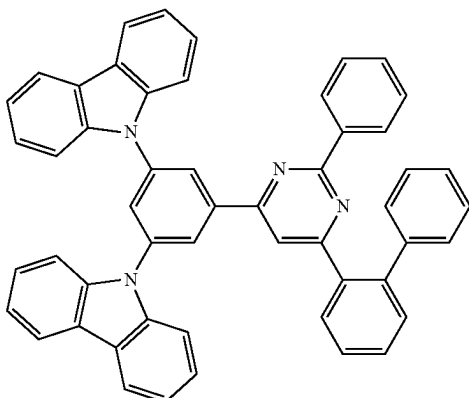
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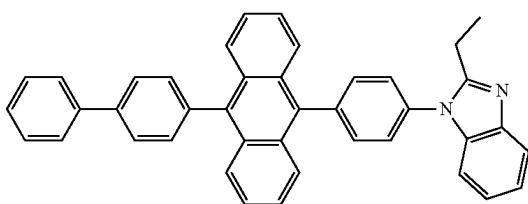
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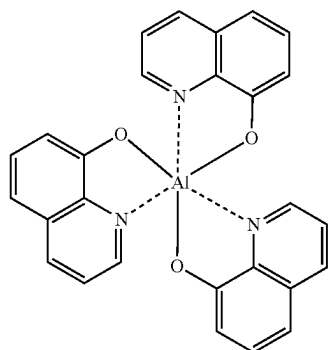
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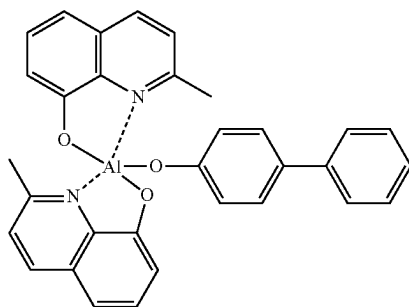
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Alq₃

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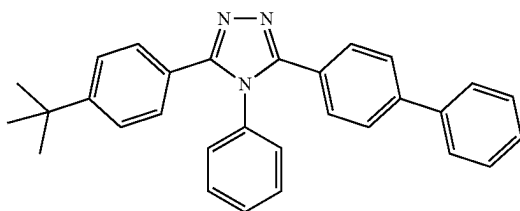
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BAQ

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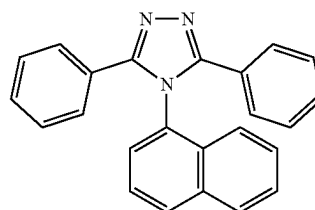
TAZ

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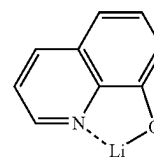
NTAZ

A thickness of the electron transport region may be in a range of about 100 Å to about 5,000 Å. For example, the thickness of the electron transport region may be in a range of about 160 Å to about 4,000 Å. When the electron transport region includes a buffer layer, a hole blocking layer, an electron control layer, an electron transport layer, or any combination thereof, a thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be in a range of about 20 Å to about 1,000 Å, and a thickness of the electron transport layer may be in a range of about 100 Å to about 1,000 Å. For example, the thickness of the buffer layer, the hole blocking layer, or the electron control layer may each independently be in a range of about 30 Å to about 300 Å. For example, the thickness of the electron transport layer may be in a range of about 150 Å to about 500 Å. When the thicknesses of the buffer layer, the hole blocking layer, the electron control layer, and/or the electron transport layer are within these ranges, satisfactory electron transporting characteristics may be obtained without a substantial increase in driving voltage.

The electron transport region (for example, the electron transport layer in the electron transport region) may further include, in addition to the materials described above, a metal-containing material.

The metal-containing material may include an alkali metal complex, an alkaline earth metal complex, or any combination thereof. A metal ion of the alkali metal complex may be a Li ion, a Na ion, a K ion, a Rb ion, or a Cs ion, and a metal ion of the alkaline earth metal complex may be a Be ion, a Mg ion, a Ca ion, a Sr ion, or a Ba ion. A ligand coordinated with the metal ion of the alkali metal complex or the alkaline earth-metal complex may include a hydroxyquinoline, a hydroxyisoquinoline, a hydroxybenzoquinoline, a hydroxyacridine, a hydroxyphenanthridine, a hydroxyphenyloxazole, a hydroxyphenylthiazole, a hydroxyphenyloxadiazole, a hydroxyphenylthiadiazole, a hydroxyphenylpyridine, a hydroxyphenylbenzimidazole, a hydroxyphenylbenzothiazole, a bipyridine, a phenanthroline, a cyclopentadiene, or any combination thereof.

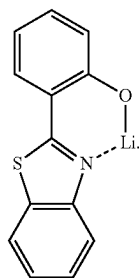
In an embodiment, the metal-containing material may include a Li complex. The Li complex may include, for example, Compound ET-D1 (LiQ) or ET-D2:



ET-D1

185

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The electron transport region may include an electron injection layer that facilitates the injection of electrons from the second electrode **150**. The electron injection layer may directly contact the second electrode **150**.

The electron injection layer may have a structure consisting of a layer consisting of a single material, a structure consisting of a layer consisting of different materials, or a multi-layered structure including layers including different materials.

The electron injection layer may include an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof.

The alkali metal may include Li, Na, K, Rb, Cs, or any combination thereof. The alkaline earth metal may include Mg, Ca, Sr, Ba, or any combination thereof. The rare earth metal may include Sc, Y, Ce, Tb, Yb, Gd, or any combination thereof.

The alkali metal-containing compound, the alkaline earth metal-containing compound, and the rare earth metal-containing compound may include oxides, halides (for example, fluorides, chlorides, bromides, or iodides), or tellurides of the alkali metal, the alkaline earth metal, and the rare earth metal, or any combination thereof.

The alkali metal-containing compound may include alkali metal oxides, such as Li_2O , Cs_2O , or K_2O , alkali metal halides, such as LiF , NaF , CsF , KF , LiI , NaI , CsI , or KI , or any combination thereof. The alkaline earth metal-containing compound may include an alkaline earth metal oxides, such as BaO , SrO , CaO , $\text{Ba}_x\text{Sr}_{1-x}\text{O}$ (x is a real number satisfying the condition of $0 < x < 1$), or the like. The rare earth metal-containing compound may include YbF_3 , ScF_3 , Sc_2O_3 , Y_2O_3 , Ce_2O_3 , GdF_3 , TbF_3 , YbI_3 , ScI_3 , TbI_3 , or any combination thereof. In embodiments, the rare earth metal-containing compound may include a lanthanide metal telluride. Examples of the lanthanide metal telluride may include LaTe , CeTe , PrTe , NdTe , PmTe , SmTe , EuTe , GdTe , TbTe , DyTe , HoTe , ErTe , TmTe , YbTe , LuTe , La_2Te_3 , Ce_2Te_3 , Pr_2Te_3 , Nd_2Te_3 , Pm_2Te_3 , Sm_2Te_3 , Eu_2Te_3 , Gd_2Te_3 , Tb_2Te_3 , Dy_2Te_3 , Ho_2Te_3 , Er_2Te_3 , Tm_2Te_3 , Yb_2Te_3 , and Lu_2Te_3 .

The alkali metal complex, the alkaline earth-metal complex, and the rare earth metal complex may include one of ions of the alkali metal, ions of the alkaline earth metal, and ions of the rare earth metal and a ligand bonded to the metal ion, for example, hydroxyquinoline, hydroxyisoquinoline, hydroxybenzoquinoline, hydroxyacridine, hydroxyphenanthridine, hydroxyphenyloxazole, hydroxyphenylthiazole, hydroxyphenyloxadiazole, hydroxyphenylthiadiazole, hydroxyphenylpyridine, hydroxyphenyl benzimidazole,

186

hydroxyphenylbenzothiazole, bipyridine, phenanthroline, cyclopentadiene, or any combination thereof.

The electron injection layer may consist of an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth metal complex, a rare earth metal complex, or any combination thereof, as described above. In embodiments, the electron injection layer may further include an organic material (for example, a compound represented by Formula 601).

In an embodiment, the electron injection layer may consist of an alkali metal-containing compound (for example, an alkali metal halide); or an alkali metal-containing compound (for example, an alkali metal halide), and an alkali metal, an alkaline earth metal, a rare earth metal, or any combination thereof. In an embodiment, the electron injection layer may be a $\text{KI}:\text{Yb}$ co-deposited layer, an $\text{RbI}:\text{Yb}$ co-deposited layer, or the like.

When the electron injection layer further includes an organic material, an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal-containing compound, an alkaline earth metal-containing compound, a rare earth metal-containing compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or any combination thereof may be homogeneously or non-homogeneously dispersed in a matrix including the organic material.

A thickness of the electron injection layer may be in a range of about 1 Å to about 100 Å. For example, the thickness of the electron injection layer may be in a range of about 3 Å to about 90 Å. When the thickness of the electron injection layer is within this range, satisfactory electron injection characteristics may be obtained without a substantial increase in driving voltage.

[Second Electrode **150**]

The second electrode **150** may be disposed on the interlayer **130** having such a structure. The second electrode **150** may be a cathode, which is an electron injection electrode, and as the material for the second electrode **150**, a metal, an alloy, an electrically conductive compound, or any combination thereof, each having a low work function, may be used.

The second electrode **150** may include lithium (Li), silver (Ag), magnesium (Mg), aluminum (Al), aluminum-lithium (Al—Li), calcium (Ca), magnesium-indium (Mg—In), magnesium-silver (Mg—Ag), ytterbium (Yb), silver-ytterbium (Ag—Yb), ITO, IZO, or any combination thereof. The second electrode **150** may be a transmissive electrode, a semi-transmissive electrode, or a reflective electrode.

The second electrode **150** may have a single-layered structure or a multi-layered structure including two or more layers.

[Capping Layer]

A first capping layer may be located outside the first electrode **110**, and/or a second capping layer may be located outside the second electrode **150**. For example, the light-emitting device **10** may have a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, and the second electrode **150** are stacked in this stated order, a structure in which the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are stacked in this stated order, or a structure in which the first capping layer, the first electrode **110**, the interlayer **130**, the second electrode **150**, and the second capping layer are stacked in this stated order.

187

Light generated in an emission layer of the interlayer **130** of the light-emitting device **10** may be extracted toward the outside through the first electrode **110** (which may be a semi-transmissive electrode or a transmissive electrode) and through the first capping layer. Light generated in an emission layer of the interlayer **130** of the light-emitting device **10** may be extracted toward the outside through the second electrode **150** (which may be a semi-transmissive electrode or a transmissive electrode) and through the second capping layer.

The first capping layer and the second capping layer may each increase external luminescence efficiency according to the principle of constructive interference. Accordingly, the light extraction efficiency of the light-emitting device **10** may be increased, so that the luminescence efficiency of the light-emitting device **10** may be improved.

Each of the first capping layer and the second capping layer may include a material having a refractive index (at a wavelength of about 589 nm) equal to or greater than about 1.6.

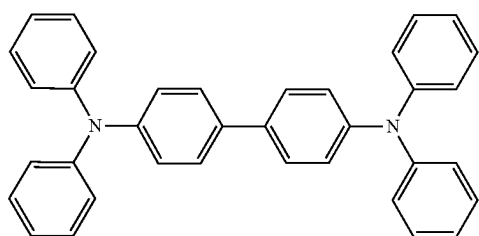
The first capping layer and the second capping layer may each independently be an organic capping layer including an organic material, an inorganic capping layer including an inorganic material, or an organic-inorganic composite capping layer including an organic material and an inorganic material.

At least one of the first capping layer and the second capping layer may each independently include carbocyclic compounds, heterocyclic compounds, amine group-containing compounds, porphyrin derivatives, phthalocyanine derivatives, a naphthalocyanine derivatives, alkali metal complexes, alkaline earth metal complexes, or any combination thereof. The carbocyclic compound, the heterocyclic compound, and the amine group-containing compound may each independently be optionally substituted with a substituent containing O, N, S, Se, Si, F, Cl, Br, I, or any combination thereof.

In an embodiment, at least one of the first capping layer and the second capping layer may each independently include an amine group-containing compound.

In an embodiment, at least one of the first capping layer and the second capping layer may each independently include a compound represented by Formula 201, a compound represented by Formula 202, or any combination thereof.

In embodiments, at least one of the first capping layer and the second capping layer may each independently include one of Compounds HT28 to HT33, one of Compounds CP1 to CP6, β -NPB, or any combination thereof:

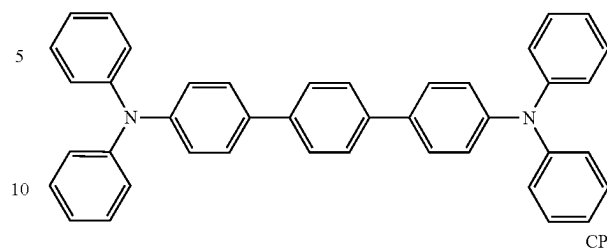


CP1

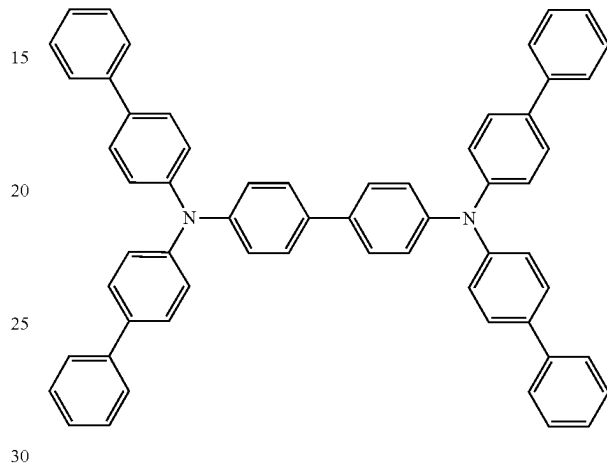
188

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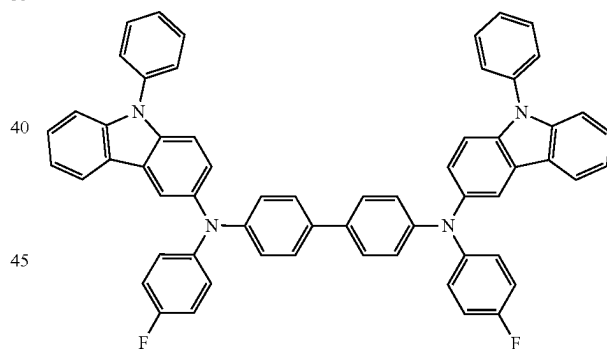
CP2



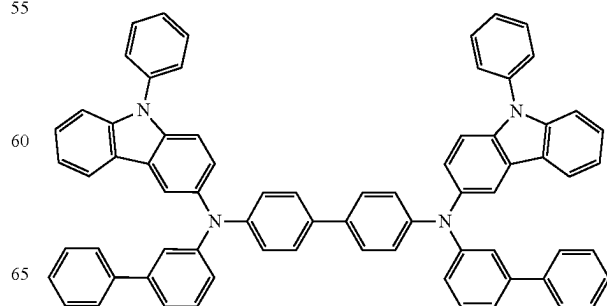
CP3



CP4

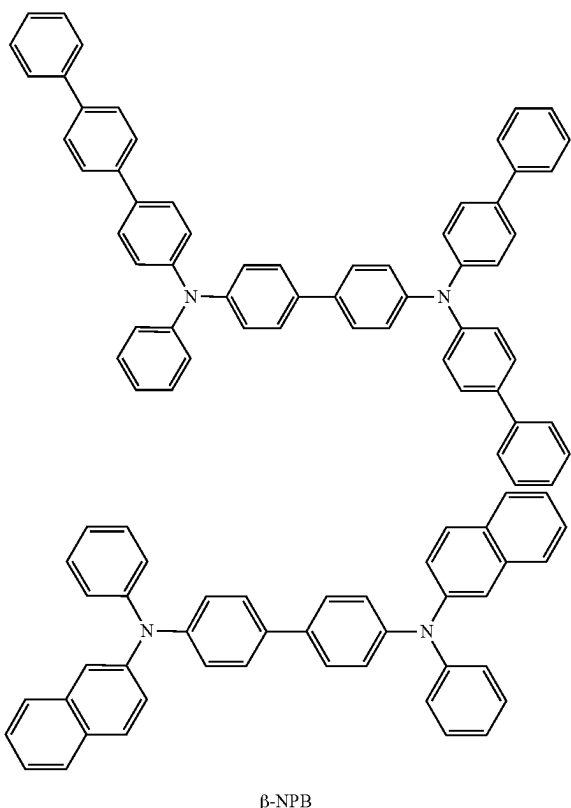


CP5



189

-continued



[Film]

The organometallic compound represented by Formula 1 may be included in various films. Accordingly, according to another aspect, a film including the organometallic compound represented by Formula 1 may be provided. The film may be, for example, an optical member (or a light controlling means) (for example, a color filter, a color conversion member, a capping layer, a light extraction efficiency enhancement layer, a selective light absorbing layer, a polarizing layer, or a quantum dot-containing layer), a light-blocking member (for example, a light reflective layer or a light absorbing layer), or a protective member (for example, an insulating layer or a dielectric layer).

[Electronic Apparatus]

The light-emitting device may be included in various electronic apparatuses. In an embodiment, the electronic apparatus including the light-emitting device may be a light-emitting apparatus, an authentication apparatus, or the like.

The electronic apparatus (for example, a light-emitting apparatus) may further include, in addition to the light-emitting device, a color filter, a color conversion layer, or a color filter and a color conversion layer. The color filter and/or the color conversion layer may be located in at least one traveling direction of light emitted from the light-emitting device. In an embodiment, the light emitted from the light-emitting device may be blue light or white light. The light-emitting device may be the same as described above. In an embodiment, the color conversion layer may include quantum dots. The quantum dot may be, for example, a quantum dot as described herein.

190

The electronic apparatus may include a first substrate. The first substrate may include subpixels, the color filter may include color filter areas respectively corresponding to the subpixels, and the color conversion layer may include color conversion areas respectively corresponding to the subpixels.

A pixel-defining layer may be located among the subpixels to define each of the subpixels.

The color filter may further include color filter areas and light-shielding patterns located among the color filter areas, and the color conversion layer may include color conversion areas and light-shielding patterns located among the color conversion areas.

The color filter areas (or the color conversion areas) may include a first area emitting first-color light, a second area emitting second-color light, and/or a third area emitting third-color light, and the first-color light, the second-color light, and/or the third-color light may have different maximum emission wavelengths from one another. In an embodiment, the first-color light may be red light, the second-color light may be green light, and the third-color light may be blue light. In an embodiment, the color filter areas (or the color conversion areas) may include quantum dots. For example, the first area may include a red quantum dot, the second area may include a green quantum dot, and the third area may not include a quantum dot. The quantum dot may be the same as described in the specification. Each of the first area, the second area, and/or the third area may further include a scatterer.

In an embodiment, the light-emitting device may emit first light, the first area may absorb the first light to emit first first-color light, the second area may absorb the first light to emit second first-color light, and the third area may absorb the first light to emit third first-color light. In this regard, the first first-color light, the second first-color light, and the third first-color light may have different maximum emission wavelengths from one another. For example, the first light may be blue light, the first first-color light may be red light, the second first-color light may be green light, and the third first-color light may be blue light.

The electronic apparatus may further include a thin-film transistor in addition to the light-emitting device as described above. The thin-film transistor may include a source electrode, a drain electrode, and an active layer, wherein any one of the source electrode and the drain electrode may be electrically connected to any one of the first electrode and the second electrode of the light-emitting device.

The thin-film transistor may further include a gate electrode, a gate insulating film, or the like.

The active layer may include crystalline silicon, amorphous silicon, organic semiconductor, oxide semiconductor, or the like.

The electronic apparatus may further include a sealing portion for sealing the light-emitting device. The sealing portion and/or the color conversion layer may be located between the color filter and the light-emitting device. The sealing portion may allow light from the light-emitting device to be extracted to the outside, and may simultaneously prevent ambient air and/or moisture from penetrating into the light-emitting device. The sealing portion may be a sealing substrate including a transparent glass substrate or a plastic substrate. The sealing portion may be a thin-film encapsulation layer including at least one of an organic layer and/or an inorganic layer. When the sealing portion is a thin film encapsulation layer, the electronic apparatus may be flexible.

191

Various functional layers may be additionally located on the sealing portion, in addition to the color filter and/or the color conversion layer, according to the use of the electronic apparatus. Examples of the functional layers may include a touch screen layer, a polarizing layer, an authentication apparatus, and the like. The touch screen layer may be a pressure-sensitive touch screen layer, a capacitive touch screen layer, or an infrared touch screen layer. The authentication apparatus may be, for example, a biometric authentication apparatus that authenticates an individual by using biometric information of a living body (for example, fingertips, pupils, etc.).

The authentication apparatus may further include, in addition to the light-emitting device, a biometric information collector.

The electronic apparatus may be applied to various displays, such as light sources, lighting, personal computers (for example, a mobile personal computer), mobile phones, digital cameras, electronic organizers, electronic dictionaries, electronic game machines, medical instruments (for example, electronic thermometers, sphygmomanometers, blood glucose meters, pulse measurement devices, pulse wave measurement devices, electrocardiogram displays, ultrasonic diagnostic devices, or endoscope displays), fish finders, various measuring instruments, meters (for example, meters for a vehicle, an aircraft, and a vessel), projectors, and the like.

[Description of FIGS. 2 and 3]

FIG. 2 is a schematic cross-sectional view of an electronic apparatus according to an embodiment of the disclosure.

The electronic apparatus of FIG. 2 includes a substrate 100, a thin-film transistor (TFT), a light-emitting device, and an encapsulation portion 300 that seals the light-emitting device.

The substrate 100 may be a flexible substrate, a glass substrate, or a metal substrate. A buffer layer 210 may be located on the substrate 100. The buffer layer 210 may prevent penetration of impurities through the substrate 100 and may provide a flat surface on the substrate 100.

A TFT may be located on the buffer layer 210. The TFT may include an active layer 220, a gate electrode 240, a source electrode 260, and a drain electrode 270.

The active layer 220 may include an inorganic semiconductor such as silicon or polysilicon, an organic semiconductor, or an oxide semiconductor, and may include a source region, a drain region, and a channel region.

A gate insulating film 230 for insulating the active layer 220 from the gate electrode 240 may be located on the active layer 220, and the gate electrode 240 may be located on the gate insulating film 230.

An interlayer insulating film 250 may be located on the gate electrode 240. The interlayer insulating film 250 may be located between the gate electrode 240 and the source electrode 260 to insulate the gate electrode 240 from the source electrode 260 and between the gate electrode 240 and the drain electrode 270 to insulate the gate electrode 240 from the drain electrode 270.

The source electrode 260 and the drain electrode 270 may be located on the interlayer insulating film 250. The interlayer insulating film 250 and the gate insulating film 230 may be formed to expose the source region and the drain region of the active layer 220, and the source electrode 260 and the drain electrode 270 may be in contact with the exposed portions of the source region and the drain region of the active layer 220.

The TFT is electrically connected to a light-emitting device to drive the light-emitting device, and is covered by

192

a passivation layer 280. The passivation layer 280 may include an inorganic insulating film, an organic insulating film, or any combination thereof. A light-emitting device is provided on the passivation layer 280. The light-emitting device may include a first electrode 110, an interlayer 130, and a second electrode 150.

The first electrode 110 may be located on the passivation layer 280. The passivation layer 280 may not completely cover the drain electrode 270 and may expose a portion of the drain electrode 270, and the first electrode 110 may be electrically connected to the exposed portion of the drain electrode 270.

A pixel defining layer 290 containing an insulating material may be located on the first electrode 110. The pixel defining layer 290 may expose a portion of the first electrode 110, and the interlayer 130 may be formed in the exposed portion of the first electrode 110. The pixel defining layer 290 may be a polyimide or polyacrylic organic film. Although not shown in FIG. 2, at least some layers of the interlayer 130 may extend beyond the upper portion of the pixel defining layer 290 to be provided in the form of a common layer.

The second electrode 150 may be disposed on the interlayer 130, and a capping layer 170 may be additionally formed on the second electrode 150. The capping layer 170 may be formed to cover the second electrode 150.

The encapsulation portion 300 may be located on the capping layer 170. The encapsulation portion 300 may be located on a light-emitting device to protect the light-emitting device from moisture and/or oxygen. The encapsulation portion 300 may include: an inorganic film including silicon nitride (SiN_x), silicon oxide (SiO_x), indium tin oxide, indium zinc oxide, or any combination thereof; an organic film including polyethylene terephthalate, polyethylene naphthalate, polycarbonate, polyimide, polyethylene sulfonate, polyoxymethylene, polyarylate, hexamethyldisiloxane, an acrylic resin (for example, polymethyl methacrylate, polyacrylic acid, or the like), an epoxy-based resin (for example, aliphatic glycidyl ether (AGE), or the like), or any combination thereof; or any combination of the inorganic film and the organic film.

FIG. 3 is a schematic cross-sectional view of an electronic apparatus according to an embodiment of the disclosure.

The electronic apparatus of FIG. 3 is the same as the electronic apparatus of FIG. 2, except that a light-shielding pattern 500 and a functional region 400 are additionally located on the encapsulation portion 300. The functional region 400 may be a color filter area, a color conversion area, or a combination of the color filter area and the color conversion area. In an embodiment, the light-emitting device included in the electronic apparatus of FIG. 3 may be a tandem light-emitting device.

[Manufacture Method]

Respective layers included in the hole transport region, the emission layer, and respective layers included in the electron transport region may be formed in a certain region by using one or more suitable methods selected from vacuum deposition, spin coating, casting, Langmuir-Blodgett (LB) deposition, ink-jet printing, laser-printing, and laser-induced thermal imaging.

When respective layers included in the hole transport region, the emission layer, and respective layers included in the electron transport region are formed by vacuum deposition, the deposition may be performed at a deposition temperature of about 100° C. to about 500° C., a vacuum degree of about 10^{-8} torr to about 10^{-3} torr, and a deposition speed of about 0.01 Å/sec to about 100 Å/sec, depending on

a material to be included in a layer to be formed and the structure of a layer to be formed.

Definitions of Terms

The term “C₃-C₆₀ carbocyclic group” as used herein may be a cyclic group consisting only of carbon as a ring-forming atom and having 3 to 60 carbon atoms, and the term “C₁-C₆₀ heterocyclic group” as used herein may be a cyclic group that has 1 to 60 carbon atoms and further has, in addition to carbon, at least one heteroatom as a ring-forming atom. The C₃-C₆₀ carbocyclic group and the C₁-C₆₀ heterocyclic group may each be a monocyclic group consisting of one ring or a polycyclic group in which two or more rings are condensed with each other. In an embodiment, a C₁-C₆₀ heterocyclic group may have 3 to 61 ring-forming atoms.

The term “cyclic group” as used herein may include the C₃-C₆₀ carbocyclic group and the C₁-C₆₀ heterocyclic group.

The term “ π electron-rich C₃-C₆₀ cyclic group” as used herein may be a cyclic group that has three to sixty carbon atoms and may not include *—N=* as a ring-forming moiety, and the term “ π electron-deficient nitrogen-containing C₁-C₆₀ cyclic group” as used herein may be a heterocyclic group that has one to sixty carbon atoms and may include *—N=* as a ring-forming moiety.

In an embodiment,

the C₃-C₆₀ carbocyclic group may be a T1 group or a condensed cyclic group in which two or more T1 groups are condensed with each other (for example, a cyclopentadiene group, an adamantane group, a norbornane group, a benzene group, a pentalene group, a naphthalene group, an azulene group, an indacene group, an acenaphthylene group, a phenalene group, a phenanthrene group, an anthracene group, a fluoanthrene group, a triphenylene group, a pyrene group, a chrysene group, a perylene group, a pentaphene group, a heptalene group, a naphthacene group, a picene group, a hexacene group, a pentacene group, a rubicene group, a coronene group, an ovalene group, an indene group, a fluorene group, a spiro-bifluorene group, a benzofluorene group, an indenophenanthrene group, or an indenoanthracene group),

the C₁-C₆₀ heterocyclic group may be a T2 group T2, a condensed cyclic group in which two or more T2 groups are condensed with each other, or a condensed cyclic group in which at least one T2 group and at least one T1 group are condensed with each other (for example, a pyrrole group, a thiophene group, a furan group, an indole group, a benzindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoinolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.),

benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.),

the π electron-rich C₃-C₆₀ cyclic group may be a T1 group, a condensed cyclic group in which two or more T1 groups are condensed with each other, a T3 group T3, a condensed cyclic group in which two or more T3 groups are condensed with each other, or a condensed cyclic group in which at least one T3 group and at least one T1 group are condensed with each other (for example, the C₃-C₆₀ carbocyclic group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, a thiophene group, a furan group, an indole group, a benzindole group, a naphthoindole group, an isoindole group, a benzoisoindole group, a naphthoisoindole group, a benzosilole group, a benzothiophene group, a benzofuran group, a carbazole group, a dibenzosilole group, a dibenzothiophene group, a dibenzofuran group, an indenocarbazole group, an indolocarbazole group, a benzofurocarbazole group, a benzothienocarbazole group, a benzosilolocarbazole group, a benzoinolocarbazole group, a benzocarbazole group, a benzonaphthofuran group, a benzonaphthothiophene group, a benzonaphthosilole group, a benzofurodibenzofuran group, a benzofurodibenzothiophene group, a benzothienodibenzothiophene group, etc.),

the π electron-deficient nitrogen-containing C₁-C₆₀ cyclic group may be a T4 group, a condensed cyclic group in which two or more T4 groups are condensed with each other, a condensed cyclic group in which at least one T4 group and at least one T1 group are condensed with each other, a condensed cyclic group in which at least one T4 group and at least one T3 group are condensed with each other, or a condensed cyclic group in which at least one T4 group, at least one T1 group, and at least one T3 group are condensed with one another (for example, a pyrazole group, an imidazole group, a triazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, a benzopyrazole group, a benzimidazole group, a benzoxazole group, a benzoisoxazole group, a benzothiazole group, a benzoisothiazole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a quinoline group, an isoquinoline group, a benzoquinoline group, a benzoisoquinoline group, a quinoxaline group, a benzoquinoxaline group, a quinazoline group, a benzoquinazoline group, a phenanthroline group, a cinnoline group, a phthalazine group, a naphthyridine group, an imidazopyridine group, an imidazopyrimidine group, an imidazotriazine group, an imidazopyrazine group, an imidazopyridazine group, an azacarbazole group, an azafluorene group, an azadibenzosilole group, an azadibenzothiophene group, an azadibenzofuran group, etc.),

wherein the T1 group may be a cyclopropane group, a cyclobutane group, a cyclopentane group, a cyclohexane group, a cycloheptane group, a cyclooctane group, a cyclobutene group, a cyclopentene group, a cyclopentadiene group, a cyclohexene group, a cyclohexadiene group, a cycloheptene group, an adamantane group, a norbornane (or a bicyclo[2.2.1]heptane) group, a norbornene group, a bicyclo[1.1.1]pentane group, a bicyclo[2.1.1]hexane group, a bicyclo[2.2.2]octane group, or a benzene group,

the T2 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, a borole group, a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, a tetrazine group, a pyrrolidine group, an imidazolidine group, a dihydropyrrole group, a piperidine group, a tetrahydropyridine group, a dihydropyridine group, a hexahydropyrimidine group, a tetrahydropyrimidine group, a dihydropyrimidine group, a piperazine group, a tetrahydropyrazine group, a dihydropyrazine group, a tetrahydropyridazine group, or a dihydropyridazine group,

the T3 group may be a furan group, a thiophene group, a 1H-pyrrole group, a silole group, or a borole group, and the T4 group may be a 2H-pyrrole group, a 3H-pyrrole group, an imidazole group, a pyrazole group, a triazole group, a tetrazole group, an oxazole group, an isoxazole group, an oxadiazole group, a thiazole group, an isothiazole group, a thiadiazole group, an azasilole group, an azaborole group, a pyridine group, a pyrimidine group, a pyrazine group, a pyridazine group, a triazine group, or a tetrazine group.

The terms "cyclic group", " C_3 - C_{60} carbocyclic group", " C_1 - C_{60} heterocyclic group", " π electron-rich C_3 - C_{60} cyclic group", or " π electron-deficient nitrogen-containing C_1 - C_{60} cyclic group" as used herein may be a group condensed to any cyclic group or a polyvalent group (for example, a divalent group, a trivalent group, a tetravalent group, etc.), depending on the structure of a formula in connection with which the terms are used. In an embodiment, "a benzene group" may be a benzo group, a phenyl group, a phenylene group, or the like, which may be easily understood by one of ordinary skill in the art according to the structure of a formula including the "benzene group."

Examples of the monovalent C_3 - C_{60} carbocyclic group and the monovalent C_1 - C_{60} heterocyclic group may include a C_3 - C_{10} cycloalkyl group, a C_1 - C_{10} heterocycloalkyl group, a C_3 - C_{10} cycloalkenyl group, a C_1 - C_{10} heterocycloalkenyl group, a C_6 - C_{60} aryl group, a C_1 - C_{60} heteroaryl group, a monovalent non-aromatic condensed polycyclic group, and a monovalent non-aromatic condensed heteropolycyclic group, and examples of the divalent C_3 - C_{60} carbocyclic group and the monovalent C_1 - C_{60} heterocyclic group may include a C_3 - C_{10} cycloalkylene group, a C_1 - C_{10} heterocycloalkylene group, a C_3 - C_{10} cycloalkenylene group, a C_1 - C_{10} heterocycloalkenylene group, a C_6 - C_{60} arylene group, a C_1 - C_{60} heteroarylene group, a divalent non-aromatic condensed polycyclic group, and a substituted or unsubstituted divalent non-aromatic condensed heteropolycyclic group.

The term " C_1 - C_{60} alkyl group" as used herein may be a linear or branched aliphatic hydrocarbon monovalent group

that has 1 to 60 carbon atoms, and examples thereof may include a methyl group, an ethyl group, an n-propyl group, an isopropyl group, an n-butyl group, a sec-butyl group, an isobutyl group, a tert-butyl group, an n-pentyl group, a tert-pentyl group, a neopentyl group, an isopentyl group, a sec-pentyl group, a 3-pentyl group, a sec-isopentyl group, an n-hexyl group, an isohexyl group, a sec-hexyl group, a tert-hexyl group, an n-heptyl group, an isoheptyl group, a sec-heptyl group, a tert-heptyl group, an n-octyl group, an isooctyl group, a sec-octyl group, a tert-octyl group, an n-nonyl group, an isononyl group, a sec-nonyl group, a tert-nonyl group, an n-decyl group, an isodecyl group, a sec-decyl group, and a tert-decyl group. The term " C_1 - C_{60} alkylene group" as used herein may be a divalent group having a same structure as the C_1 - C_{60} alkyl group.

The term " C_2 - C_{60} alkenyl group" as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon double bond in the middle or at the terminus of the C_2 - C_{60} alkyl group, and examples thereof may include an ethenyl group, a propenyl group, and a butenyl group. The term " C_2 - C_{60} alkenylene group" as used herein may be a divalent group having a same structure as the C_2 - C_{60} alkenyl group.

The term " C_2 - C_{60} alkynyl group" as used herein may be a monovalent hydrocarbon group having at least one carbon-carbon triple bond in the middle or at the terminus of the C_2 - C_{60} alkyl group, and examples thereof may include an ethynyl group and a propynyl group. The term " C_2 - C_{60} alkynylene group" as used herein may be a divalent group having a same structure as the C_2 - C_{60} alkynyl group.

The term " C_1 - C_{60} alkoxy group" as used herein may be a monovalent group represented by $-O(A_{101})$ (wherein A_{101} is a C_1 - C_{60} alkyl group), and examples thereof may include a methoxy group, an ethoxy group, and an isopropoxy group.

The term " C_3 - C_{10} cycloalkyl group" as used herein may be a monovalent saturated hydrocarbon cyclic group having 3 to 10 carbon atoms, and examples thereof may include a cyclopropyl group, a cyclobutyl group, a cyclopentyl group, a cyclohexyl group, a cycloheptyl group, a cyclooctyl group, an adamantanyl group, a norbornanyl group (or bicyclo[2.2.1]heptyl group), a bicyclo[1.1.1]pentyl group, a bicyclo[2.1.1]hexyl group, and a bicyclo[2.2.2]octyl group. The term " C_3 - C_{10} cycloalkylene group" as used herein may be a divalent group having a same structure as the C_3 - C_{10} cycloalkyl group.

The term " C_1 - C_{10} heterocycloalkyl group" as used herein may be a monovalent cyclic group that further includes, in addition to a carbon atom, at least one heteroatom as a ring-forming atom and has 1 to 10 carbon atoms, and examples thereof may include a 1,2,3,4-oxatriazolidinyl group, a tetrahydrofuranyl group, and a tetrahydrothiophenyl group. The term " C_1 - C_{10} heterocycloalkylene group" as used herein may be a divalent group having a same structure as the C_1 - C_{10} heterocycloalkyl group.

The term " C_3 - C_{10} cycloalkenyl group" as used herein may be a monovalent cyclic group that has 3 to 10 carbon atoms and at least one carbon-carbon double bond in the ring thereof and no aromaticity, and examples thereof may include a cyclopentenyl group, a cyclohexenyl group, and a cycloheptenyl group. The term " C_3 - C_{10} cycloalkenylene group" as used herein may be a divalent group having a same structure as the C_3 - C_{10} cycloalkenyl group.

The term " C_1 - C_{10} heterocycloalkenyl group" as used herein may be a monovalent cyclic group that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, 1 to 10 carbon atoms, and at least one

double bond in the cyclic structure thereof. Examples of the C₁-C₁₀ heterocycloalkenyl group may include a 4,5-dihydro-1,2,3,4-oxatriazolyl group, a 2,3-dihydrofuranyl group, and a 2,3-dihydrothiophenyl group. The term "C₁-C₁₀ heterocycloalkenylene group" as used herein may be a divalent group having a same structure as the C₁-C₁₀ heterocycloalkenyl group.

The term "C₆-C₆₀ aryl group" as used herein may be a monovalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms, and the term "C₆-C₆₀ arylene group" as used herein may be a divalent group having a carbocyclic aromatic system having 6 to 60 carbon atoms. Examples of the C₆-C₆₀ aryl group may include a phenyl group, a pentalenyl group, a naphthyl group, an azulenyl group, an indacenyl group, an acenaphthyl group, a phenalenyl group, a phenanthrenyl group, an anthracenyl group, a fluoranthenyl group, a triphenylenyl group, a pyrenyl group, a chrysenyl group, a perylenyl group, a pentaphenyl group, a heptalenyl group, a naphthacenyl group, a picenyl group, a hexacenyl group, a pentacenyl group, a rubicenyl group, a coronenyl group, and an ovalenyl group. When the C₆-C₆₀ aryl group and the C₆-C₆₀ arylene group each include two or more rings, the rings may be condensed with each other.

The term "C₁-C₆₀ heteroaryl group" as used herein may be a monovalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. The term "C₁-C₆₀ heteroarylene group" as used herein may be a divalent group having a heterocyclic aromatic system that has, in addition to a carbon atom, at least one heteroatom as a ring-forming atom, and 1 to 60 carbon atoms. Examples of the C₁-C₆₀ heteroaryl group may include a pyridinyl group, a pyrimidinyl group, a pyrazinyl group, a pyridazinyl group, a triazinyl group, a quinolinyl group, a benzoquinolinyl group, an isoquinolinyl group, a benzoisoquinolinyl group, a quinoxalinyl group, a benzoquinoxalinyl group, a quinazolinyl group, a benzoquinazolinyl group, a cinnolinyl group, a phenanthrolinyl group, a phthalazinyl group, and a naphthyridinyl group. When the C₁-C₆₀ heteroaryl group and the C₁-C₆₀ heteroarylene group each include two or more rings, the rings may be condensed with each other.

The term "monovalent non-aromatic condensed polycyclic group" as used herein may be a monovalent group (for example, having 8 to 60 carbon atoms) having two or more rings condensed to each other, only carbon atoms as ring-forming atoms, and no aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed polycyclic group may include an indenyl group, a fluorenyl group, a spiro-bifluorenyl group, a benzofluorenyl group, an indenophenanthrenyl group, and an indeno anthracenyl group. The term "divalent non-aromatic condensed polycyclic group" as used herein may be a divalent group having a same structure as a monovalent non-aromatic condensed polycyclic group.

The term "monovalent non-aromatic condensed heteropolycyclic group" as used herein may be a monovalent group (for example, having 1 to 60 carbon atoms) having two or more rings condensed to each other, at least one heteroatom other than carbon atoms, as a ring-forming atom, and non-aromaticity in its entire molecular structure. Examples of the monovalent non-aromatic condensed heteropolycyclic group may include a pyrrolyl group, a thiophenyl group, a furanyl group, an indolyl group, a benzoindolyl group, a naphthoindolyl group, an isoindolyl group, a benzoisoindolyl group, a naphthoisoindolyl group, a benzosilolyl group,

a benzothiophenyl group, a benzofuranyl group, a carbazoyl group, a dibenzosilolyl group, a dibenzothiophenyl group, a dibenzofuranyl group, an azacarbazoyl group, an azafluorenyl group, an azadibenzosilolyl group, an azadibenzothiophenyl group, an azadibenzofuranyl group, a pyrazolyl group, an imidazolyl group, a triazolyl group, a tetrazolyl group, an oxazolyl group, an isoxazolyl group, a thiazolyl group, an isothiazolyl group, an oxadiazolyl group, a thiadiazolyl group, a benzopyrazolyl group, a benzimidazolyl group, a benzoxazolyl group, a benzothiazolyl group, a benzoxadiazolyl group, a benzothiadiazolyl group, an imidazopyridinyl group, an imidazopyrimidinyl group, an imidazotriazinyl group, an imidazopyrazinyl group, an imidazopyridazinyl group, an indeno carbazoyl group, an indolocarbazoyl group, a benzofurocarbazoyl group, a benzothienocarbazoyl group, a benzosilolocarbazoyl group, a benzoinolocarbazoyl group, a benzocarbazoyl group, a benzonaphthofuranyl group, a benzonaphthothiophenyl group, a benzonaphtho silolyl group, a benzofurodibenzofuranyl group, a benzofurodibenzothiophenyl group, and a benzothienodibenzothiophenyl group. The term "divalent non-aromatic condensed heteropolycyclic group" as used herein may be a divalent group having a same structure as a monovalent non-aromatic condensed heteropolycyclic group.

The term "C₆-C₆₀ aryloxy group" as used herein may be represented by —O(A₁₀₂) (wherein A₁₀₂ is a C₆-C₆₀ aryl group), and the term "C₆-C₆₀ arylthio group" as used herein may be represented by —S(A₁₀₃) (wherein A₁₀₃ is a C₆-C₆₀ aryl group).

The term "C₇-C₆₀ aryl alkyl group" as used herein may be represented by —(A₁₀₄)(A₁₀₅) (where A₁₀₄ may be a C₁-C₅₄ alkylene group, and A₁₀₅ may be a C₆-C₅₉ aryl group), and the term "C₂-C₆₀ heteroaryl alkyl group" as used herein may be represented by —(A₁₀₆)(A₁₀₇) (where A₁₀₆ may be a C₁-C₅₉ alkylene group, and A₁₀₇ may be a C₁-C₅₉ heteroaryl group).

The term "R_{10a}" as used herein may be:

deuterium (-D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group;

a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, or a C₁-C₆₀ alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C₃-C₆₀ carbocyclic group, a C₁-C₆₀ heterocyclic group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₁)(Q₁₂), —B(Q₁₁)(Q₁₂), —C(=O)(Q₁₁), —S(=O)₂(Q₁₁), —P(=O)(Q₁₁)(Q₁₂), or any combination thereof;

a C₃-C₆₀ carbocyclic group, a C₁-C₆₀ heterocyclic group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryloxy group, or a C₁-C₆₀ heteroarylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₆₀ carbocyclic group, a C₁-C₆₀ heterocyclic group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), —P(=O)(Q₂₁)(Q₂₂), or any combination thereof; or

—Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), or —P(=O)(Q₃₁)(Q₃₂).

199

Q₁₁ to Q₁₃, Q₂₁ to Q₂₃, and Q₃₁ to Q₃₃ as used herein may each independently be: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C₁-C₆₀ alkyl group; a C₂-C₆₀ alkenyl group; a C₂-C₆₀ alkynyl group; a C₁-C₆₀ alkoxy group; a C₃-C₆₀ carbocyclic group or a C₁-C₆₀ heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C₁-C₆₀ alkyl group, a C₁-C₆ alkoxy group, a phenyl group, a biphenyl group, or any combination thereof.

The term “heteroatom” as used herein may be any atom other than a carbon atom or a hydrogen atom. Examples of the heteroatom may include O, S, N, P, Si, B, Ge, Se, or any combination thereof.

The term “the third-row transition metal” as used herein may include hafnium (Hf), tantalum (Ta), tungsten (W), rhenium (Re), osmium (Os), iridium (Ir), platinum (Pt), gold (Au), etc.

The term “Ph” as used herein refers to a phenyl group, the term “Me” as used herein refers to a methyl group, the term “Et” as used herein refers to an ethyl group, the term “ter-Bu” or “Bu” as used herein refers to a tert-butyl group, and the term “OMe” as used herein refers to a methoxy group.

The term “biphenyl group” as used herein may be “a phenyl group substituted with a phenyl group.” For example, the “biphenyl group” may be a substituted phenyl group having a C₆-C₆₀ aryl group as a substituent.

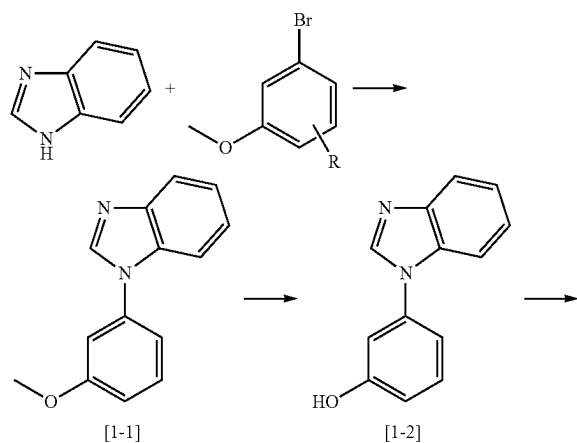
The term “terphenyl group” as used herein may be “a phenyl group substituted with a biphenyl group”. For example, the “terphenyl group” may be a substituted phenyl group having, as a substituent, a C₆-C₆₀ aryl group substituted with a C₆-C₆₀ aryl group.

In the specification, *, **, and *** as used herein, unless defined otherwise, each represent a binding site to a neighboring atom in a corresponding formula or moiety.

Hereinafter, a compound according to embodiments and a light-emitting device according to embodiments will be described in detail with reference to Examples. The wording “B was used instead of A” used in describing Examples refers to that an identical molar equivalent of B was used in place of A.

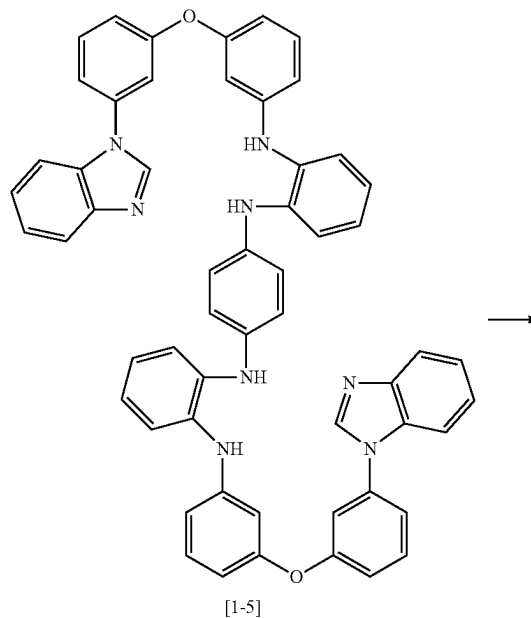
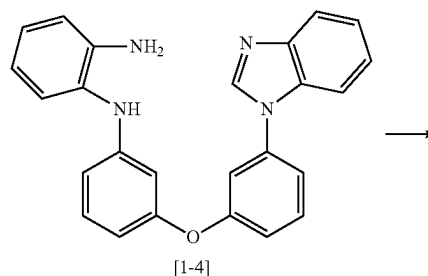
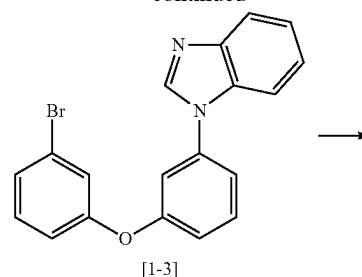
EXAMPLES

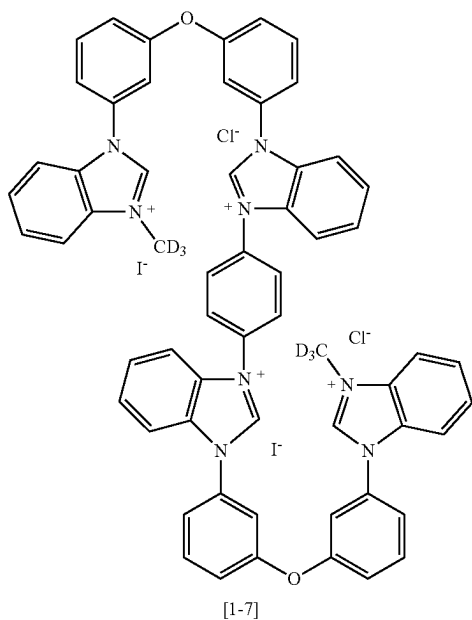
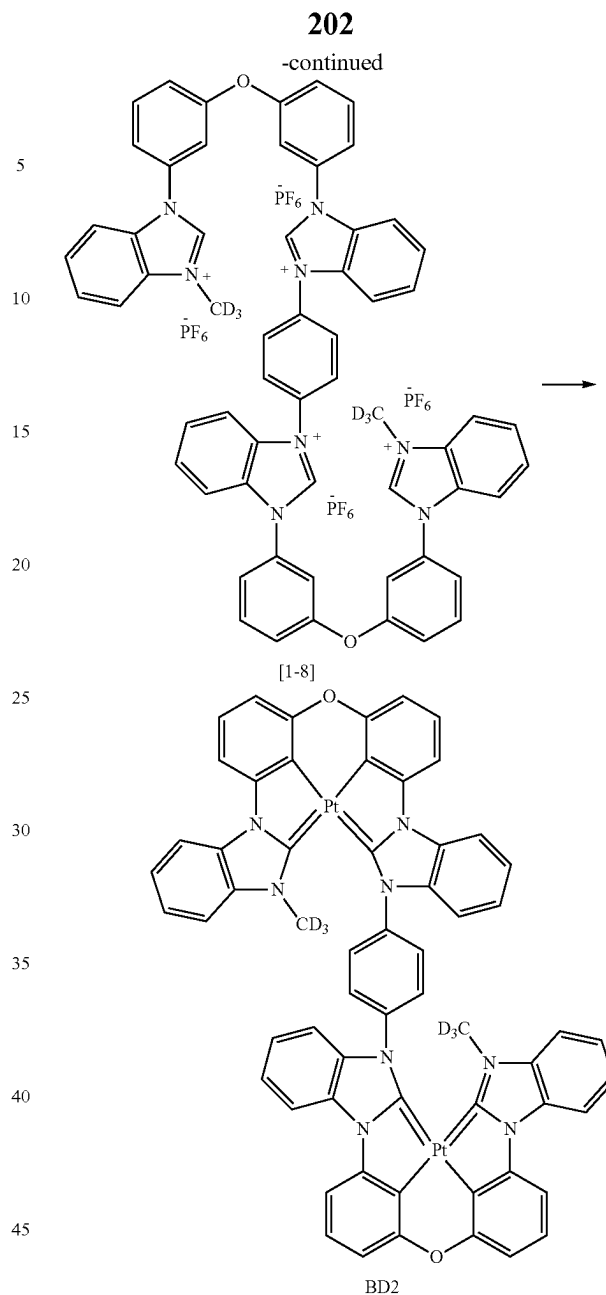
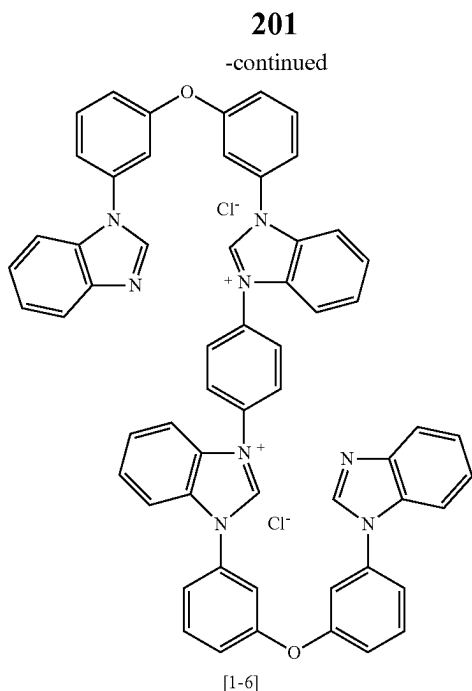
Synthesis Example 1: Synthesis of Compound BD2



200

-continued





(1) Synthesis of Intermediate [1-1]

21.0 g (62 mmol) of benzimidazole, 20.2 g (74 mmol) of 3-bromoanisole, 26.4 g (124 mmol) of potassium phosphate tribasic, 1,180 mg (6.2 mmol) of iodo copper, and 760 mg (6.2 mmol) of picolinic acid were added to a reaction vessel, and suspended in 650 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 21.2 g (40 mmol) of Intermediate [1-1].

(2) Synthesis of Intermediate [1-2]

21.2 g (40 mmol) of Intermediate [1-1] was added to a reaction vessel, and suspended in 400 mL of HCl/AcOH (volume ratio of 4:1). The reaction temperature was raised to

203

120° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 15.3 g (32 mmol) of Intermediate [1-2].

(3) Synthesis of Intermediate [1-3]

15.3 g (32 mmol) of Intermediate [1-2], 12.1 g (35 mmol) of 1,3-dibromobenzene, 18.5 g (70 mmol) of potassium phosphate tribasic, 560 mg (3.2 mmol) of iodo copper, and 420 mg (3.5 mmol) of picolinic acid were added to a reaction vessel, and suspended in 400 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.5 g (28 mmol) of Intermediate [1-3].

(4) Synthesis of Intermediate [1-4]

14.5 g (28 mmol) of Intermediate [1-3], 3.0 g (28 mmol) of benzene-1,2-diamine, 460 mg (0.5 mmol) of tris(dibenzylideneacetone)dipalladium, 480 mg (1.0 mmol) of XPhos, and 5.0 g (52 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 260 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.2 g (19 mmol) of Intermediate [1-4].

(5) Synthesis of Intermediate [1-5]

14.2 g (19 mmol) of Intermediate [1-4], 2.1 g (9 mmol) of 1,4-dibromobenzene, 430 mg (0.4 mmol) of tris(dibenzylideneacetone)dipalladium, 450 mg (0.8 mmol) of XPhos, and 4.5 g (40 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 250 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.1 g (5 mmol) of Intermediate [1-5].

(6) Synthesis of Intermediate [1-6]

3.1 g (5 mmol) of Intermediate [1-5], 20 mL (250 mmol) of triethyl orthoformate, and 3.5 g (35 mmol) of a HCl 35 wt % solution were added to a reaction vessel, heated, and stirred at 80° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and a residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.2 g (4 mmol) of Intermediate [1-6].

(7) Synthesis of Intermediates [1-7] and [1-8]

3.2 g (4 mmol) of Intermediate [1-6] and 0.6 g (16 mmol) of iodomethane were added to a reaction vessel, and suspended in 110 mL of dichloromethane. The reaction temperature was raised to 50° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the

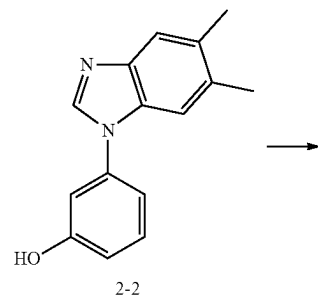
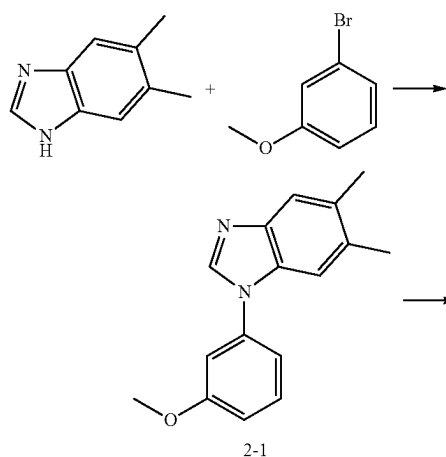
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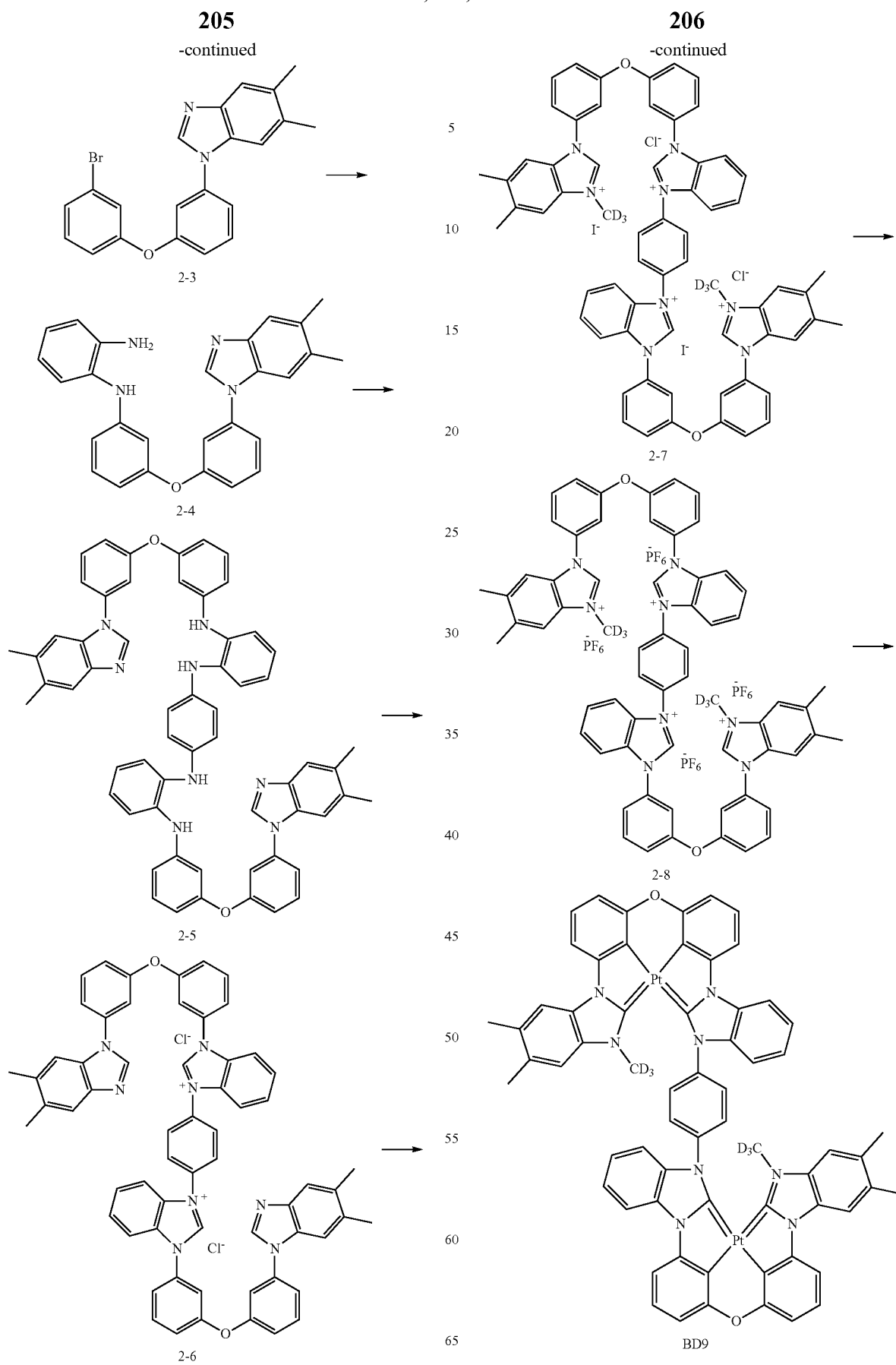
reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.8 g (4 mmol) of Intermediate [1-7]. Immediately afterwards, 2.3 g (14 mmol) of ammonium hexafluorophosphate was added to a reaction vessel, and suspended in a solution of methanol and water at a ratio of 2:1. The reaction mixture was stirred at room temperature for 12 hours. The resulting solid was filtered, and separated by column chromatography to obtain 3.5 g (3 mmol) of Intermediate [1-8].

(8) Synthesis of Compound BD2

3.5 g (3 mmol) of Intermediate [1-8], 1.0 g (4 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 0.4 g (6 mmol) of sodium acetate were suspended in 100 mL of dioxane. The reaction mixture was heated, and stirred at 110° C. for 72 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 0.7 g (1 mmol) of Compound BD2.

Synthesis Example 2: Synthesis of Compound BD9





(1) Synthesis of Intermediate [2-1]

21.0 g (60 mmol) of 3,4-dimethyl benzimidazole, 20.2 g (72 mmol) 3-bromoanisole, 26.4 g (124 mmol) of potassium phosphate tribasic, 1,180 mg (6.2 mmol) of iodo copper, and 760 mg (6.2 mmol) of picolinic acid were added to a reaction vessel, and suspended in 650 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 22.3 g (42 mmol) of Intermediate [2-1].

(2) Synthesis of Intermediate [2-2]

22.3 g (42 mmol) of Intermediate [2-1] was added to a reaction vessel, and suspended in 400 mL of HCl/AcOH (volume ratio of 4:1). The reaction temperature was raised to 120° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.7 g (32 mmol) of Intermediate [2-2].

(3) Synthesis of Intermediate [2-3]

14.7 g (32 mmol) of Intermediate [2-2], 12.1 g (35 mmol) of 1,3-dibromobenzene, 18.5 g (70 mmol) of potassium phosphate tribasic, 560 mg (3.2 mmol) of iodo copper, and 420 mg (3.5 mmol) of picolinic acid were added to a reaction vessel, and suspended in 400 mL of dimethyl sulfoxide. The reaction mixture was heated, and stirred at 160° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.1 g (25 mmol) of Intermediate [2-3].

(4) Synthesis of Intermediate [2-4]

14.1 g (25 mmol) of Intermediate [2-3], 3.0 g (28 mmol) of benzene-1,2-diamine, 460 mg (0.5 mmol) of tris(dibenzylideneacetone)dipalladium, 480 mg (1.0 mmol) of XPhos, and 5.0 g (52 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 260 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 14.0 g (18 mmol) of Intermediate [2-4].

(5) Synthesis of Intermediate [2-5]

14.0 g (18 mmol) of Intermediate [2-4], 2.1 g (9 mmol) of 1,4-dibromobenzene, 430 mg (0.4 mmol) of tris(dibenzylideneacetone)dipalladium, 450 mg (0.8 mmol) of XPhos, and 4.5 g (40 mmol) of sodium tert-butoxide were added to a reaction vessel, and suspended in 250 mL of toluene. The reaction temperature was raised to 110° C., and the reaction mixture was stirred for 6 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A

residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.0 g (4 mmol) of Intermediate [2-5].

(6) Synthesis of Intermediate [2-6]

3.0 g (4 mmol) of Intermediate [2-5], 20 mL (250 mmol) of triethyl orthoformate, and 3.5 g (35 mmol) of a HCl 35 wt % solution were added to a reaction vessel, heated, and stirred at 80° C. for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and a residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 2.9 g (3 mmol) of Intermediate [2-6].

(7) Synthesis of Intermediates [2-7] and [2-8]

2.9 g (3 mmol) of Intermediate [2-6] and 0.6 g (16 mmol) of iodomethane were added to a reaction vessel, and suspended in 110 mL of dichloromethane. The reaction temperature was raised to 50° C., and the reaction mixture was stirred for 12 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 3.8 g (4 mmol) of Intermediate [2-7]. Immediately afterwards, 2.3 g (14 mmol) of ammonium hexafluorophosphate was added to a reaction vessel, and suspended in a solution of methanol and water at a ratio of 2:1. The reaction mixture was stirred at room temperature for 12 hours. The resulting solid was filtered, and separated by column chromatography to obtain 3.6 g (3 mmol) of Intermediate [2-8].

(8) Synthesis of Compound BD9

3.6 g (3 mmol) of Intermediate [2-8], 1.0 g (4 mmol) of dichloro(1,5-cyclooctadiene)platinum, and 0.4 g (6 mmol) of sodium acetate were suspended in 100 mL of dioxane. The reaction mixture was heated, and stirred at 110° C. for 72 hours. After completion of the reaction, the reaction mixture was cooled to room temperature, and extracted with ethyl acetate. An organic layer extracted therefrom was washed with a saturated aqueous sodium chloride solution, and dried using sodium sulfate. A residue obtained by removing the solvent therefrom was separated by column chromatography to obtain 0.6 g (1 mmol) of Compound BD9.

Synthesis Example 3: Synthesis of Compounds BD1 to BD104

Compounds BD1 to BD104 were synthesized in substantially the same manner as Compound BD2, except that a different starting material from that of Compound BD2 was used.

¹H NMR and MS/FAB of the compounds synthesized according to Synthesis Examples are shown in Table 1.

TABLE 1

Compound	No.	¹ H NMR (CDCl ₃ , 400 MHz)	MS/FAB	
			found	calc.
BD2		6.57 (d, 2H), 6.78 (d, 2H), 7.13-7.32 (m, 16H), 7.35-7.47 (m, 12H)	1299.11	1299.23
BD9		1.35 (s, 6H), 1.39 (s, 6H), 6.56 (d, 2H), 6.79 (d, 2H), 7.13-7.34 (m, 12H), 7.35-7.48 (m, 12H)	1355.21	1355.33
BD17		6.57 (d, 2H), 6.78 (d, 2H), 7.13-7.32 (m, 16H), 7.33-7.42 (m, 10H)	1299.12	1299.17

209

TABLE 1-continued

Compound	No.	¹ H NMR (CDCl ₃ , 400 MHz)	MS/FAB	
			found	calc.
BD25		6.58 (d, 2H), 6.79 (d, 2H), 7.13-7.33 (m, 16H), 7.33-7.42 (m, 10H)	1299.05	1299.17
BD45		1.35 (s, 12H), 1.39 (s, 12H), 6.58 (d, 2H), 6.81 (d, 2H), 7.12-7.34 (m, 12H), 7.35-7.49 (m, 12H)	1411.40	1411.44

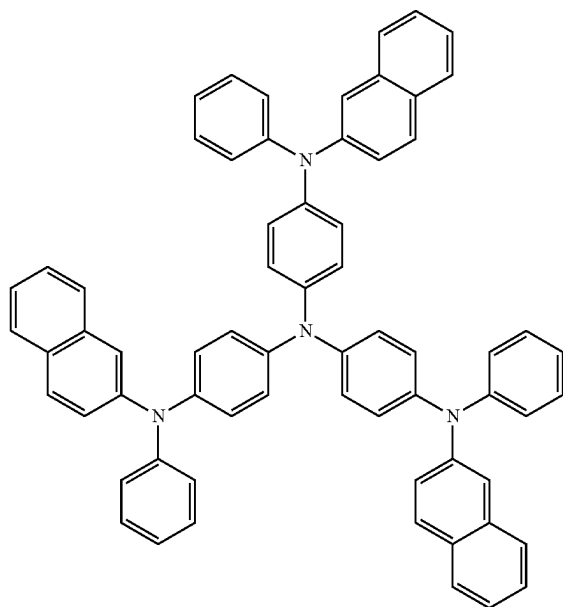
Example 1

As an anode, a glass substrate with 15 Ωcm² (1,200 Å) ITO thereon, which was manufactured by Corning Inc., was cut to a size of 50 mm×50 mm×0.7 mm, and the glass substrate was sonicated by using isopropyl alcohol and pure water for 5 minutes each, and ultraviolet (UV) light was irradiated for 30 minutes thereto and ozone was exposed thereto for cleaning. The resultant glass substrate was loaded onto a vacuum deposition apparatus.

2-TNATA, which is a compound in the art, was vacuum-deposited on the substrate to form a hole injection layer having a thickness of 600 Å, and 4,4'-bis[N-(1-naphthyl)-N-phenylamino]biphenyl (hereinafter, referred to as NPB) as a hole transport compound was vacuum-deposited thereon to form a hole transport layer having a thickness of 300 Å.

mCBP as a host and Compound BD2 as a dopant in an amount of 10 wt % were co-deposited on the hole transport layer to form an emission layer having a thickness of 300 Å.

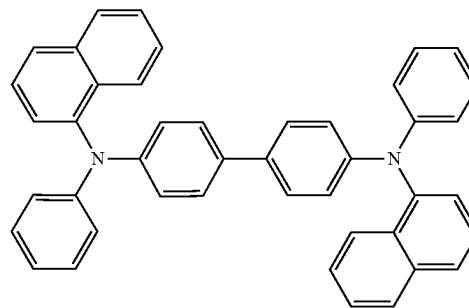
TSPO1 was vacuum-deposited on the emission layer to form a hole blocking layer having a thickness of 50 Å. Alq₃ was deposited on the hole blocking layer to form an electron transport layer having a thickness of 300 Å, LiF, which is a halogenated alkali metal, was deposited on the electron transport layer to form an electron injection layer having a thickness of 10 Å, and Al was vacuum-deposited on the electron injection layer to form a LiF/Al cathode electrode having a thickness of 3,000 Å, thereby completing the manufacture of a light-emitting device.



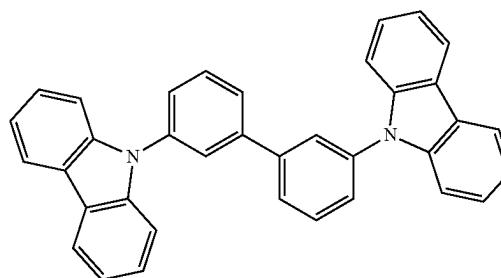
2-TNATA

210

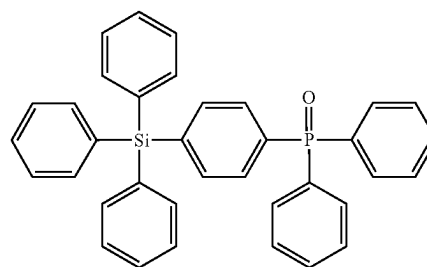
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NPB



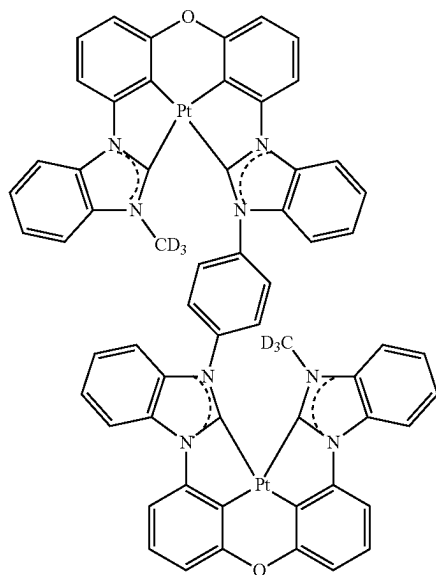
mCBP



TSPO1

Examples 2 to 5 and Comparative Examples 1 and 2

Light-emitting devices were manufactured in the same manner as in Example 1, except that in forming an emission layer, for use as a dopant, corresponding compounds shown in Table 2 were used instead of Compound BD2.

211**212**

-continued

BD2

BD17

5

10

15

20

25

30

35

40

BD9

BD25

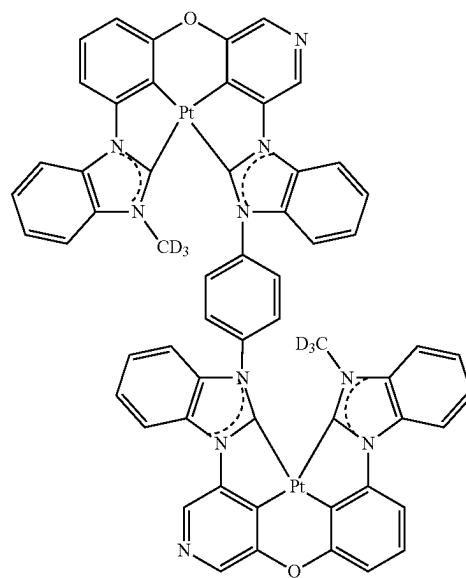
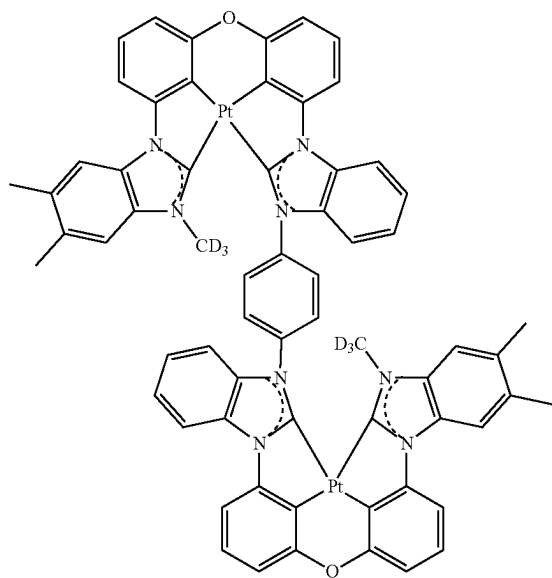
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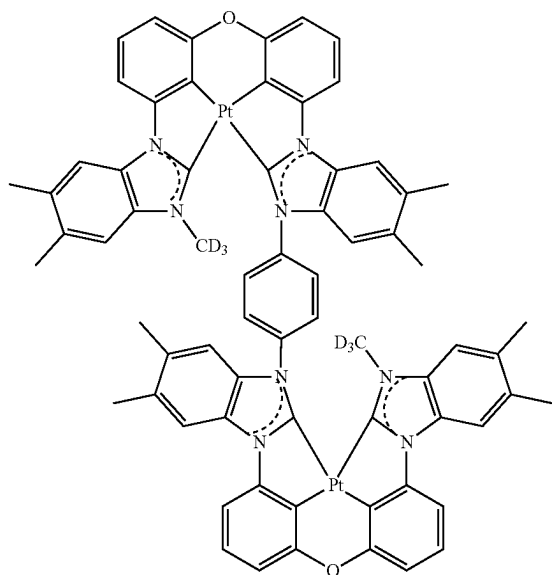
65



213

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BD45



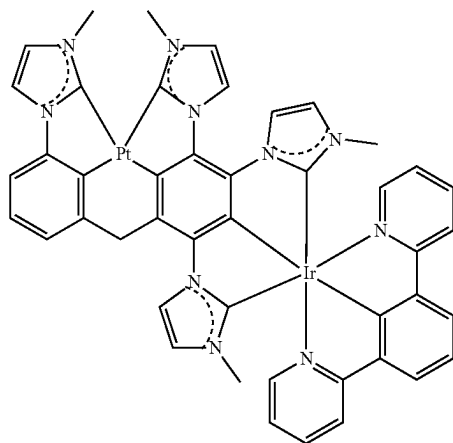
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15

20

CE1 25



30

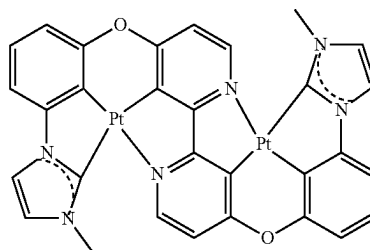
35

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214

-continued

CE2



Evaluation Example

The driving voltage, current density, luminance, efficiency, emission color, and emission wavelength of the light-emitting devices manufactured according to Examples 1 to 5 and Comparative Examples 1 to 3 were measured by using Keithley SMU 236 and a luminance meter PR650, and results thereof are shown in Table 2.

TABLE 2

	Dopant in emission layer	Driving voltage (V)	Current density (mA/cm ²)	Luminance (cd/m ²)	Efficiency (cd/A)	Emission color	Emission wavelength (nm)
Example 1	BD2	4.76	50	4032	8.32	Blue	457
Example 2	BD9	4.80	50	4019	8.85	Blue	458
Example 3	BD17	4.97	50	4011	9.12	Bluish green	463
Example 4	BD25	4.95	50	4005	9.01	Bluish green	465
Example 5	BD45	4.90	50	3990	8.26	Bluish green	462
Comparative Example 1	CE1	5.22	50	3910	7.93	Bluish green	464
Comparative Example 2	CE2	5.29	50	3950	7.75	Bluish green	467

215

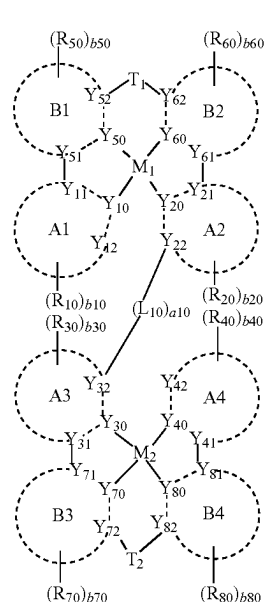
Referring to Table 2, it was found that the light-emitting devices according to an embodiment have a lower driving voltage and higher luminance and efficiency than the light-emitting devices of Comparative Examples.

A light-emitting device including the organometallic compound according to an embodiment may have a low driving voltage, high luminance and efficiency, and a long lifespan.

Embodiments have been disclosed herein, and although terms are employed, they are used and are to be interpreted in a generic and descriptive sense only and not for purpose of limitation. In some instances, as would be apparent by one of ordinary skill in the art, features, characteristics, and/or elements described in connection with an embodiment may be used singly or in combination with features, characteristics, and/or elements described in connection with other embodiments unless otherwise specifically indicated. Accordingly, it will be understood by those of ordinary skill in the art that various changes in form and details may be made without departing from the spirit and scope of the disclosure as set forth in the claims.

What is claimed is:

1. An organometallic compound represented by Formula 1:



[Formula 1]

wherein in Formula 1,

M₁ and M₂ are each independently platinum (Pt), palladium (Pd), copper (Cu), silver (Ag), gold (Au), rhodium (Rh), iridium (Ir), ruthenium (Ru), osmium (Os), titanium (Ti), zirconium (Zr), hafnium (Hf), europium (Eu), terbium (Tb), or thulium (Tm),

A1 to A4 and B1 to B4 are each independently a C₅-C₆₀ carbocyclic group or a C₁-C₆₀ heterocyclic group,

Y₁₀ to Y₁₂, Y₂₀ to Y₂₂, Y₃₀ to Y₃₂, Y₄₀ to Y₄₂, Y₅₀ to Y₅₂, Y₆₀ to Y₆₂, Y₇₀ to Y₇₂, and Y₈₀ to Y₈₂ are each independently C or N,

T₁ and T₂ are each independently a single bond,

—O—, *—S—*, *—C(R₁)(R₂)—*, *—C(R₁)=*, *—C(R₁)—*, *—C(R₁)=C(R₂)—*, *—C(=O)—*, *—C(=S)—*, *—C≡C—*, *—B

216

(R₁)—*, *—N(R₁)—*, *—P(R₁)—*, *—Si(R₁)(R₂)—*, or *—Ge(R₁)(R₂)—*,

L₁₀ is a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a},

a10 is 1, 2, 3, 4, or 5,

R₁, R₂, R₁₀, R₂₀, R₃₀, R₄₀, R₅₀, R₆₀, R₇₀, and R₈₀ are each independently hydrogen, deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, an amidino group, a hydrazino group, a hydrazono group, a C₁-C₂₀ alkyl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₂₀ alkoxy group unsubstituted or substituted with at least one R_{10a}, a C₃-C₁₀ cycloalkyl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₁₀ heterocycloalkyl group unsubstituted or substituted with at least one R_{10a}, a C₃-C₁₀ cycloalkenyl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₁₀ heterocycloalkenyl group unsubstituted or substituted with at least one R_{10a}, a C₆-C₆₀ aryl group unsubstituted or substituted with at least one R_{10a}, a C₆-C₆₀ aryloxy group unsubstituted or substituted with at least one R_{10a}, a C₆-C₆₀ arylthio group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heteroaryl group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heteroaryloxy group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heteroarylthio group unsubstituted or substituted with at least one R_{10a}, a monovalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R_{10a}, a monovalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R_{10a}, —Si(Q₄₁)(Q₄₂)(Q₄₃), —N(Q₄₁)(Q₄₂), —B(Q₄₁)(Q₄₂), —C(=O)(Q₄₁), —S(=O)₂(Q₄₁), or —P(=O)(Q₄₁)(Q₄₂),

two neighboring substituents among R₁, R₂, R₁₀, R₂₀, R₃₀, R₄₀, R₅₀, R₆₀, R₇₀, and R₈₀ are optionally linked to each other to form a C₃-C₆₀ carbocyclic group unsubstituted or substituted with at least one R_{10a} or a C₁-C₆₀ heterocyclic group unsubstituted or substituted with at least one R_{10a},

b10, b20, b30, b40, b50, b60, b70, and b80 are each independently 1, 2, 3, 4, 5, 6, 7, or 8,

* and *' each indicate a binding site to a neighboring atom, and

R_{10a} is:

deuterium (—D), —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, or a nitro group;

a C₁-C₆ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, or a C₁-C₆₀ alkoxy group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C₃-C₆₀ carbocyclic group, a C₁-C₆₀ heterocyclic group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, —Si(Q₁₁)(Q₁₂)(Q₁₃), —N(Q₁₁)(Q₁₂), —B(Q₁₁)(Q₁₂), —C(=O)(Q₁₁), —S(=O)₂(Q₁₁), —P(=O)(Q₁₁)(Q₁₂), or a combination thereof;

a C₃-C₆₀ carbocyclic group, a C₁-C₆₀ heterocyclic group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryloxy group, or a C₁-C₆₀ heteroarylthio group, each unsubstituted or substituted with deuterium, —F, —Cl, —Br, —I, a hydroxyl group, a cyano group, a nitro group, a C₁-C₆₀ alkyl group, a C₂-C₆₀ alkenyl group, a C₂-C₆₀ alkynyl group, a C₁-C₆₀ alkoxy group, a C₃-C₆₀ carbocyclic group, a C₁-C₆₀ heterocyclic

217

clic group, a C₆-C₆₀ aryloxy group, a C₆-C₆₀ arylthio group, a C₁-C₆₀ heteroaryloxy group, a C₁-C₆₀ heteroarylthio group, —Si(Q₂₁)(Q₂₂)(Q₂₃), —N(Q₂₁)(Q₂₂), —B(Q₂₁)(Q₂₂), —C(=O)(Q₂₁), —S(=O)₂(Q₂₁), —P(=O)(Q₂₁)(Q₂₂), or a combination thereof; or
—Si(Q₃₁)(Q₃₂)(Q₃₃), —N(Q₃₁)(Q₃₂), —B(Q₃₁)(Q₃₂), —C(=O)(Q₃₁), —S(=O)₂(Q₃₁), or —P(=O)(Q₃₁)(Q₃₂),

wherein Q₁₁ to Q₁₃, Q₂₁ to Q₂₃, Q₃₁ to Q₃₃, and Q₄₁ to Q₄₃ are each independently: hydrogen; deuterium; —F; —Cl; —Br; —I; a hydroxyl group; a cyano group; a nitro group; a C₁-C₆₀ alkyl group; a C₂-C₆₀ alkenyl group; a C₂-C₆₀ alkynyl group; a C₁-C₆₀ alkoxy group; or a C₃-C₆₀ carbocyclic group or a C₁-C₆₀ heterocyclic group, each unsubstituted or substituted with deuterium, —F, a cyano group, a C₁-C₆₀ alkyl group, a C₁-C₆ alkoxy group, a phenyl group, a biphenyl group, or a combination thereof.

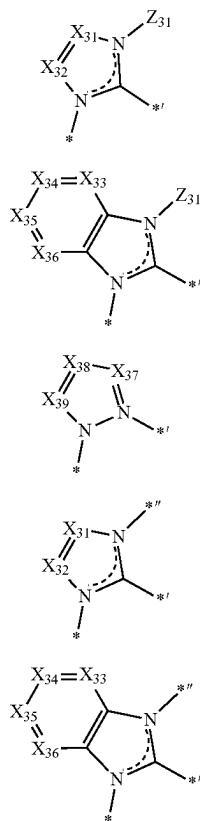
2. The organometallic compound of claim 1, wherein M₁ and M₂ are each independently Pt or Pd.

3. The organometallic compound of claim 1, wherein A1 to A4 are each independently:

a 5-membered heterocyclic group, or
a polycyclic C₁-C₃₀ heterocyclic group comprising a 5-membered heterocyclic group.

4. The organometallic compound of claim 1, wherein A1 and A4 are each independently a group represented by one of Formulae 3-1 to 3-3, and

A2 and A3 are each independently a group represented by one of Formulae 3-4 and 3-5:



218

wherein in Formulae 3-1 to 3-5,

X₃₁ to X₃₉ are each independently C(Z₃₂) or N, Z₃₁ and Z₃₂ are each independently the same as described in connection with R₁₀ in Formula 1, and

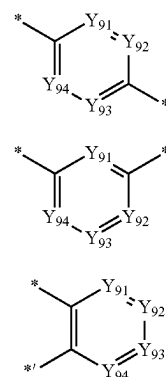
*, **, and *** each indicate a binding site to a neighboring atom.

5. The organometallic compound of claim 1, wherein a bond between M₁ and Y₁₀ is a coordinate bond, a bond between M₁ and Y₂₀ is a coordinate bond, a bond between M₂ and Y₃₀ is a coordinate bond, a bond between M₂ and Y₄₀ is a coordinate bond, a bond between M₁ and Y₅₀ is a covalent bond, a bond between M₁ and Y₆₀ is a covalent bond, a bond between M₂ and Y₇₀ is a covalent bond, and a bond between M₂ and Y₈₀ is a covalent bond.

6. The organometallic compound of claim 1, wherein T₁ and T₂ are each independently *—O—* or *—S—*.

7. The organometallic compound of claim 1, wherein L₁₀ is a C₁-C₂₀ alkylene group unsubstituted or substituted with at least one R_{10a}, a C₂-C₂₀ alkenylene group unsubstituted or substituted with at least one R_{10a}, a C₂-C₂₀ alkynylene group unsubstituted or substituted with at least one R_{10a}, a C₃-C₁₀ cycloalkylene group unsubstituted or substituted with at least one R_{10a}, a C₁-C₁₀ heterocycloalkylene group unsubstituted or substituted with at least one R_{10a}, a C₃-C₁₀ cycloalkenylene group unsubstituted or substituted with at least one R_{10a}, a C₁-C₁₀ heterocycloalkenylene group unsubstituted or substituted with at least one R_{10a}, a C₆-C₆₀ arylene group unsubstituted or substituted with at least one R_{10a}, a C₁-C₆₀ heteroarylene group unsubstituted or substituted with at least one R_{10a}, a divalent non-aromatic condensed polycyclic group unsubstituted or substituted with at least one R_{10a}, or a divalent non-aromatic condensed heteropolycyclic group unsubstituted or substituted with at least one R_{10a}.

8. The organometallic compound of claim 1, wherein L₁₀ is a group represented by one of Formulae 4-1 to 4-3:



wherein in Formulae 4-1 to 4-3,

Y₉₁ is C(R₉₁) or N,

Y₉₂ is C(R₉₂) or N,

Y₉₃ is C(R₉₃) or N,

Y₉₄ is C(R₉₄) or N,

R₉₁ to R₉₄ are each independently the same as described in connection with R_{10a} in Formula 1, and

*, **, and *** each indicate a binding site to a neighboring atom.

9. The organometallic compound of claim 1, wherein R₁, R₂, R₁₀, R₂₀, R₃₀, R₄₀, R₅₀, R₆₀, R₇₀, and R₈₀ are each independently:

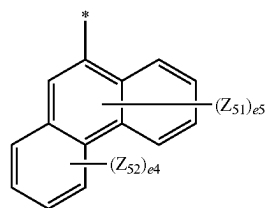
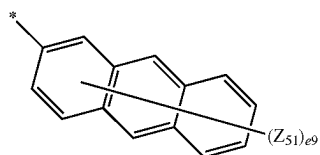
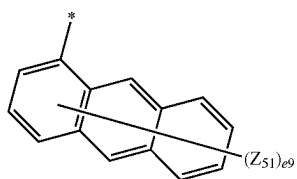
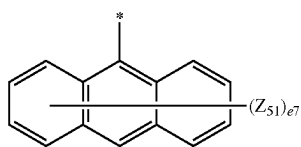
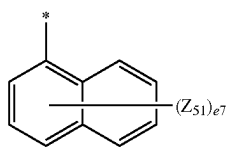
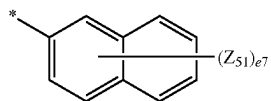
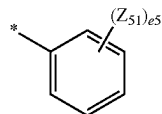
219

hydrogen, deuterium, —F, —Cl, —Br, —I, a cyano group, a C₁-C₂₀ alkyl group, or a C₁-C₂₀ alkoxy group; or

a C₁-C₂₀ alkyl group or a C₁-C₂₀ alkoxy group, each substituted with deuterium, —F, —Cl, —Br, —I, a cyano group, a phenyl group, a biphenyl group, or a combination thereof; or

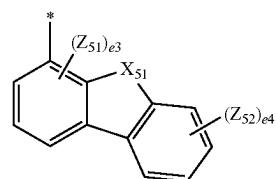
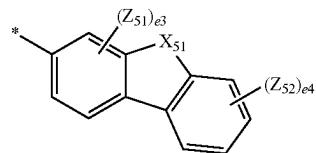
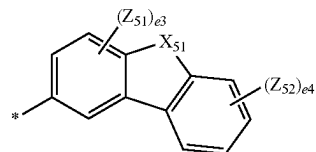
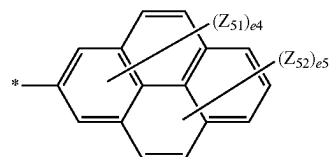
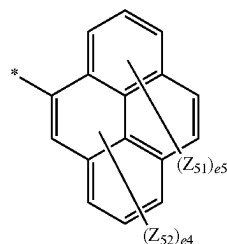
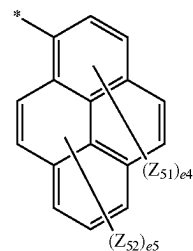
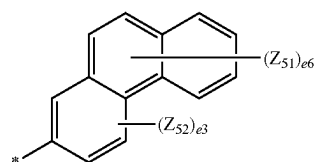
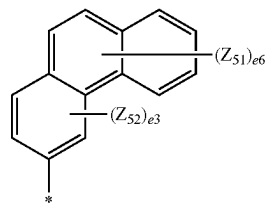
a group represented by one of Formulae 5-1 to 5-26 and 6-1 to 6-55, and

wherein neighboring two or more of R₁, R₂, R₁₀, R₂₀, R₃₀, R₄₀, R₅₀, R₆₀, R₇₀, and R₈₀ are optionally linked to each other to form a cyclopentane group, a cyclohexane group, a benzene group, a naphthylene group, a fluorene group, a pyridine group, a pyrimidine group, a triazine group, or a carbazole group:



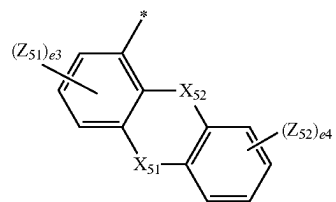
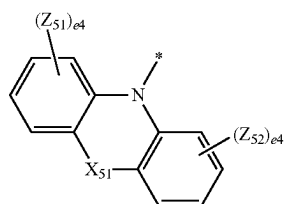
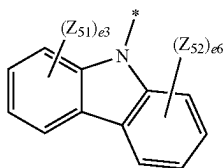
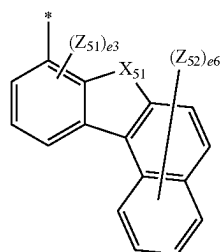
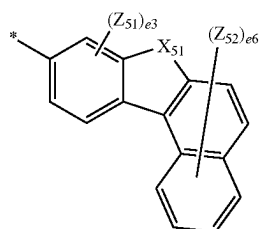
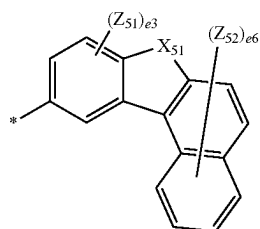
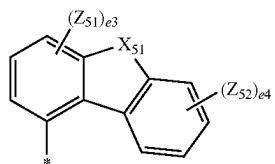
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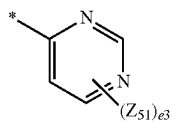
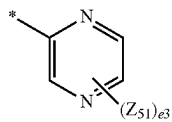
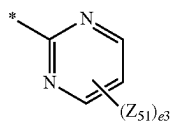
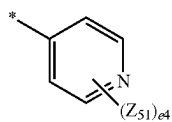
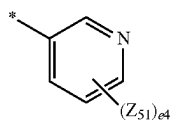
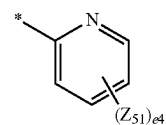
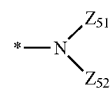
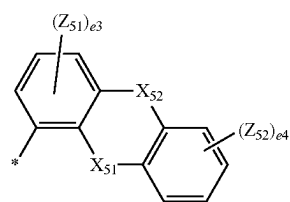
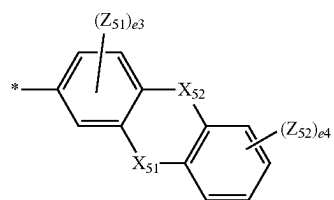
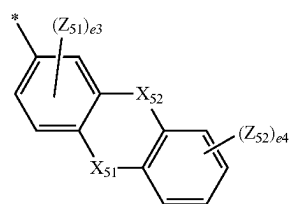


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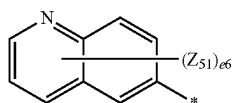
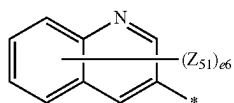
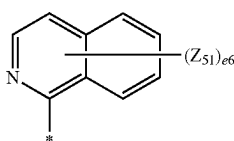
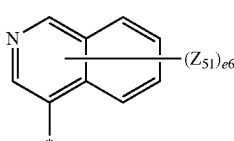
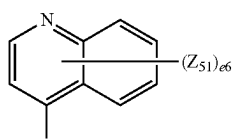
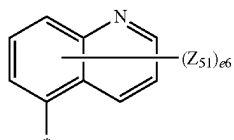
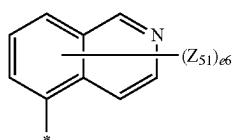
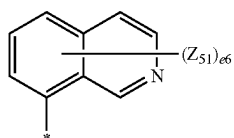
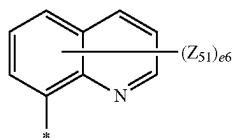
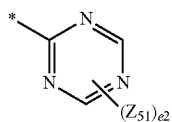
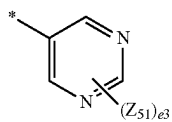
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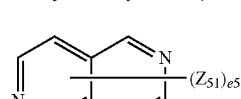
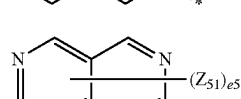
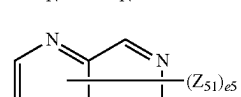
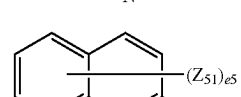
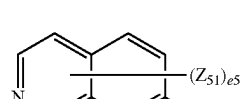
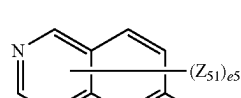
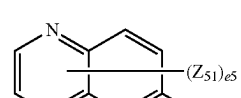
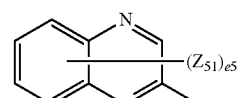
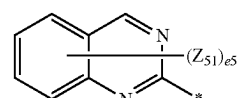
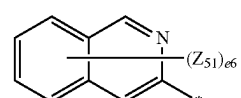
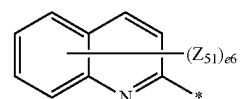
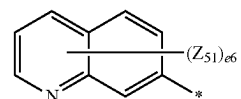
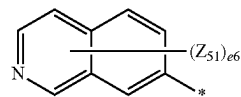
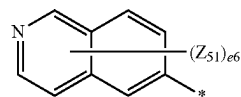


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**224**

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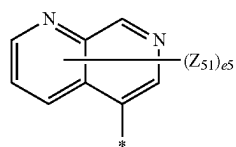
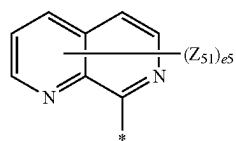
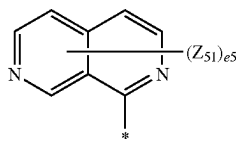
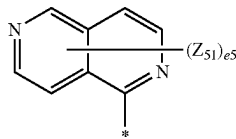
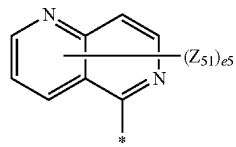
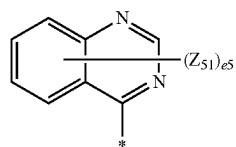
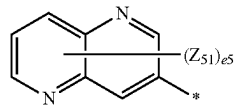
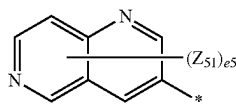
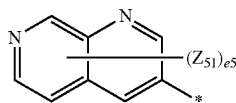
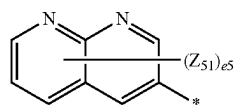
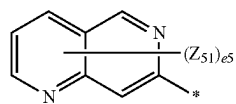
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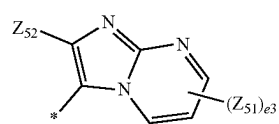
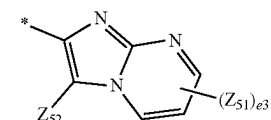
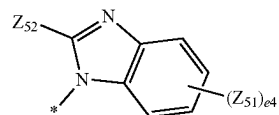
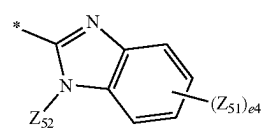
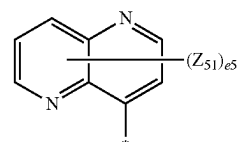
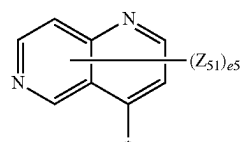
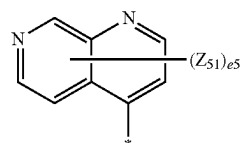
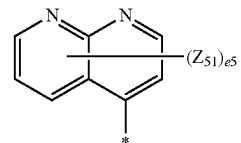
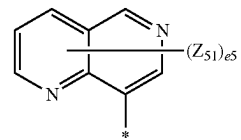
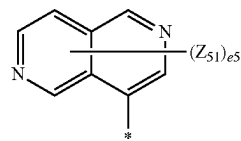
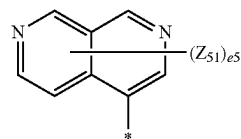
6-31

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**226**

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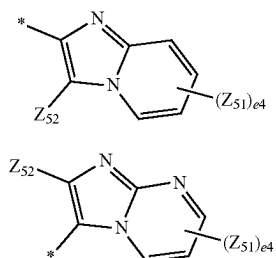
6-51

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6-53

227

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wherein in Formulae 5-1 to 5-26 and 6-1 to 6-55,
 X_{51} and X_{52} are each independently O, S, C(Z_{53})(Z_{54}),
 N(Z_{53}), or Si(Z_{53})(Z_{54}),

Z_{51} to Z_{54} are each independently hydrogen, deuterium,
 —F, —Cl, —Br, —I, a hydroxyl group, a cyano group,
 a nitro group, an amidino group, a hydrazino group, a
 hydrazono group, a C_1 - C_{20} alkyl group, a C_2 - C_{20} alkenyl
 group, a C_2 - C_{20} alkynyl group, a C_1 - C_{20} alkoxy
 group, a phenyl group, a biphenyl group, a terphenyl
 group, a naphthyl group, a fluorenyl group, a spiro-
 bifluorenyl group, a phenanthrenyl group, an anthracenyl
 group, a triphenylenyl group, a pyridinyl group, a
 pyrimidinyl group, a carbazolyl group, or a triazinyl
 group,

e2 is 1 or 2,

e3 is 1, 2, or 3,

e4 is 1, 2, 3, or 4,

e5 is 1, 2, 3, 4, or 5,

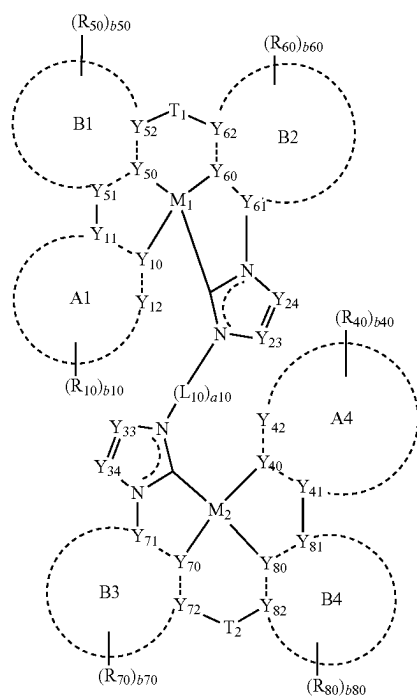
e6 is 1, 2, 3, 4, 5, or 6,

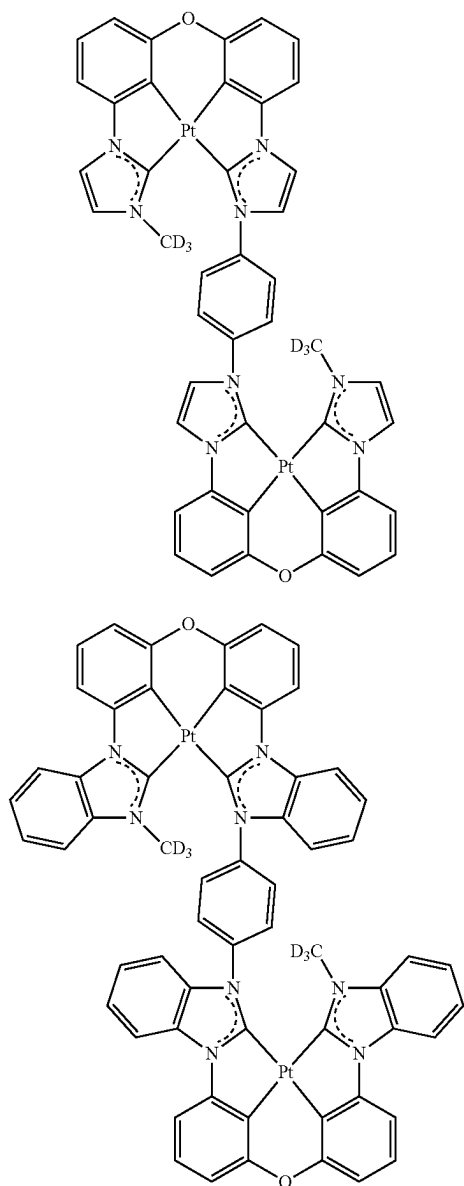
e7 is 1, 2, 3, 4, 5, 6, or 7,

e9 is 1, 2, 3, 4, 5, 6, 7, 8, or 9, and

* indicates a binding site to a neighboring atom.

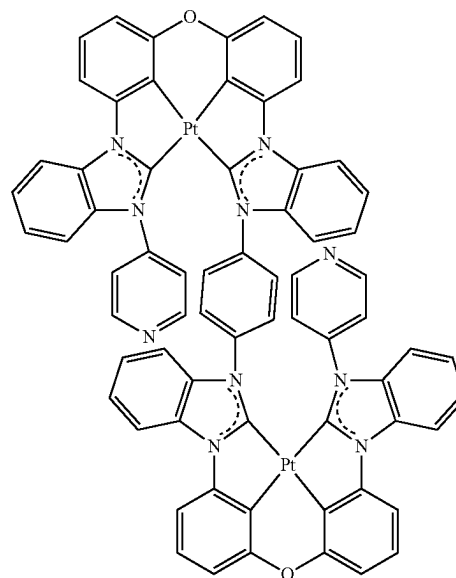
10. The organometallic compound of claim 1, wherein the
 organometallic compound is represented by one of Formulae
 11-1 to 11-3:



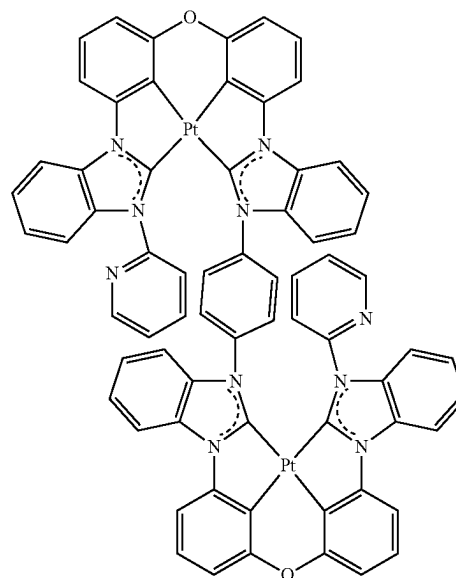
229 Y_{33} is C(R₃₃) or N, Y_{34} is C(R₃₄) or N, Y_{35} is C(R₃₅) or N, Y_{36} is C(R₃₆) or N, Y_{37} is C(R₃₇) or N, Y_{38} is C(R₃₈) or N, R_{23} to R_{28} are each independently the same as described in connection with R_{20} in Formula 1, and R_{33} to R_{38} are each independently the same as described in connection with R_{30} in Formula 1.**11.** The organometallic compound of claim 1, wherein the organometallic compound is electrically neutral.**12.** The organometallic compound of claim 1, wherein the organometallic compound is one selected from Compounds BD1 to BD104:**230**

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BD3

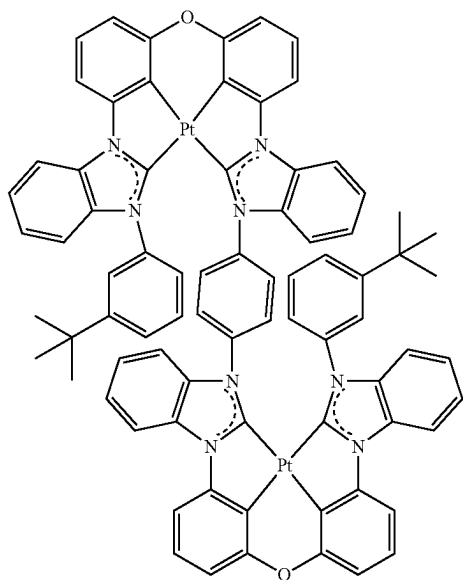


BD4



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232

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BD5

BD7

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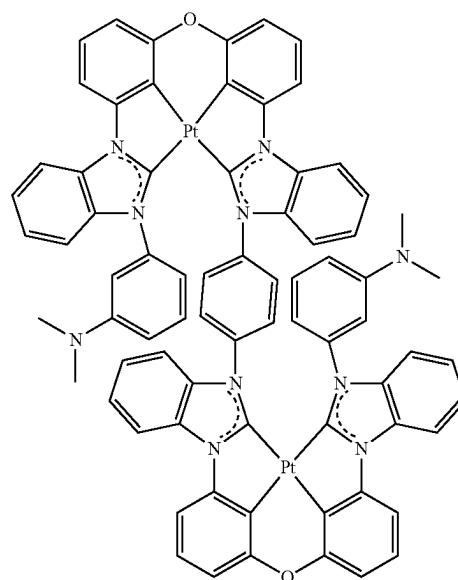
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BD6

BD8

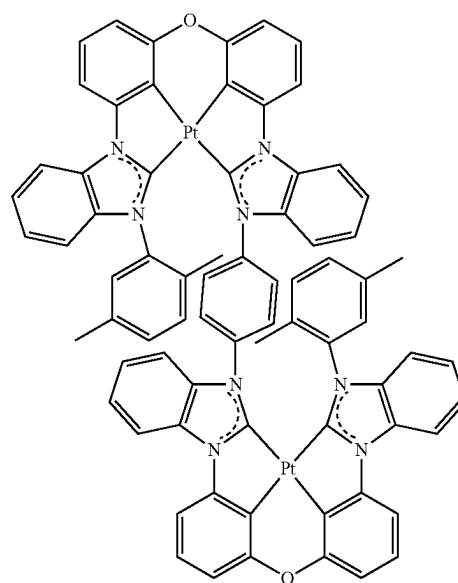
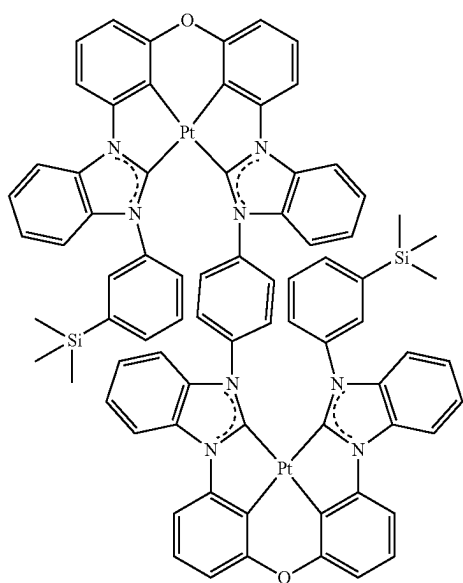
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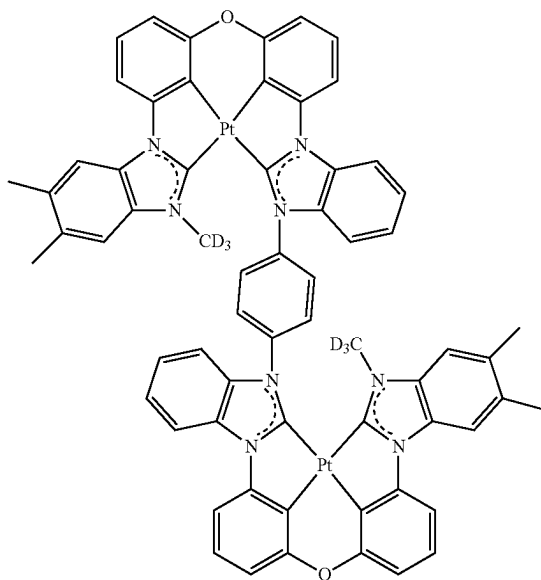
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233

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BD9

**234**

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BD11

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BD10

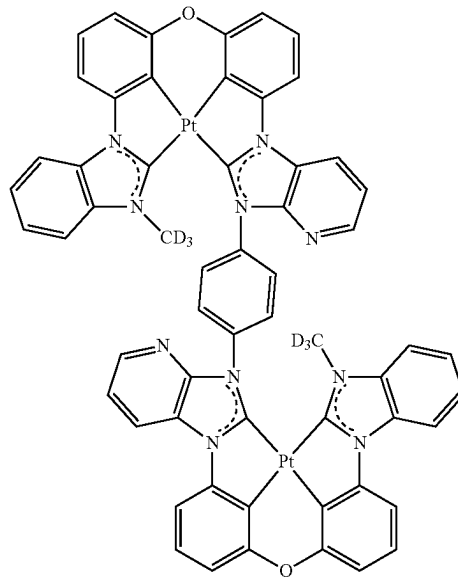
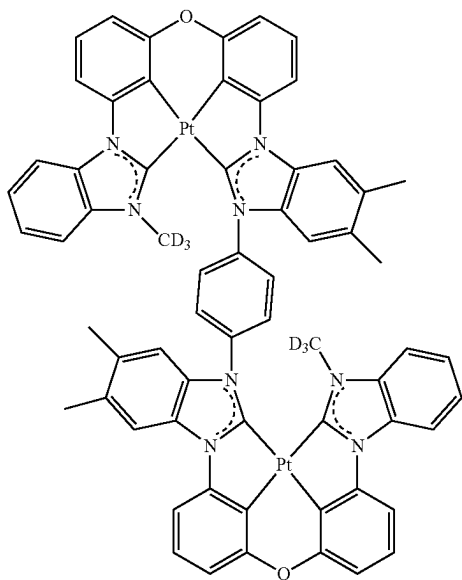
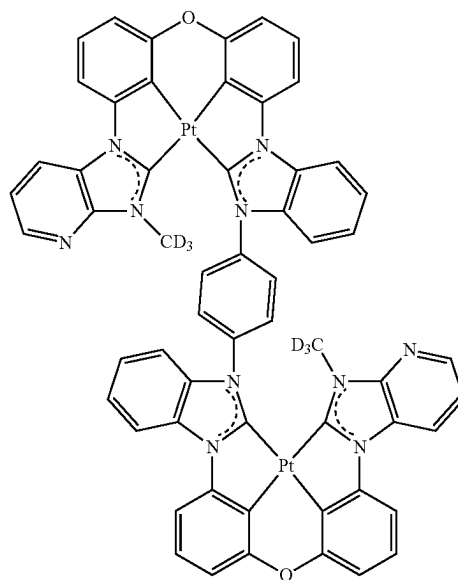
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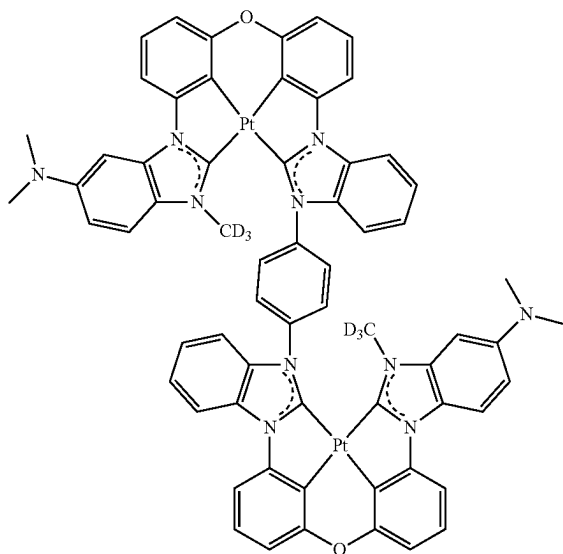
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235

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BD13

**236**

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BD15

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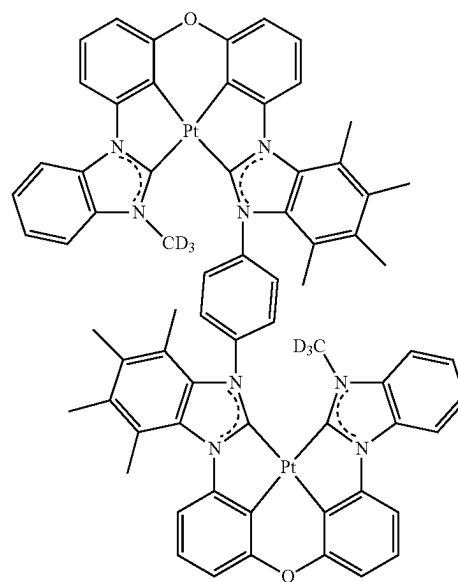
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BD14

BD16

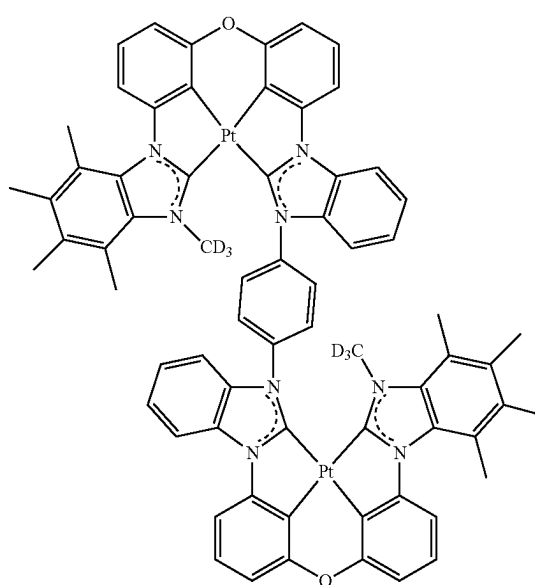
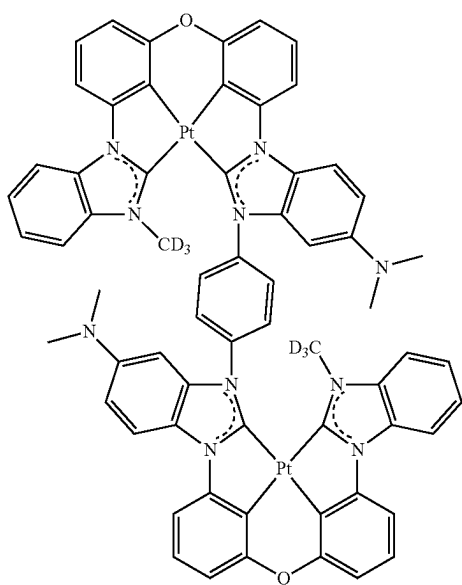
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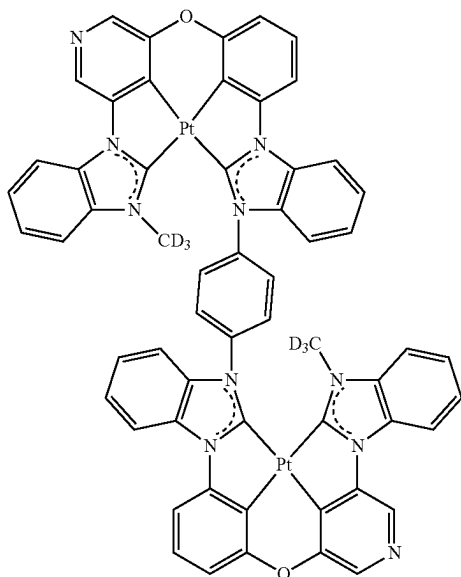
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237

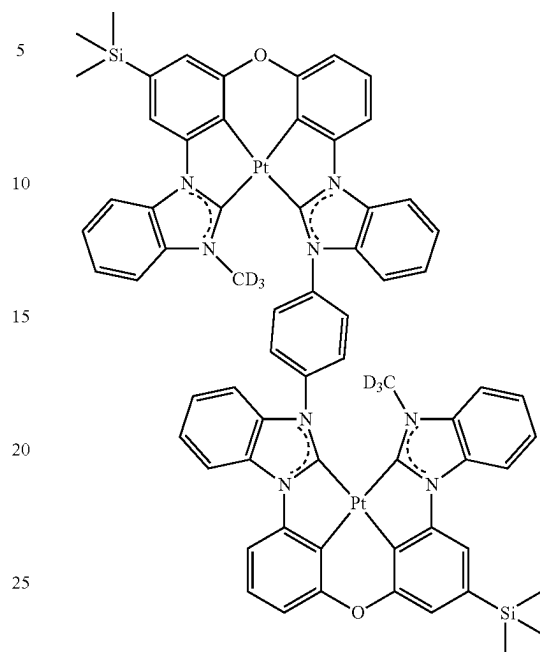
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**238**

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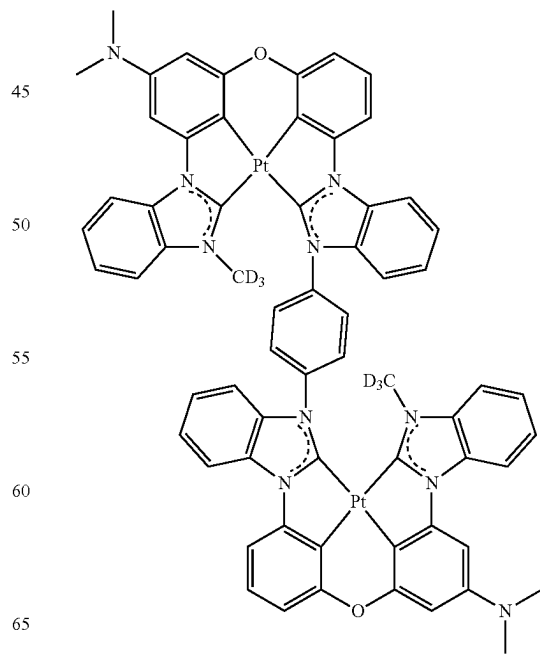
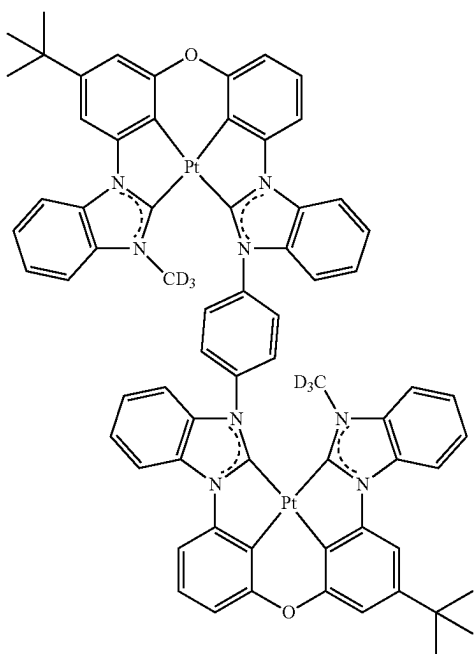
BD17

BD19



BD18

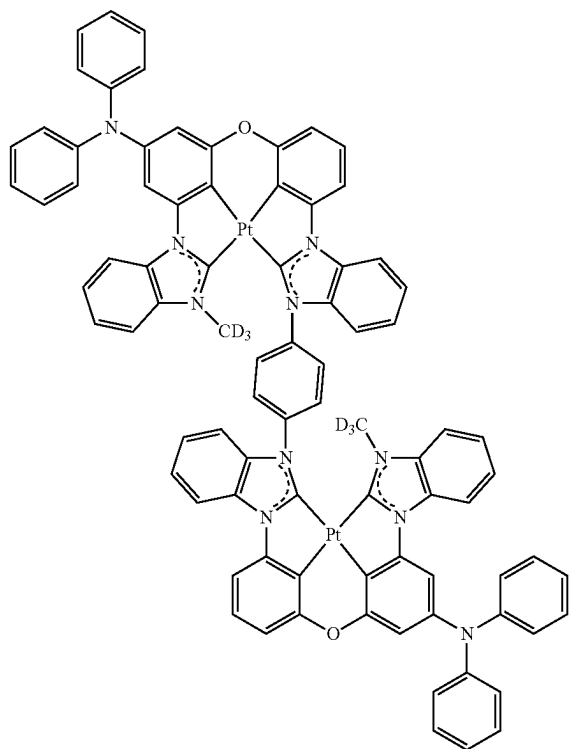
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239

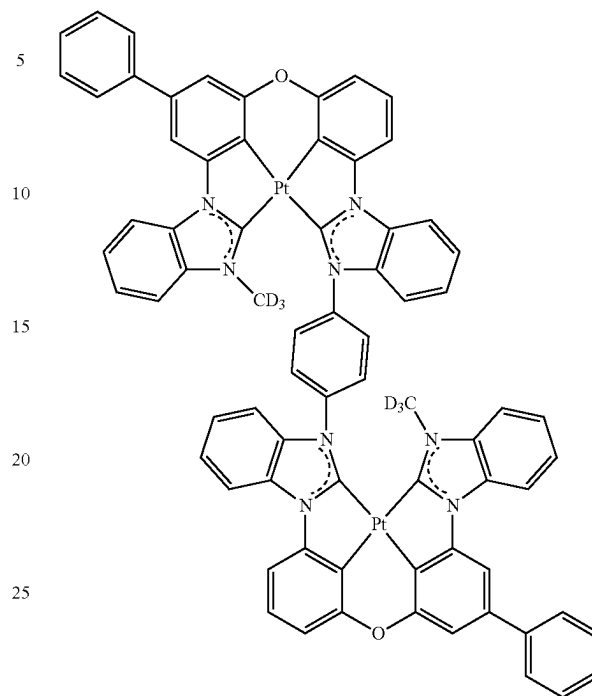
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BD21

**240**

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BD23



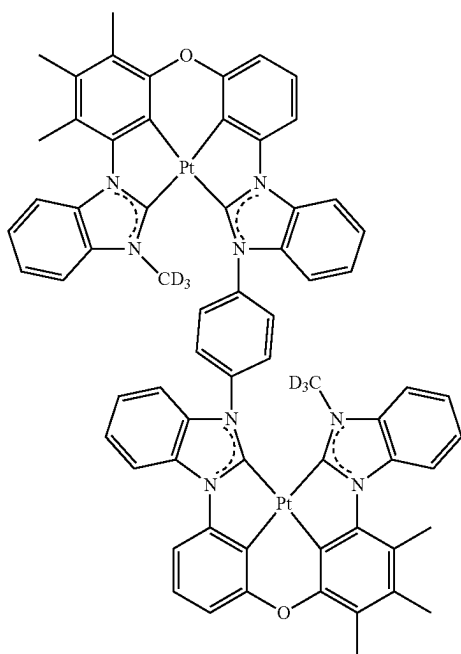
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BD22

BD24



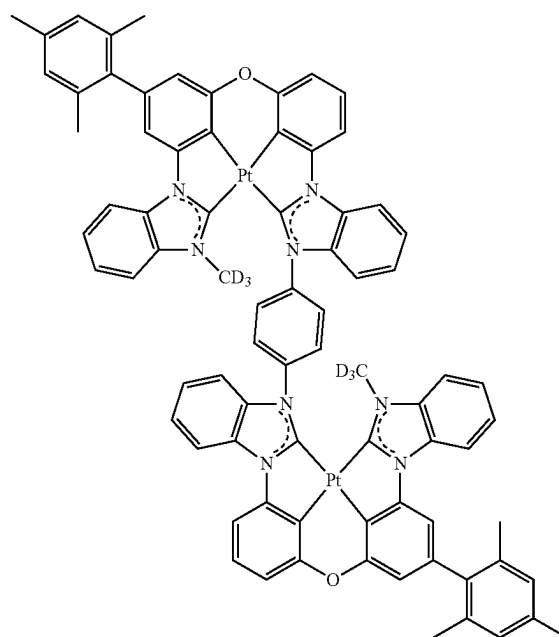
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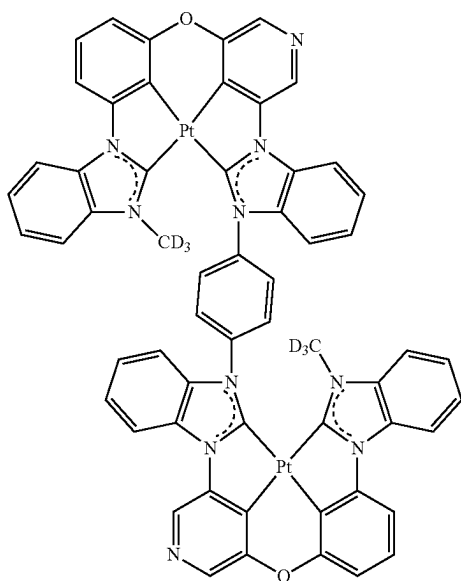
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241

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**242**

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BD25

BD27

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BD26

BD28

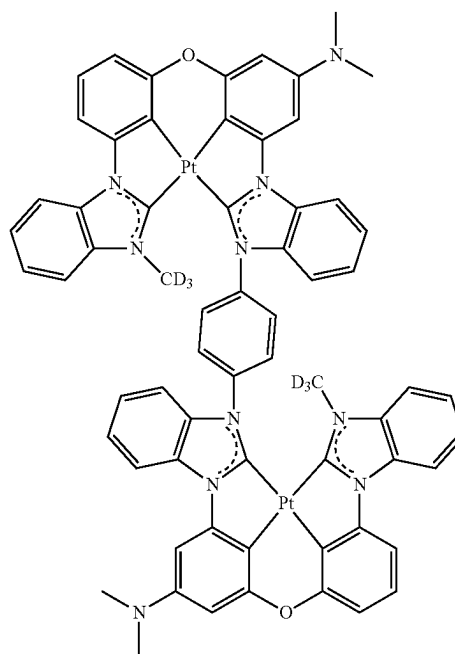
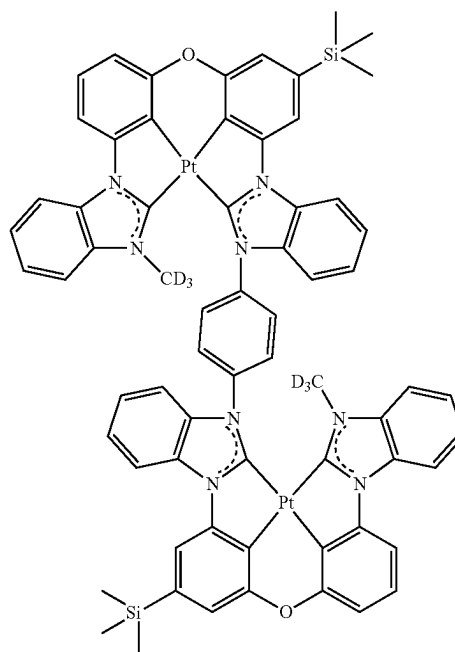
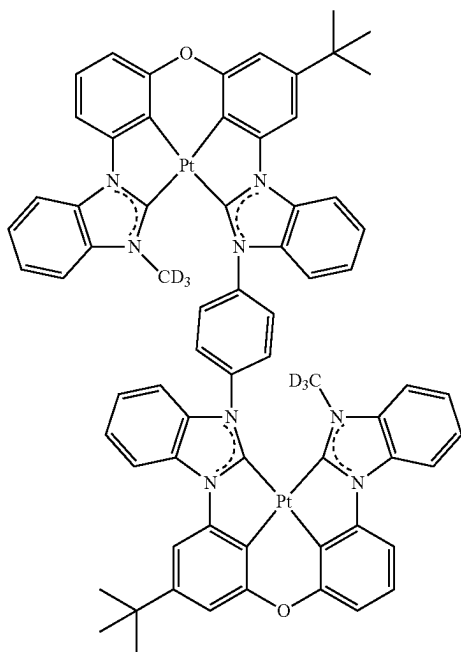
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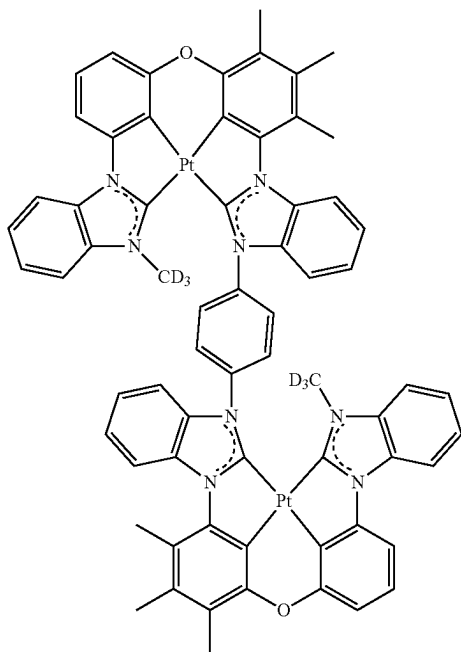
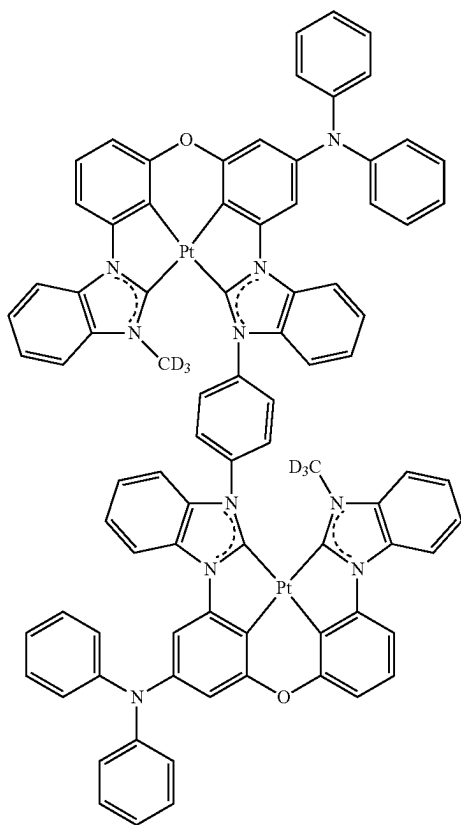
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243

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**244**

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BD29

BD31

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BD30

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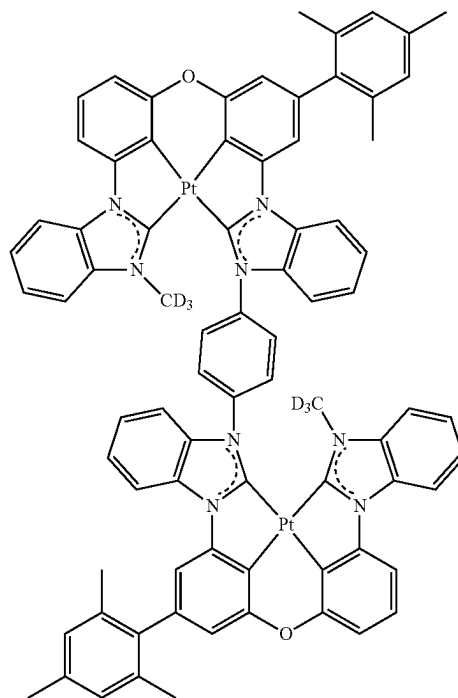
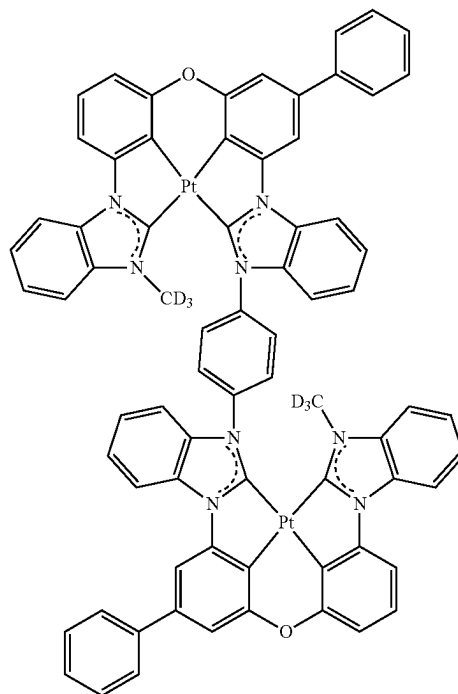
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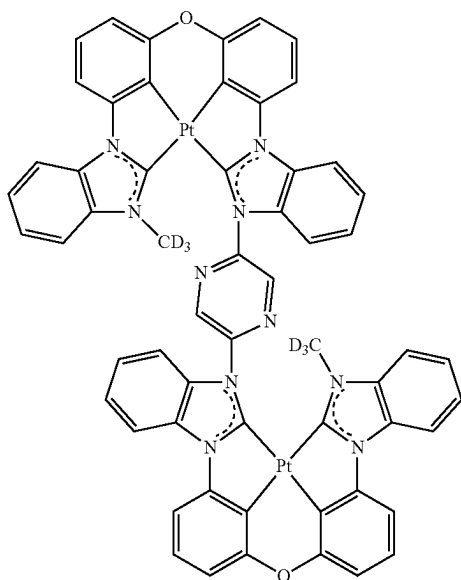
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BD32



245

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BD33

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BD34

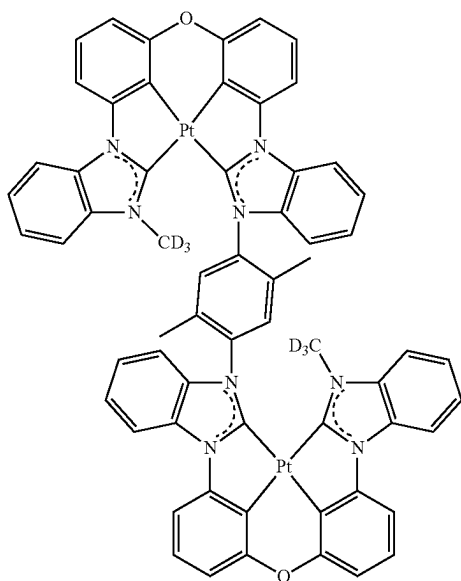
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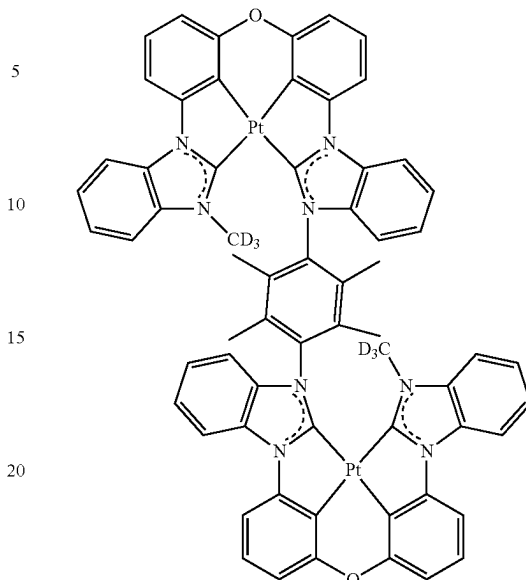
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**246**

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BD35

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BD36

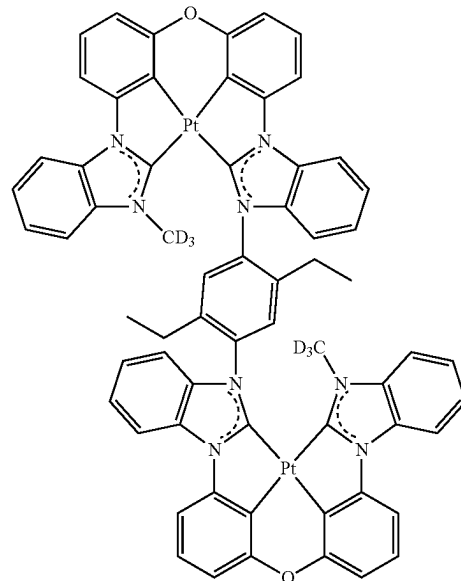
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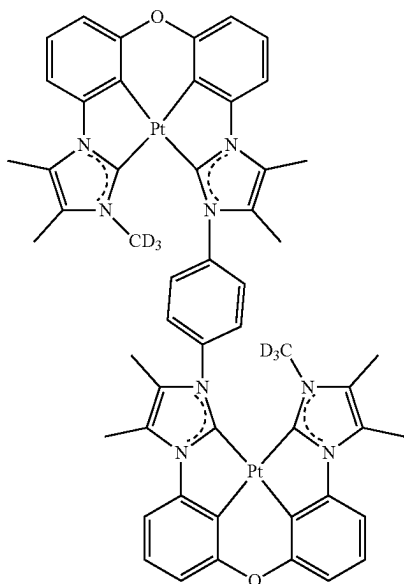
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247

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**248**

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BD37

BD39

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BD38

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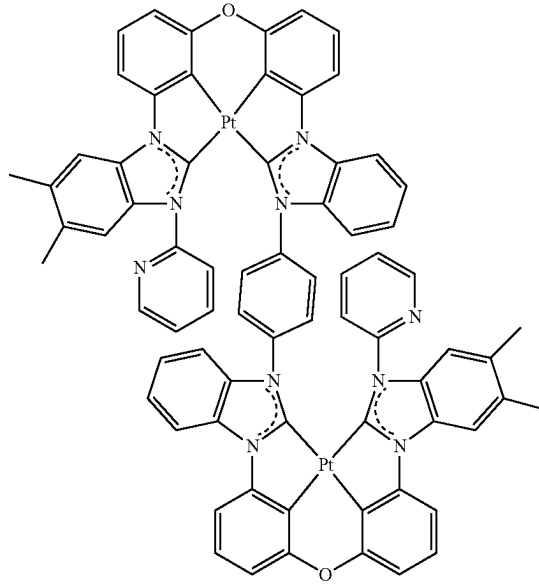
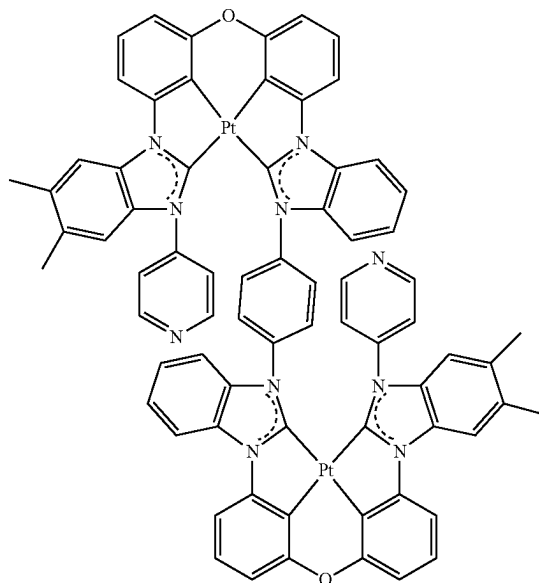
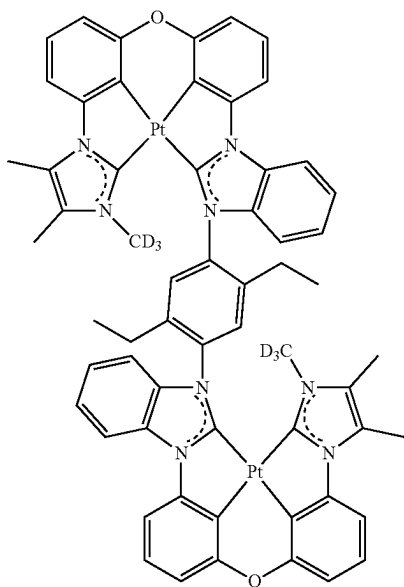
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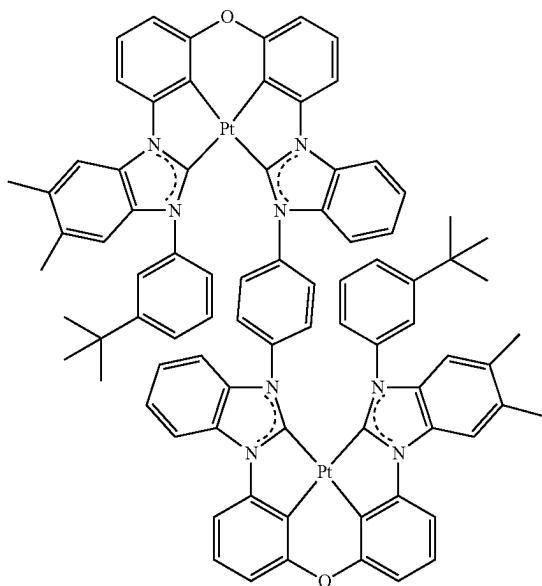
BD40



249

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BD41

**250**

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BD43

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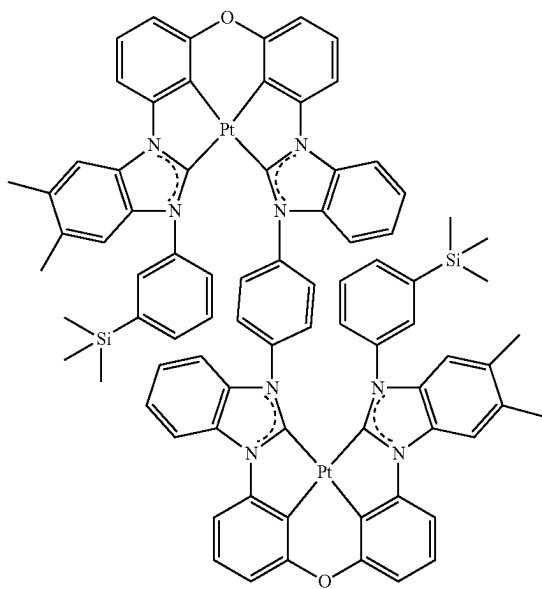
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BD42



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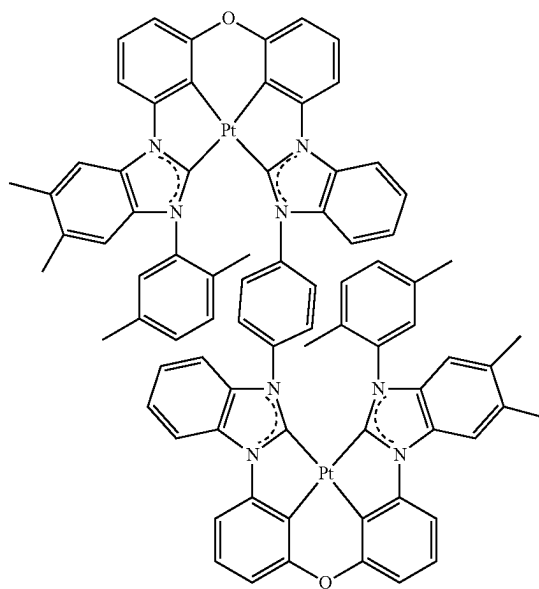
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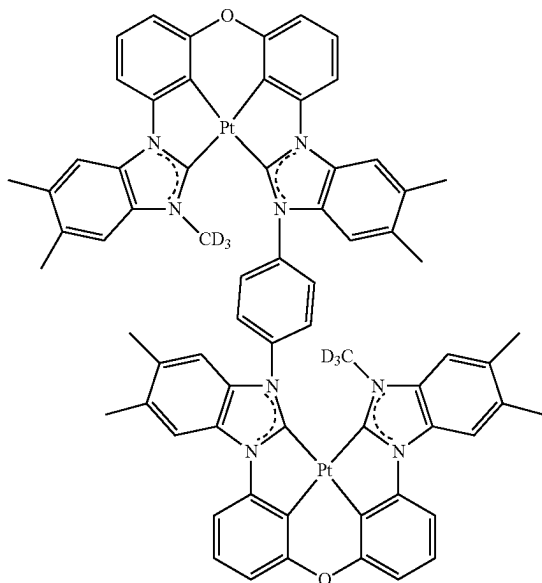
BD44



251

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BD45



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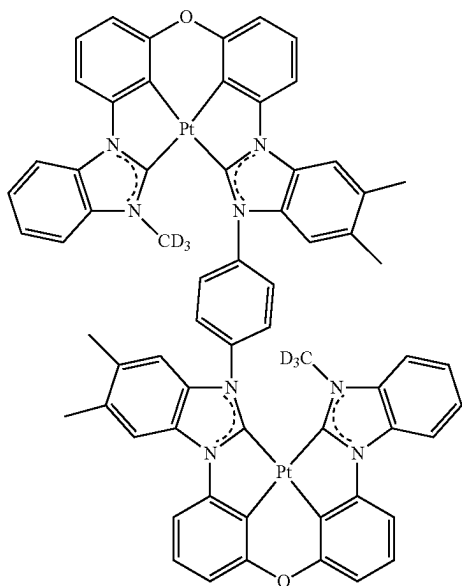
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BD46



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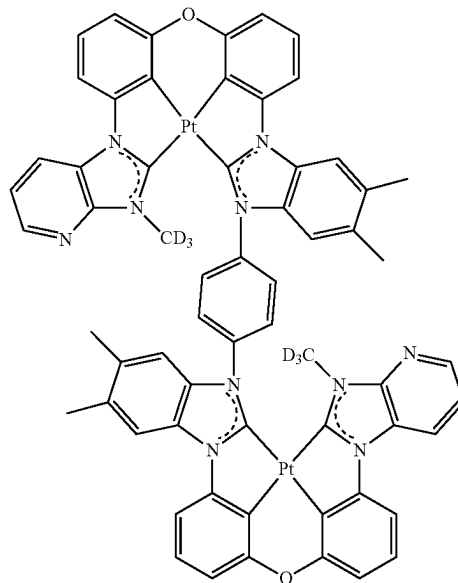
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252

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BD47



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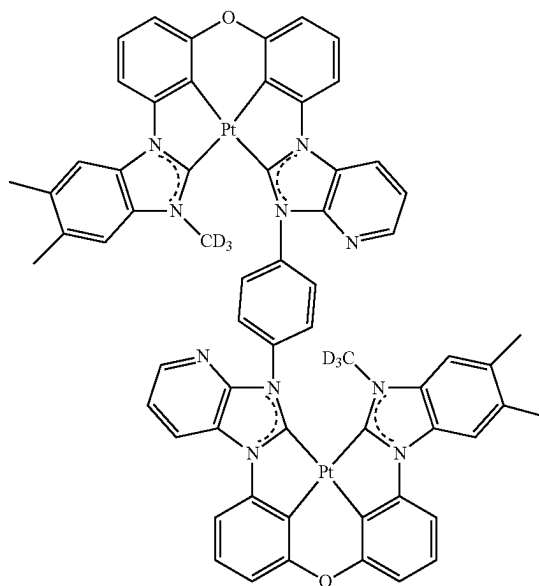
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BD48



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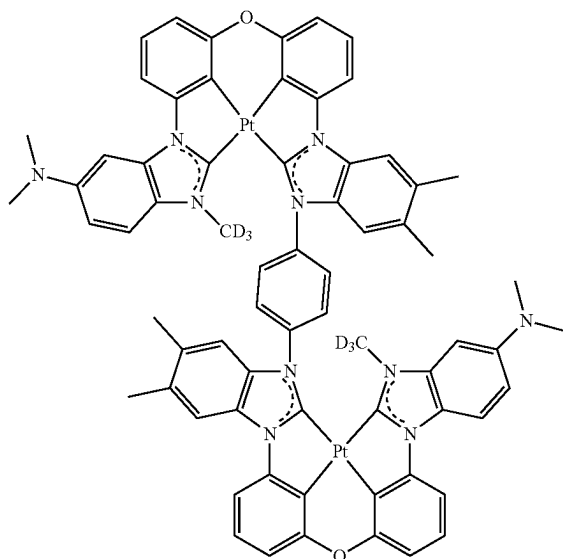
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253

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BD49

**254**

-continued

BD51

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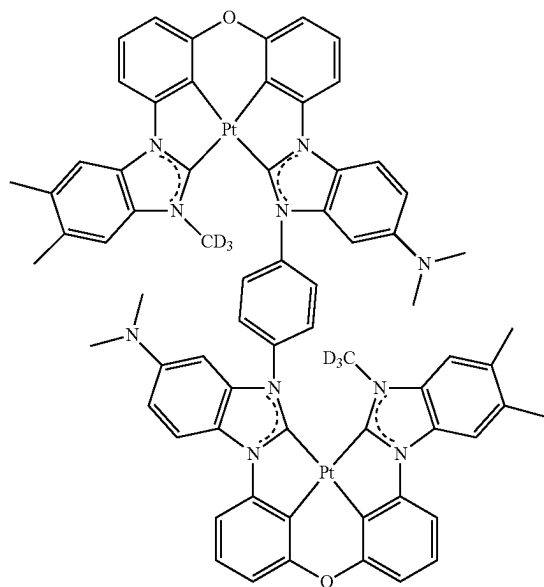
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BD50



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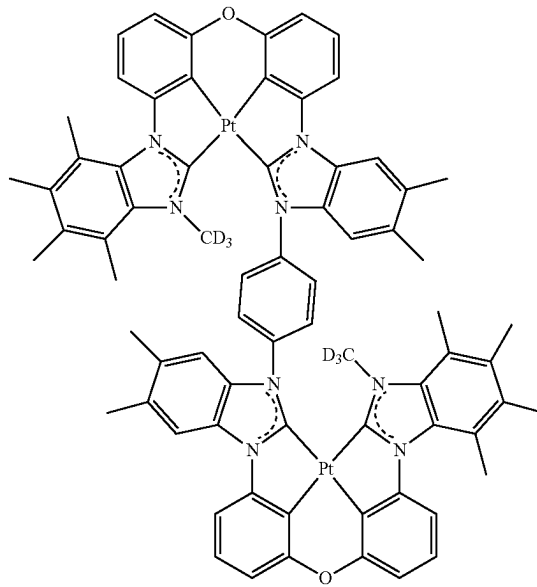
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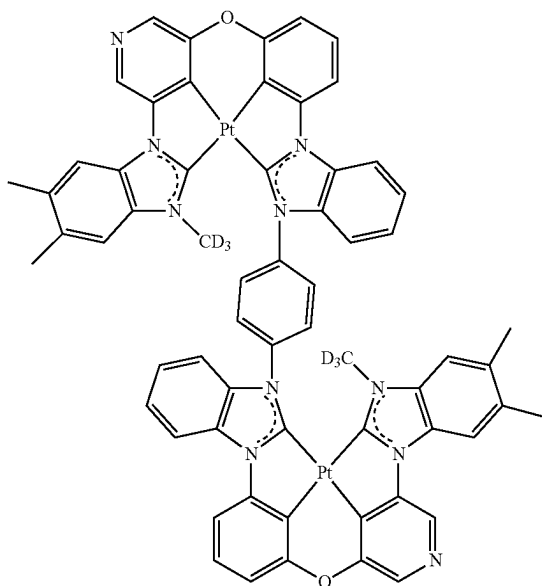
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255

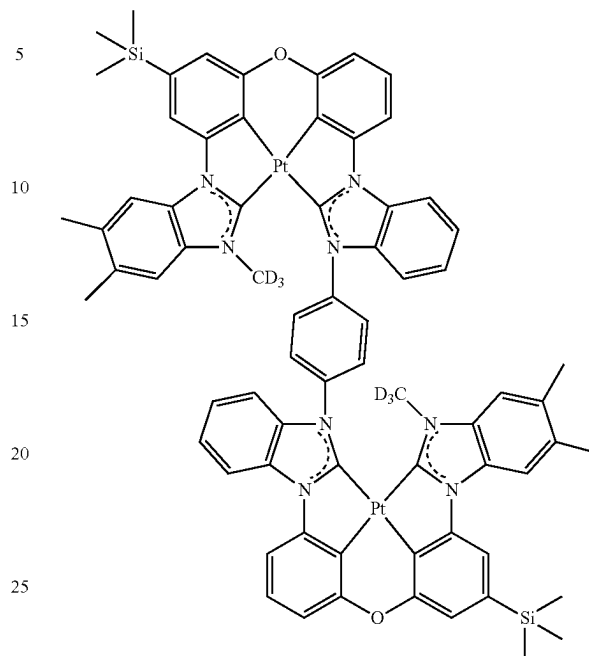
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BD53

**256**

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BD55

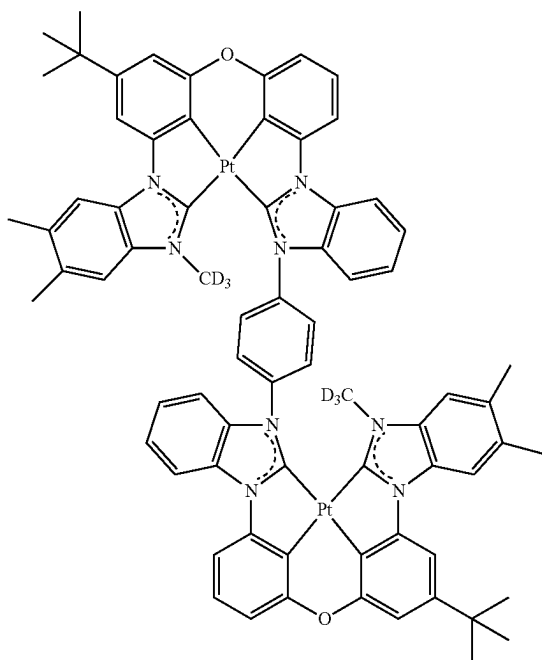


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BD54



BD56

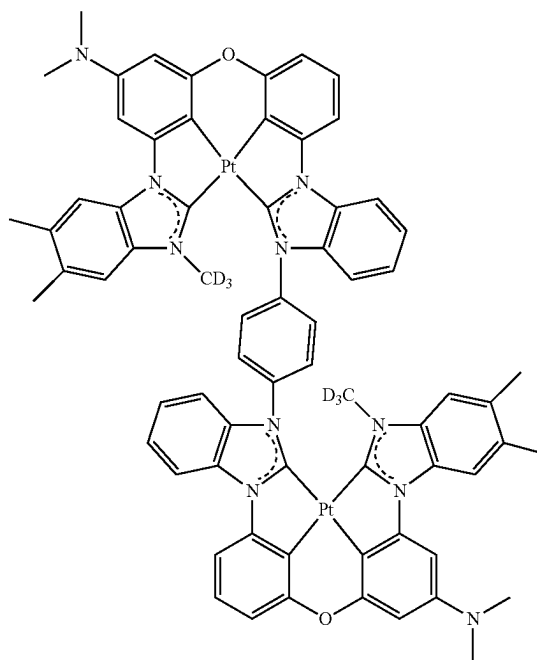
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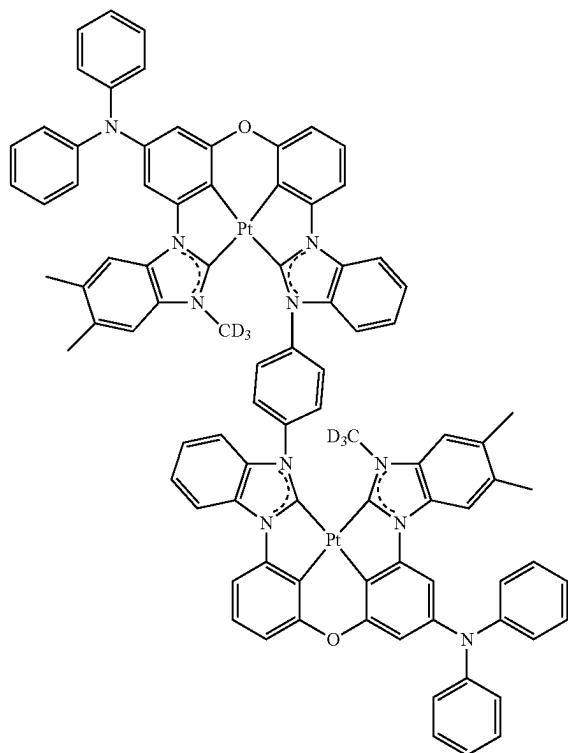
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257

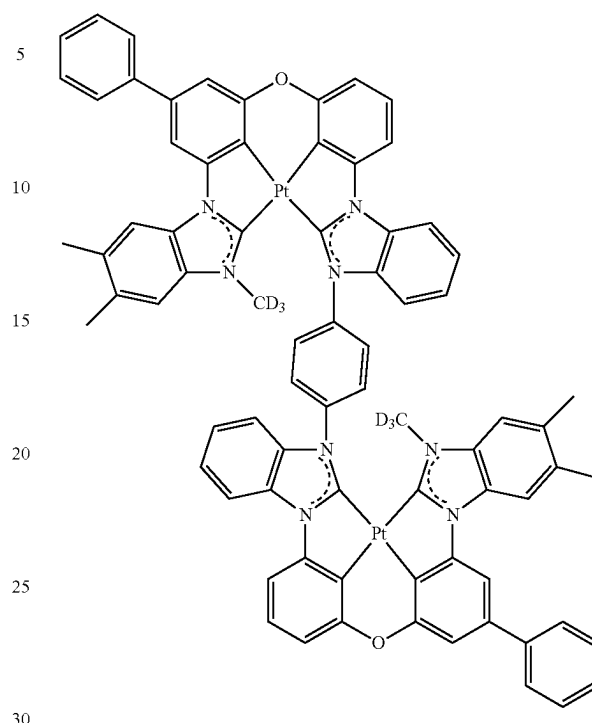
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BD57

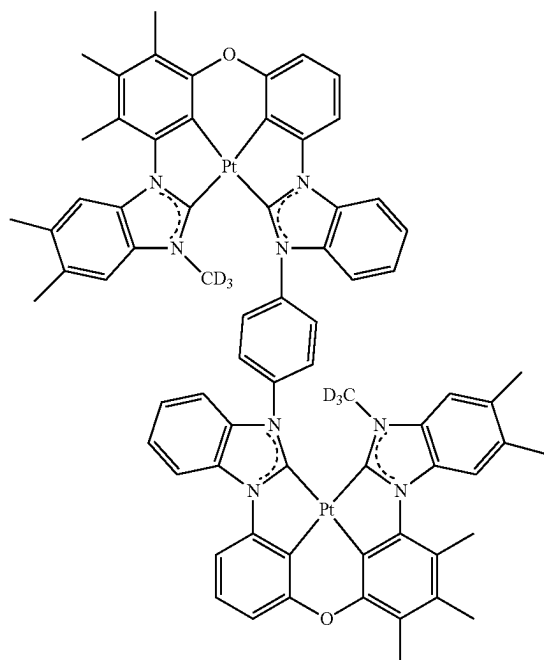
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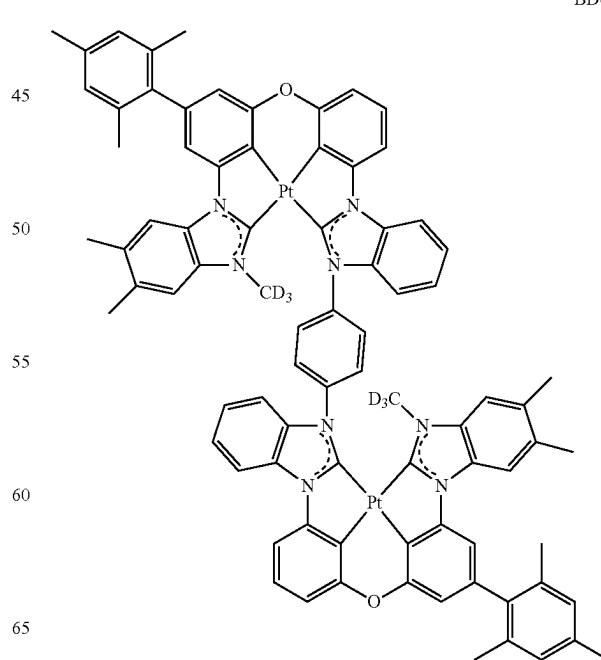
BD59



BD58



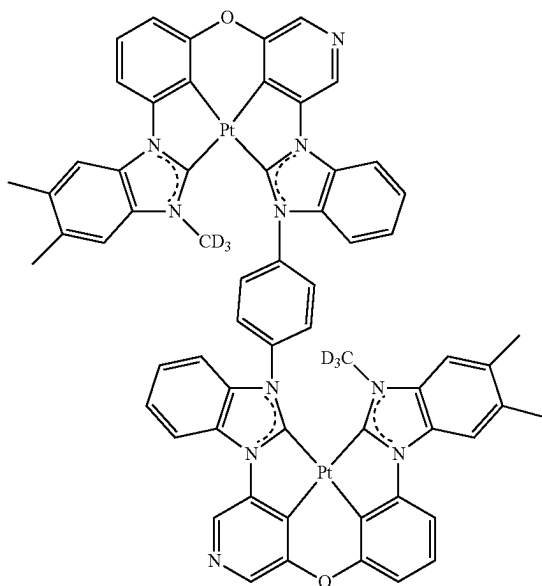
BD60



259

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BD61

**260**

-continued

BD63

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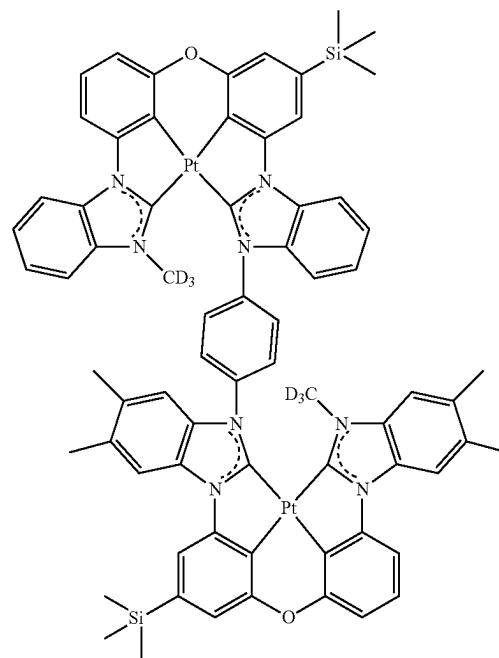
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BD62

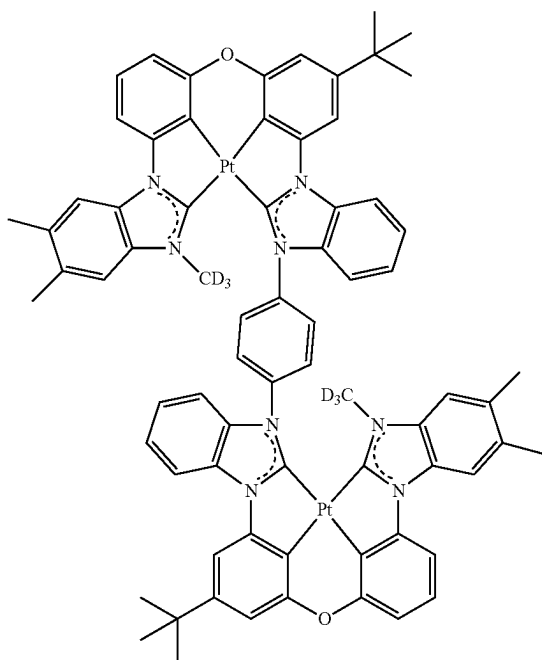


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BD64



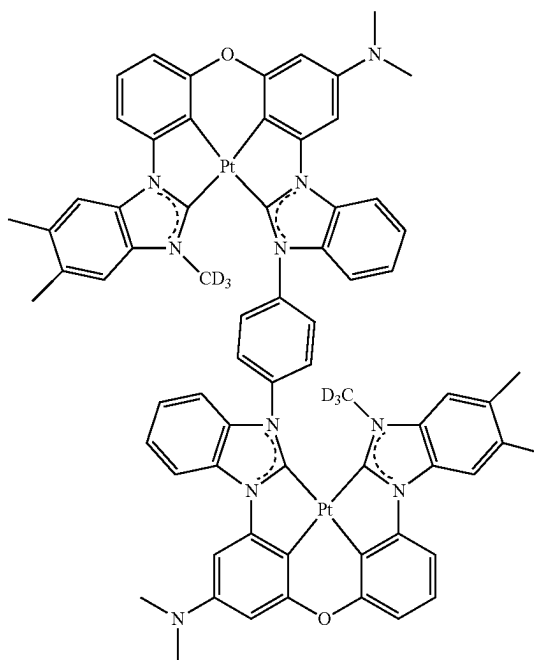
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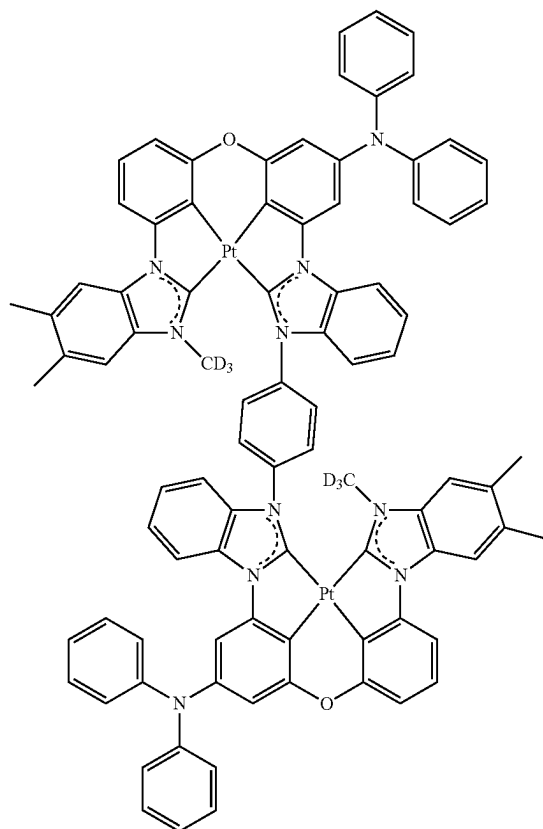
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261

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BD65

**262**

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BD67

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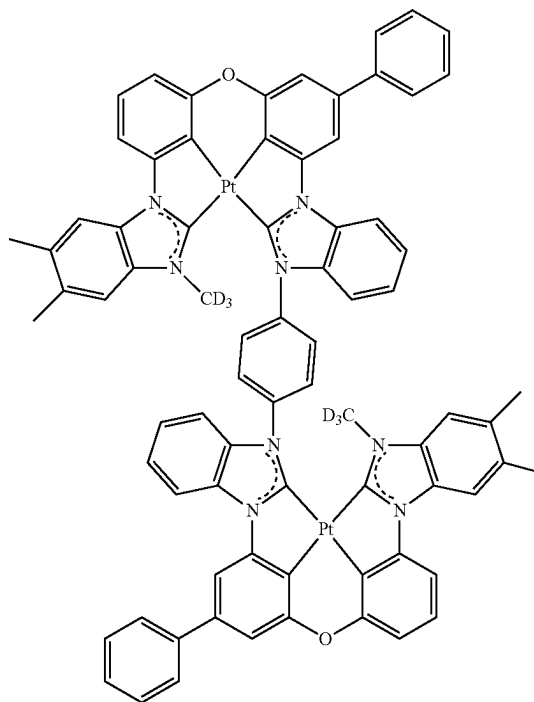
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BD66

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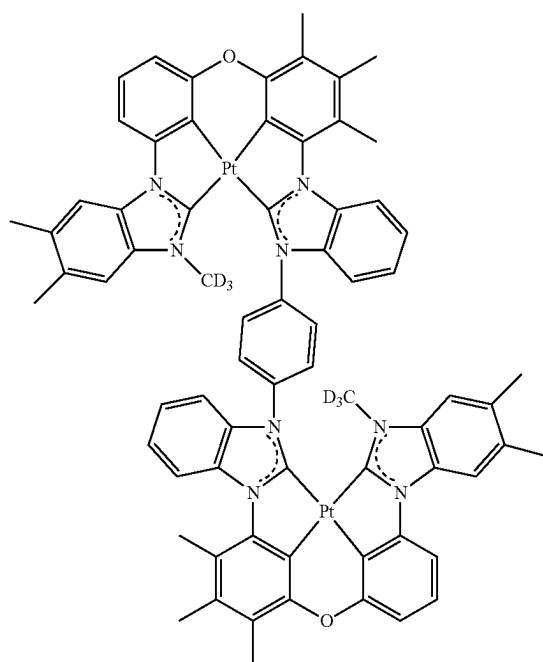
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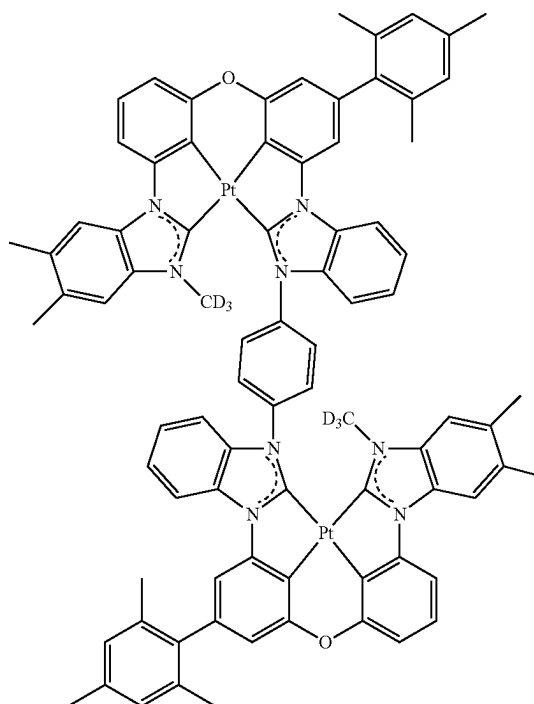
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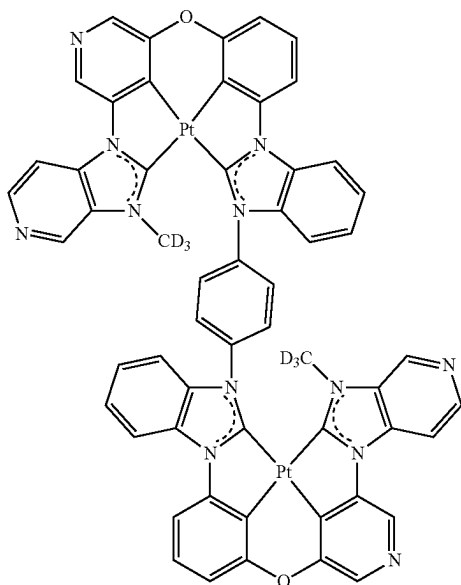


BD68



263

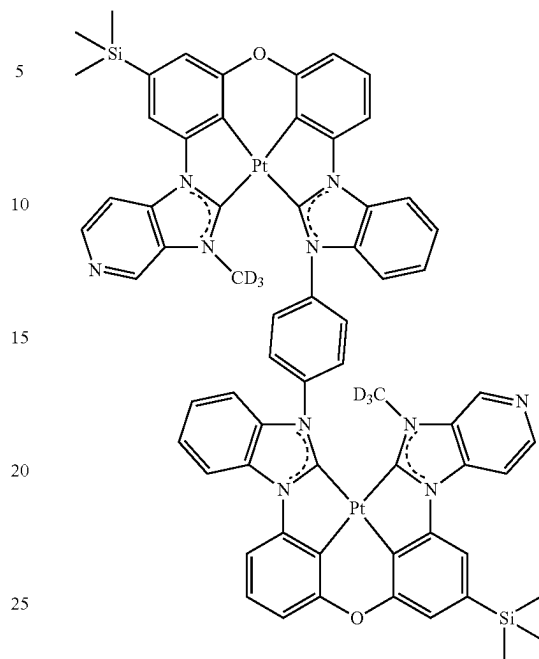
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**264**

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BD69

BD71



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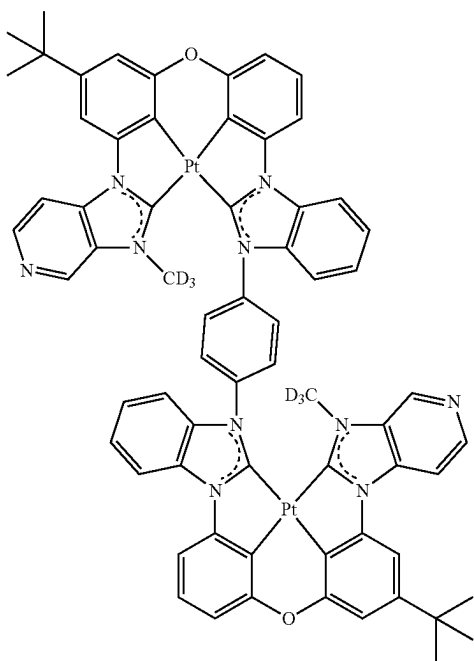
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BD70

BD72



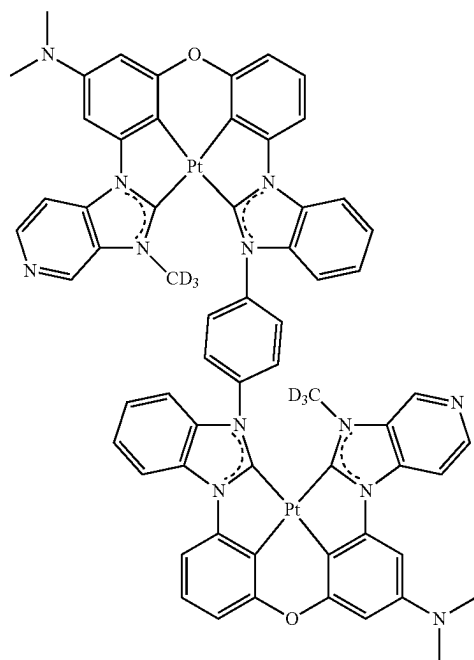
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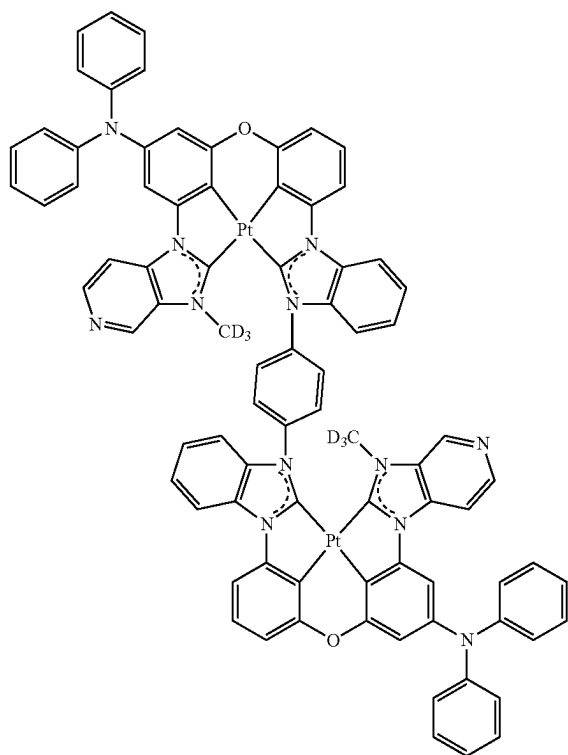
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265

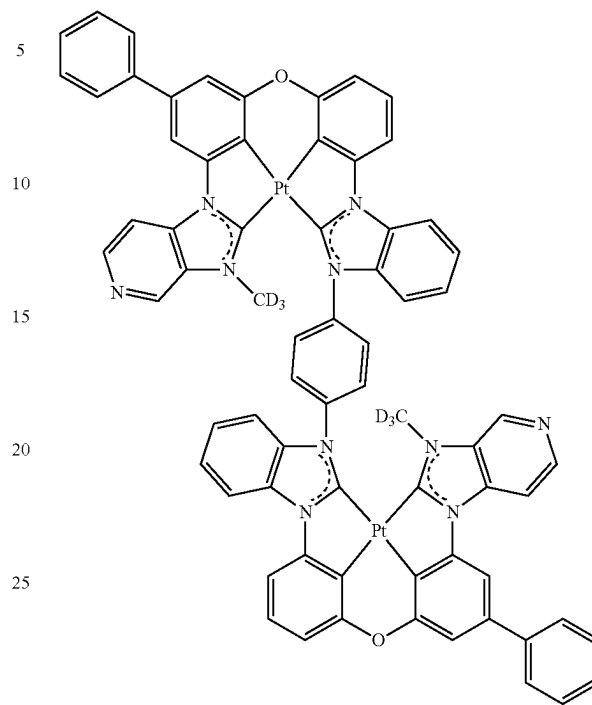
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BD73

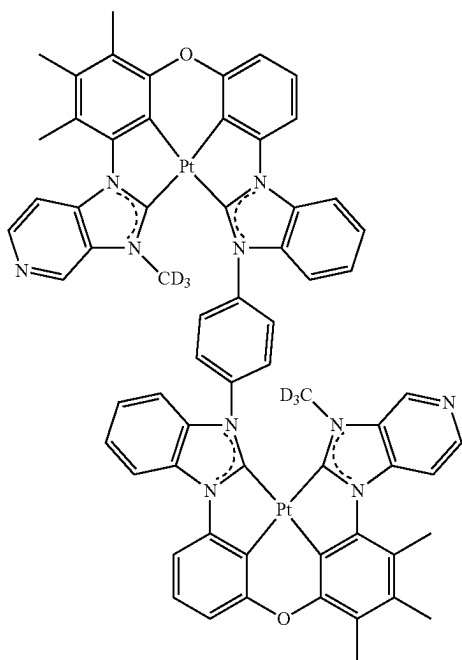
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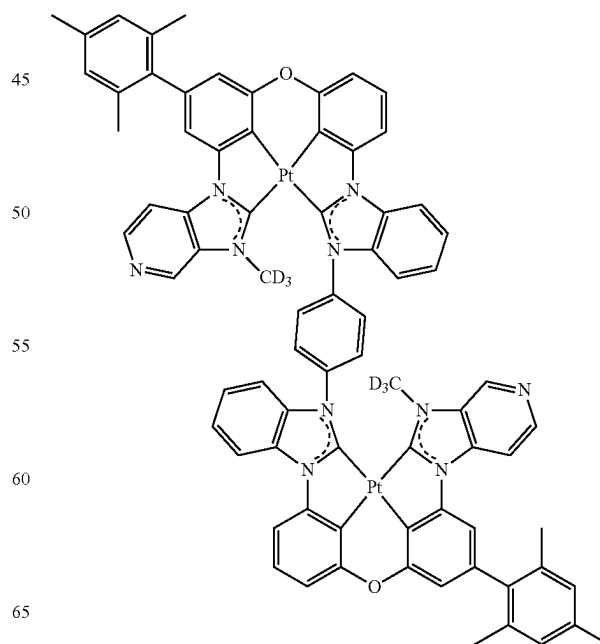
BD75



BD74

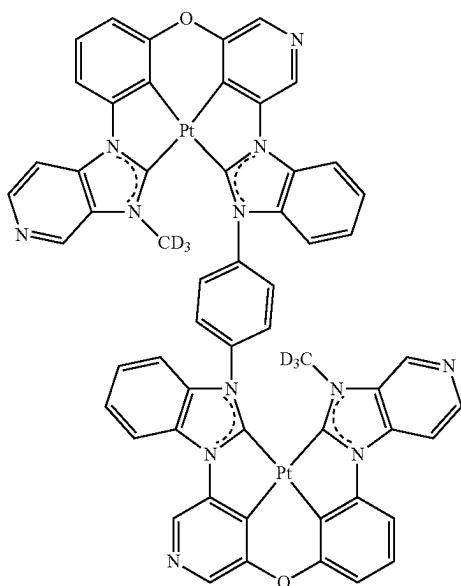


BD76



267

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BD77

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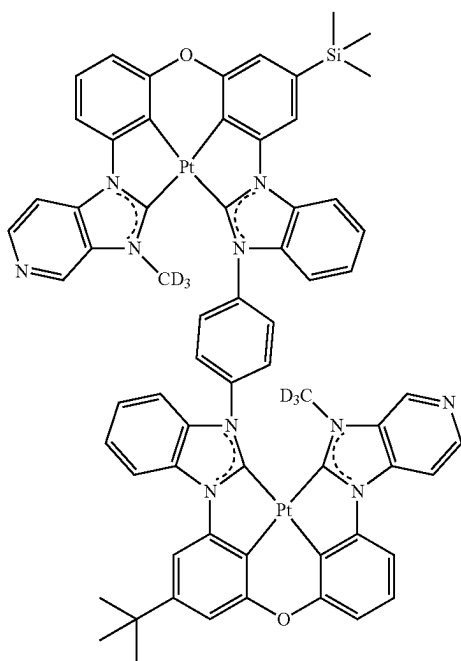
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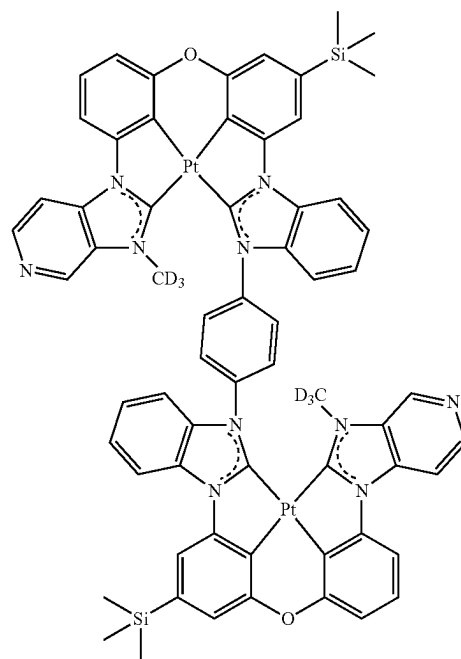
BD78



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268

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BD79

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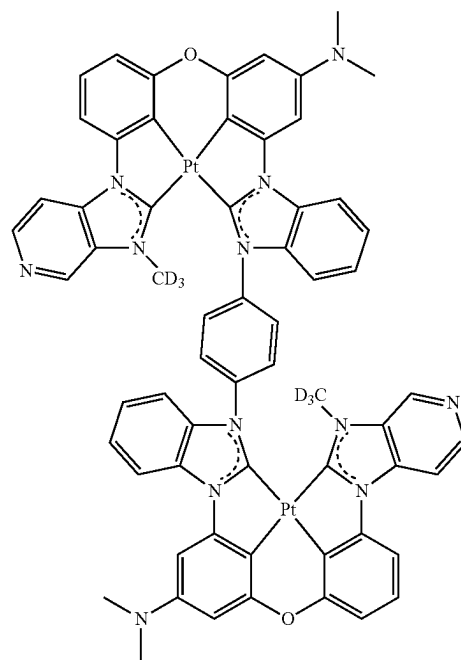
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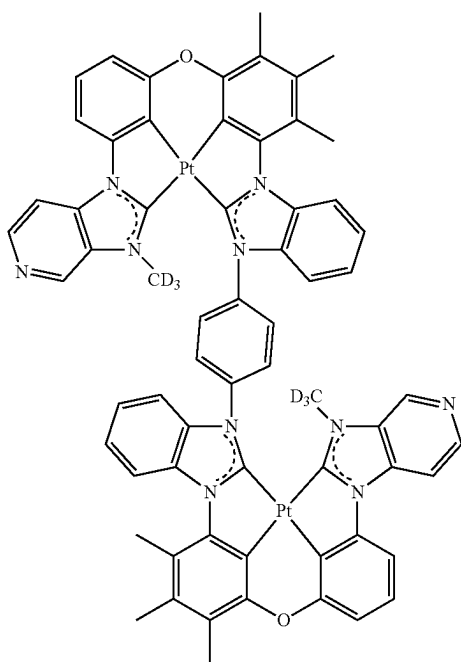
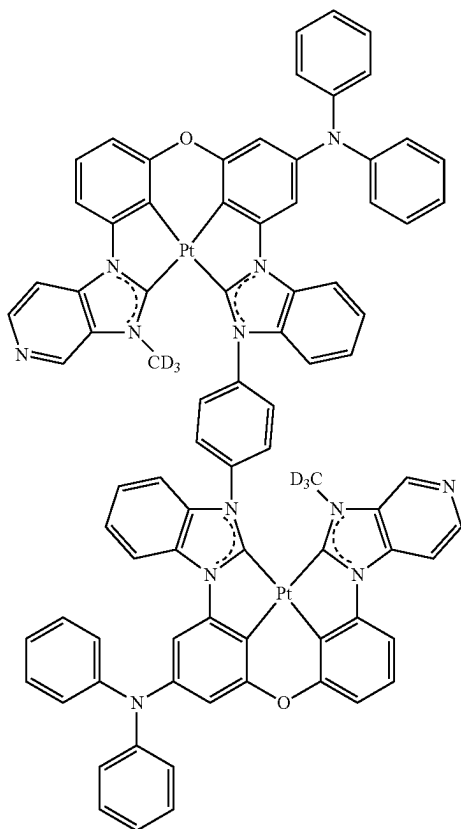
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BD80



269

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**270**

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BD81

BD83

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BD82

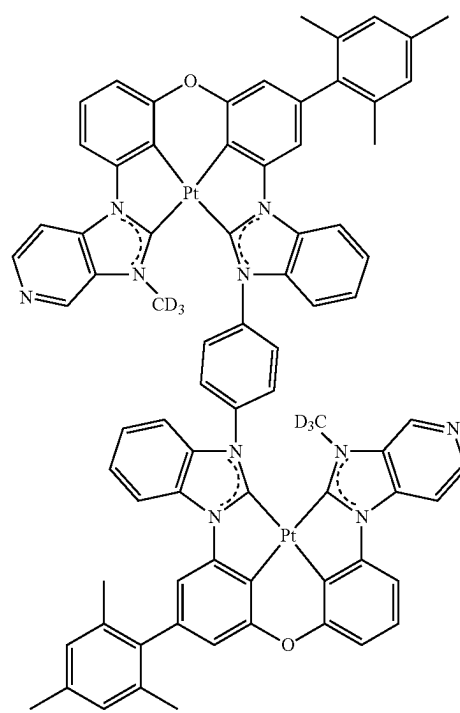
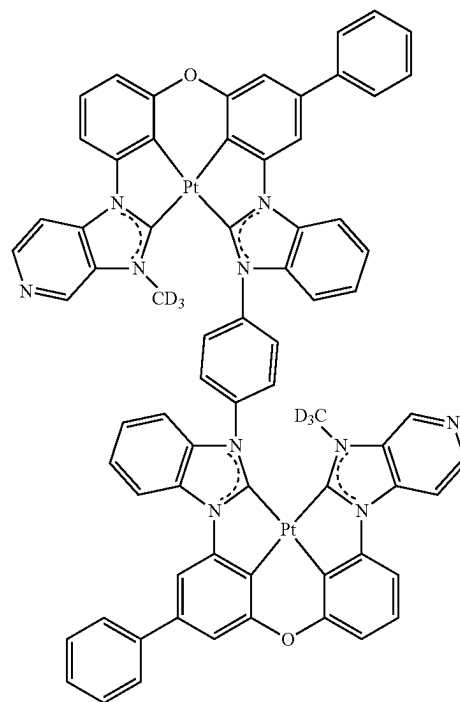
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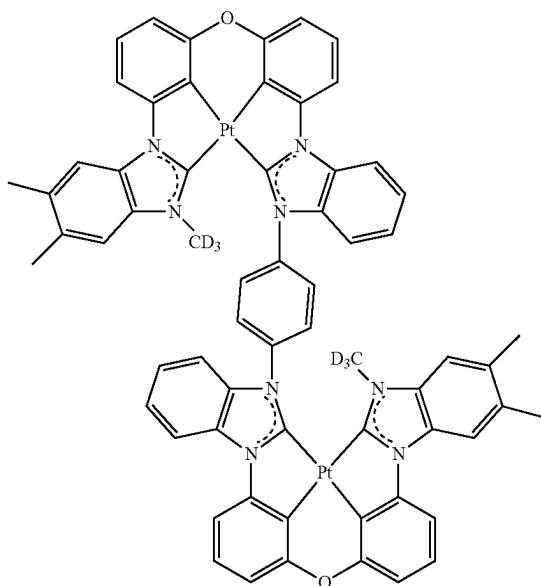


BD84

271

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BD85



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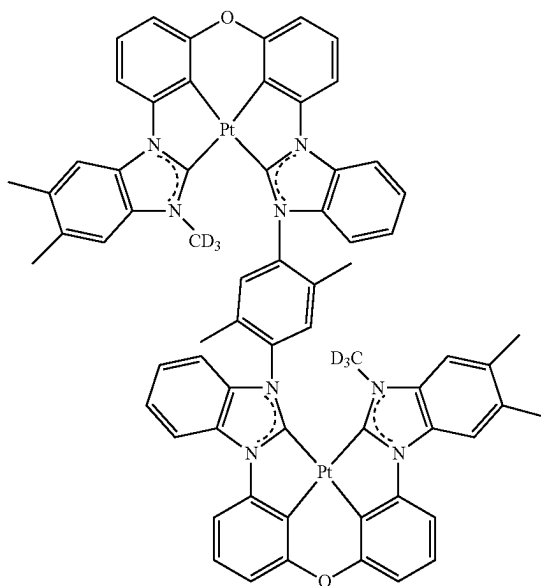
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BD86



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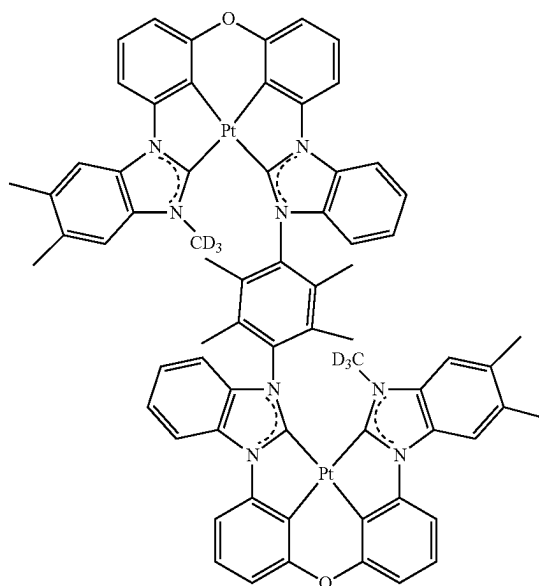
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272

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BD87



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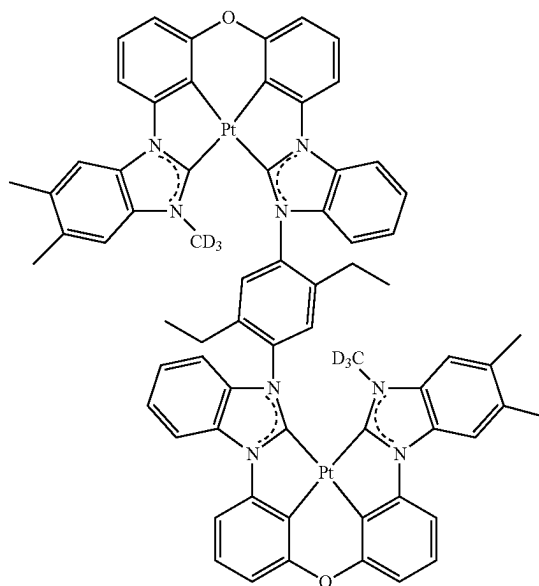
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BD88



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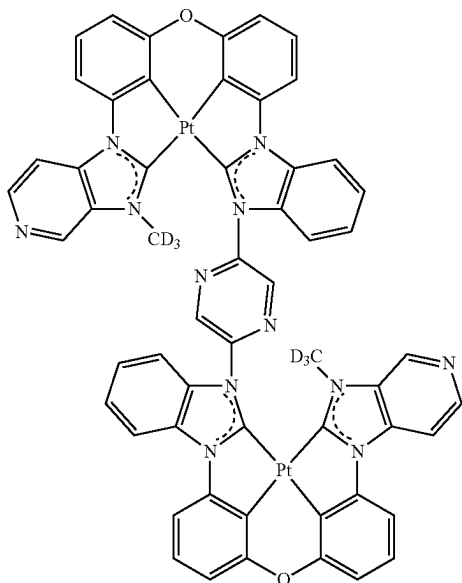
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273

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**274**

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BD89

BD91

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BD90

BD92

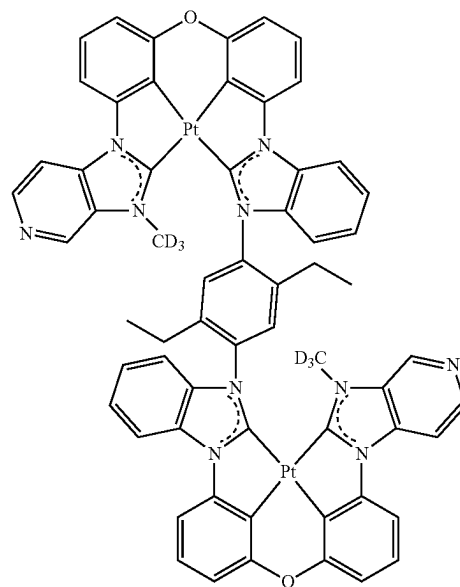
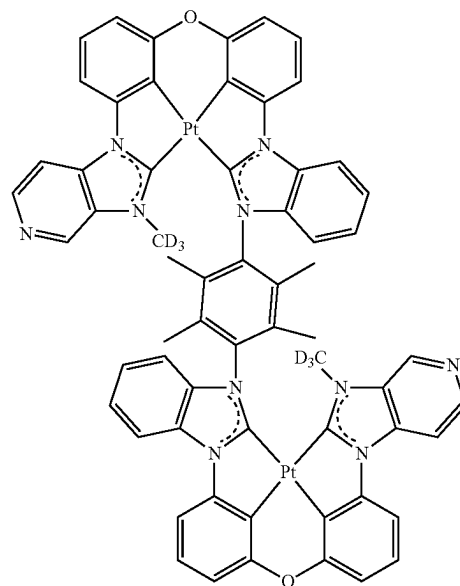
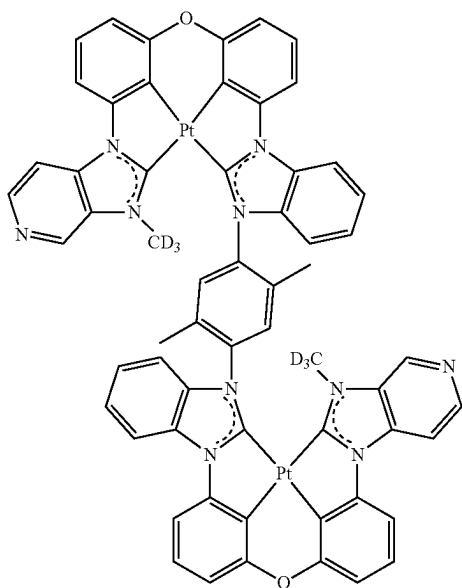
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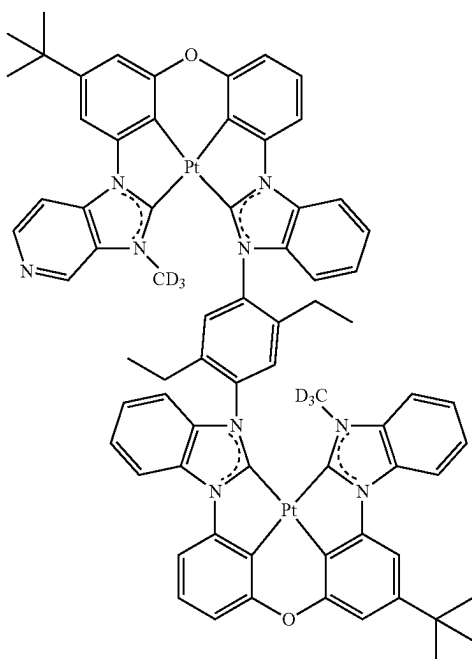
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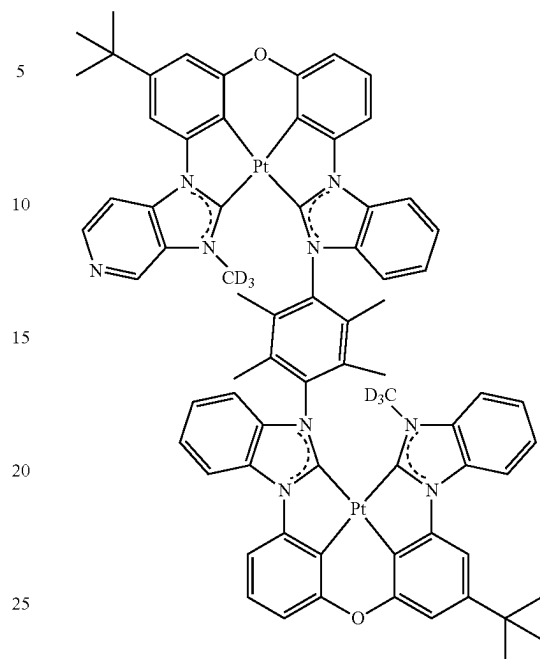
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BD93

276

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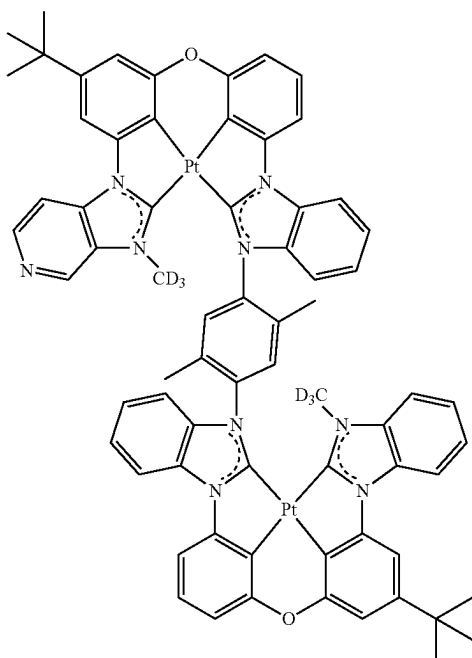
BD95

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BD94



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BD96

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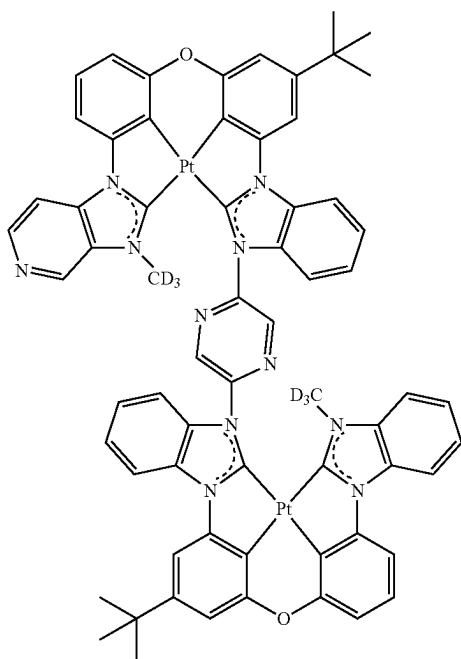
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277

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BD97

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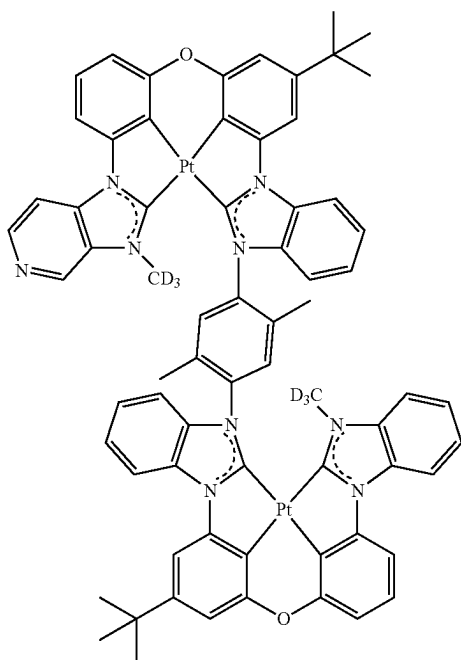
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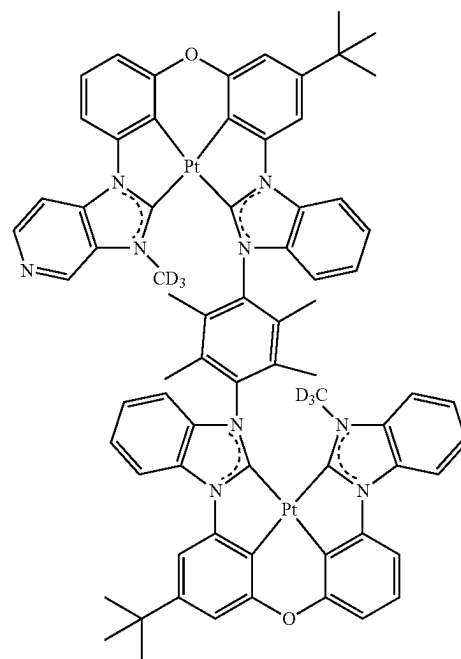
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BD98

**278**

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BD99

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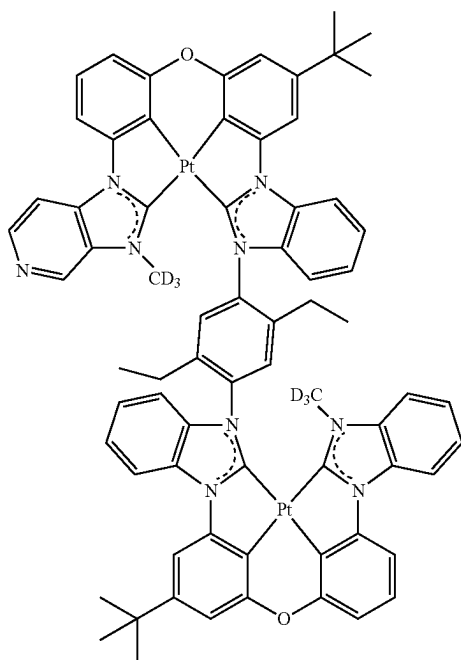
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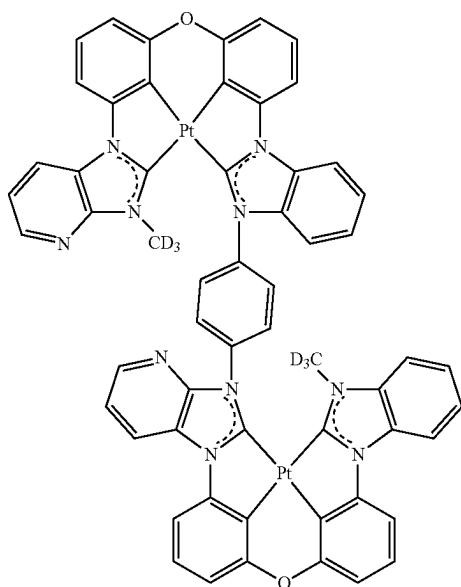
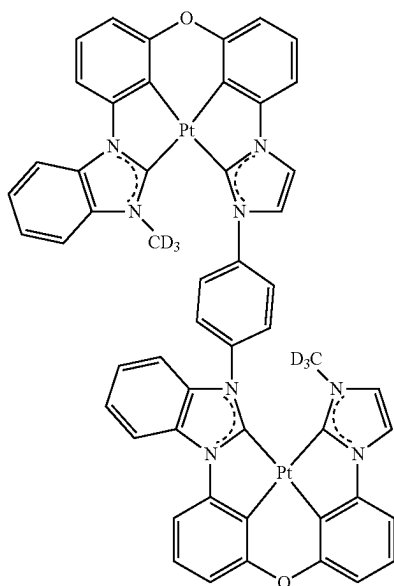
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BD100



279

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**280**

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BD101

BD103

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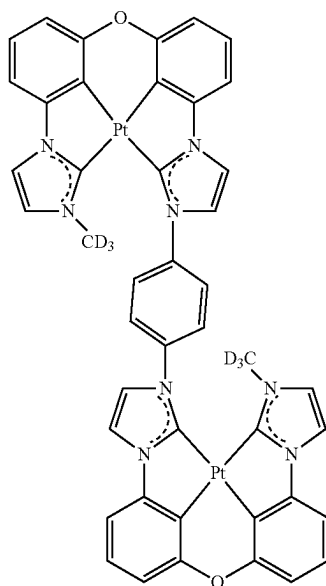
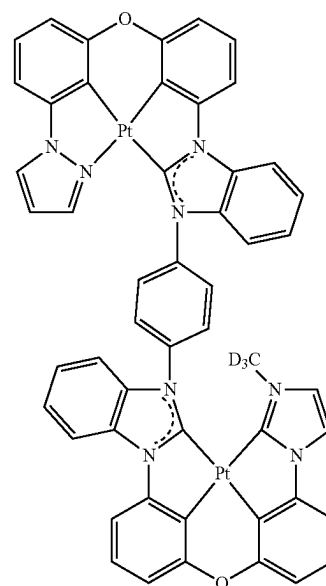
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BD102

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BD104

13. A light-emitting device comprising:

a first electrode;

a second electrode facing the first electrode;

an interlayer disposed between the first electrode and the second electrode and comprising an emission layer; and the organometallic compound of claim 1.

14. The light-emitting device of claim 13, wherein

the first electrode is an anode,

the second electrode is a cathode, and

the interlayer further comprises:

a hole transport region disposed between the first electrode and the emission layer and comprising a hole injection layer, a hole transport layer, an emission auxiliary layer, an electron blocking layer, or a combination thereof; and

an electron transport region disposed between the emission layer and the second electrode and comprising a hole blocking layer, an electron transport layer, an electron injection layer, or a combination thereof.

15. The light-emitting device of claim 13, wherein the emission layer comprises the organometallic compound.

16. The light-emitting device of claim 15, wherein the emission layer comprises a host and a dopant, and the dopant comprises the organometallic compound. 5

17. The light-emitting device of claim 14, wherein the hole transport region further comprises a p-dopant having a lowest unoccupied molecular orbital (LUMO) energy level equal to or less than about -3.5 eV.

18. The light-emitting device of claim 14, wherein 10
the electron transport region comprises the electron transport layer and the electron injection layer, and
at least one layer of the electron transport layer and the electron injection layer comprises an alkali metal, an alkaline earth metal, a rare earth metal, an alkali metal 15
compound, an alkaline earth-metal compound, a rare earth metal compound, an alkali metal complex, an alkaline earth-metal complex, a rare earth metal complex, or a combination thereof.

19. An electronic apparatus comprising the light-emitting 20
device of claim 13.

20. The electronic apparatus of claim 19, further comprising a thin-film transistor, wherein
the thin-film transistor comprises a source electrode and a drain electrode, and 25
the first electrode of the light-emitting device is electrically connected to the source electrode or the drain electrode.

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